

Gasoline Profiles and Impact of Ethanol Blending in Latin America

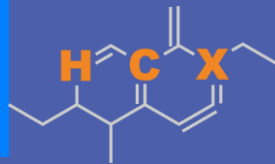
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Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

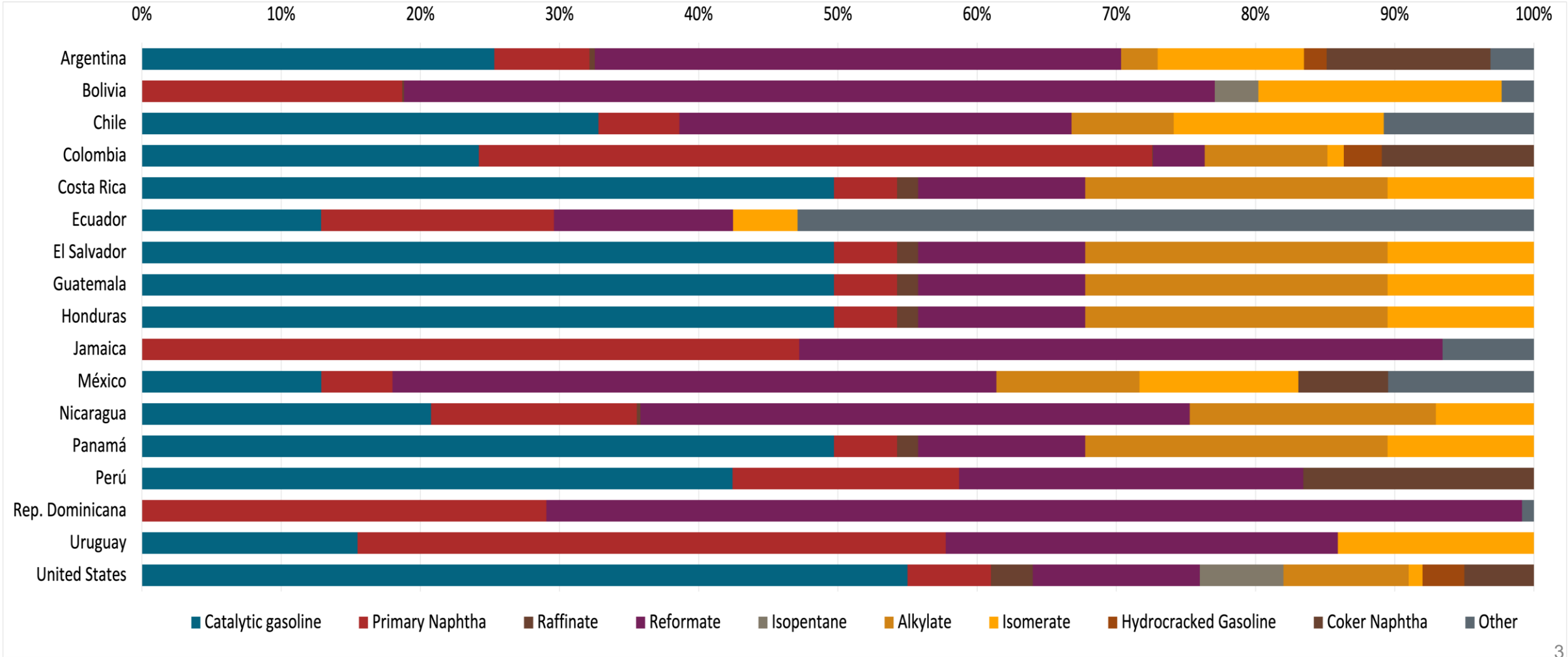
- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.



Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



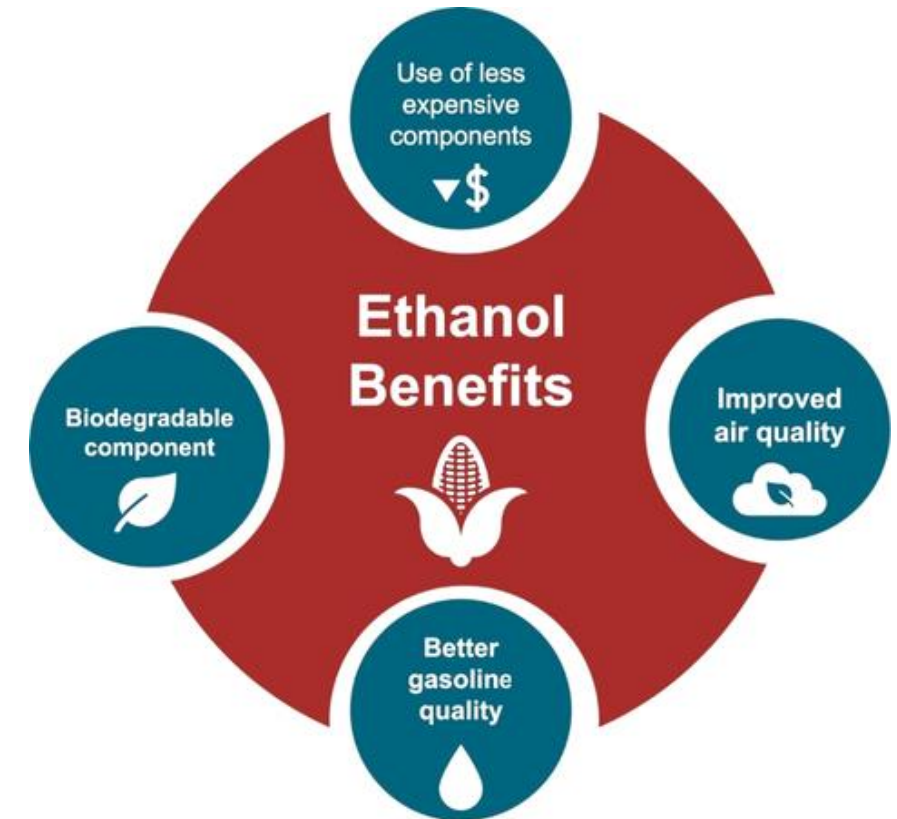
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

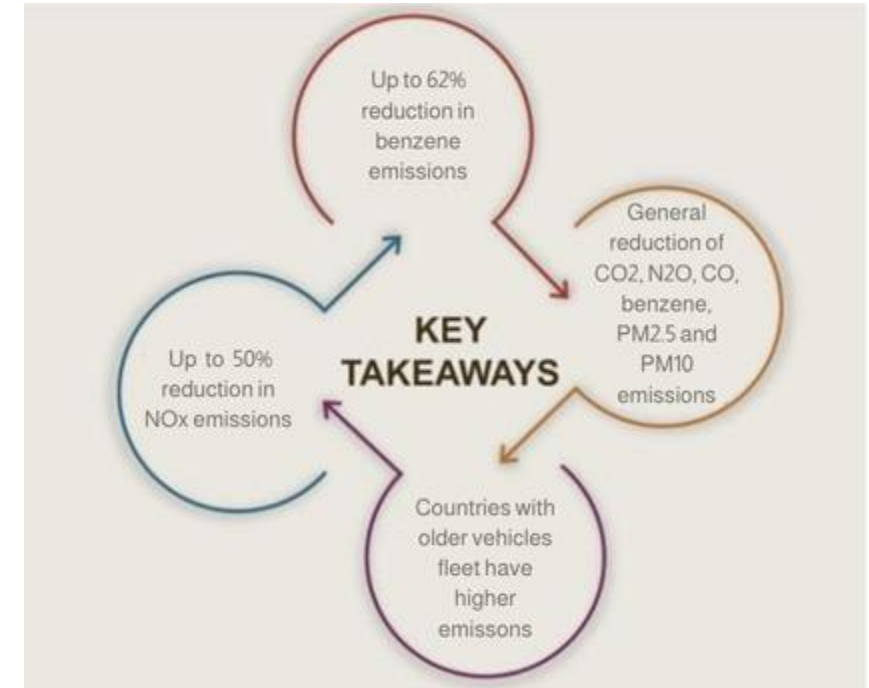
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

*Source: Bureau of transportation statistics.

Main Results



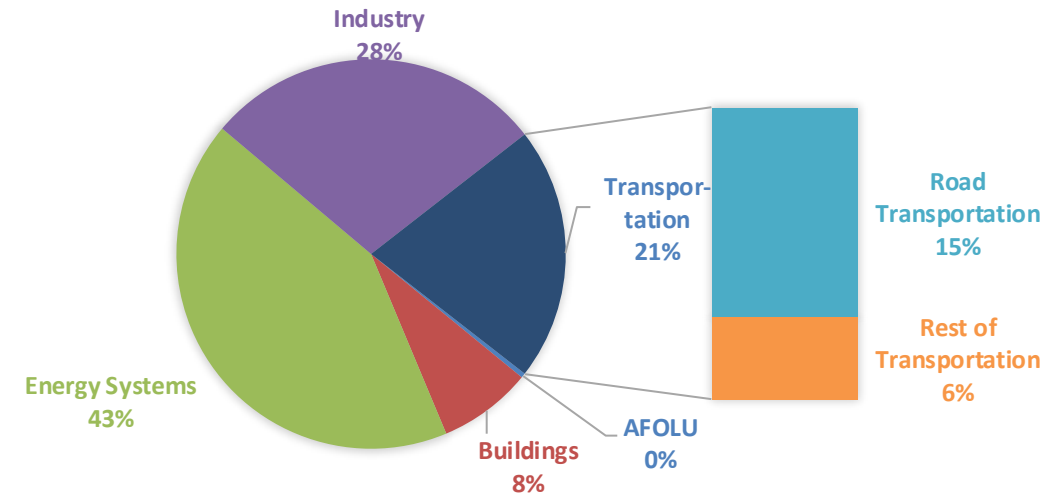
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

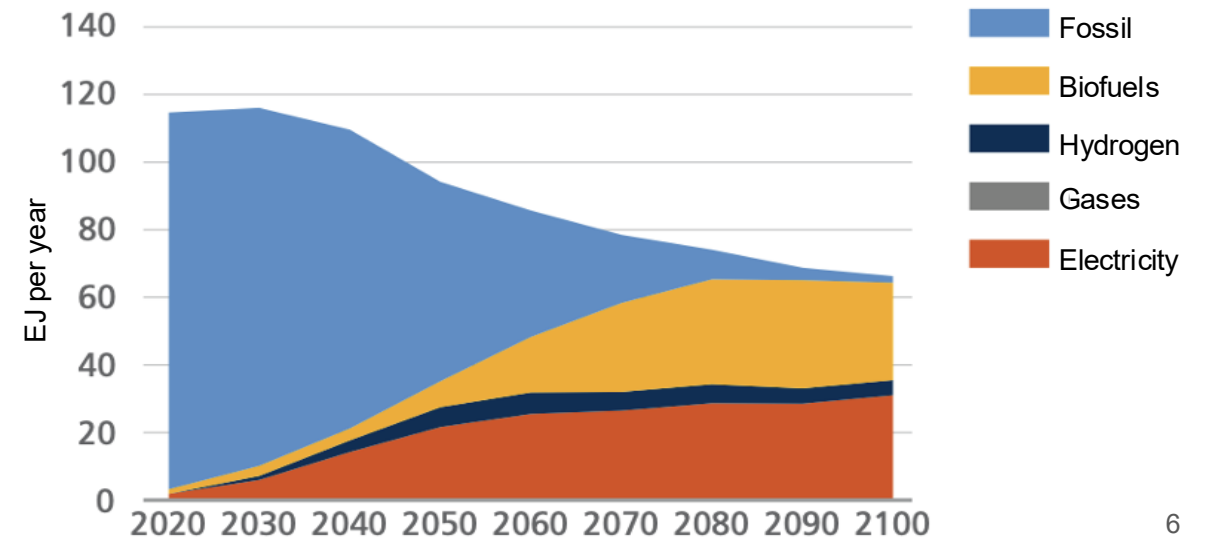
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

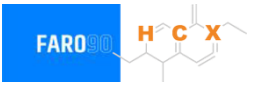
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Case Studies



1. [Argentina](#)
2. [Bolivia](#)
3. [Chile](#)
4. [Colombia](#)
5. [Costa Rica](#)
6. [Dominican Republic](#)
7. [Ecuador](#)
8. [El Salvador](#)
9. [Guatemala](#)
10. [Honduras](#)
11. [Jamaica](#)
12. [México](#)
13. [Nicaragua](#)
14. [Panamá](#)
15. [Perú](#)
16. [Uruguay](#)

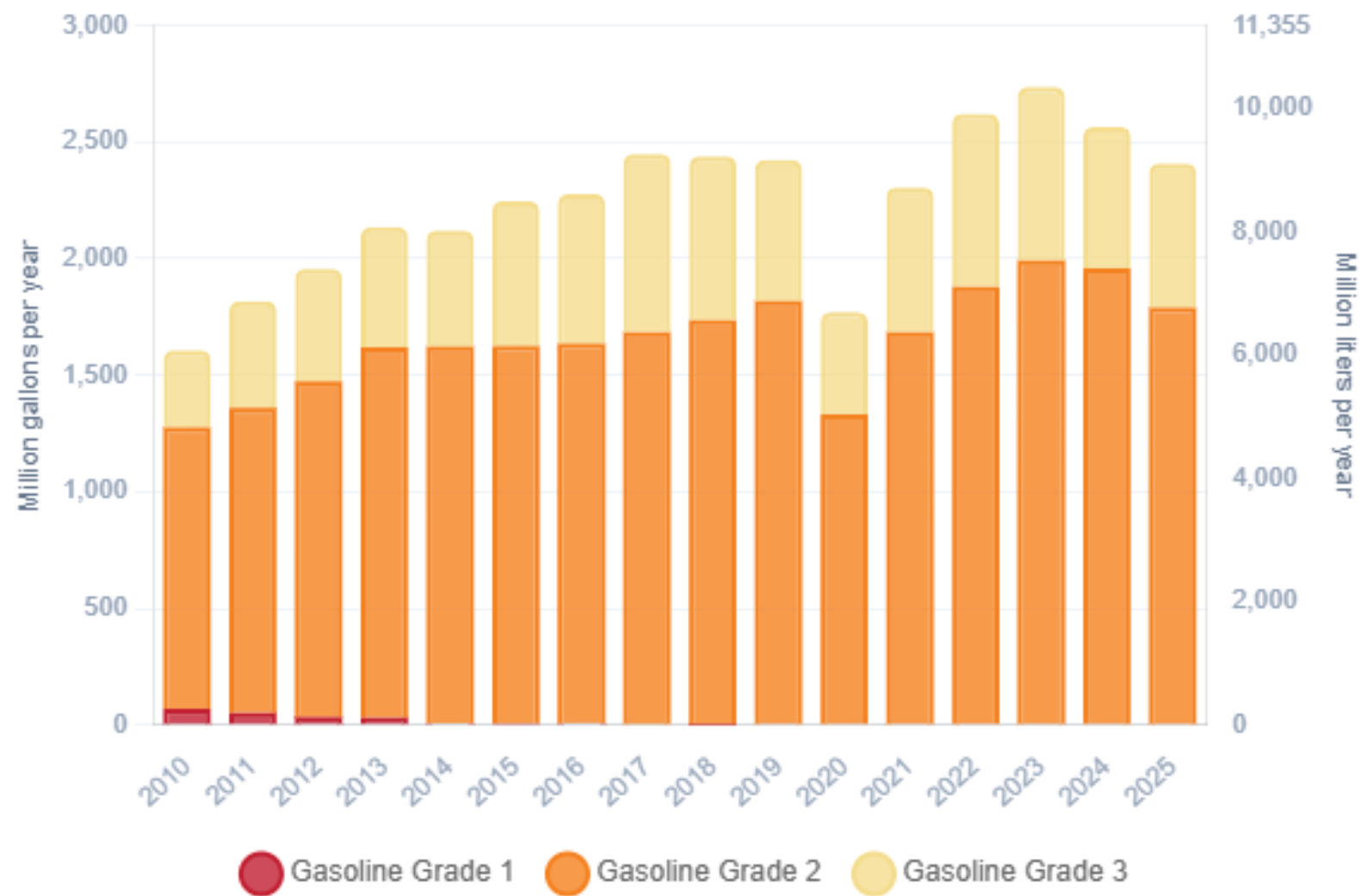
Ethanol Blending in Gasoline - Argentina



Argentina's total gasoline demand was 2,408 million gallons in 2025 (9,117 million liters), with a market share of approximately 74.2% as Nafta Súper / Grado 2 (RON 93 min – AKI 87) and 25.8% as Nafta Ultra / Grado 3 (RON 97 min – AKI 92). Octane delivered to market is usually higher than the minimum required. The demand for gasoline is supplied mostly by local production, requiring imports of less than 75 million liters per year average.

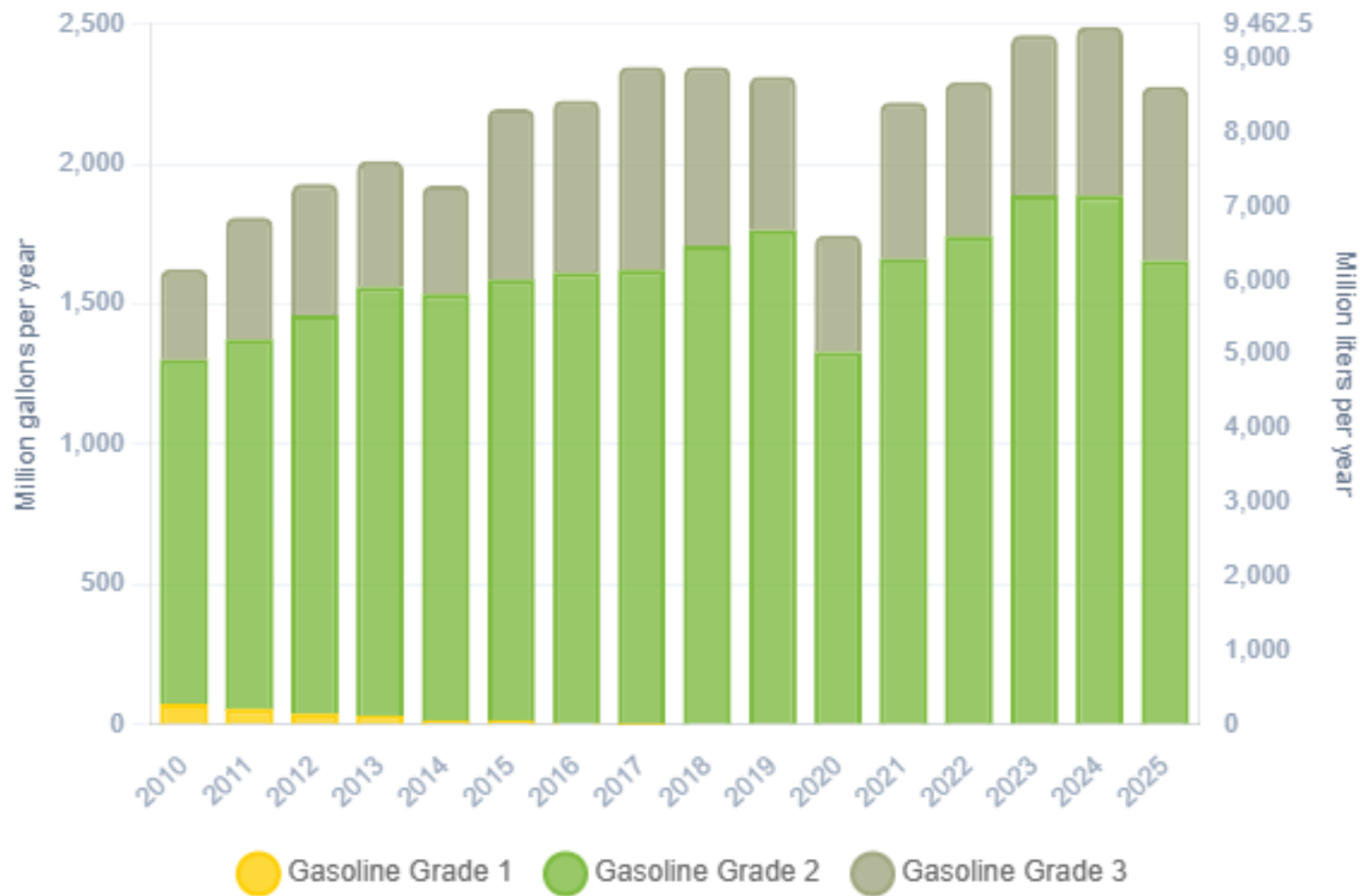
Argentina's policy is self-sufficiency and production and supply of ethanol for gasoline blend. Ethanol concentration varies according to local production. Currently, the concentration limit is 15% v/v. (Resolution 2026-79-APN-SE#MEC). Ethanol production is equal to consumption with some exports and imports from other countries.

Gasoline Demand in Argentina



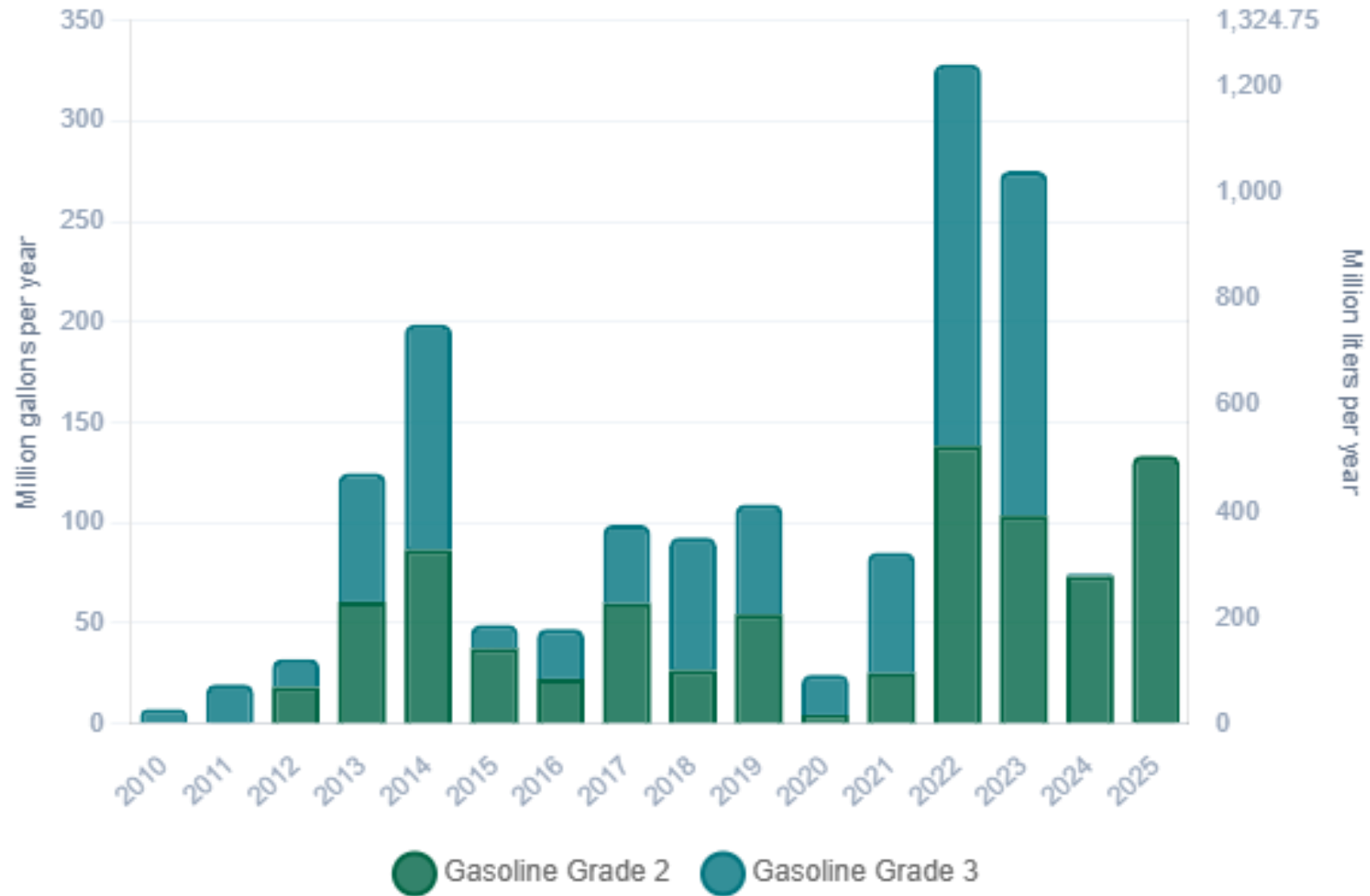
Source: Ministerio de Economía, 2025

Gasoline Production in Argentina



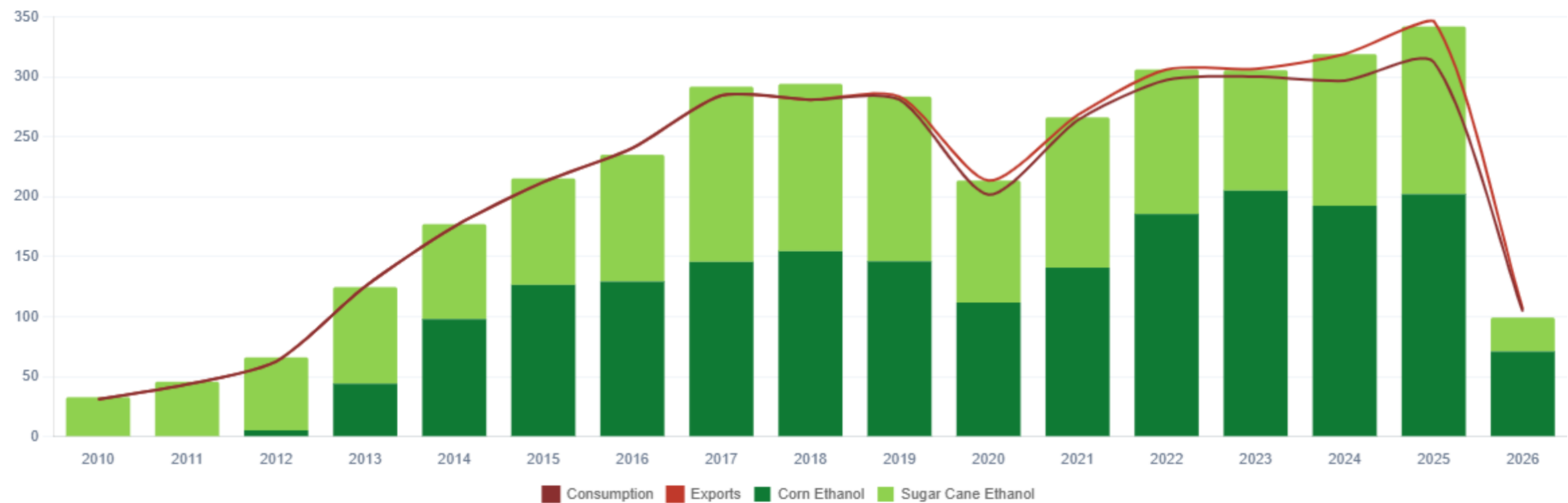
Source: Ministerio de Economía, 2025

Gasoline Imports in Argentina



Source: Ministerio de Economía, 2025

Ethanol Balance in Argentina

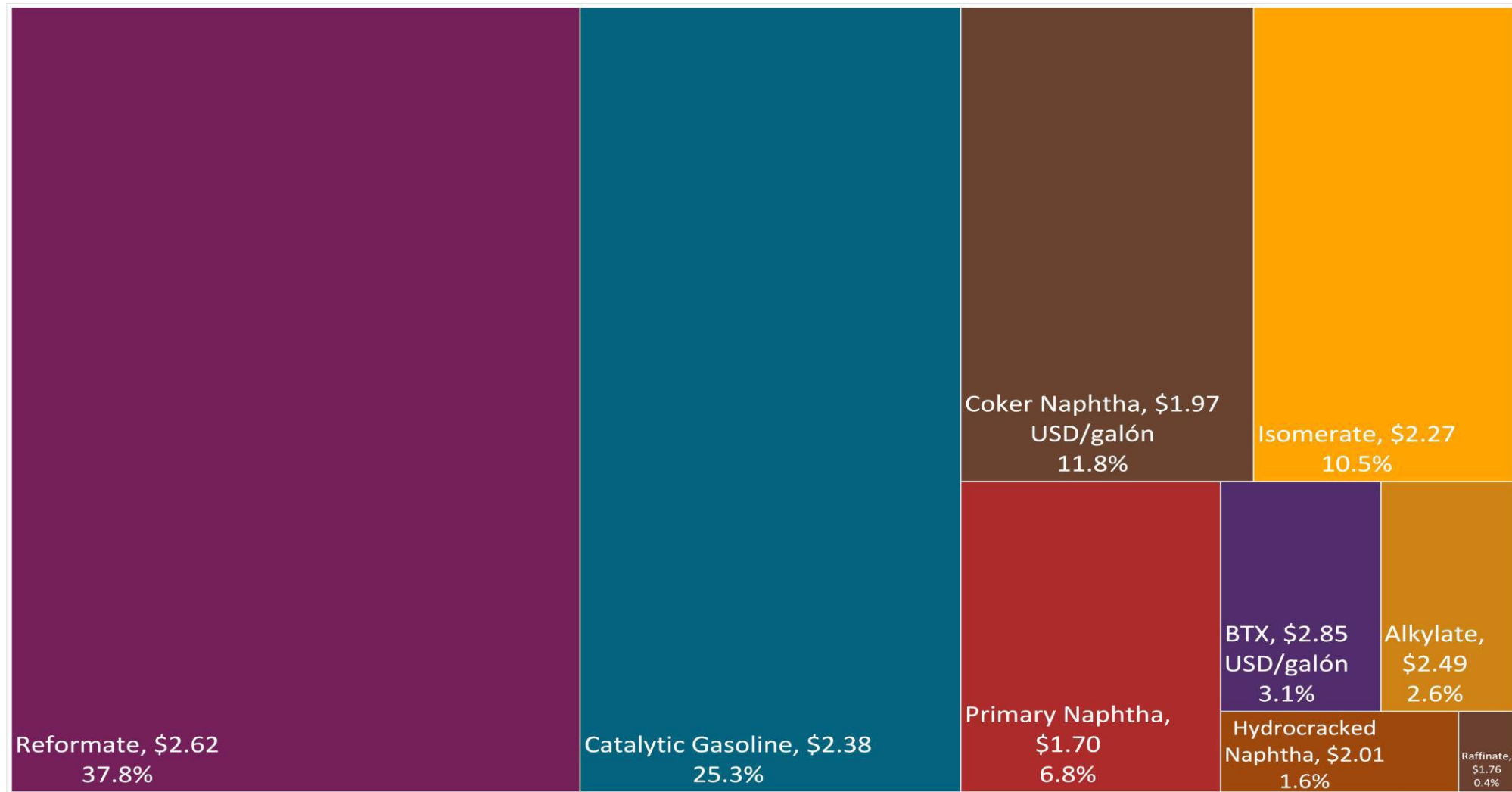


Source: Ministerio de Economía, 2025

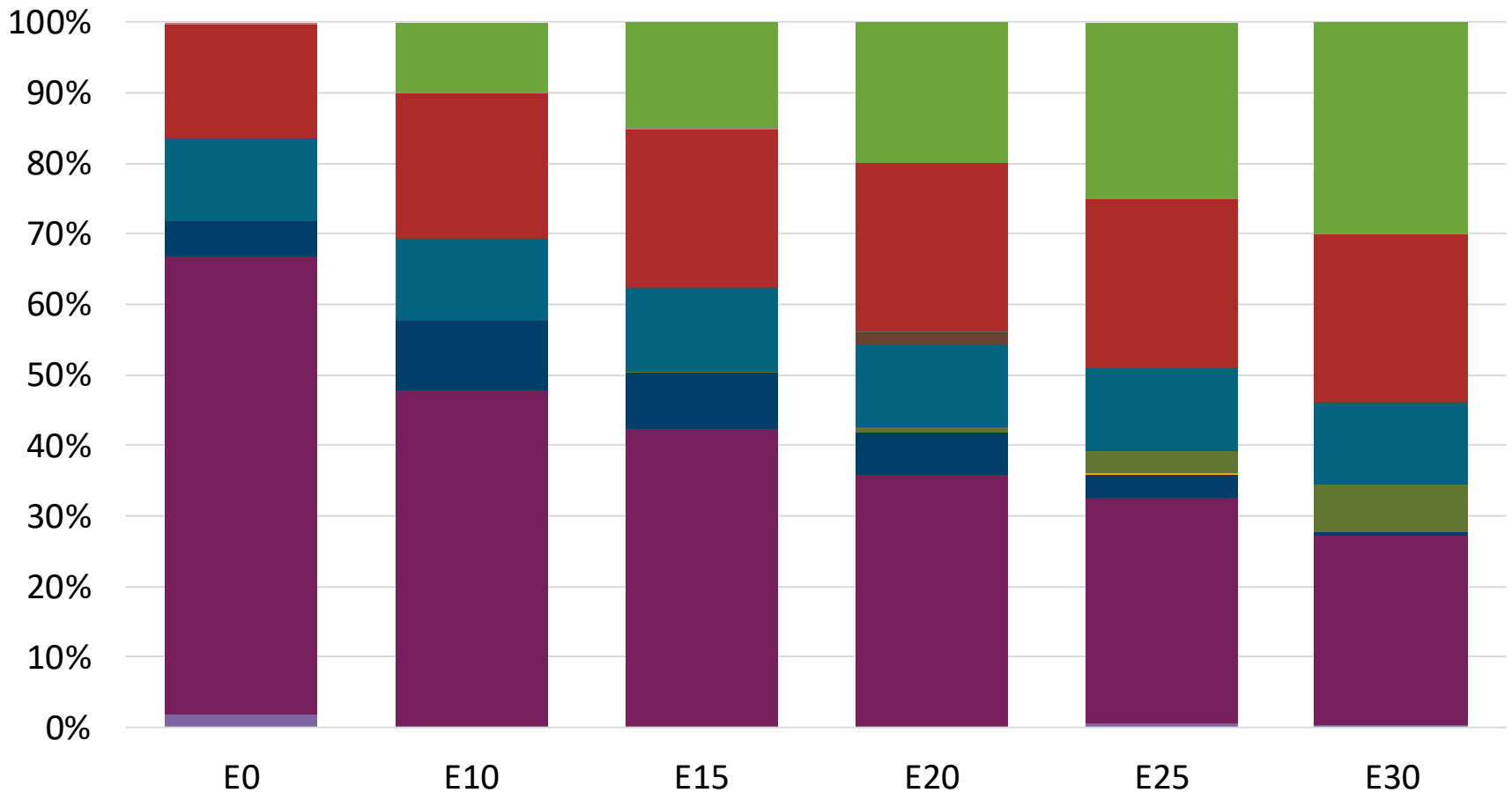
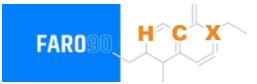
Gasoline Quality in Argentina

Name	Resolution 576/2019 and Resolution 492/2023						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019						2017			
Applicability	North Zone		Central Zone		South Zone		All countries			
Selected Grade	Gasoline Grade 2	Gasoline Grade 3	Gasoline Grade 2	Gasoline Grade 3	Gasoline Grade 2	Gasoline Grade 3	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 40 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2.5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 93	> 97	> 93	> 97	> 93	> 97	> 95	> 95	> 98	> 98
MON	> 83	> 85	> 83	> 85	> 83	> 85	> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 150 mg/kg	< 10 mg/kg	< 150 mg/kg	< 10 mg/kg	< 150 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	< 4.5 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<> 12 - 14 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 35 - 70 kPa	<> 45 - 80 kPa	<> 45 - 80 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 45 - 80 kPa	<> 45 - 80 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa	<> 55 - 90 kPa				
RVP 37.8°C (Transition)			<> 45 - 80 kPa	<> 45 - 80 kPa						
MTBE	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	< 15 %v/v	-	-	-	-
Ethers 5 or more C Atoms	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	< 22 %v/v	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Available Blending Components

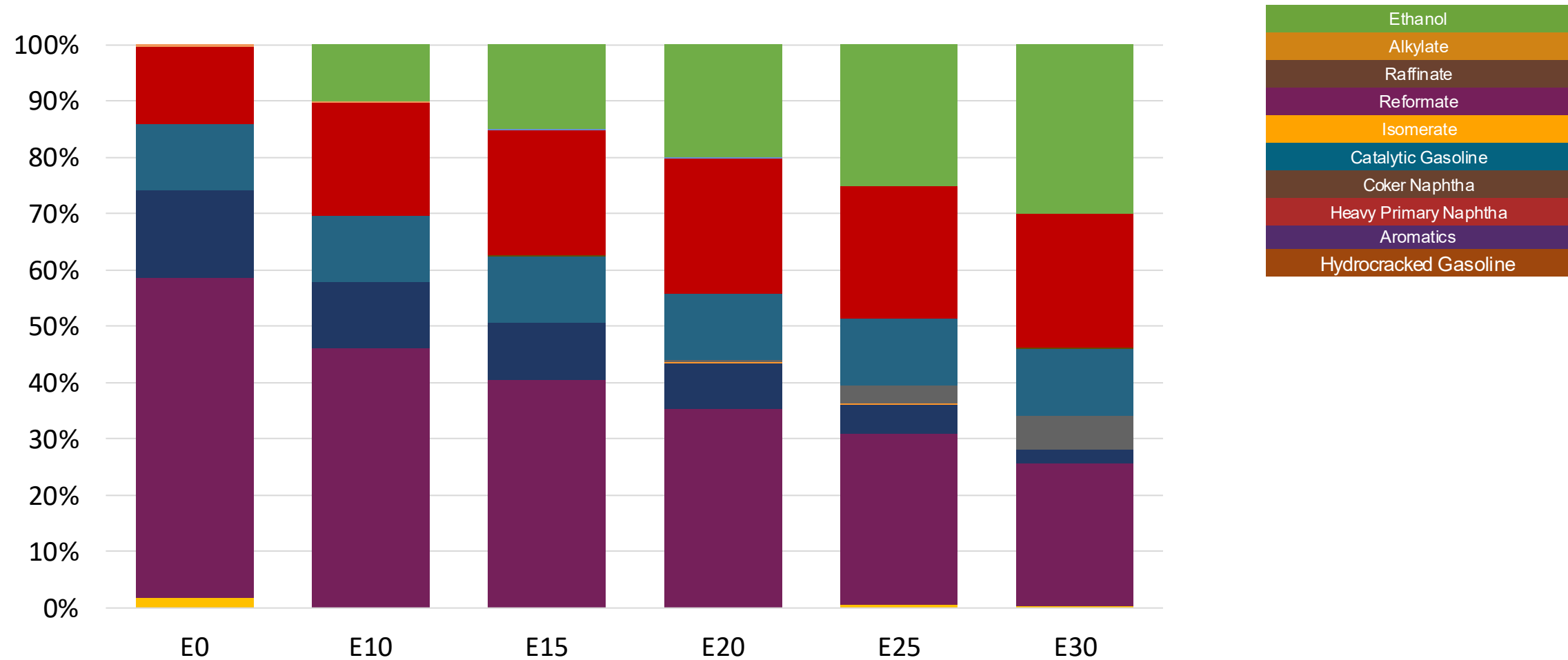
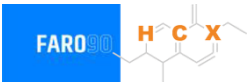


Argentina – Nafta Grado 2 – North & Central Zones– Constant Octane



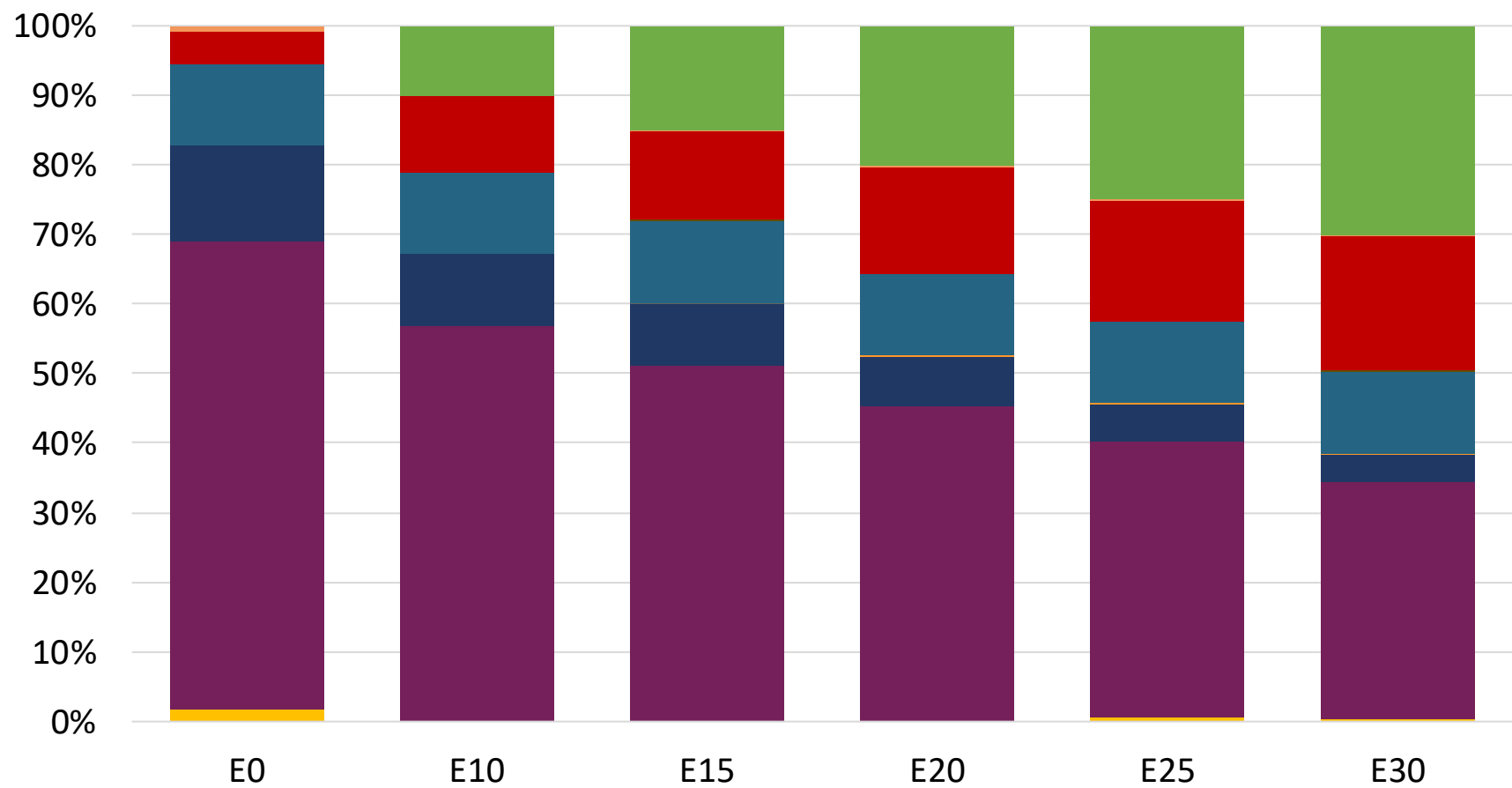
Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US\$/gal)	\$ 2.07	\$ 1.83	\$ 1.77	\$ 1.70	\$ 1.64	1.58

Argentina – Nafta Grado 2 – South Zone – Constant Octane



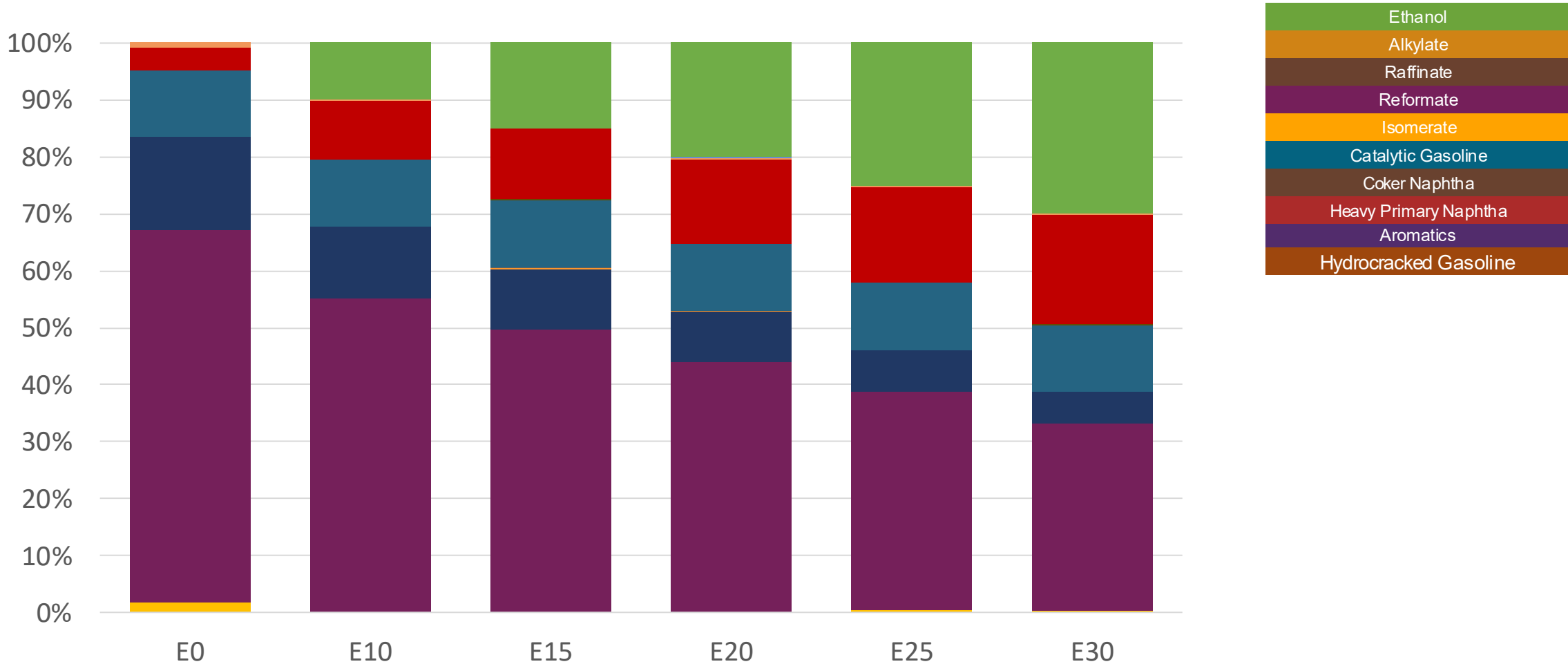
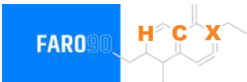
Octane (RON)	93.0					
Price (US\$/gal)	\$ 1.92	\$ 1.80	\$ 1.74	\$ 1.68	\$ 1.62	1.56

Argentina – Nafta Grado 3 – North & Central Zones– Constant Octane



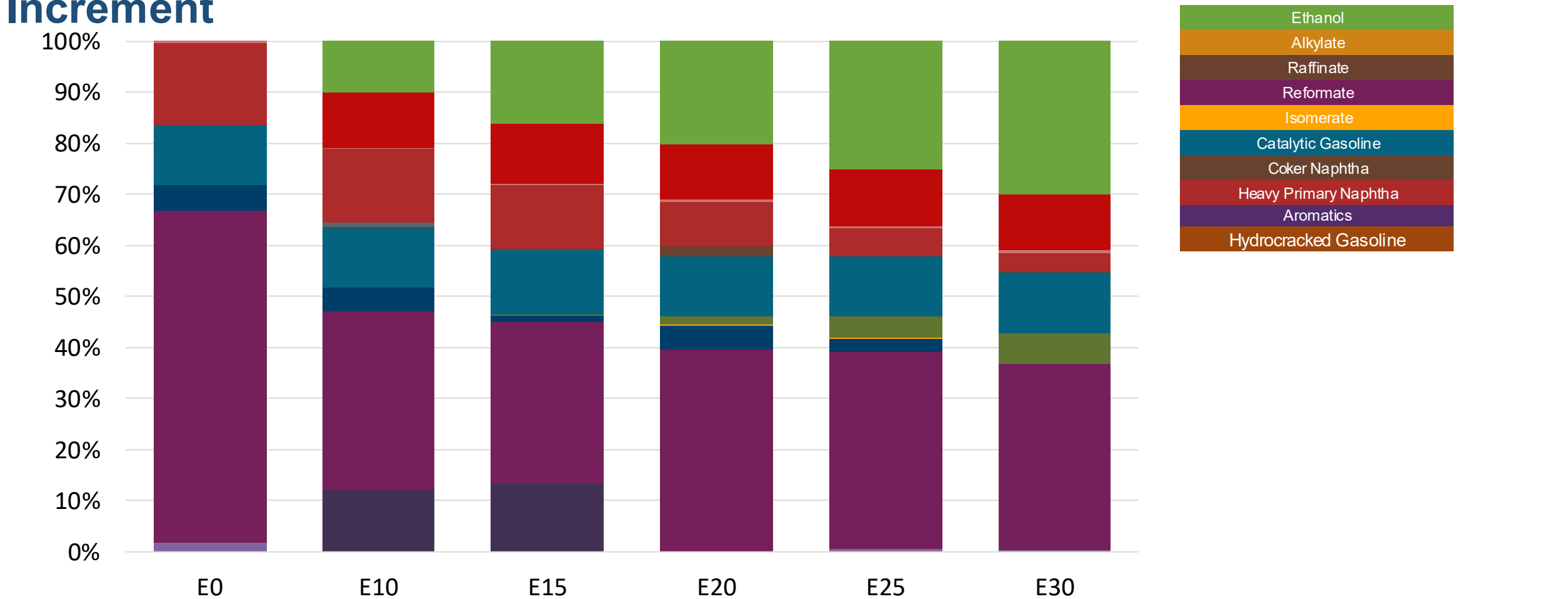
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.32	\$ 2.19	\$ 2.13	\$ 2.07	2.01	\$ 1.95

Argentina – Nafta Grado 3 – South Zone – Constant Octane



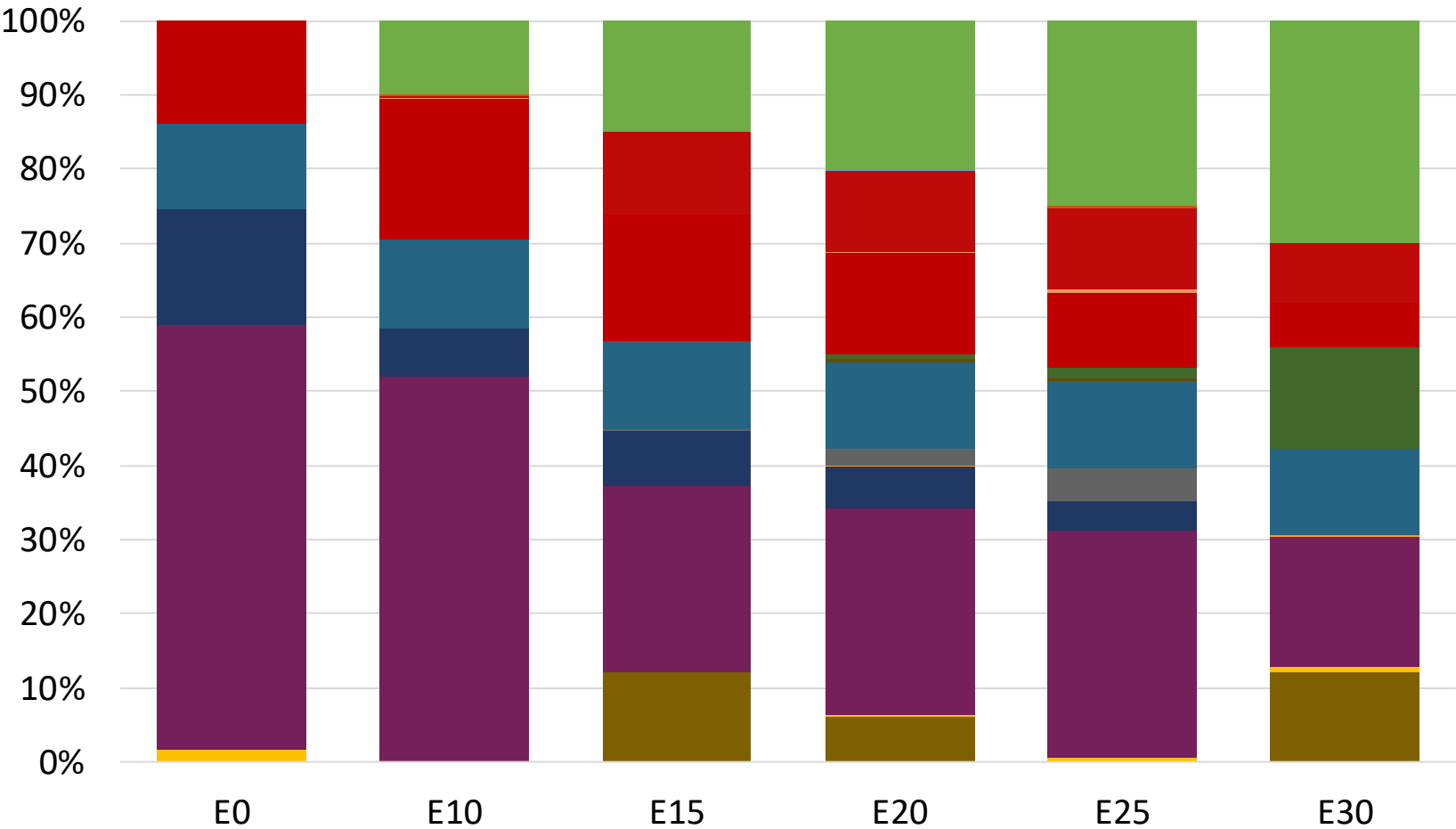
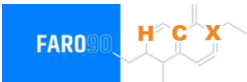
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.11	\$ 1.98	\$ 1.92	\$ 1.86	1.80	\$ 1.73

Argentina – Nafta Grado 2 – North & Central Zones– Octane Increment



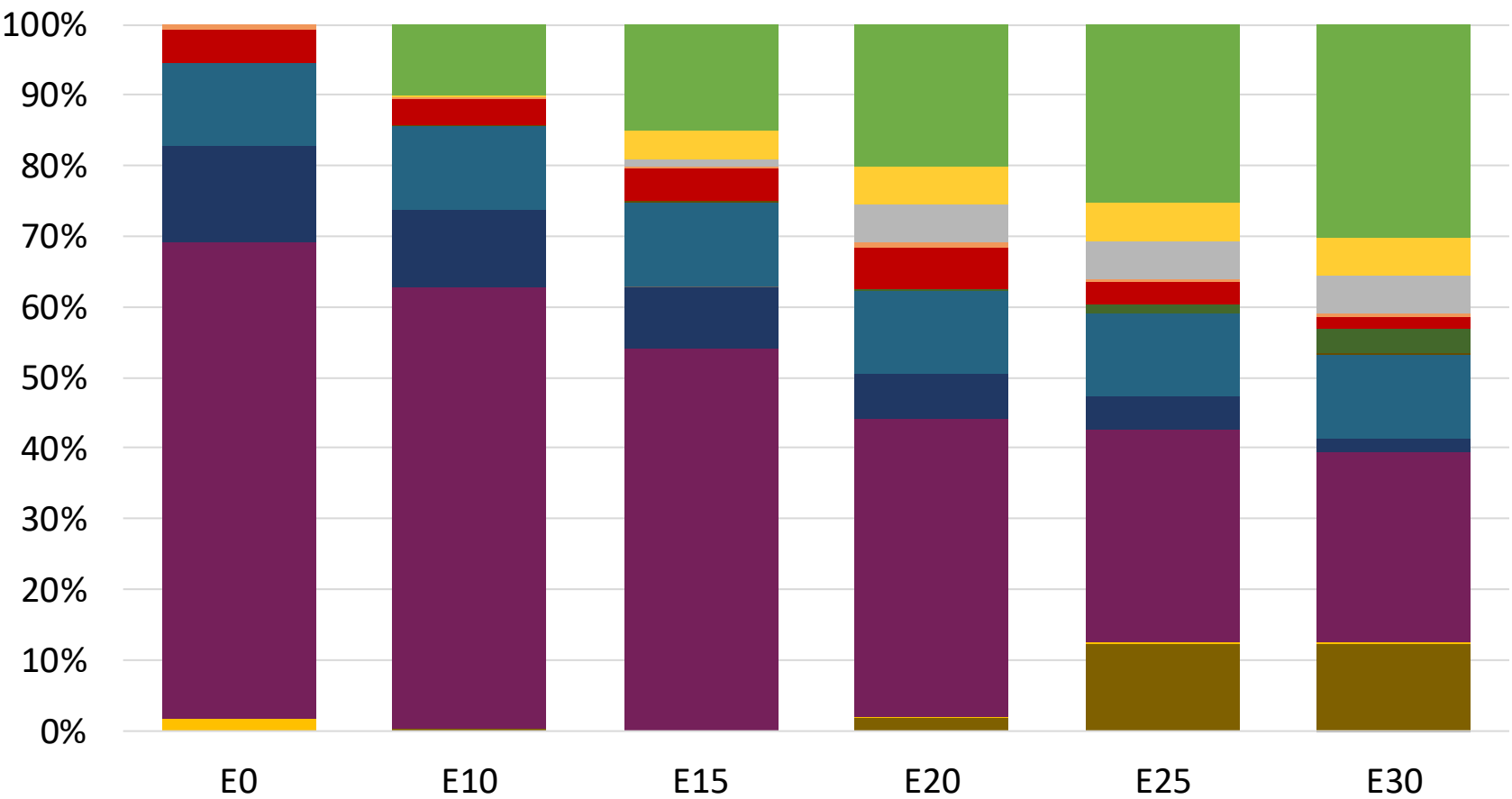
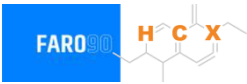
Octane (RON)	EO	E10	E15	E20	E25	E30
	93.0	95.4	96.0	99.9	101.3	102.6
Price (US\$/gal)	\$ 2.07	\$ 2.07	\$ 2.07	\$ 2.07	\$ 2.07	\$ 2.07

Argentina – Nafta Grado 2 – South Zone – Octane Increment



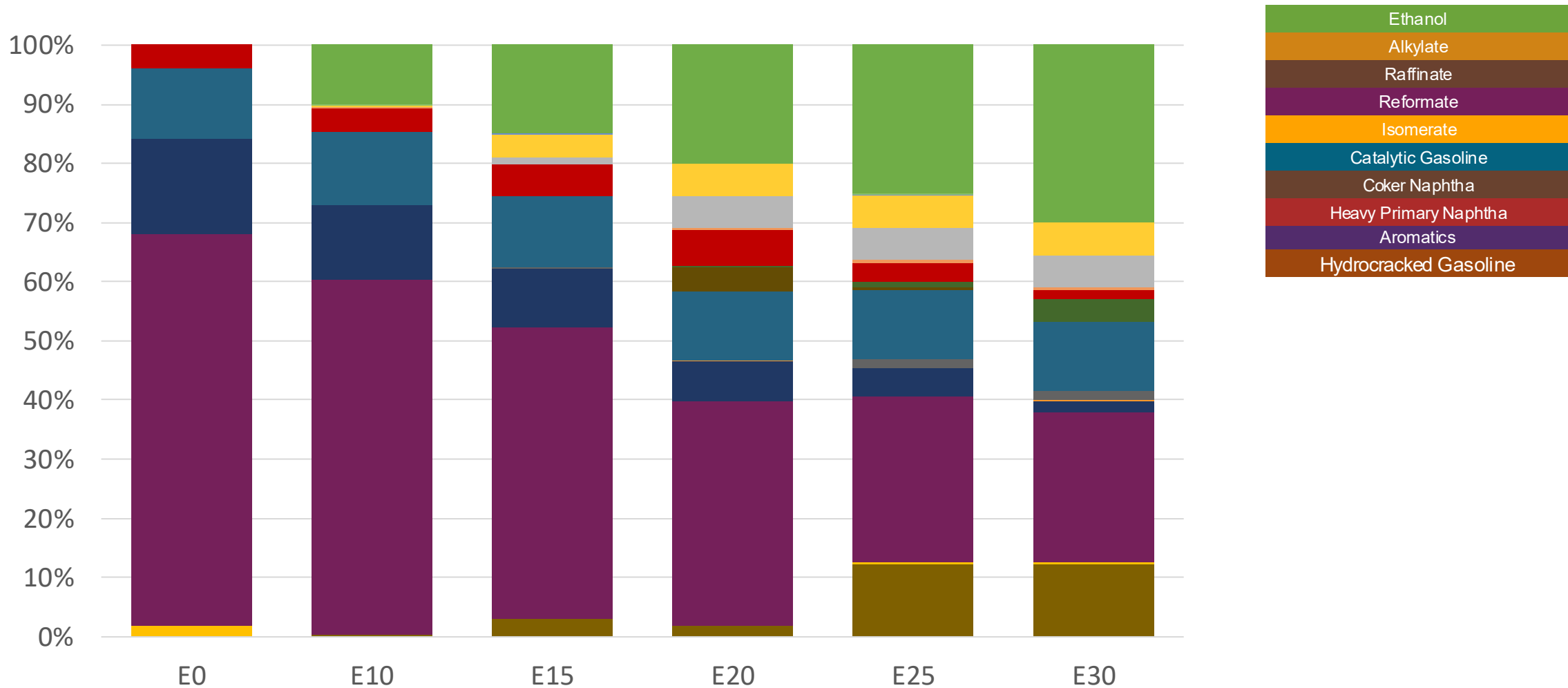
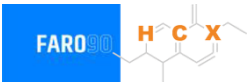
Octane (RON)	93.0	93.9	94.9	96.8	98.6	98.1
Price (US\$/gal)	\$ 1.92	\$ 1.92	\$ 1.92	\$ 1.92	\$ 1.92	\$ 1.92

Argentina – Nafta Grado 3 – North & Central Zones– Octane Increment



Octane (RON)	97.0	99.8	100.8	101.6	102.2	103.2
Price (US\$/gal)	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35	\$ 2.35

Argentina – Nafta Grado 3 – South Zone – Octane Increment



Octane (RON)	97.0	99.5	100.2	100.9	101.6	102.6
Price (US\$/gal)	\$ 2.10	\$ 2.10	\$ 2.10	\$ 2.10	\$ 2.10	\$ 2.10

Gasoline Vehicle Fleet - Argentina

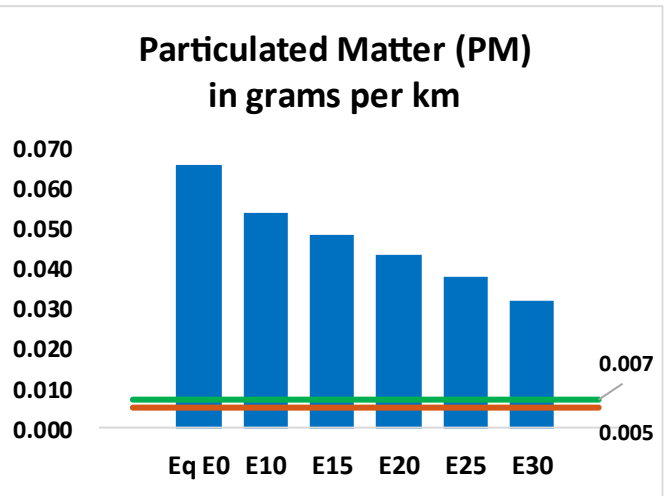
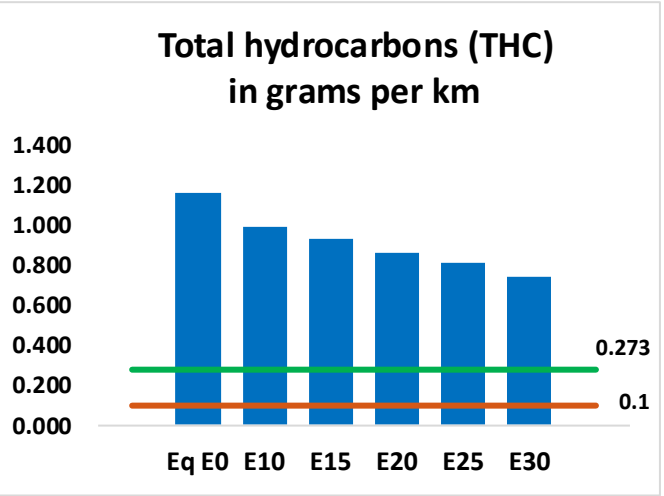
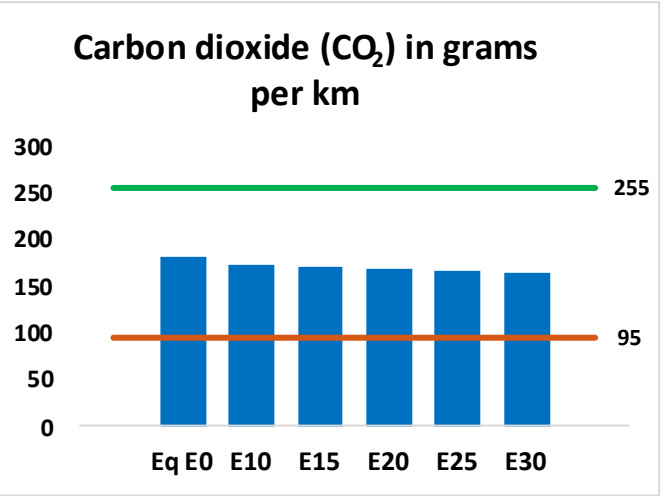
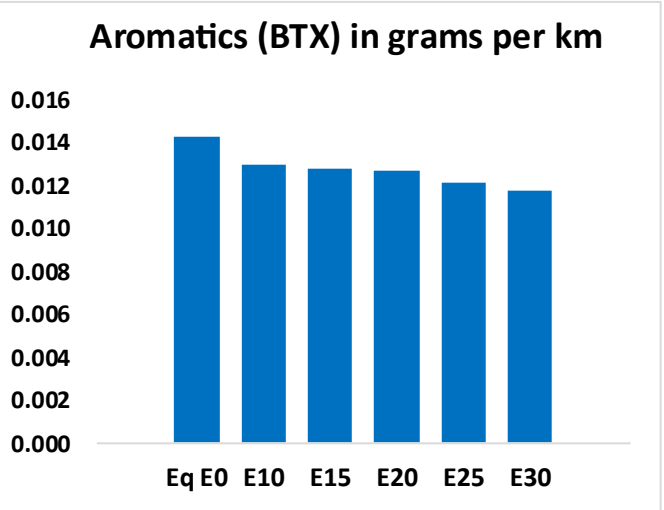
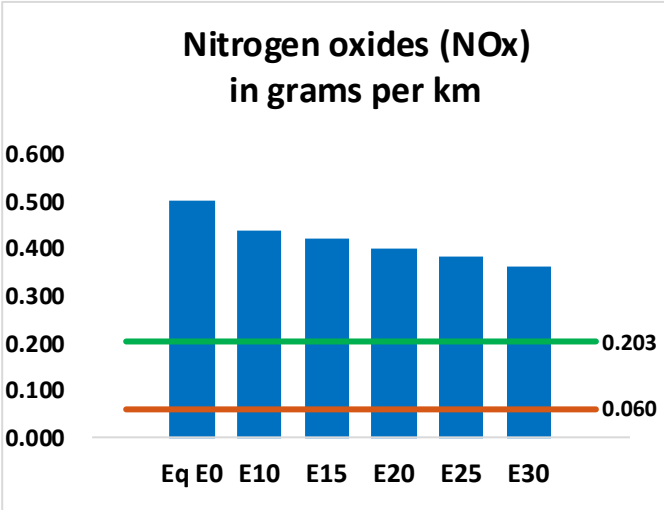
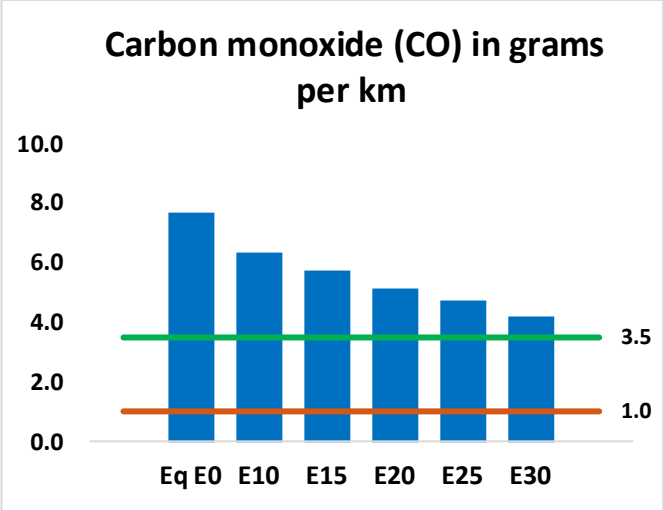
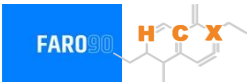


Vehicular Fleet: **17,673,409**
Average age: **12,5 years**
Motorcycles: **41.6%**

Source: Ministerio de Transporte, CAFAM, AFAC, 2025

Argentina – Vehicular Emissions

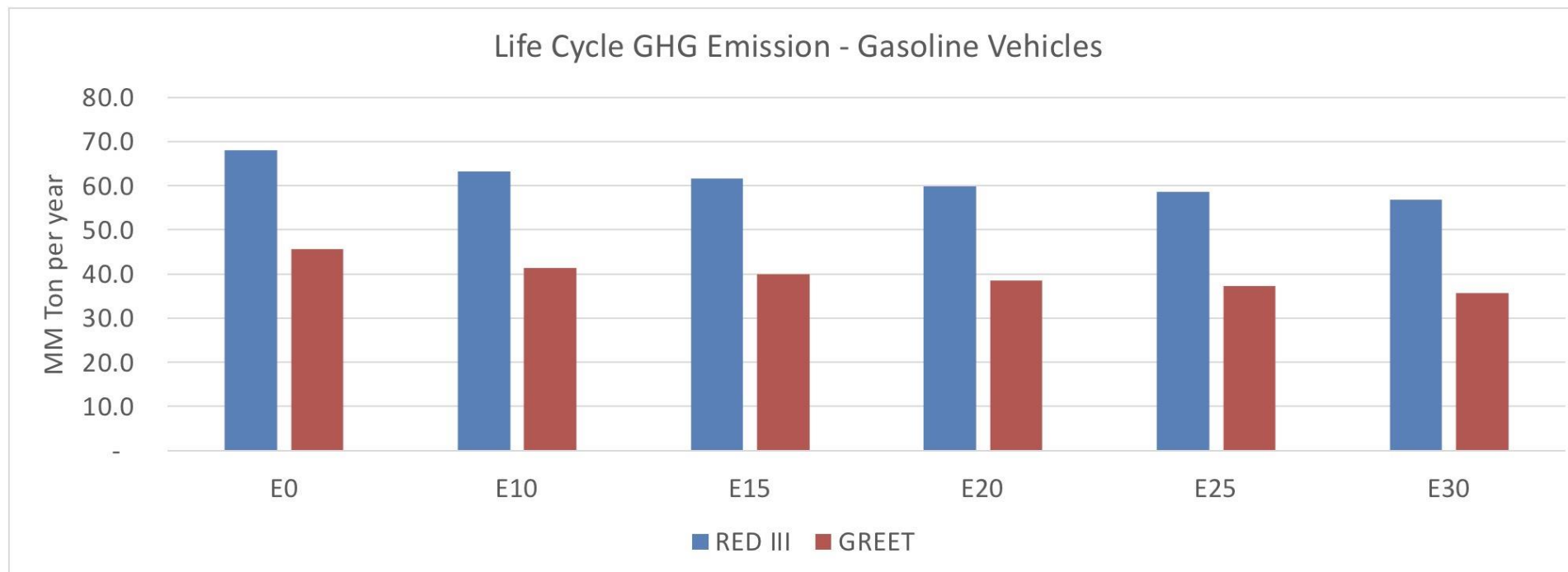
United States
Euro 6 Standard



Argentina – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,527,943	2,303,930	2,079,917	1,890,558	1,697,335	1,547,954	1,389,122	-18%	-33%
CO2	59,981,136	58,438,764	56,982,080	55,836,440	55,269,533	55,042,406	54,027,816	-5%	-8%
VOC	381,844	354,527	327,209	305,221	283,277	266,458	243,820	-14%	-26%
NOx	165,413	152,732	143,910	139,352	132,081	126,349	119,620	-13%	-20%
SOx	680	628	577	533	492	447	399	-15%	-28%
NH3	26,210	25,951	25,691	25,813	26,104	26,308	26,477	-2%	0%
N2O	1,105	1,099	1,094	1,100	1,122	1,135	1,148	-1%	2%
THC	83,875	83,875	83,875	85,133	86,979	88,572	90,082	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,780	2,613	2,446	2,319	2,195	2,101	1,971	-12%	-21%
Acetaldehyde	7,881	9,045	13,272	20,211	27,536	31,465	37,179	68%	249%
Formaldehyde	31,273	30,404	35,442	41,617	43,600	48,233	52,507	13%	39%
Benzene	4,695	4,511	4,275	4,207	4,172	3,990	3,861	-9%	-11%
PM2.5	3,991	3,628	3,265	2,946	2,633	2,299	1,948	-18%	-34%
PM10	17,625	16,023	14,421	13,011	11,631	10,152	8,604	-18%	-34%

Argentina – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/año)	359.7
Target @ 2035 (MMT/year)	197.5
Delta	162.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.4%	12.4%	15.6%	18.2%	21.7%
RED II/III	0.0%	7.1%	9.5%	11.9%	14.0%	16.6%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	2.7%	3.5%	4.4%	5.2%	6.1%
RED II/III	0.0%	3.0%	4.0%	5.0%	5.9%	7.0%

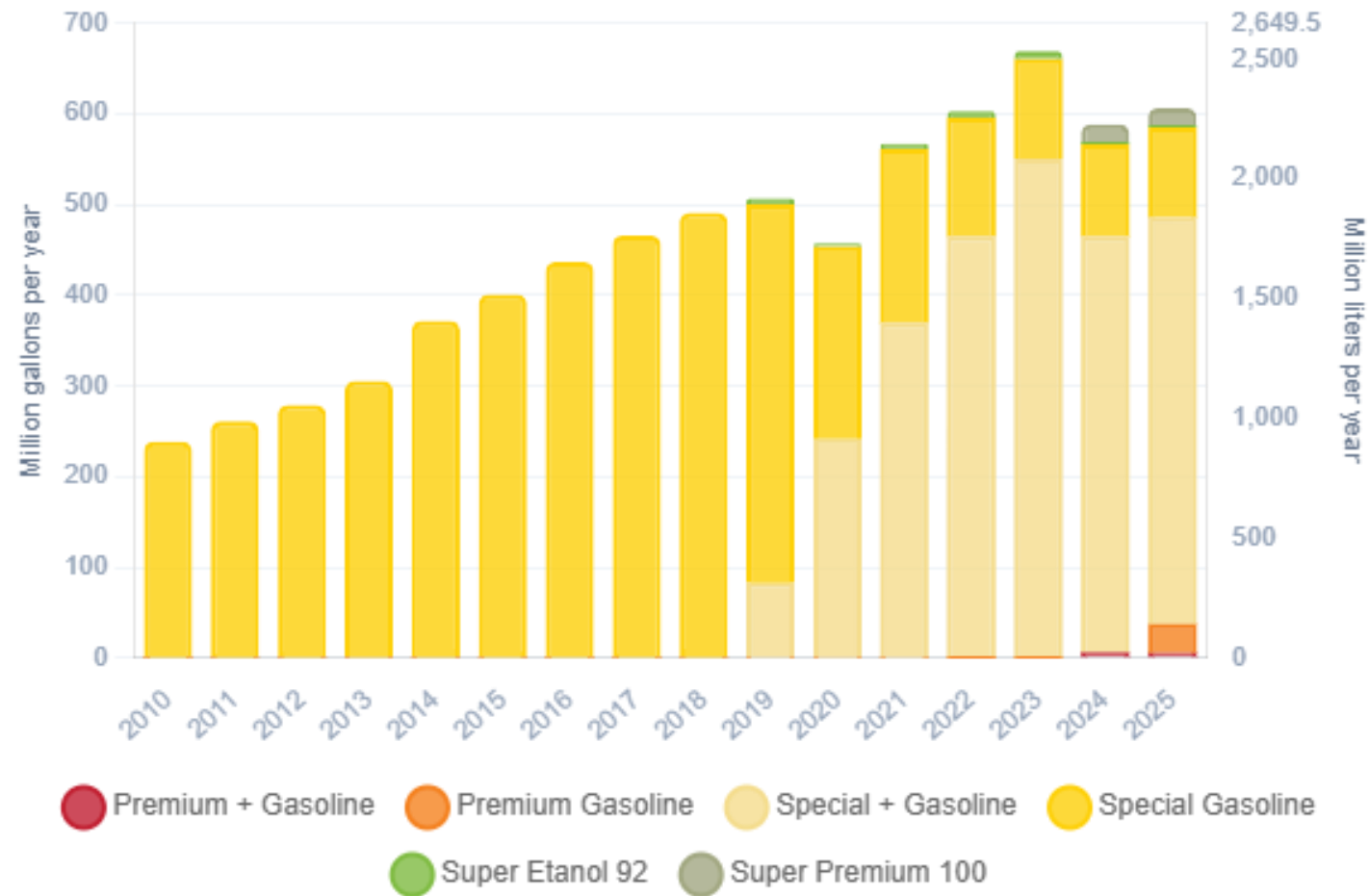
Ethanol Blending in Gasoline - Bolivia



In 2025, gasoline demand was 606 million gallons (2,295 million liters). Gasoline Imports represented 52% of demand, and are supplied from Brazil, Chile, Argentina, and the United States. Regular Gasoline (Especial) with 85 RON Octane represented 90% of demand and the Gasoline with a 95 RON or more octane, the remaining 10%. Gasoline production has decreased in the last years.

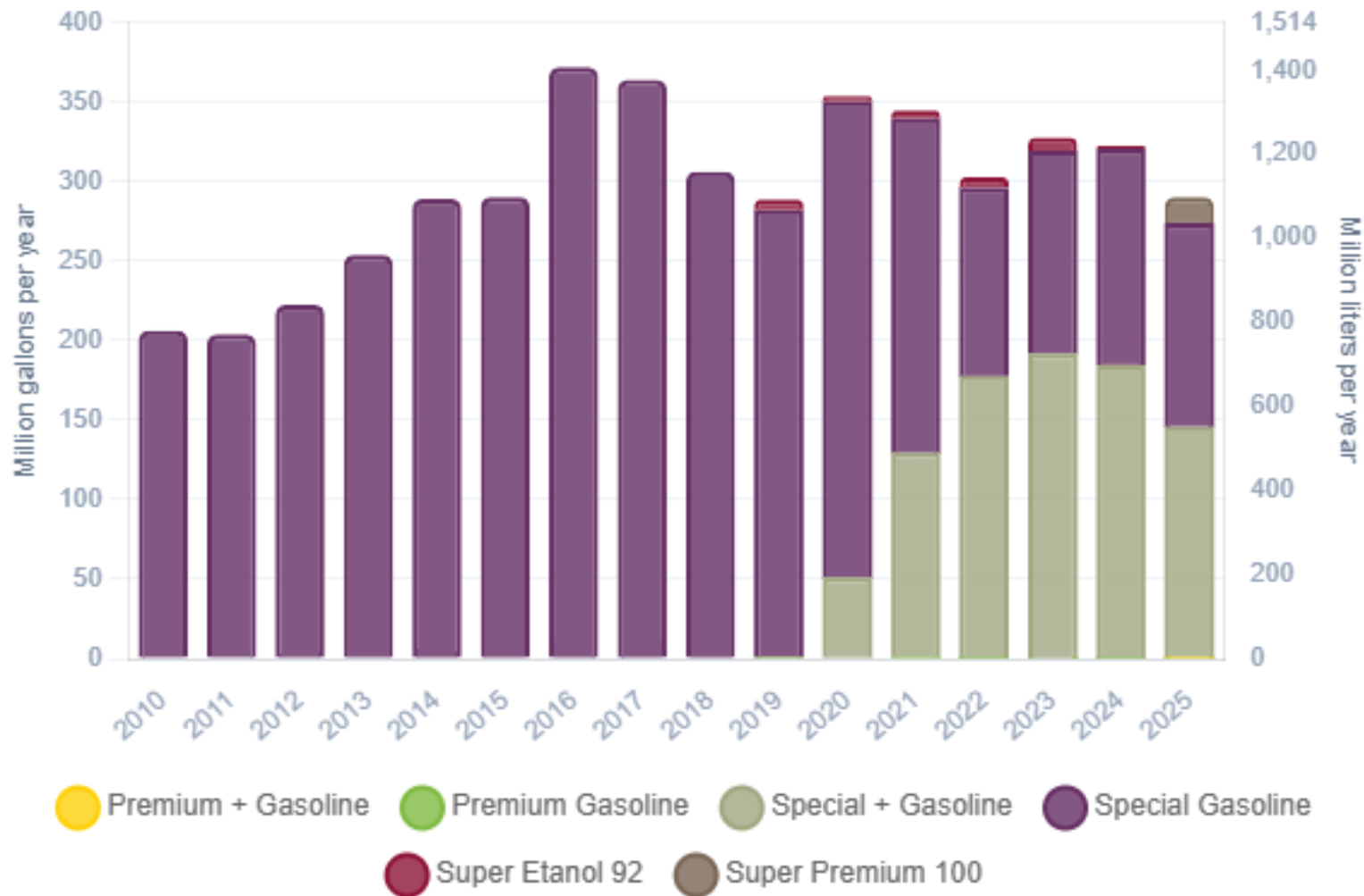
Gasoline E8 and E12 are currently marketed (Decreto Supremo No. 4718:ANH), but recently was approved blends of 25%. About 80% of gasoline demand contains ethanol. All ethanol consumed in Bolivia is produced by five local companies: Azúcar Aguaí, Unagro, Poplar Capital, Granosol, and Guabirá.

Gasoline Demand in Bolivia



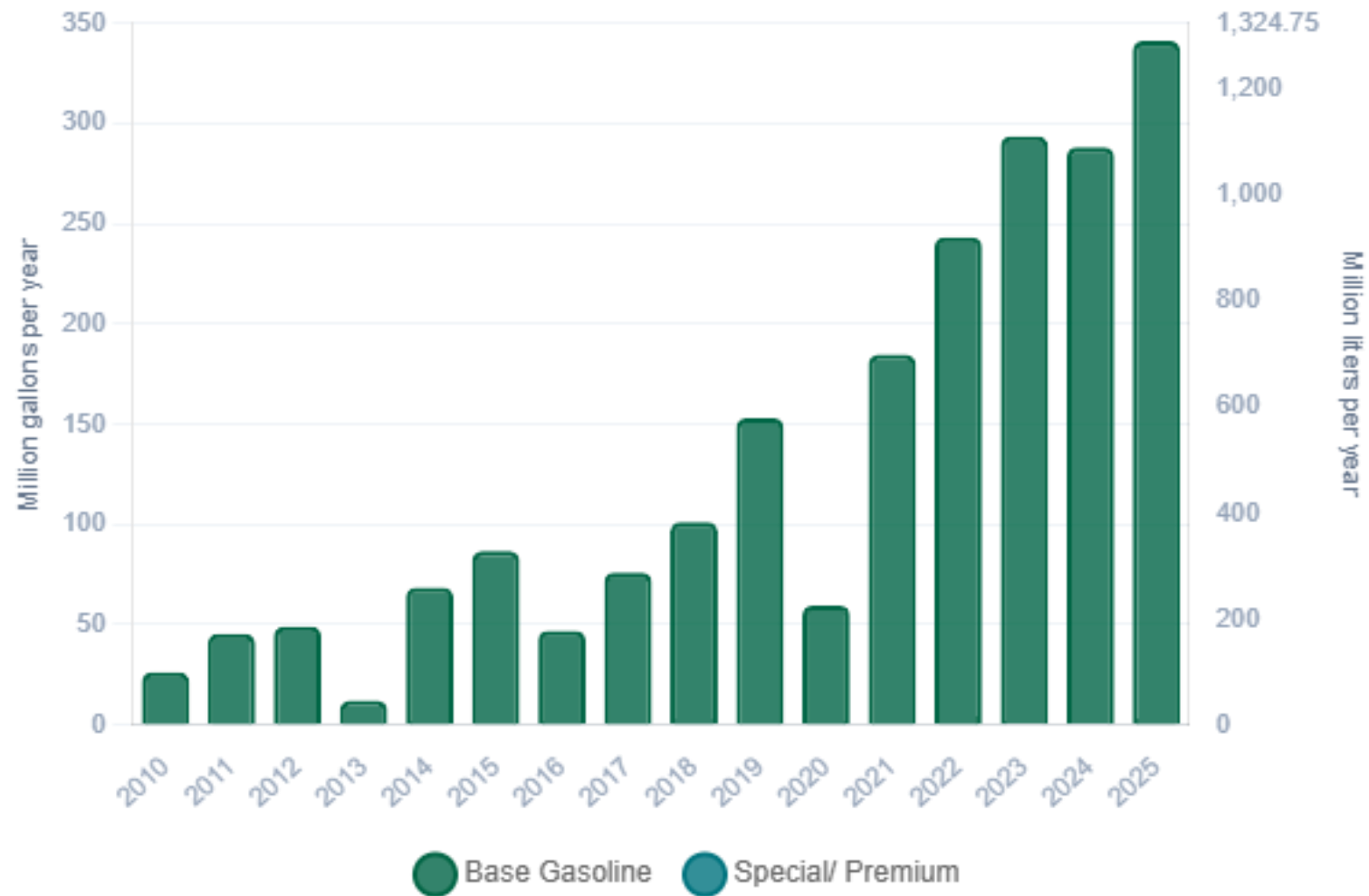
Source: ANH,INE - Bolivia

Gasoline Production in Bolivia



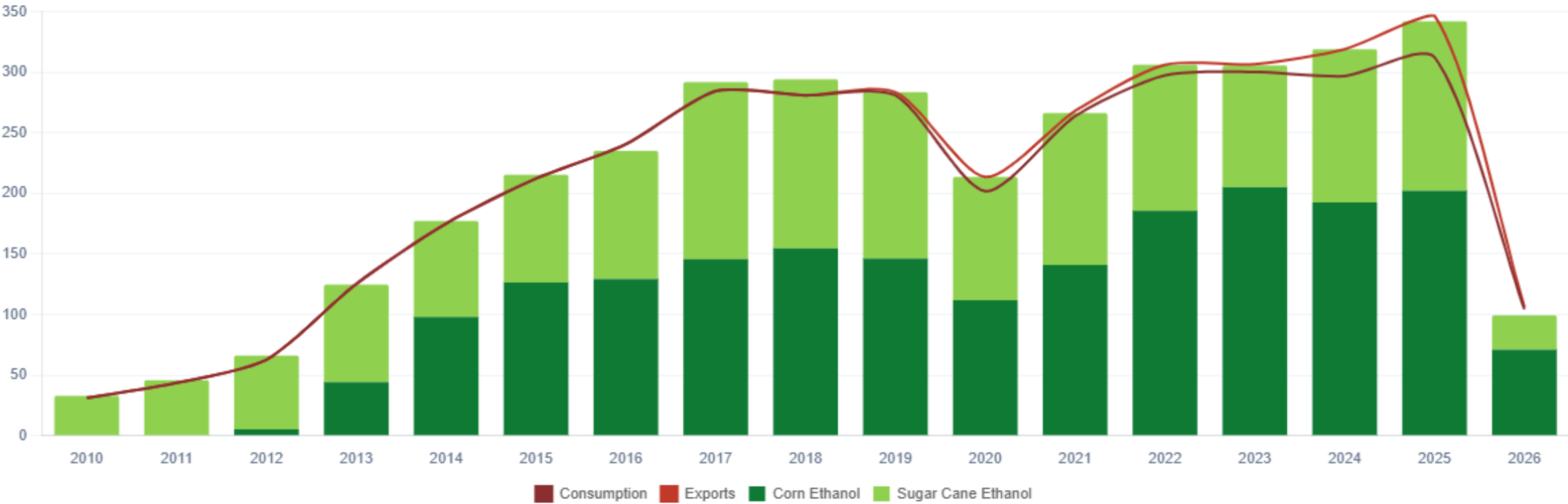
Source: ANH,INE - Bolivia

Gasoline Imports in Bolivia



Source: ANH,INE - Bolivia

Ethanol Balance in Bolivia

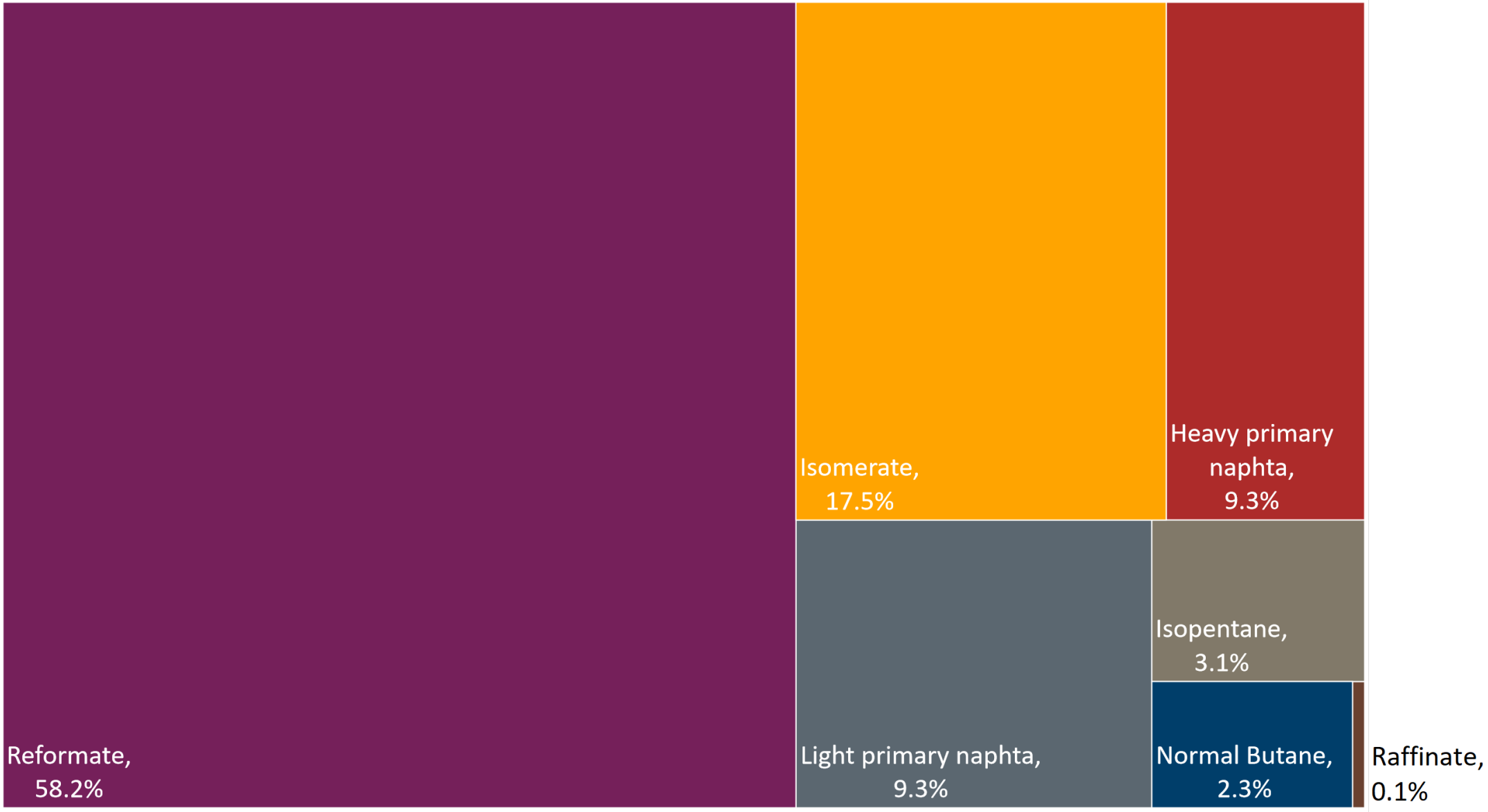
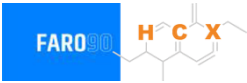


Source: ANH,INE - Bolivia

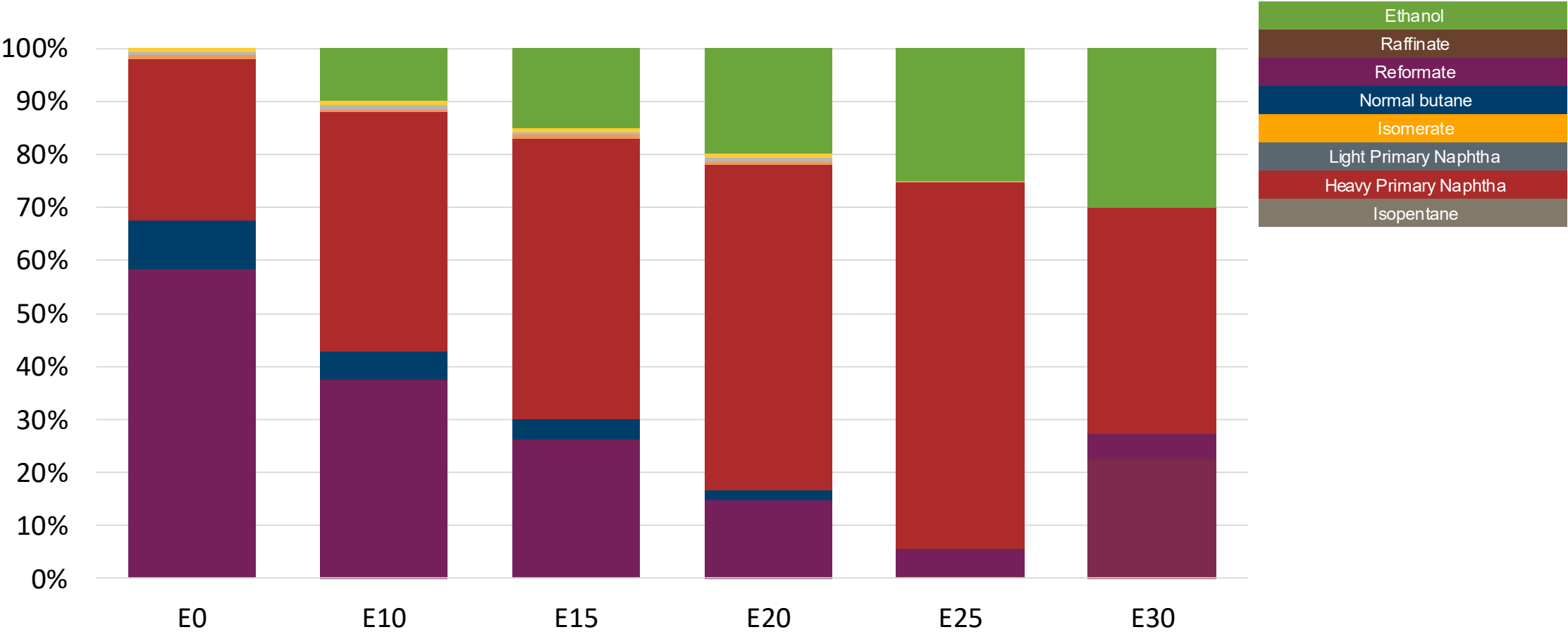
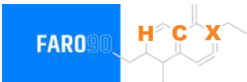
Gasoline Quality in Bolivia

Name	Supreme Decree 4718		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2022		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 3%v/v (2,7%v/v for E12 in summer)	< 3%v/v (2,7%v/v for E12 in summer)	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 42 % v/v	< 48 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 18 mg/l	< 18 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 85	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 12 %v/v	< 12 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	⇔ 7-9 psi	⇔ 7-9 psi	⇔ 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	⇔ 7-9.5 psi	⇔ 7-9.5 psi				
RVP 37.8°C (Transition)	⇔ 7-9 psi	⇔ 7-9 psi				
MTBE			-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Available Blending Components

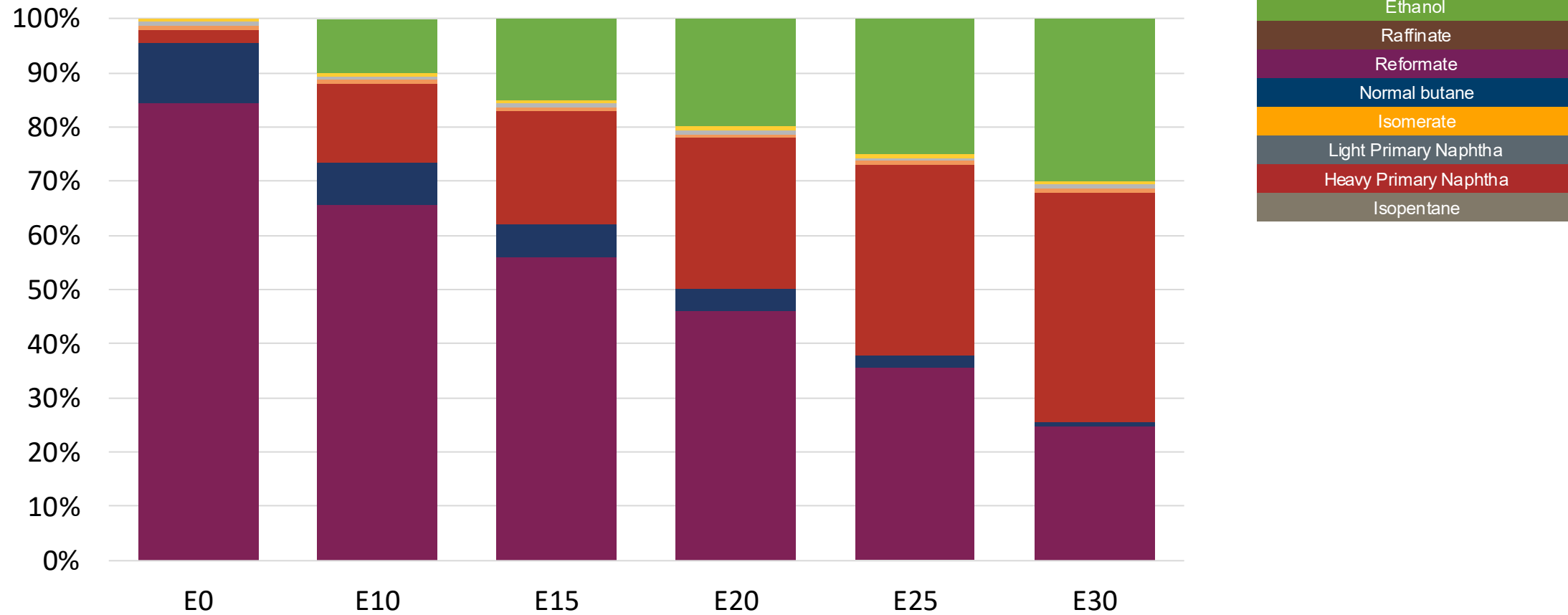


Ethanol Blending - Gasoline Especial – Constant Octane



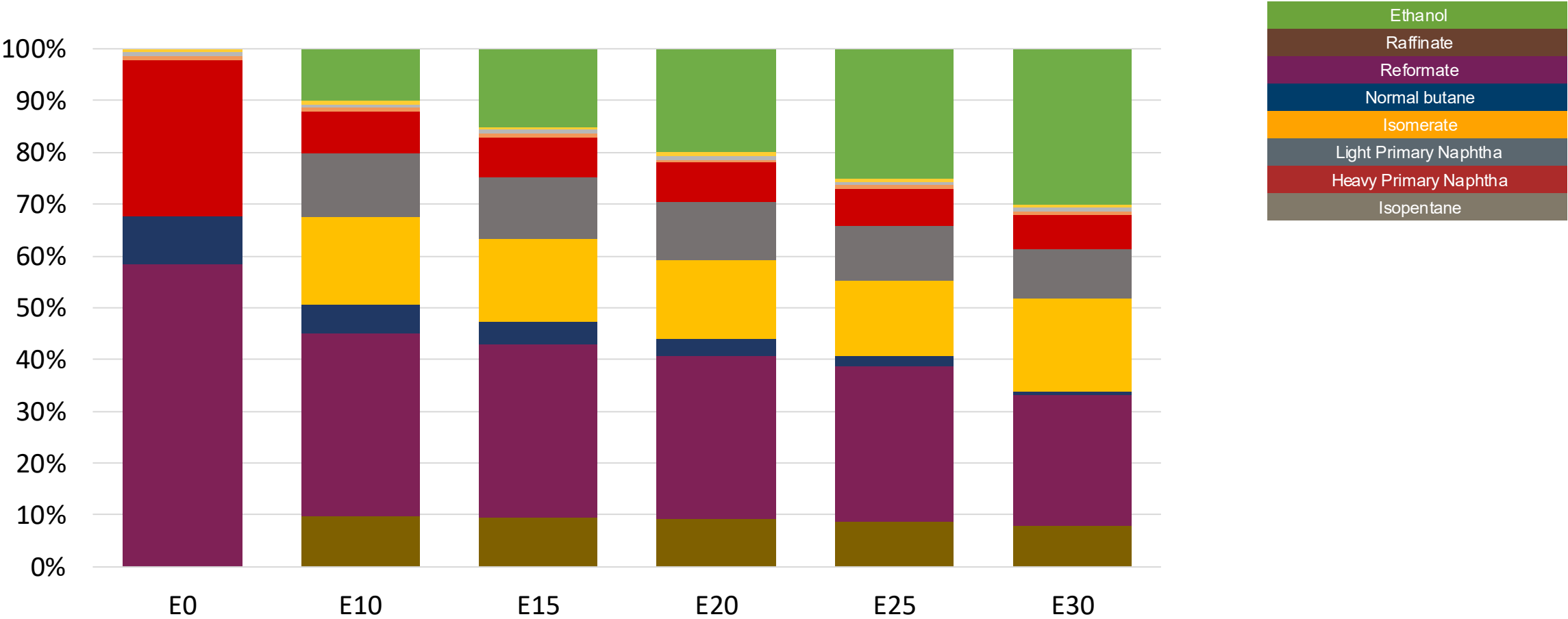
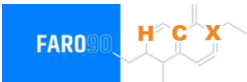
Octane (RON)	85.0	85.0	85.0	85.0	85.0	89.2
Price (US\$/gal)	\$2.05	\$1.75	\$1.58	\$1.41	\$1.23	\$1.33

Ethanol Blending – Gasoline Premium – Constant Octane



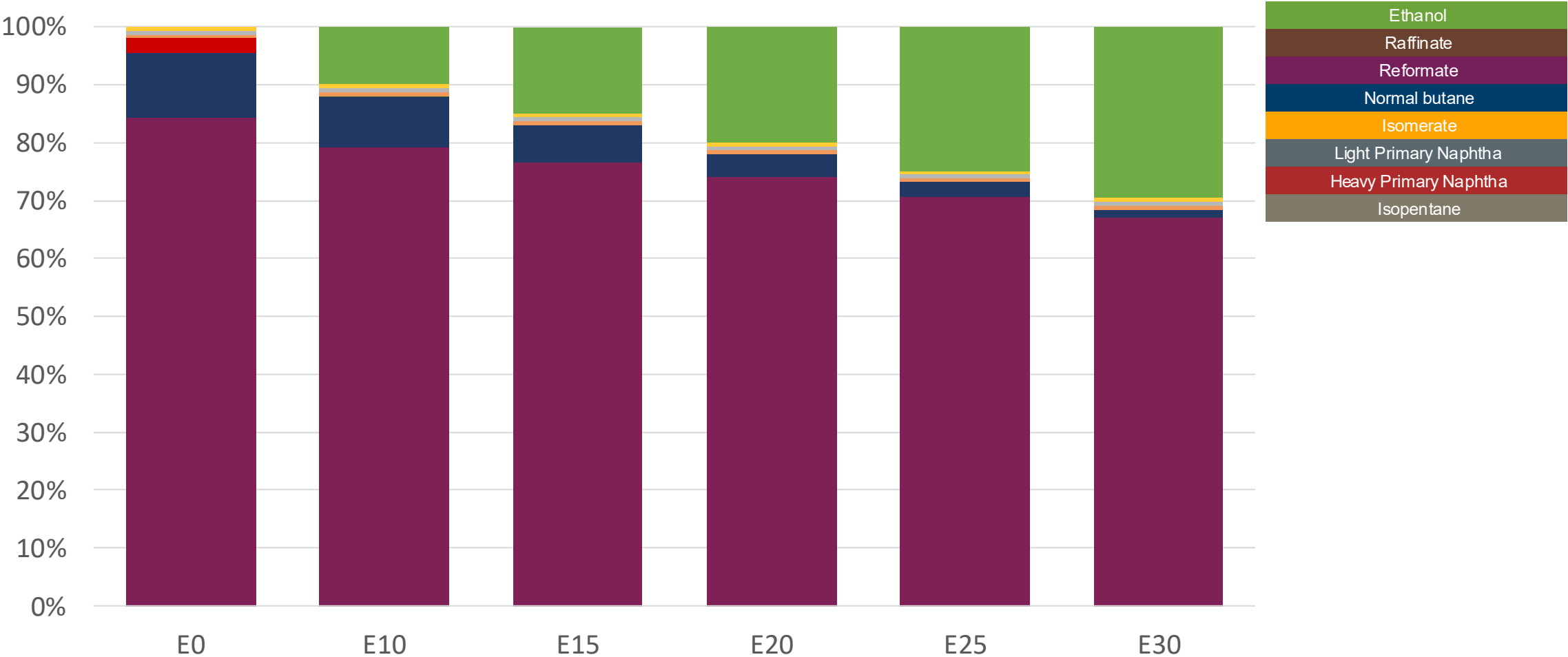
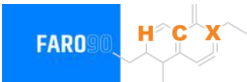
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$2.55	\$2.29	\$2.16	\$2.01	\$1.86	\$1.71

Ethanol Blending - Gasoline Especial – Octane Increment



Octane (RON)	85.0	90.2	92.8	95.3	97.7	99.9
Price (US\$/gal)	\$2.05	\$2.05	\$2.05	\$2.05	\$2.05	\$2.05

Ethanol Blending - Gasoline Premium – Octane Increment



Octane (RON)	95.0	99.7	101.6	103.4	105.4	107.7
Price (US\$/gal)	\$2.55	\$2.55	\$2.55	\$2.55	\$2.55	\$2.55

Gasoline Vehicle Fleet Bolivia

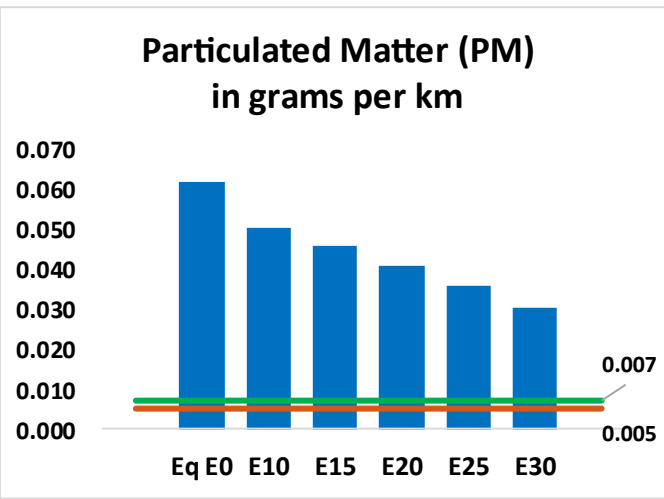
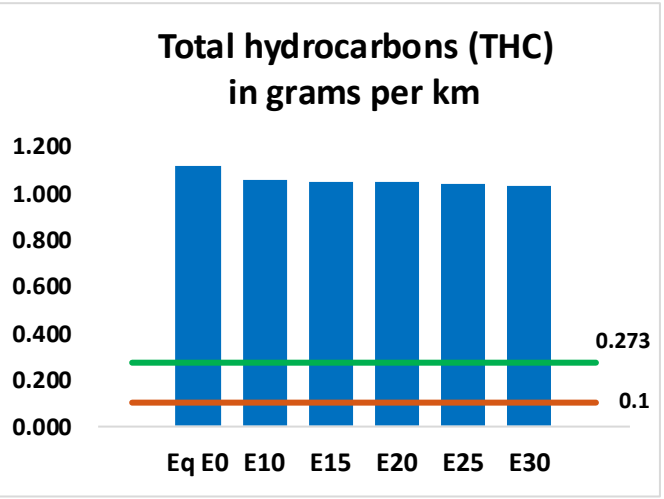
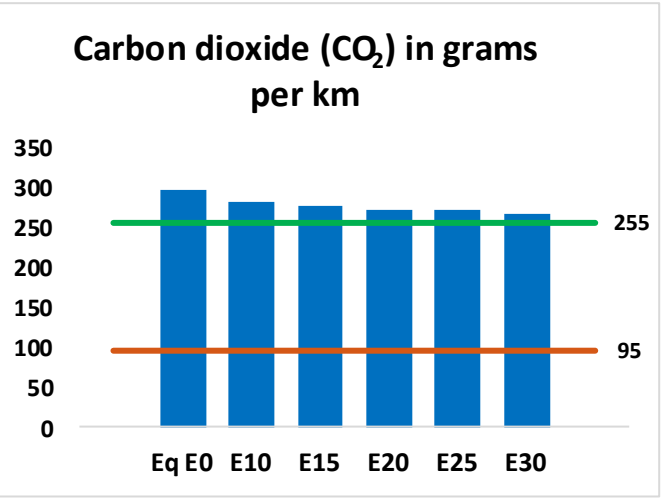
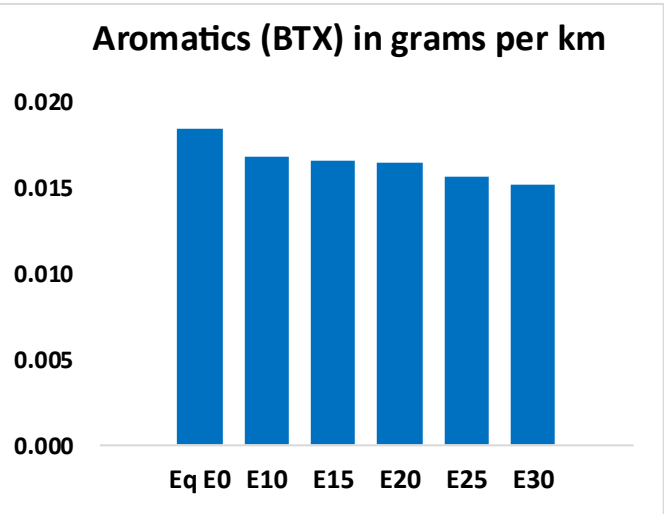
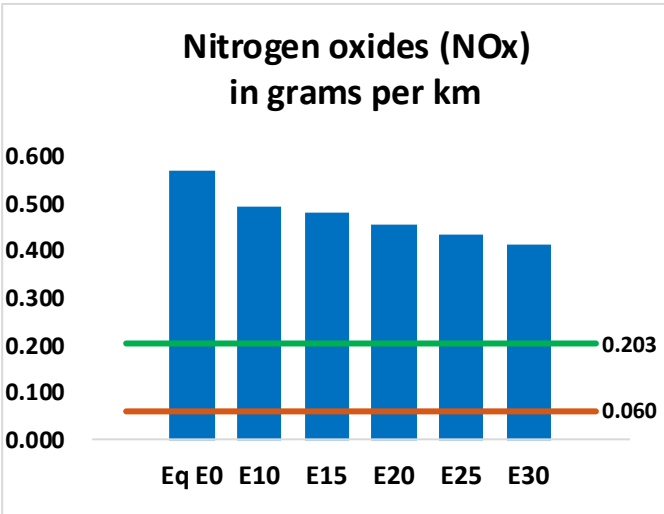
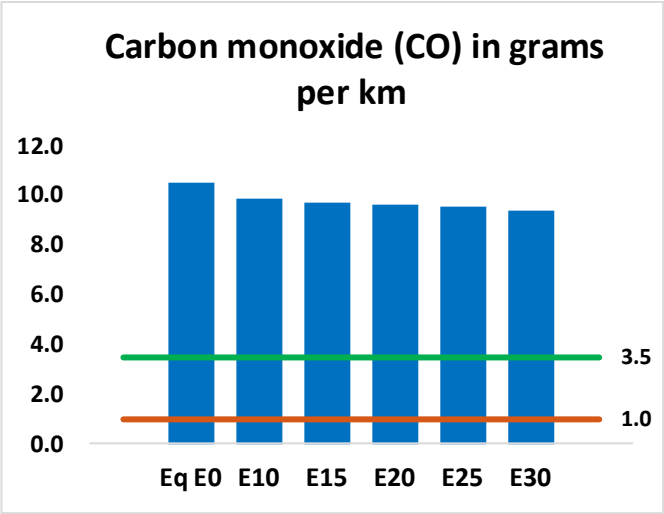
Vehicle Type	Engine	Exhaust	Recovery	Age	Number of vehicles			
Auto/Sml Truck	Multi FI-Pt	Euro V	PCV Tank	< 4 years, < 80 tkm	161,820			
				4-8 years, 80-160 tkm	228,679			
				> 8 years, > 160 tkm	177,284			
		Euro IV		> 8 years, > 160 tkm	177,284			
		Euro III		> 8 years, > 160 tkm	168,420			
		3-Ways		> 8 years, > 160 tkm	363,432			
Sml Engine	FI 4-cycle	Euro III	PCV	< 4years, < 80 tkm	294,897			
				4-8 years, 80-160 tkm	270,613			
		Euro II		4-8 years, 80-160 tkm	60,633			
				> 8 years, > 160 tkm	296,030			
Truck	FI	Euro V	PCV	< 4 years, < 80 tkm	33,765			
				4-8 years, 80-160 tkm	53,667			
				> 8 years, > 160 tkm	42,421			
		Euro IV		> 8 years, > 160 tkm	46,279			
		Euro III		> 8 years, > 160 tkm	26,996			
		3-Ways		> 8 years, > 160 tkm	269,955			

Vehicular Fleet: **2,672,176**
Gasoline Vehicle Fleet: 2,191,184
Average Age: 12 years
Motorcides: **34.5%**

Source: Instituto Nacional de Estadística – Bolivia (INE), analysis Faro 90

Bolivia – Vehicular Emissions

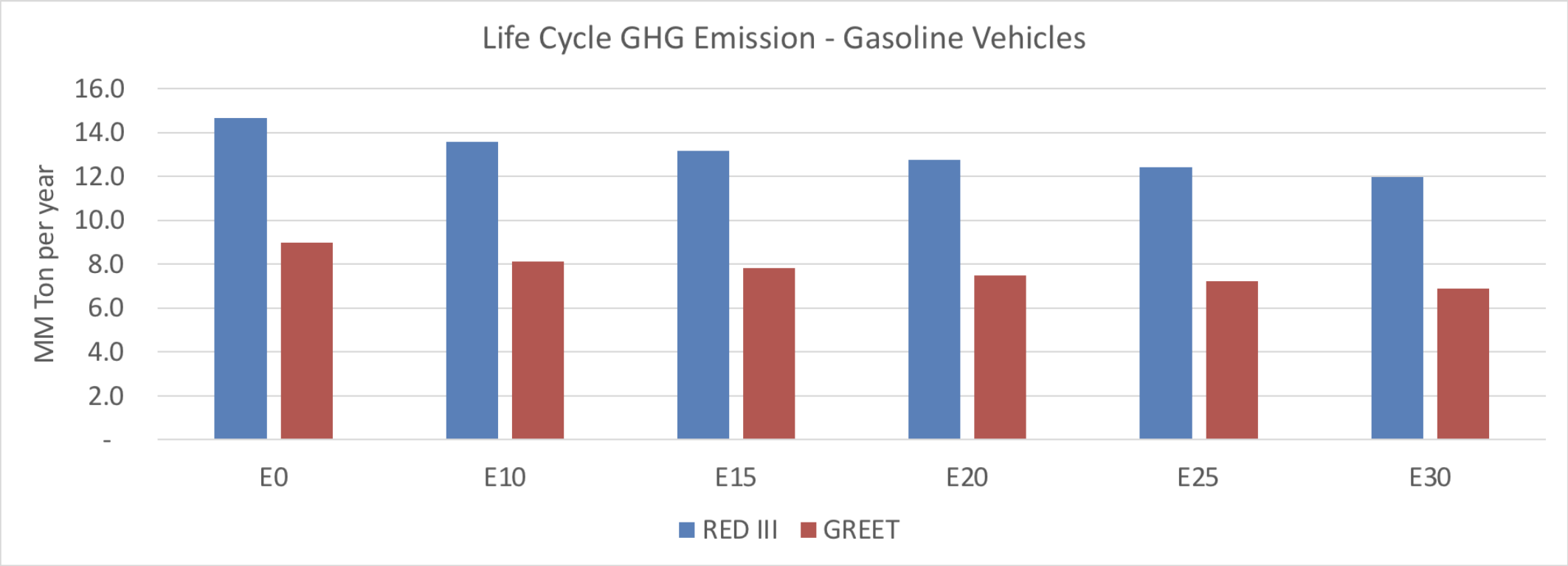
United States
Euro 6



Bolivia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	574,534	557,512	540,490	531,838	525,569	521,485	513,437	-6%	-9%
CO2	16,226,876	15,809,614	15,415,533	15,105,599	14,952,232	14,925,126	14,650,013	-5%	-8%
VOC	60,984	59,521	58,057	57,487	57,203	57,072	56,589	-5%	-6%
NOx	31,229	28,834	27,169	26,309	24,936	23,854	22,583	-13%	-20%
SOx	188	174	160	148	136	123	110	-15%	-28%
NH3	3,527	3,492	3,457	3,473	3,513	3,540	3,563	-2%	0%
N2O	835	830	827	831	848	858	867	-1%	2%
CH4	13,854	13,854	13,854	14,061	14,366	14,629	14,879	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	243	236	228	224	221	219	216	-6%	-9%
Acetaldehyde	662	760	1,115	1,698	2,313	2,643	3,123	68%	249%
Formaldehyde	2,700	2,625	3,060	3,593	3,764	4,164	4,534	13%	39%
Benzene	1,011	972	921	906	899	859	832	-9%	-11%
PM2.5	1,188	1,080	972	877	784	684	580	-18%	-34%
PM10	2,205	2,005	1,804	1,628	1,455	1,270	1,077	-18%	-34%

Bolivia – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	63.8
2035 Target (MMT/year)	44.6
Delta	19.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.8%	13.2%	16.7%	19.5%	23.4%
RED II/III	0.0%	7.4%	10.2%	13%	15.3%	18.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	4.6%	6.2%	7.9%	9.2%	11.0%
RED II/III	0.0%	5.7%	7.8%	9.9%	11.7%	14.0%

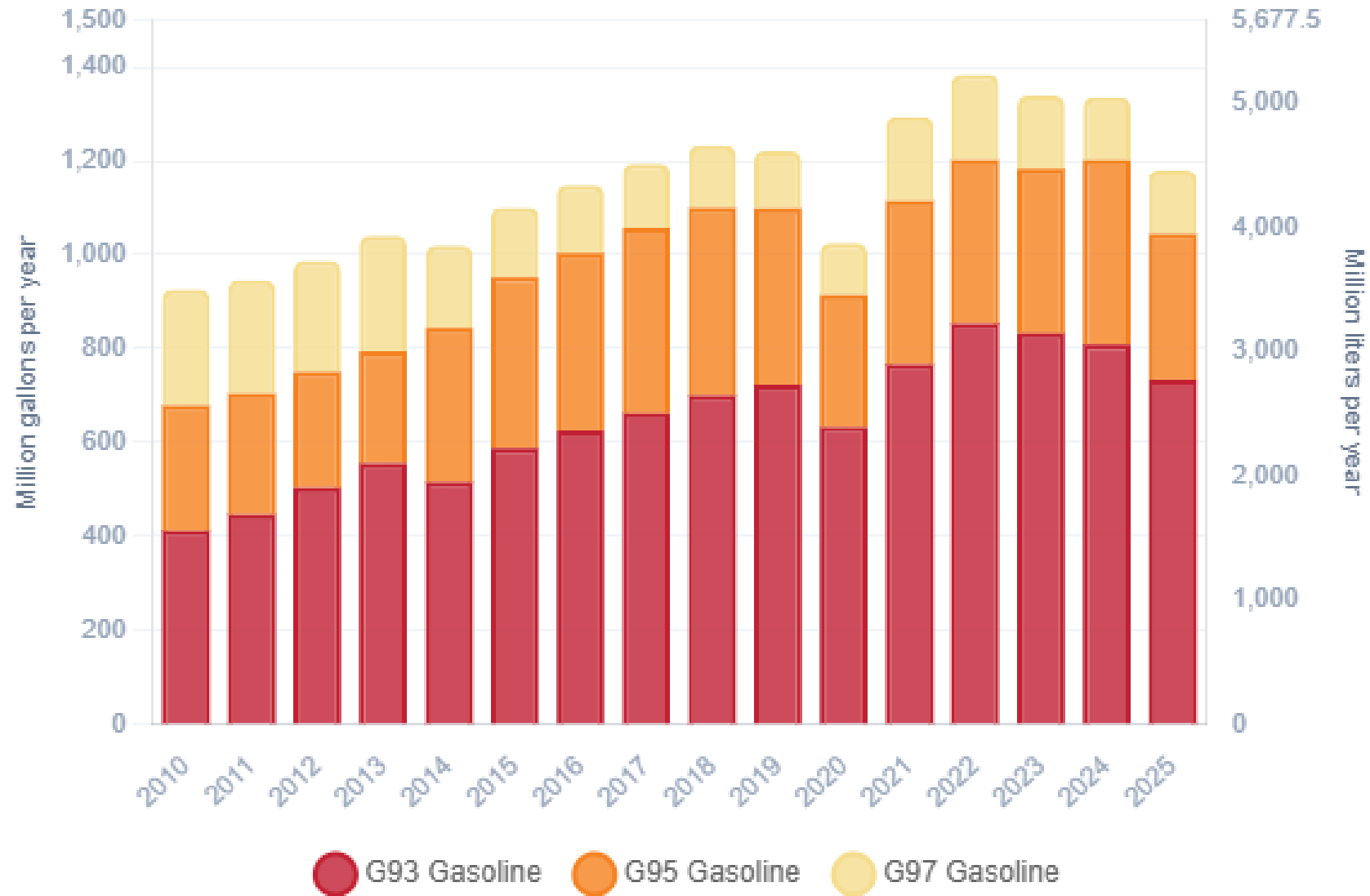
Ethanol Blending in Gasoline - Chile



Chile's gasoline consumption is currently over 1,180 million gallons per year (4,460 million liters) distributed in three grades: RON 93 (AKI 87), RON 95 (AKI 88) and RON 97 (AKI 92). There is a more stringent quality specification for Santiago Metropolitan Area. In 2025, the market share of RON 93 gasoline was 62%. Government company ENAP produces 77% of the total demand (CNE). Gasoline imports made in the country come mainly from the United States. Refineries only produce RON 93 and RON 97, gasoline RON 95 is blended from these grades in gas stations.

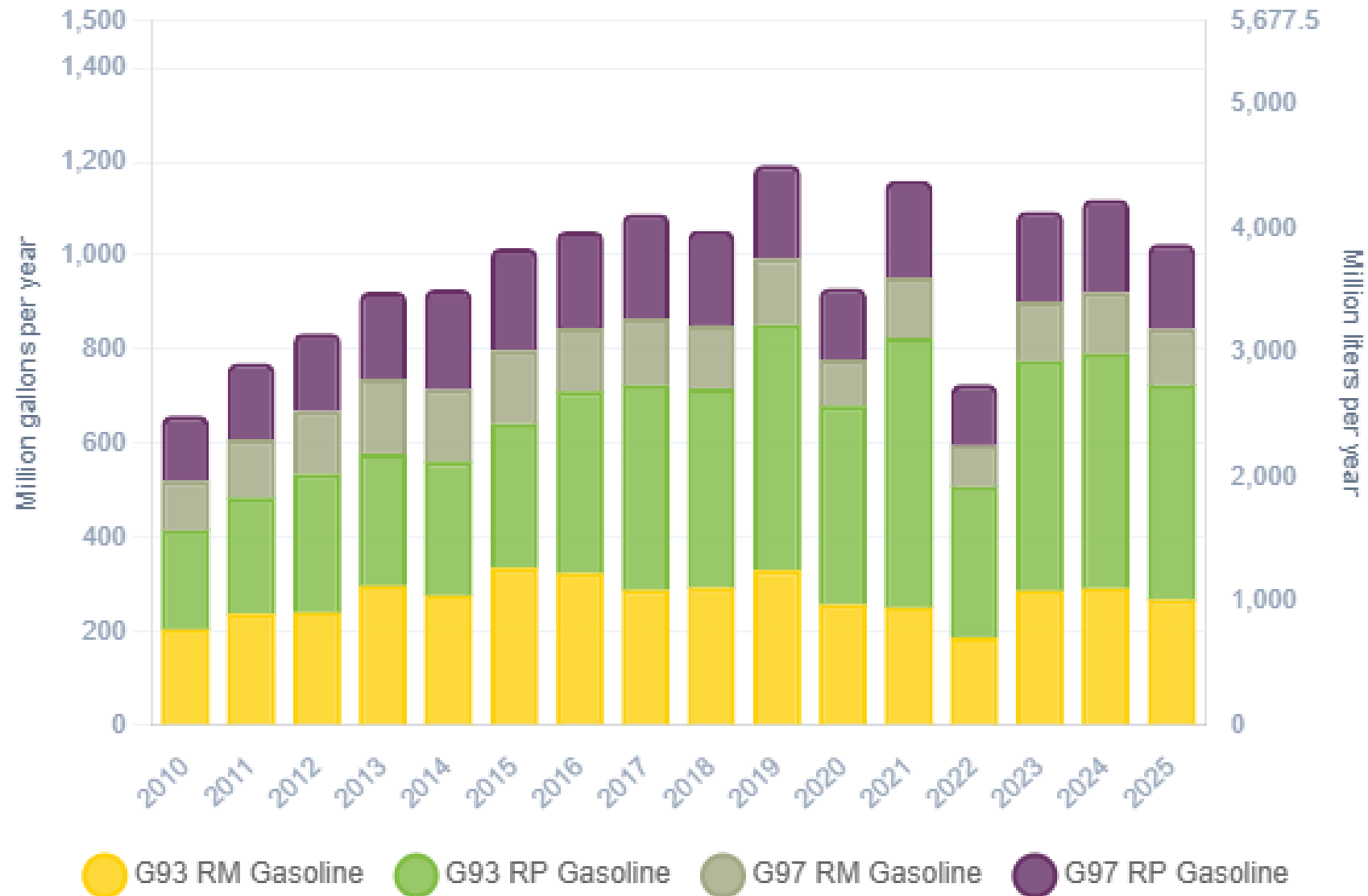
Ethanol can be blended up to 5% v/v (Decreto Supremo 56) but it is not used because of regulatory barriers. Chile does not produce or imports ethanol. The use of ethanol is being analyzed to achieve short term goals in the national economy decarbonization program.

Gasoline Demand in Chile



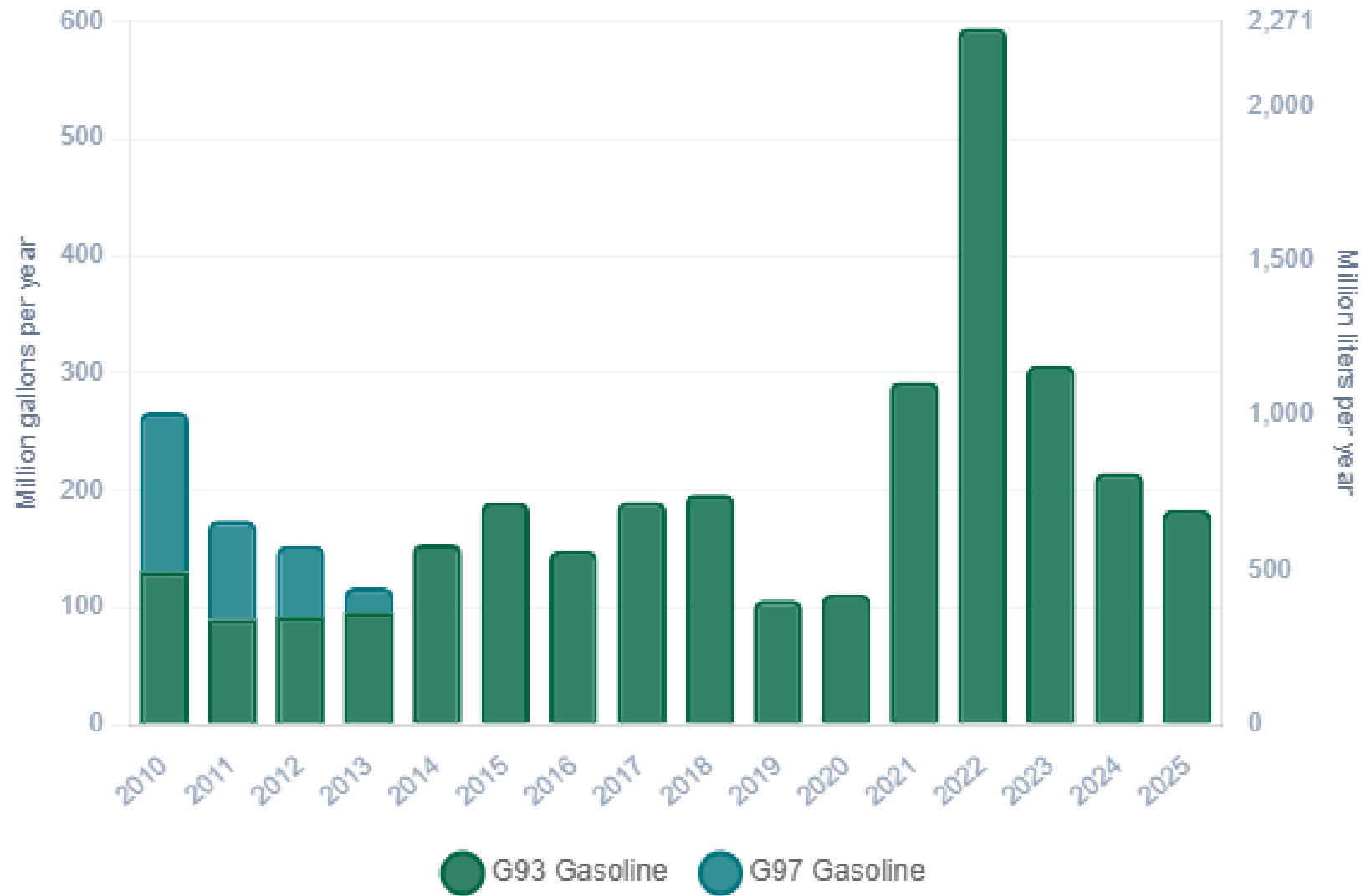
Fuente: CNE, 2025

Gasoline Production in Chile



Fuente: CNE, 2025

Gasoline Imports in Chile



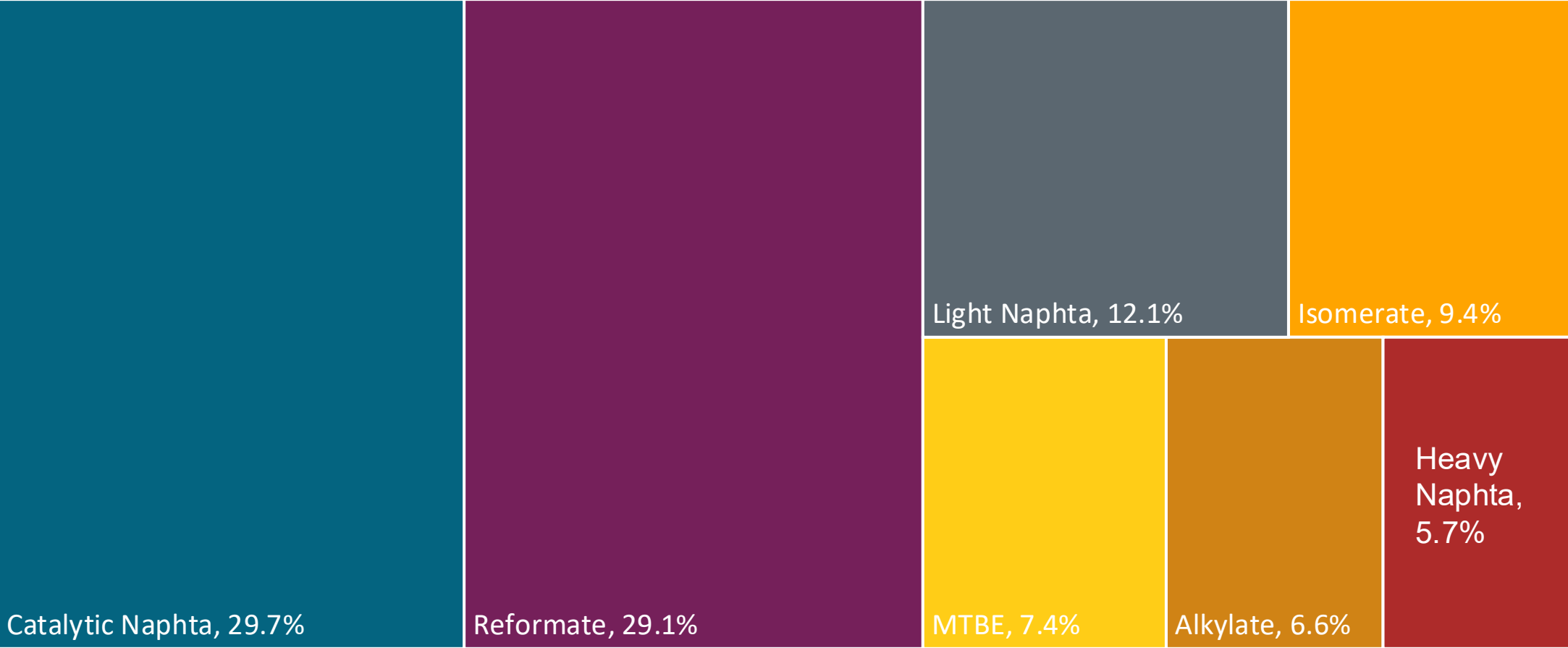
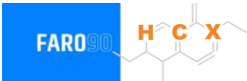
Fuente: CNE, 2025

Gasoline Quality in Chile

Name	PPDA DS31	PPDA DS31	DS 60	DS 60	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2017	2017	2012	2012	2017			
Applicability	Metropolitan Region	Metropolitan Region	Rest of the country	Rest of the country	All countries			
Selected Grade	G93	G97	G93	G97	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 38 %v/v	< 38 %v/v	< 38 %v/v	< 38 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	12 %v/v	12 %v/v	20 %v/v	20 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	Report	Report	Report	Report	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	93	97	93	97	> 95	> 95	> 98	> 98
MON	Report	Report	Report	Report	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 15 mg/kg	< 15 mg/kg	< 15 mg/kg	< 15 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2 %m/m	< 2 %m/m	< 2 %m/m	< 2 %m/m	< 2,7 % m/m	< 3,7 % m/m	< 2,7 % m/m	< 3,7 % m/m
Ethanol (EtOH)	< 5 %v/v	< 5 %v/v	< 5 %v/v	< 5 %v/v	< 5 %v/v	< 10 %v/v	< 5 %v/v	< 10 %v/v
RVP 37.8°C (Summer)	< 55 kPa	< 55 kPa	< 69 kPa, < 83 kPa (Magallanes y región Antártica)	< 69 kPa, < 83 kPa (Magallanes y región Antártica)	< 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	< 69 kPa	< 69 kPa	< 45 - 80 kPa	< 45 - 80 kPa				
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	< 22 %v/v	Based on oxygen content	< 22 %v/v

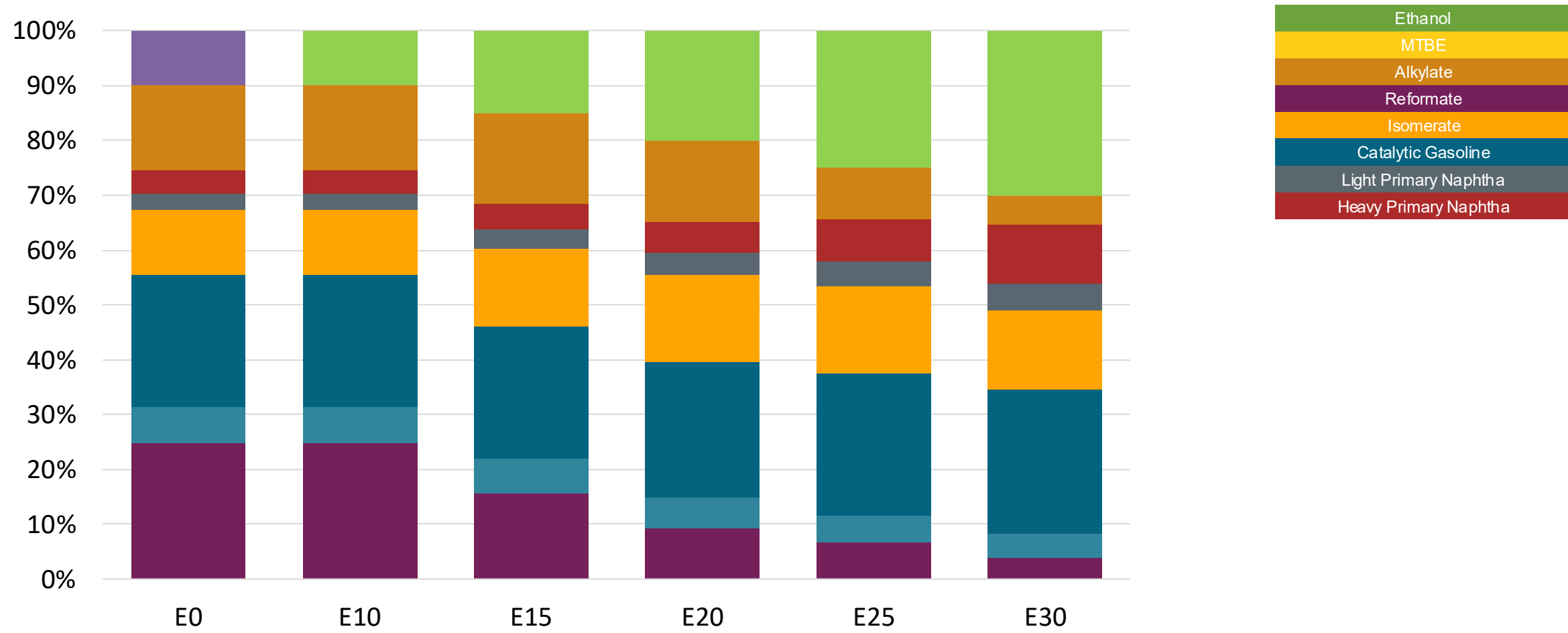
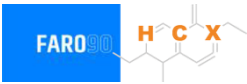
Source: ENAP, 2022

Available Blending Components



Fuente: ENAP, 2025

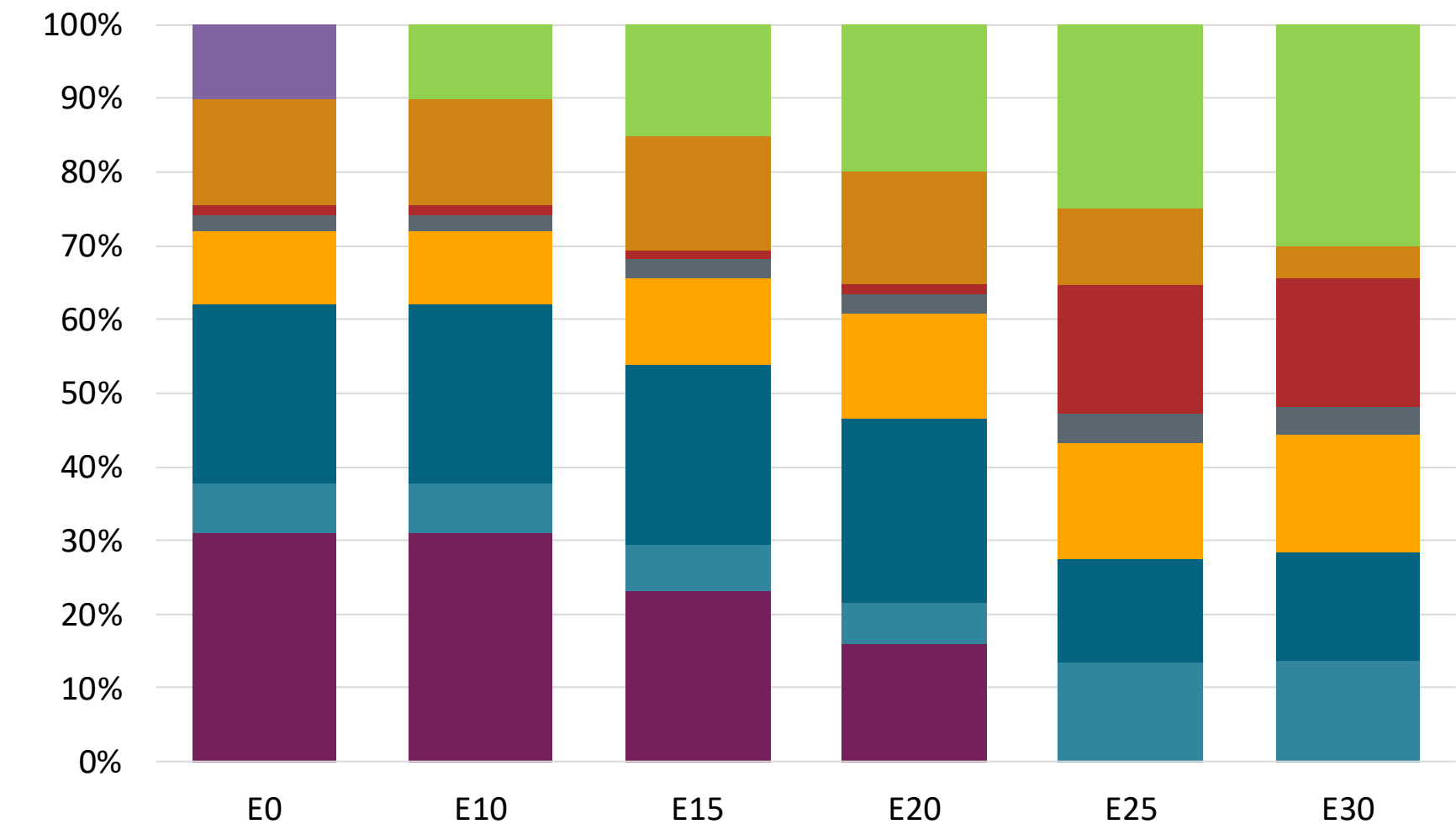
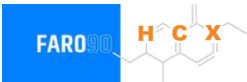
Chile – G93 RP Aconcagua – Constant Octane



Octane (RON)	93	93	93	93	93	93
Price (US\$/gal)	\$2.390	\$2.308	\$2.219	\$2.131	\$2.039	\$1.949

Source: Faro90, 2025

Chile – G97 RP Aconcagua – Constant Octane

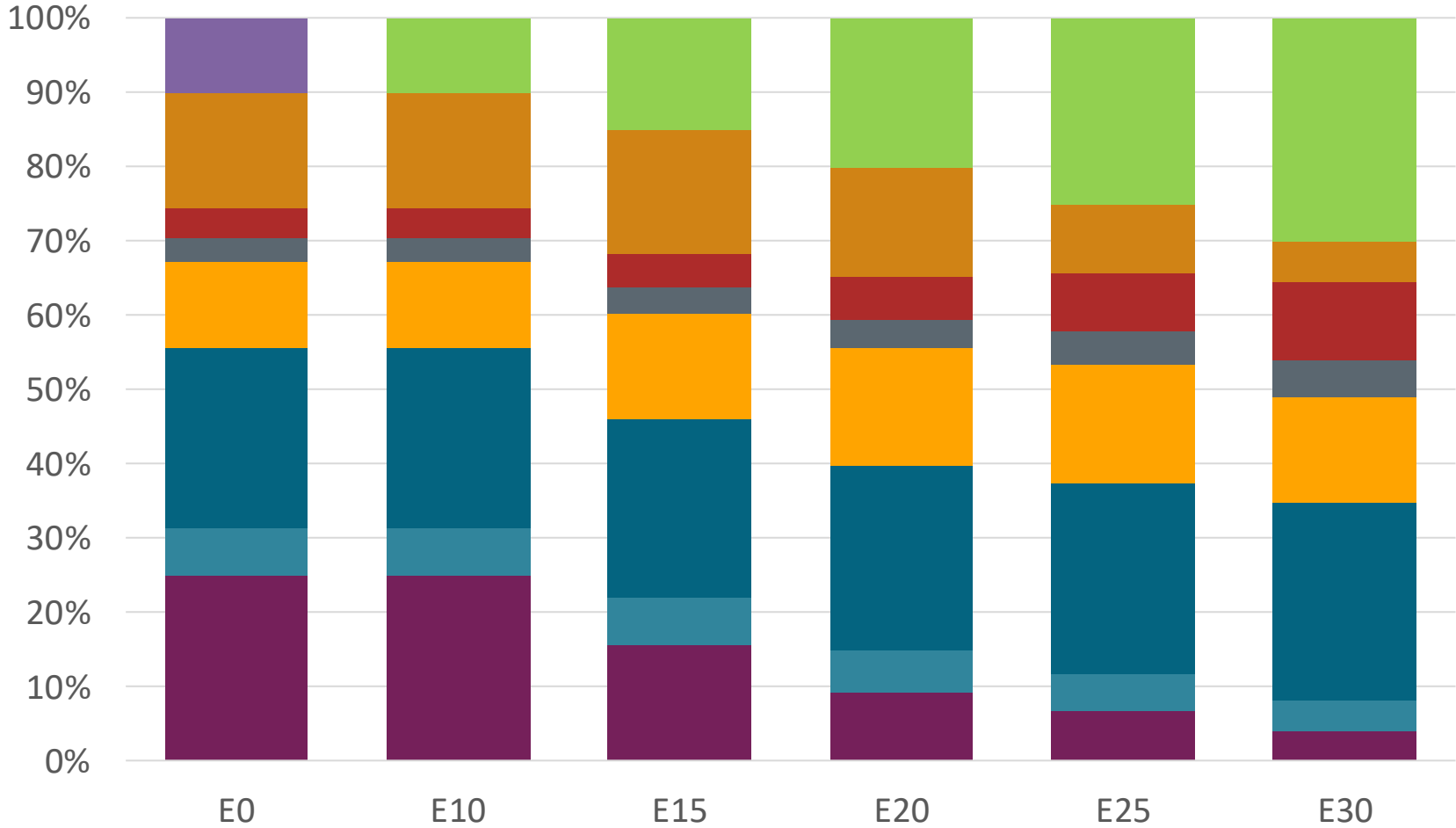
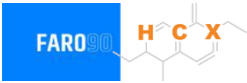


Ethanol
MTBE
Alkylate
Reformate
Isomate
Catalytic Gasoline
Light Primary Naphtha
Heavy Primary Naphtha

Octane (RON)	97	97	97	97	97	97
Price (US\$/gal)	\$2.516	\$2.434	\$2.356	\$2.276	\$1.922	1.867

Source: Faro90, 2025

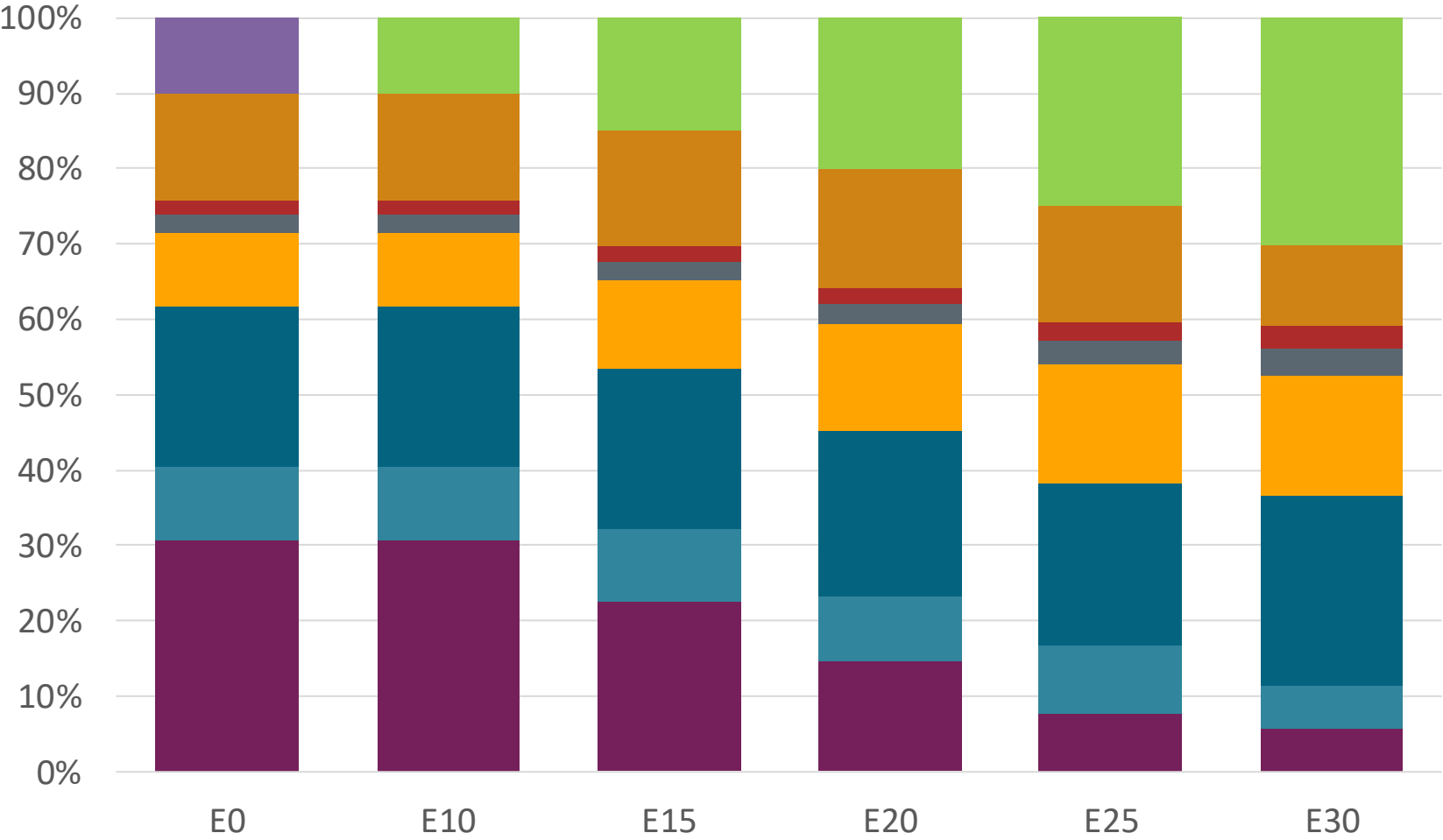
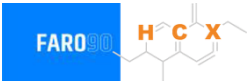
Chile – G93 RM Aconcagua – Constant Octane



Octane (RON)	93	93	93	93	93	93
Price (US\$/gal)	\$2.390	\$2.308	\$2.219	\$2.131	\$2.039	\$1.949

Source: Faro90, 2025

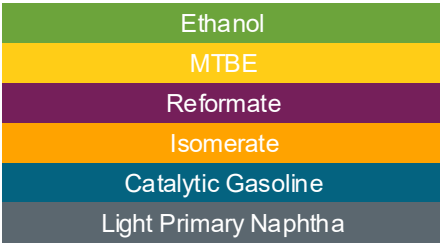
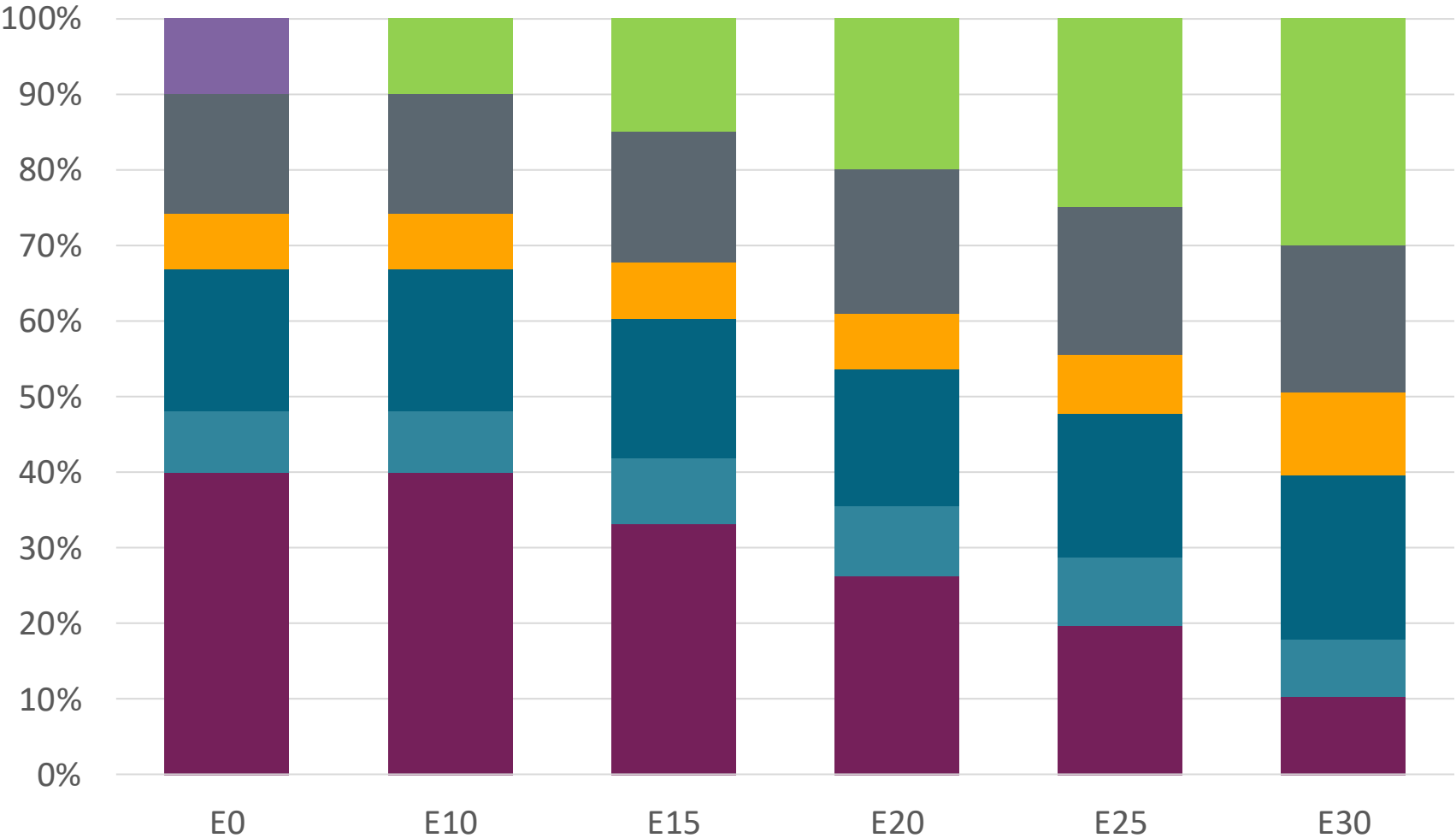
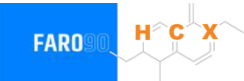
Chile – G97 RM Aconcagua – Constant Octane



Octane (RON)	97	97	97	97	97	97
Price (US\$/gal)	\$2.50	\$2.42	\$2.34	\$2.26	\$2.18	\$2.11

Source: Faro90, 2025

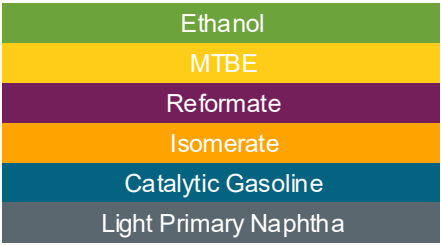
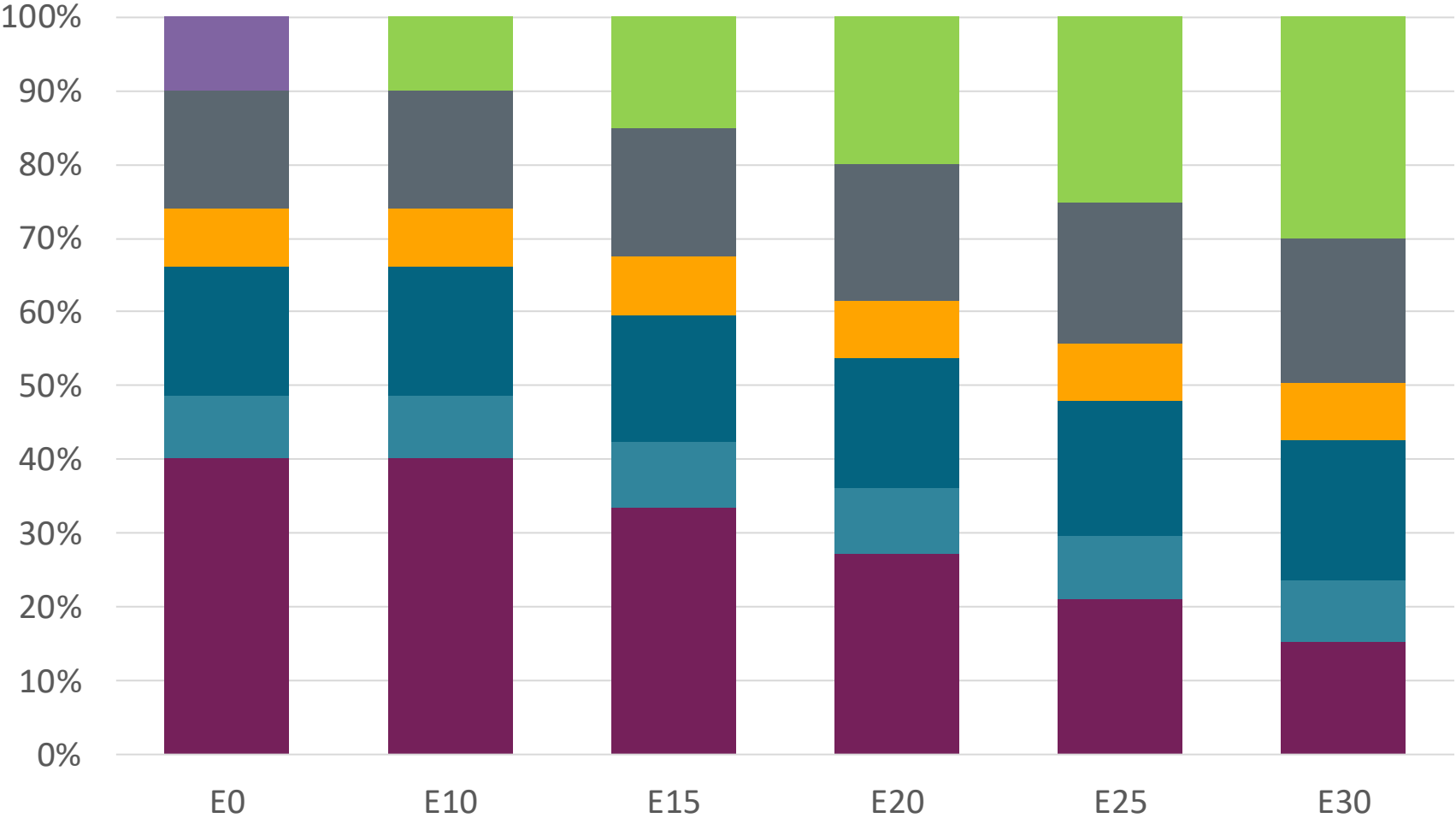
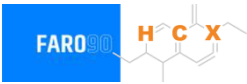
Chile – G93 RP Biobío – Constant Octane



Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US\$/gal)	\$ 2.25	\$ 2.17	\$ 2.09	\$ 2.00	\$ 1.93	\$ 1.84

Source: Faro90, 2025

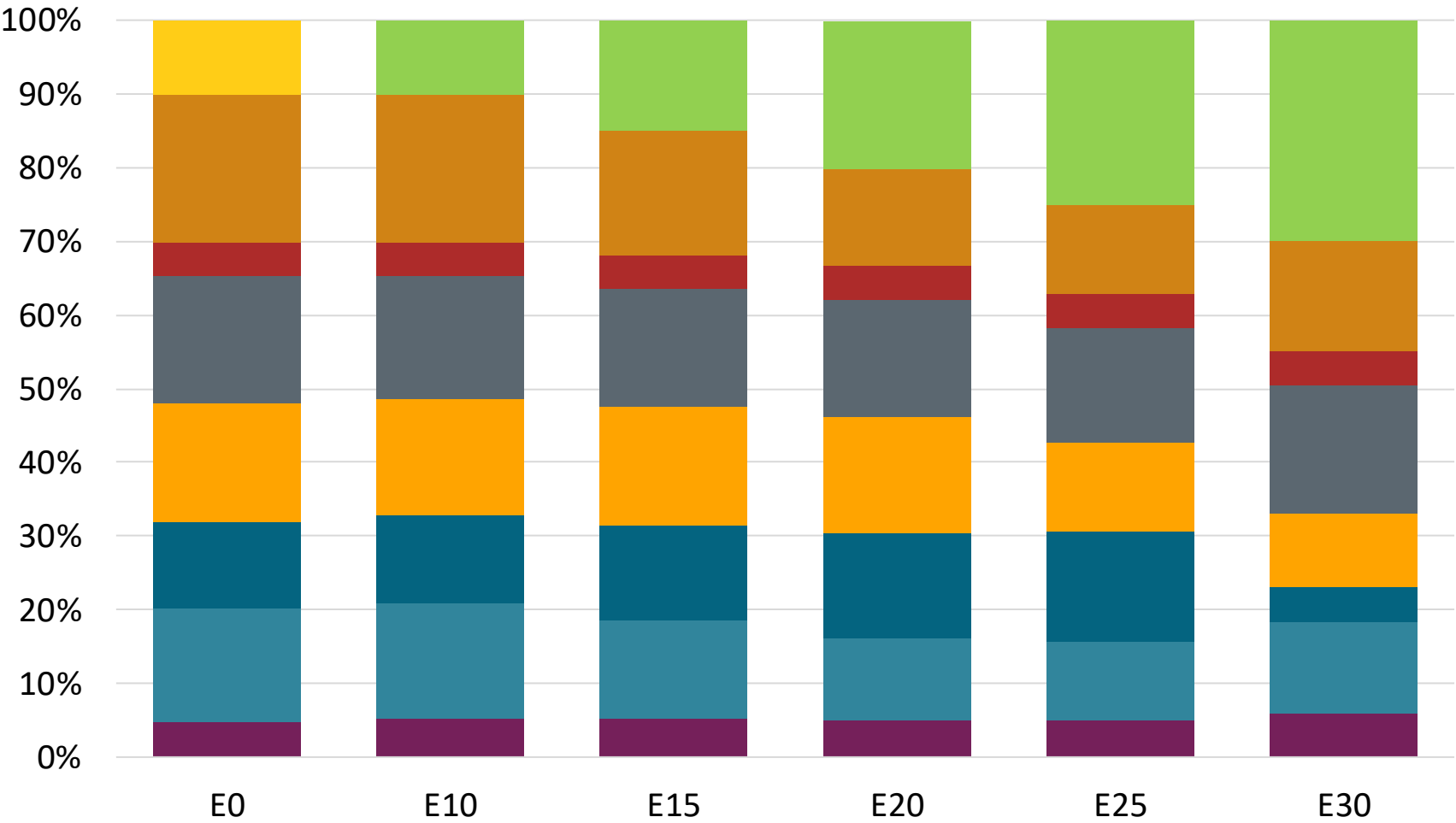
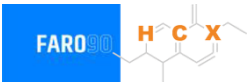
Chile – G97 RP Biobío – Constant Octane



Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	\$ 2.34	\$ 2.26	\$ 2.18	\$ 2.10	\$ 2.02	\$ 1.96

Source: Faro90, 2025

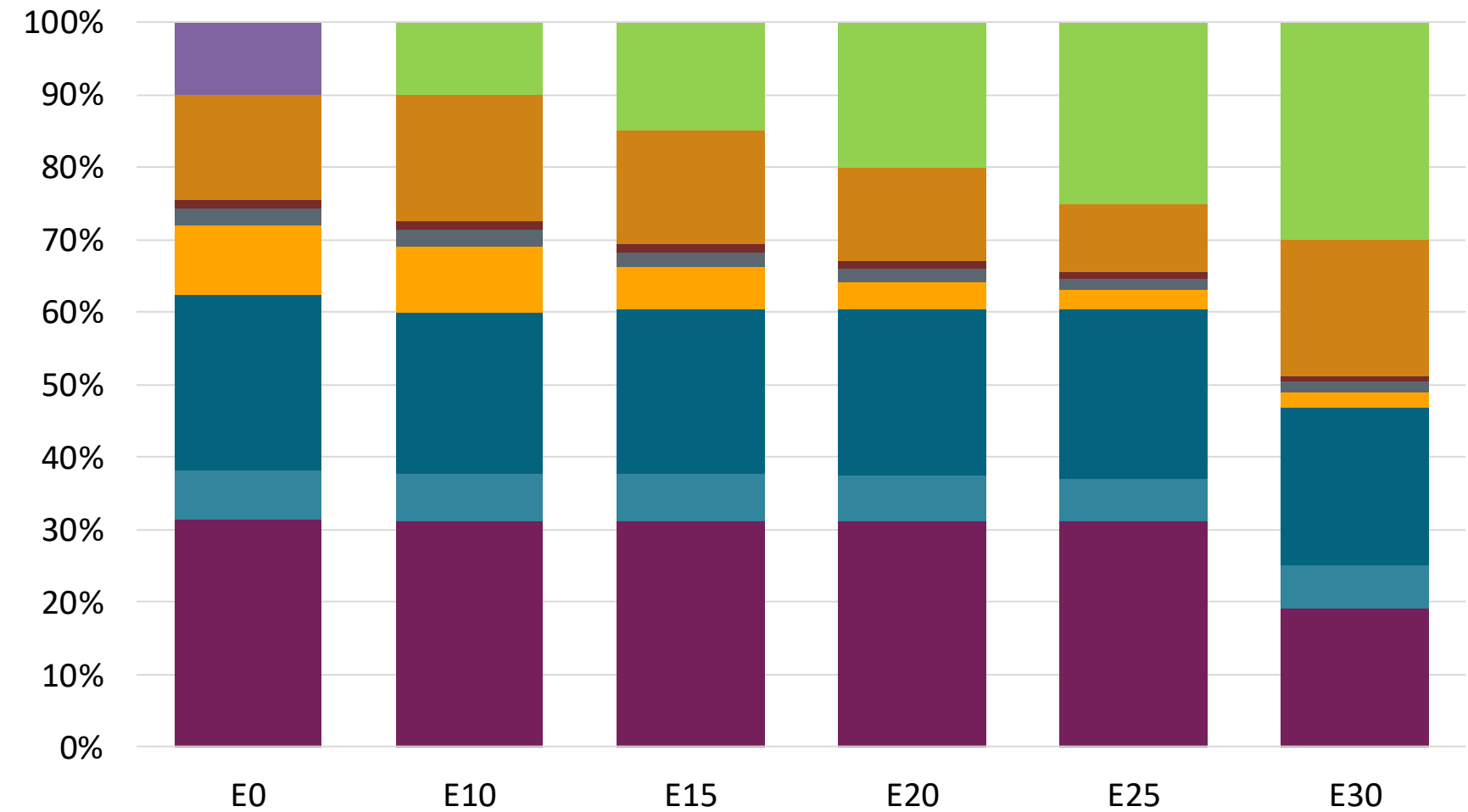
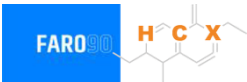
Chile – G93 RP Aconcagua – Octane Increment



Octane (RON)	E0	E10	E15	E20	E25	E30
	93.0	95.1	97.5	100.1	102.0	104.3
Price (US\$/gal)	\$ 2.19	\$ 2.19	\$ 2.19	\$ 2.19	\$ 2.19	\$ 2.19

Source: Faro90, 2025

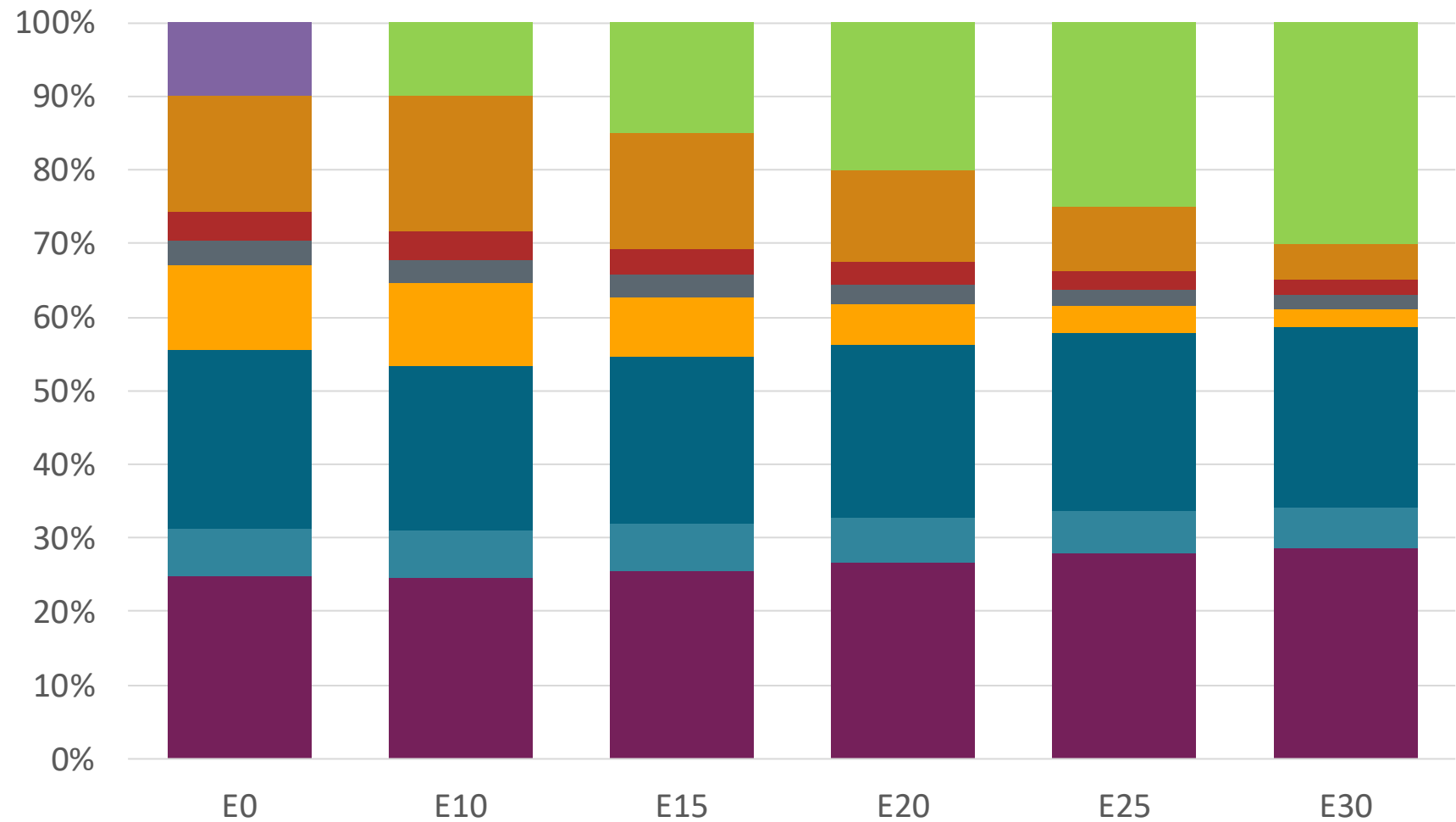
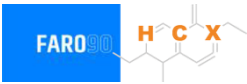
Chile – G97 RP Aconcagua – Octane Increment



Octane (RON)	97.0	100.1	102.1	104.3	106.9	109.4
Price (US\$/gal)	\$2.52	\$2.52	\$2.52	\$2.52	\$2.52	\$2.52

Source: Faro90, 2025

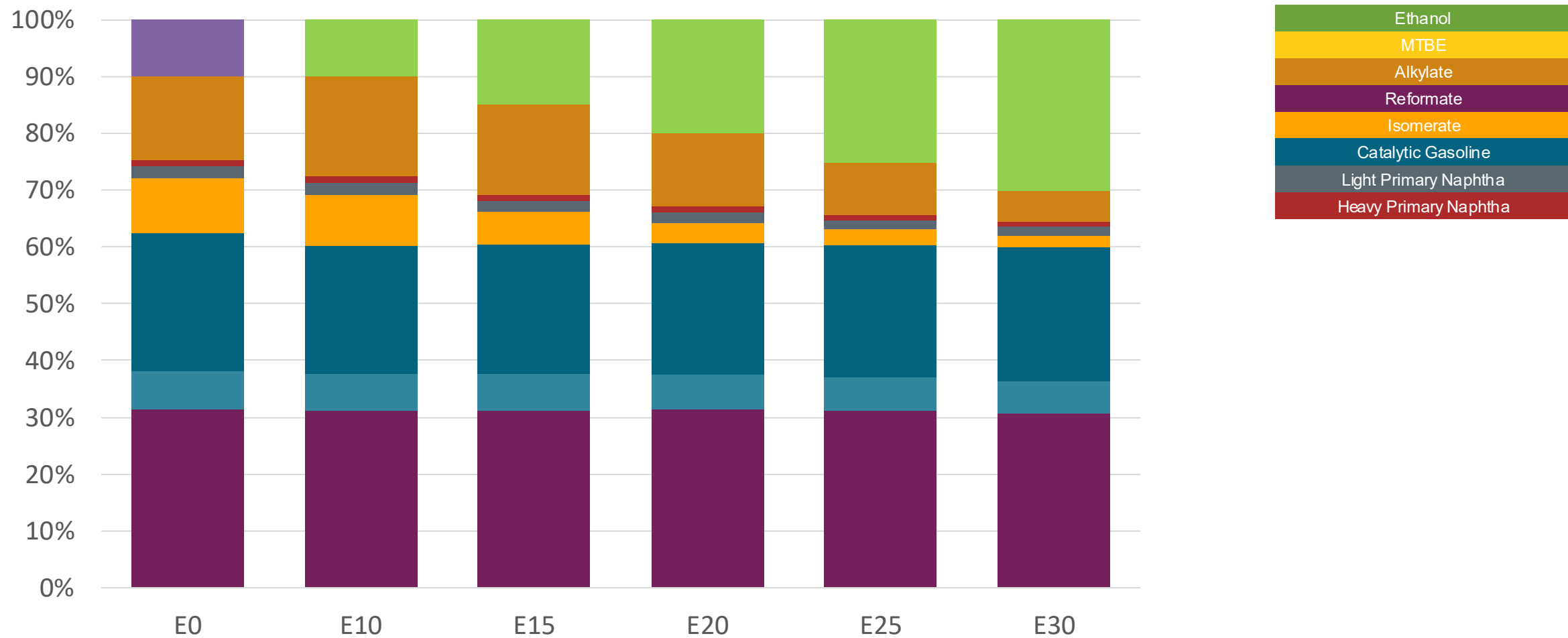
Chile – G93 RM Aconcagua – Octane Increment



Octane (RON)	93.0	96.2	98.1	100.0	102.1	104.3
Price (US\$/gal)	\$ 2.390	\$ 2.390	\$ 2.390	\$ 2.390	\$ 2.390	\$ 2.390

Source: Faro90, 2025

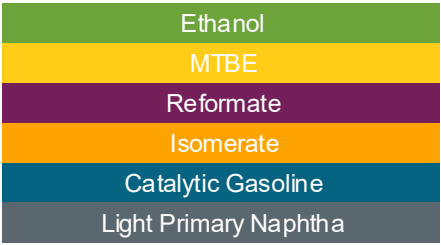
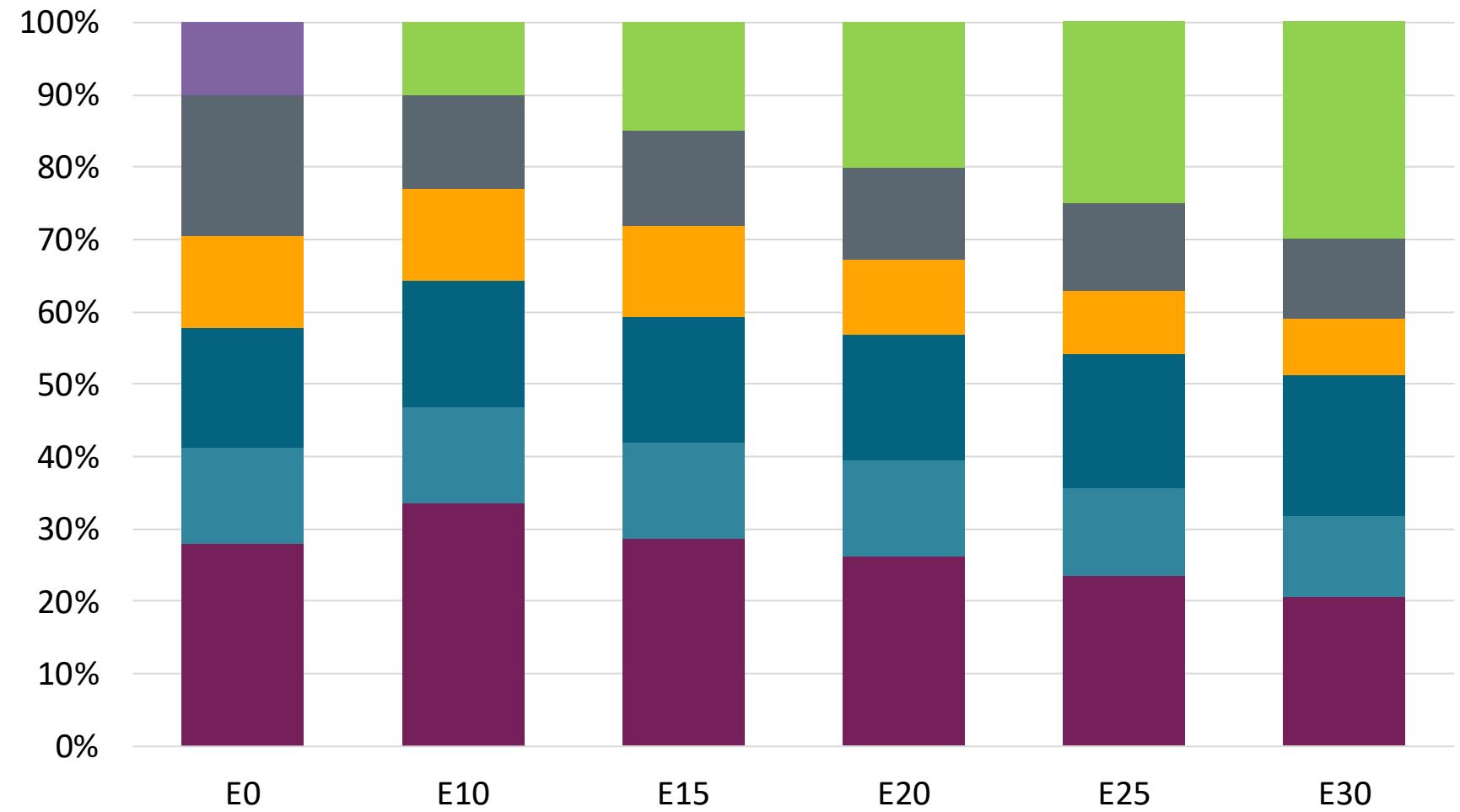
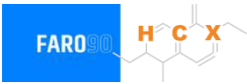
Chile – G97 RM Aconcagua – Octane Increment



Octane (RON)	97.0	100.1	102.1	104.3	106.9	109.5
Price (US\$/gal)	\$ 2.52	\$ 2.52	\$ 2.52	\$ 2.52	\$ 2.52	\$ 2.52

Source: Faro90, 2025

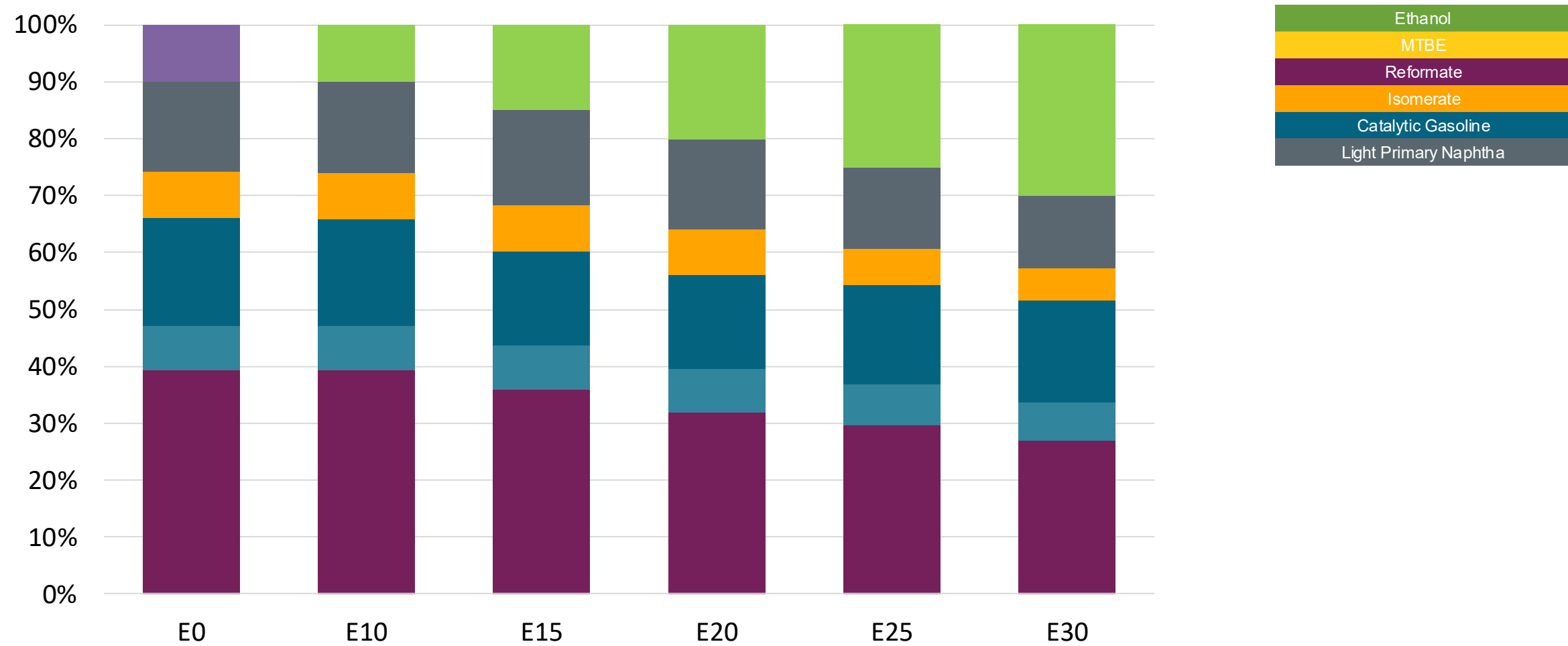
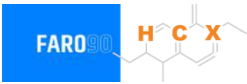
Chile – G93 RP Biobio – Octane Increment



Octane (RON)	93.0	93.0	95.9	98.2	100.4	102.7
Price (US\$/gal)	\$ 2.12	\$ 2.12	\$ 2.12	\$ 2.12	\$ 2.12	\$ 2.12

Source: Faro90, 2025

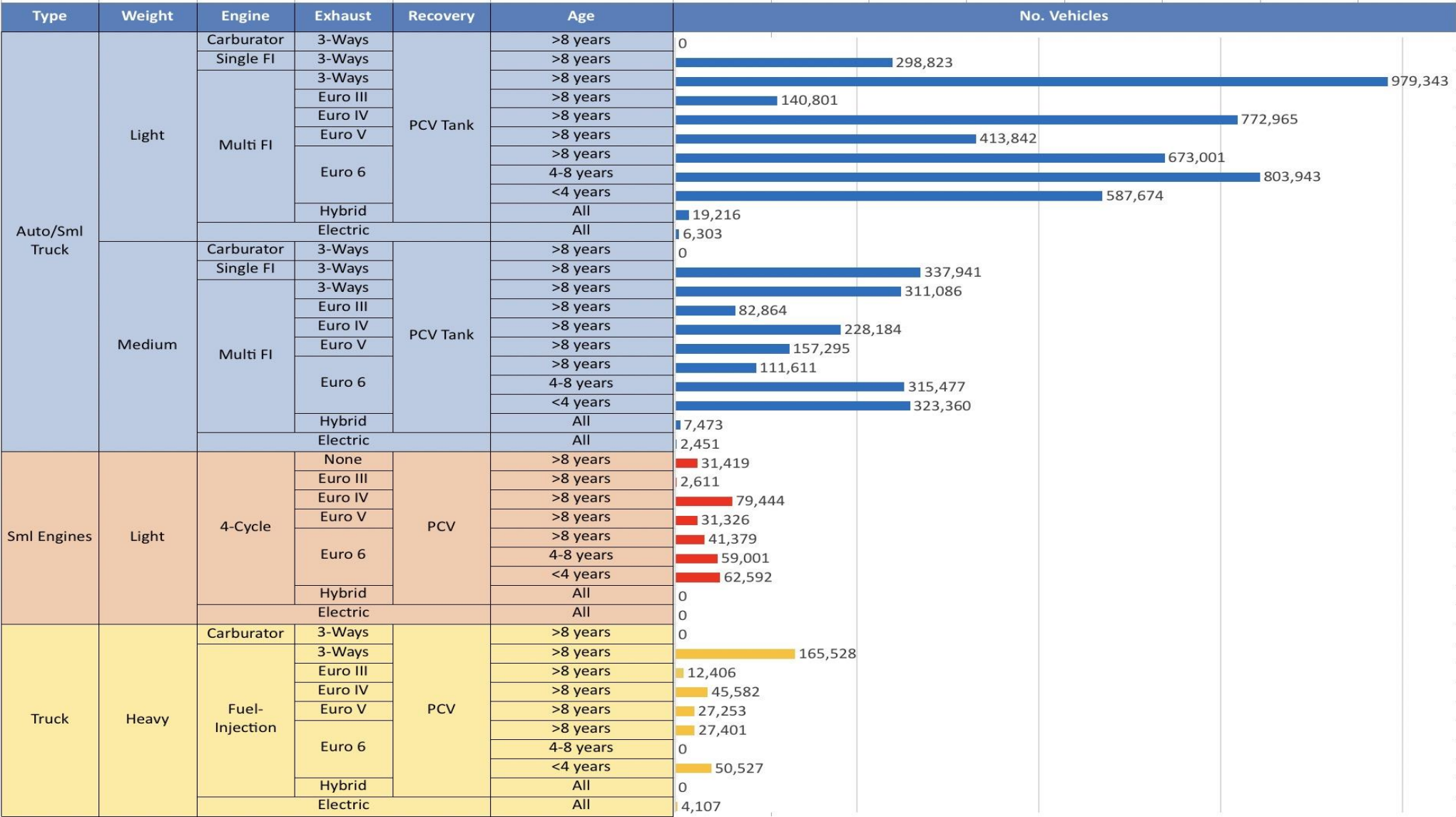
Chile – G97 RP Biobío – Octane Increment



Octane (RON)	97.0	101.0	103.7	107.1	108.9	111.1
Price (US\$/gal)	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347	\$ 2.347

Source: Faro90, 2025

Chile – Vehicular Fleet



Vehicular Fleet: 6,818,833 vehicles that use gasoline

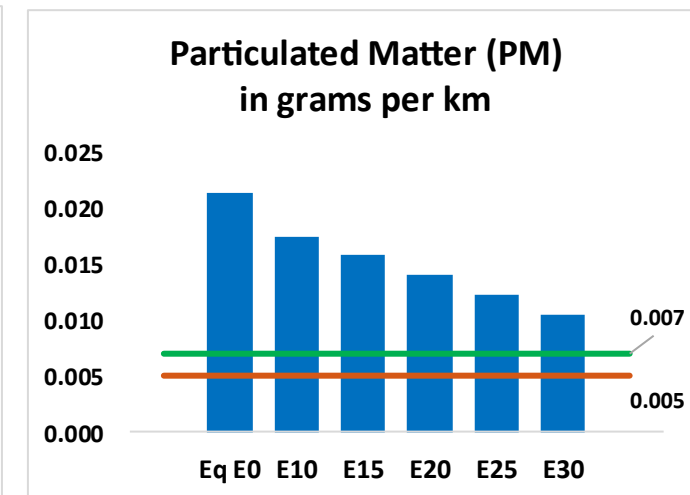
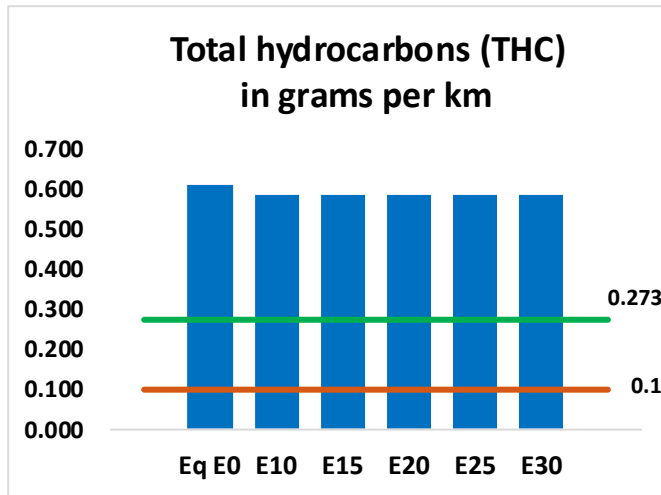
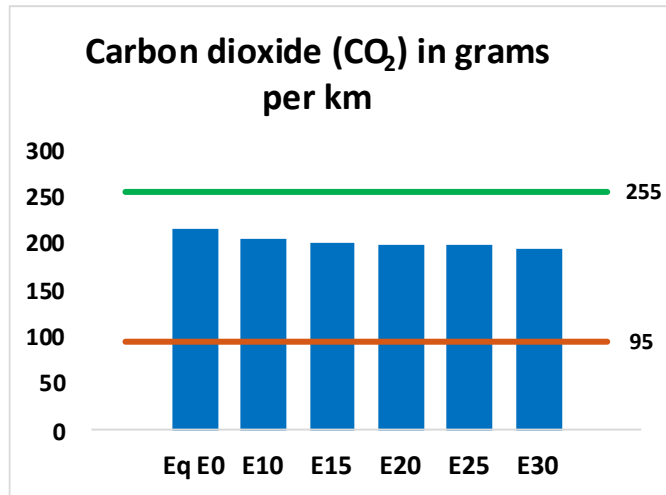
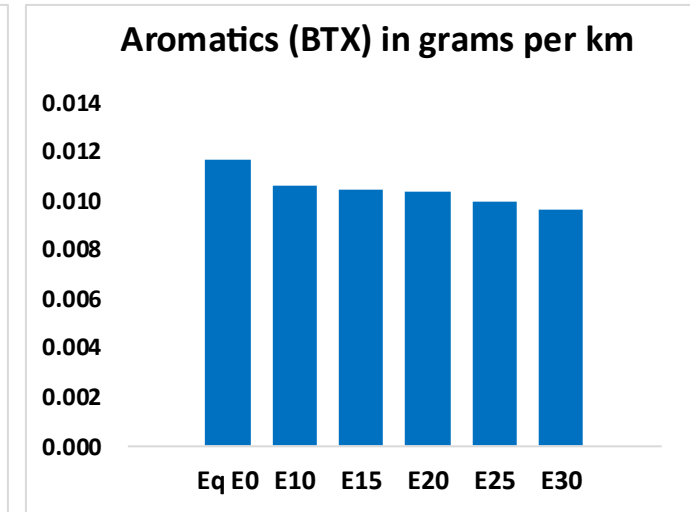
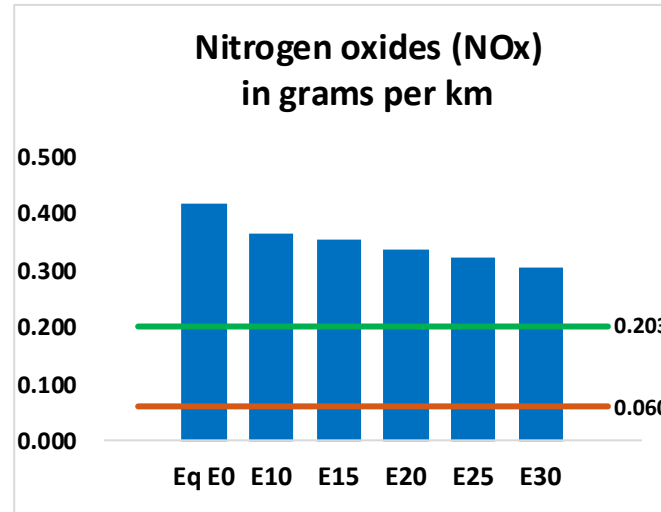
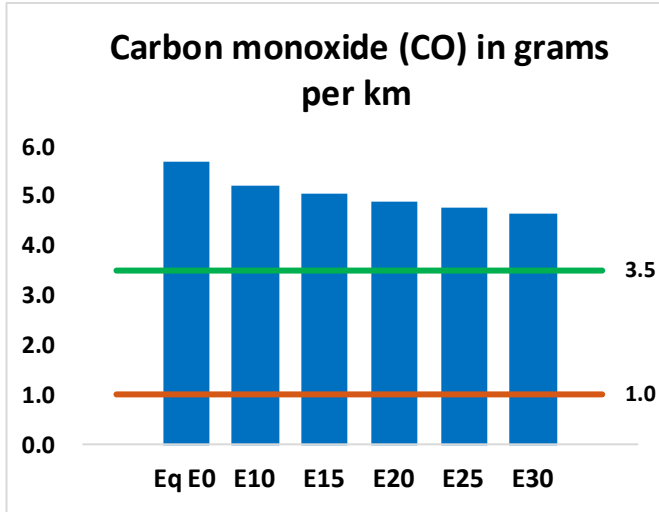
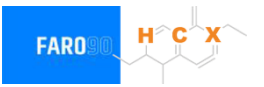
Average Age: 10 year

Source: INE, 2024

Main Technology: Motor Multi-Fuel-Injection, Euro V, PCV Tank

Chile – Gasoline Vehicular Fleet Emissions

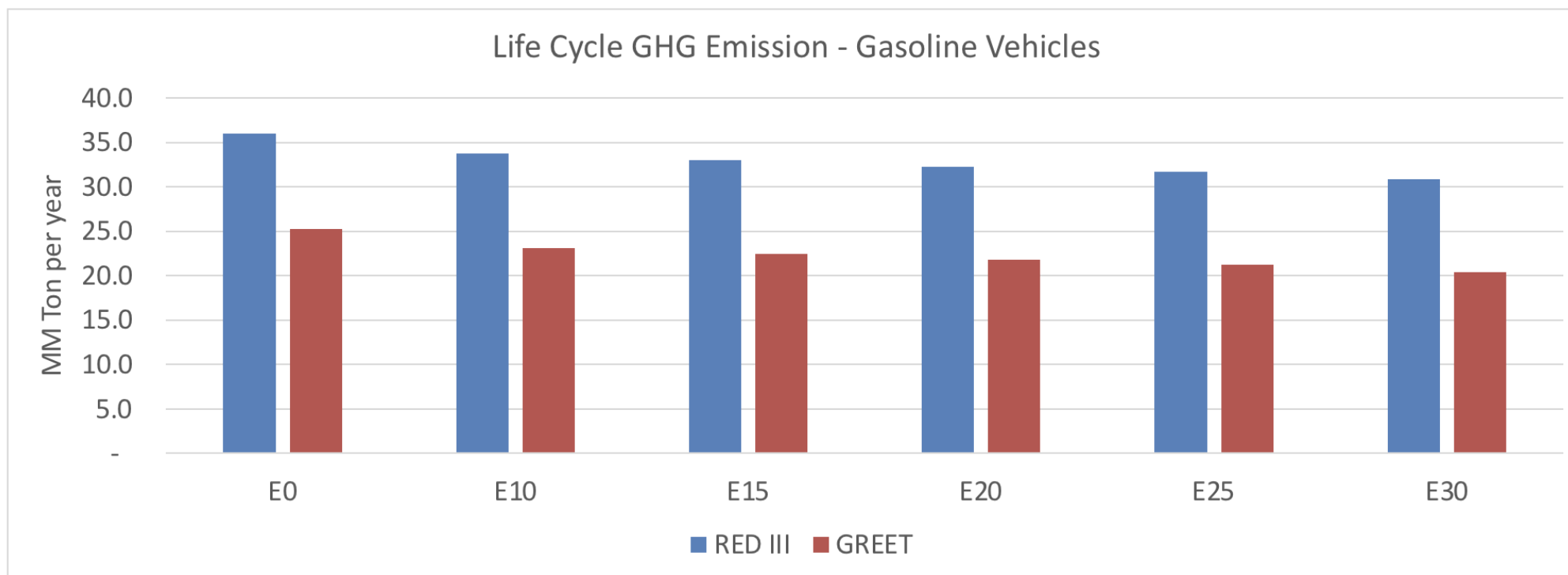
United States
Euro 6



Chile – Gasoline Vehicular Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,022,222	978,171	934,120	904,761	878,317	858,919	831,031	-9%	-14%
CO2	38,858,712	37,859,488	36,915,776	36,173,575	35,806,305	35,604,011	34,947,727	-5%	-8%
VOC	109,513	107,505	105,498	105,105	105,284	105,596	105,399	-4%	-4%
NOx	75,197	69,431	65,538	63,587	60,408	57,933	55,008	-13%	-20%
SOx	446	412	378	350	322	293	262	-15%	-28%
NH3	13,101	12,972	12,842	12,903	13,048	13,150	13,235	-2%	0%
N2O	611	607	605	608	620	627	634	-1%	2%
CH4	24,116	24,116	24,116	24,478	25,009	25,467	25,901	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	687	670	652	645	640	638	631	-5%	-7%
Acetaldehyde	1,363	1,565	2,296	3,497	4,764	5,444	6,432	68%	249%
Formaldehyde	4,080	3,966	4,624	5,429	5,688	6,292	6,850	13%	39%
Benzene	2,100	2,018	1,912	1,882	1,867	1,785	1,727	-9%	-11%
PM2.5	2,240	2,036	1,833	1,654	1,478	1,290	1,094	-18%	-34%
PM10	1,597	1,451	1,306	1,179	1,054	920	779	-18%	-34%

Chile – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	118.1
2035 Target (MMT/year)	90.1
Delta	28.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	8.5%	11.2%	13.9%	16.1%	19.2%
RED II/III	0.0%	6.3%	8.3%	10.3%	12.0%	14.2%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	7.6%	10.1%	12.5%	14.5%	17.3%
RED II/III	0.0%	8.1%	10.7%	13.3%	15.4%	18.3%

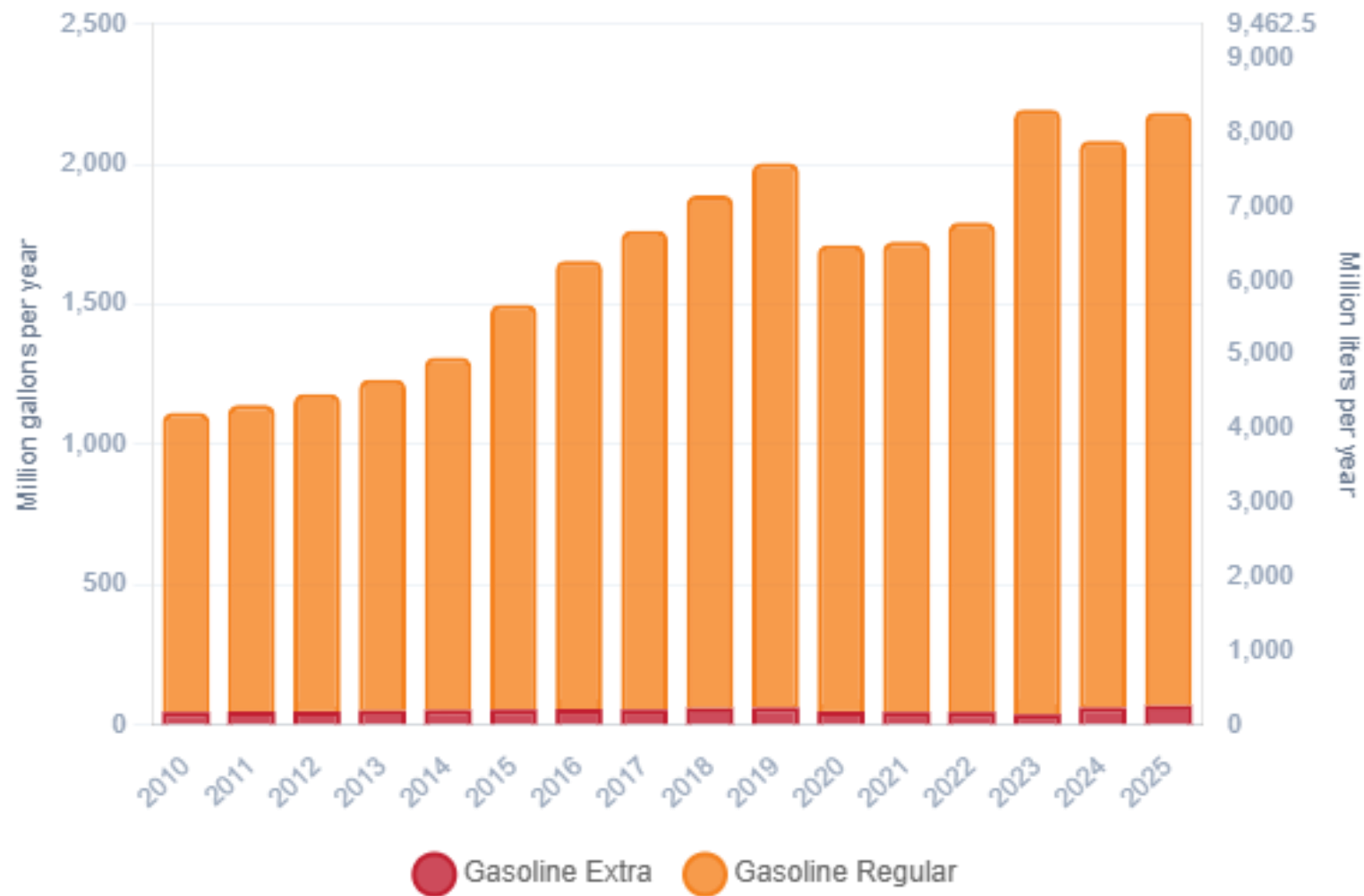
Ethanol Blending in Gasoline - Colombia



In 2025, gasoline consumption in Colombia was 2,184 million gallons (8,270 million liters). Market share was 3% for Extra gasoline E10 (RON 97 – AKI 94) and remaining 97% for Regular gasoline (regular) E10 (RON 89 – AKI 84). Local production supplies 43% of total demand. United States is the main source of gasoline imports.

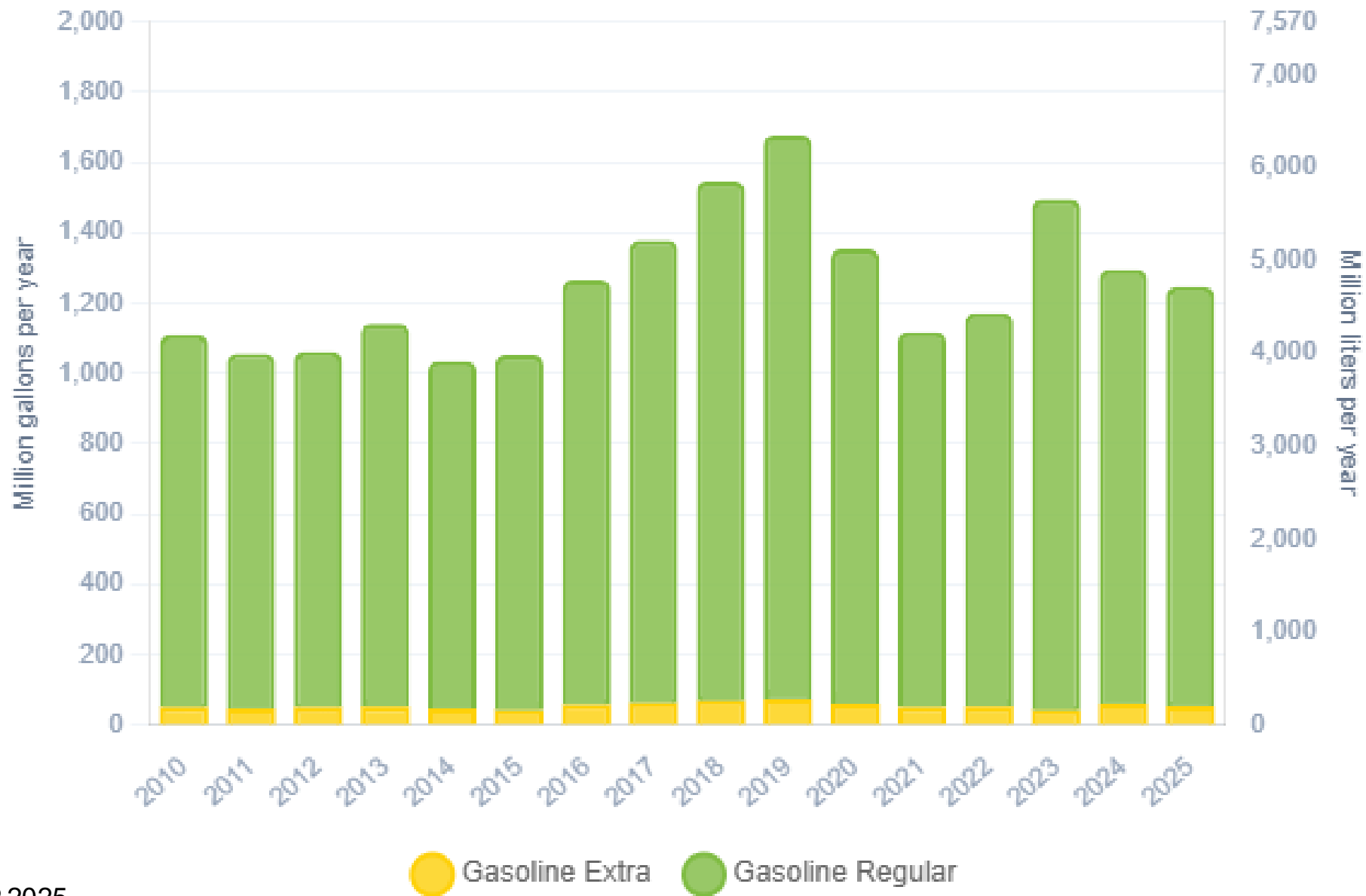
Ethanol blends are obligatory in cities with more than 500,000 habitants. Resolution 40444 of 2023 authorizes the blending of up to 10% ethanol in the country and allows for lower percentages in particular cases. E10 is the predominant grade in the country except on the border with Venezuela. The ethanol supply comes from imports from the United States and from local production.

Gasoline Demand in Colombia



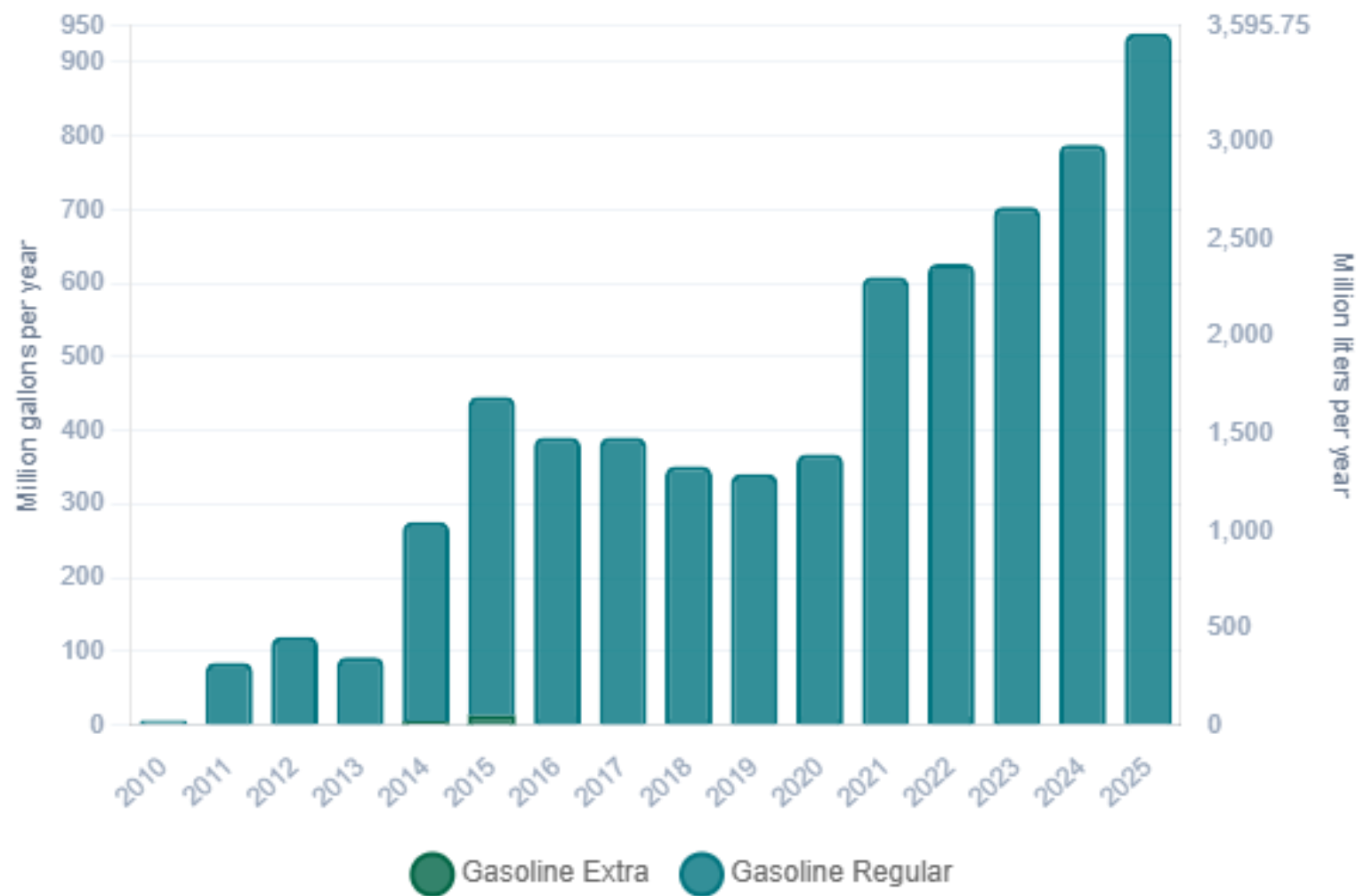
Source: ACP,2025

Gasoline Production in Colombia



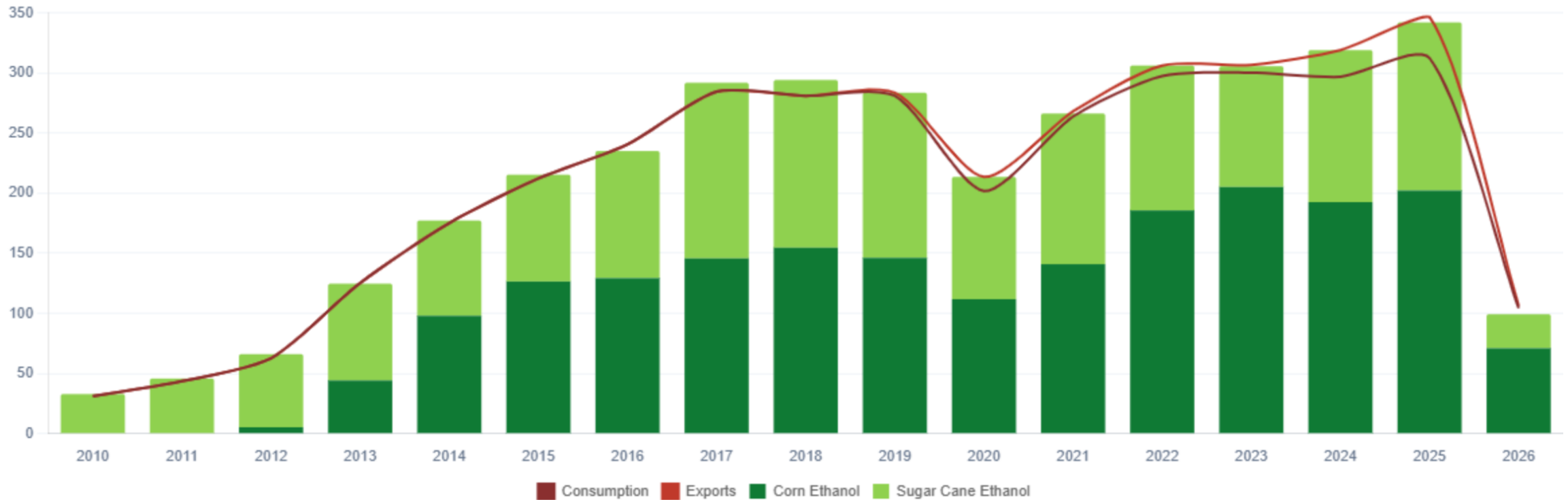
Source: ACP,2025

Gasoline Imports in Colombia



Source: ACP,2025

Ethanol Balance in Colombia



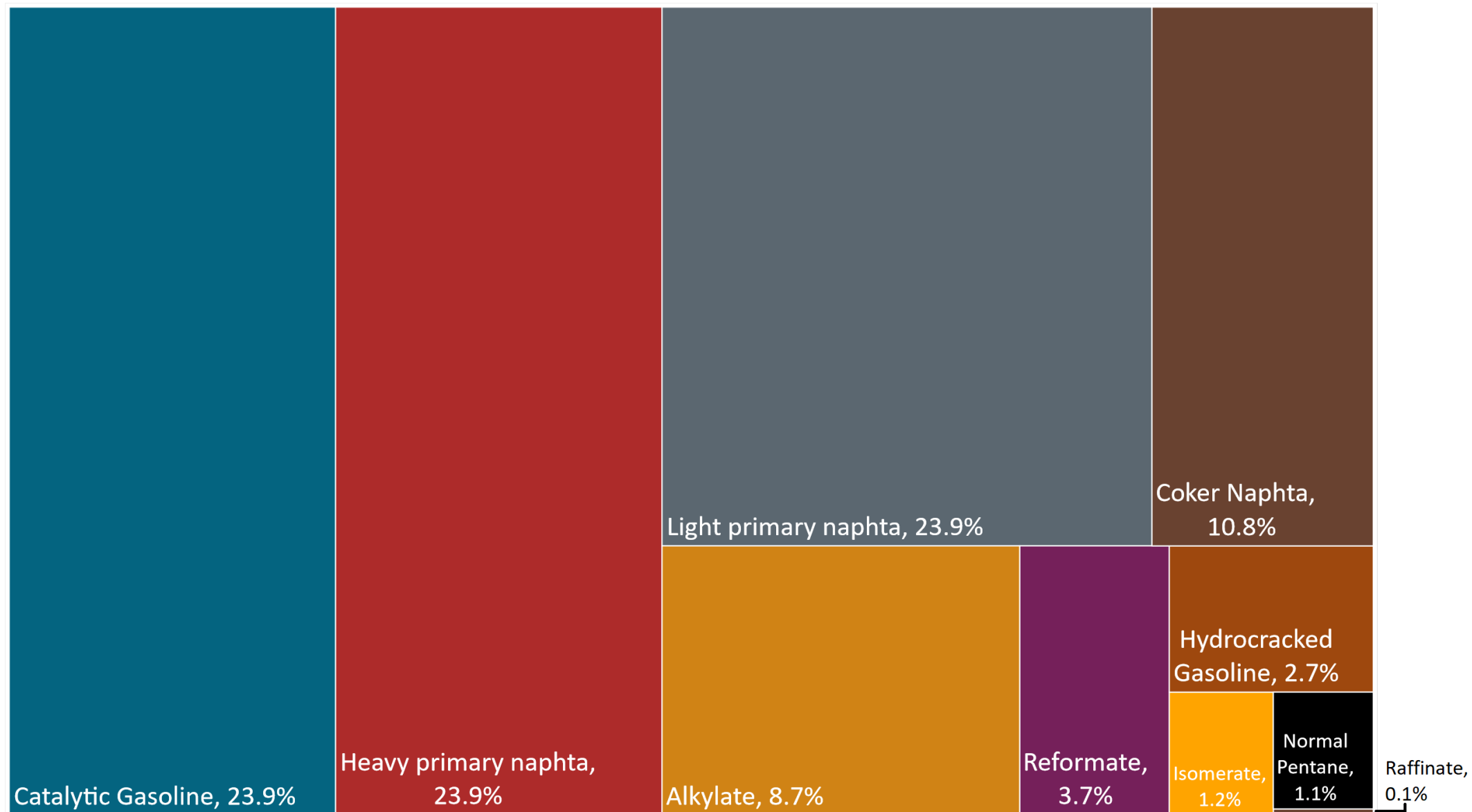
Source: Asocaña - Balance Azucarero Colombiano, 2024; EIA, 2024

Gasoline Quality in Colombia

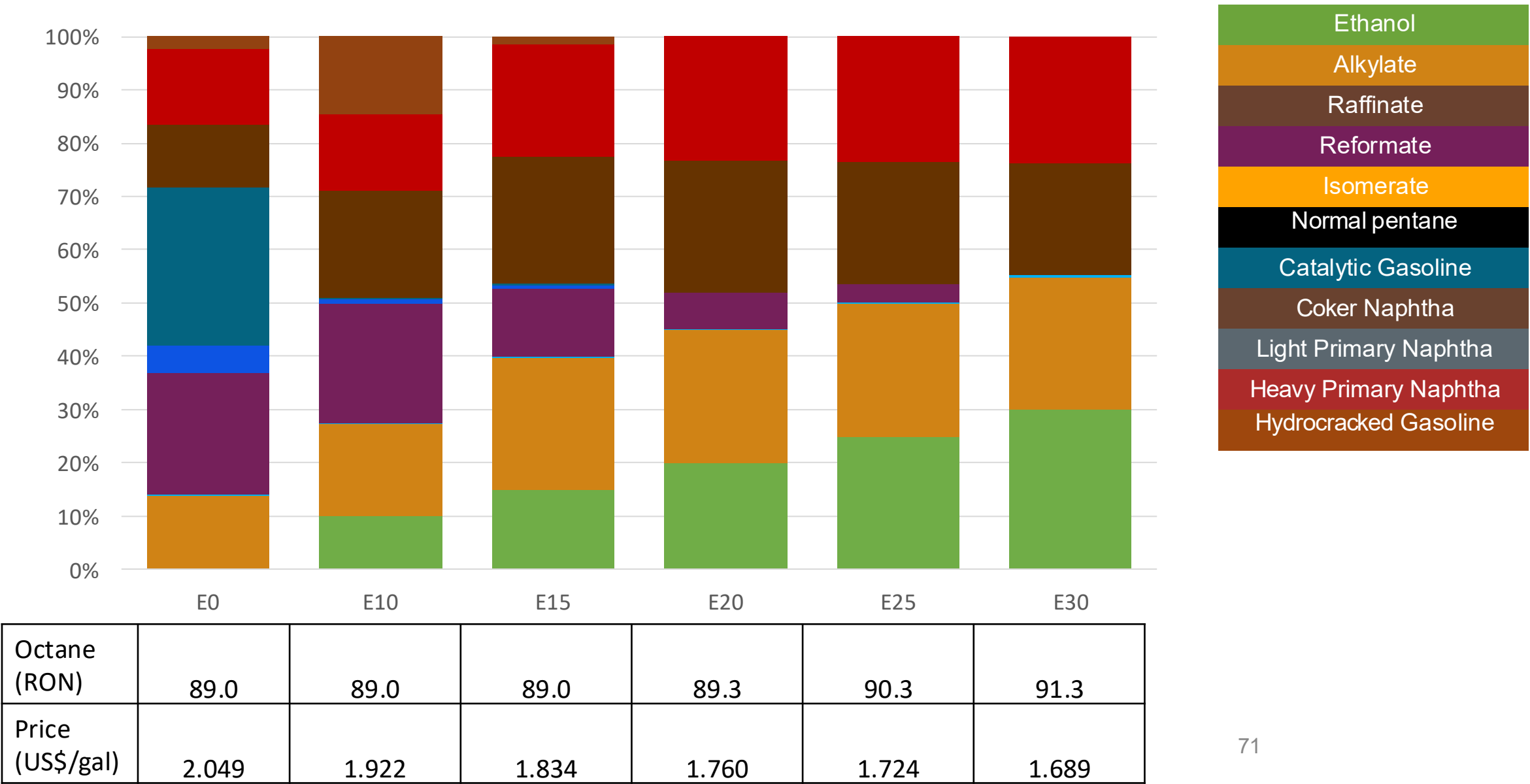
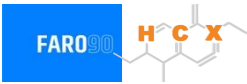
Name	Resolution 40103 de 2021				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	07/01/2023				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Extra	Gasoline Regular E10	Gasoline Extra E10	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1,0 %v/v	< 2,0 %v/v	< 0,9 %v/v	< 1,8 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 28 %v/v	< 35 %v/v	< 25 %v/v	< 31,5 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	-	-	-	-	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 84	> 93	> 89	> 97	> 95	> 95	> 98	> 98
MON	-	-	-	-	> 85	> 88	> 85	> 88
AKI	> 81	> 91	> 84	> 94				
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content			< 3,7 %m/m	< 3,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<> 9,5 - 10,5 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 65 kPa	<> 65 kPa	<> 65 kPa	<> 65 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: Ministerio de Energía y Minas, 2021

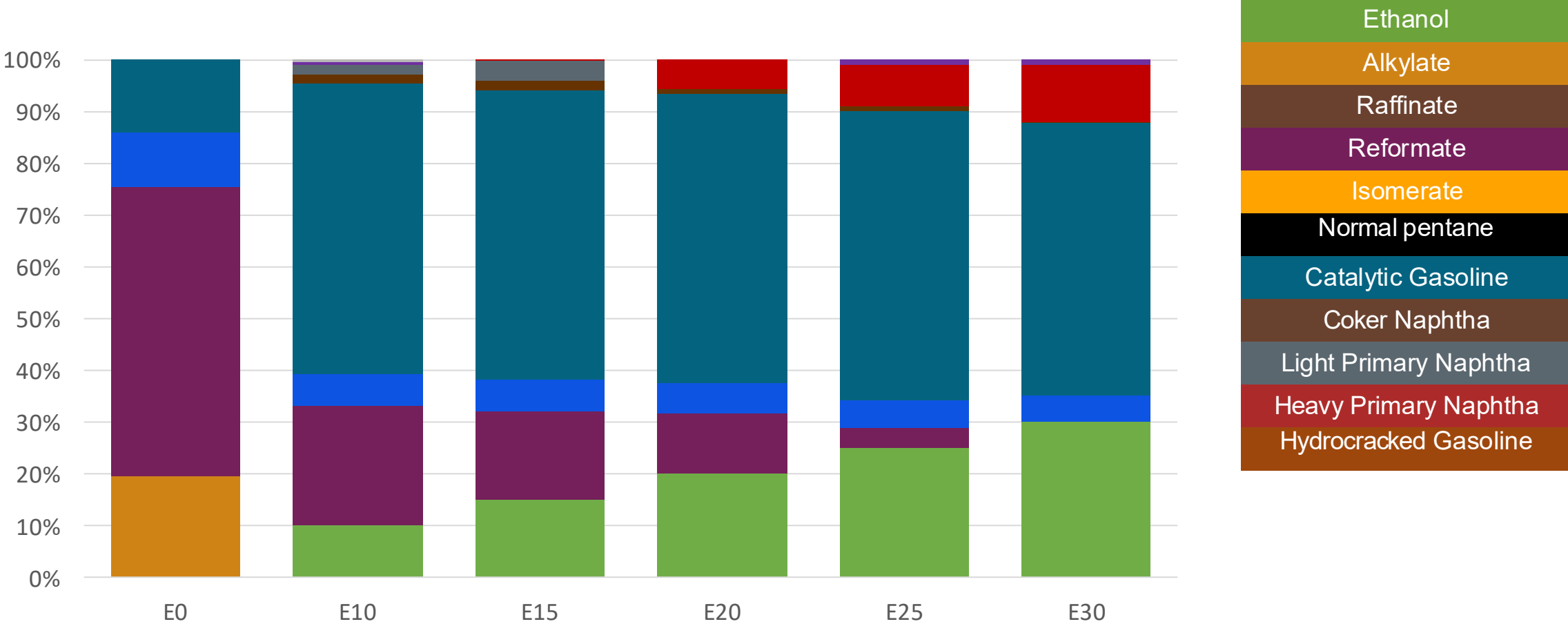
Available Blending Components



Colombia – Regular Gasoline – Constant Octane

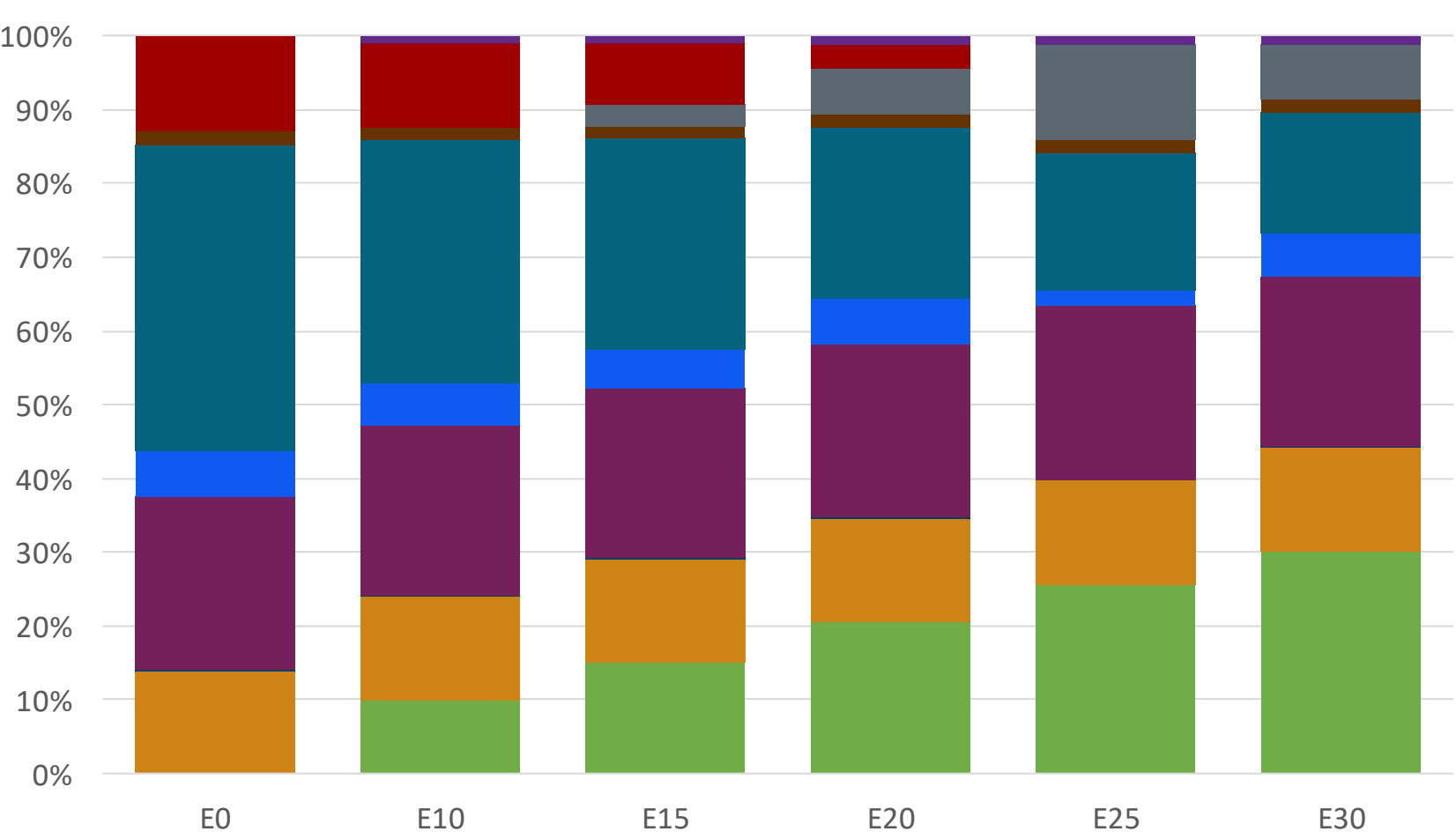


Colombia – Extra Gasoline – Constant Octane



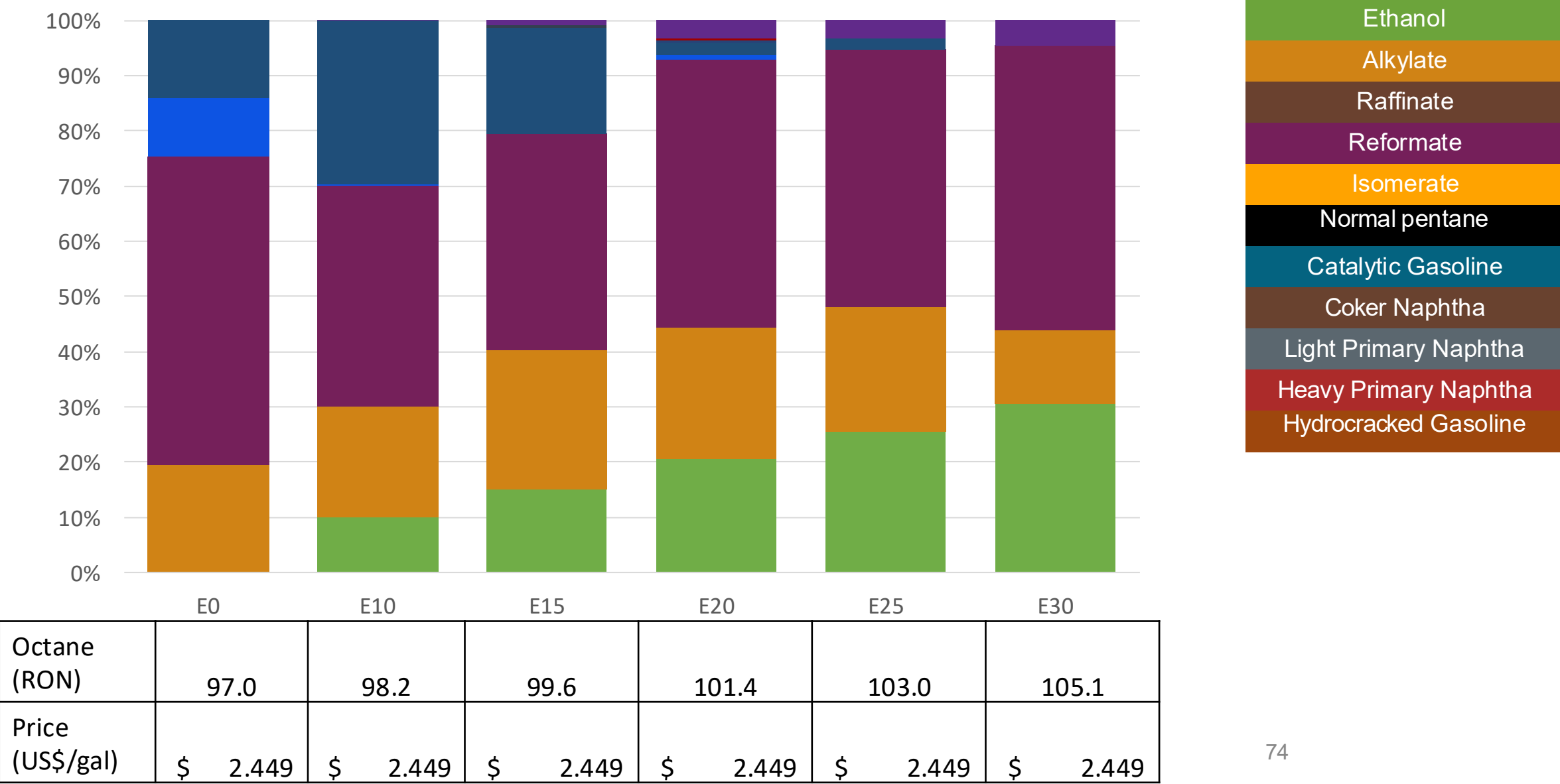
Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US\$/gal)	2.449	2.240	2.142	2.053	1.974	1.890

Colombia – Regular Gasoline – Increased Octane

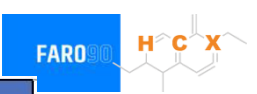


Octane (RON)	90.1	93.5	95.2	97.2	97.9	100.7
Price (US\$/gal)	2.117	2.117	2.117	2.117	2.117	2.117

Colombia – Extra Gasoline – Increased Octane



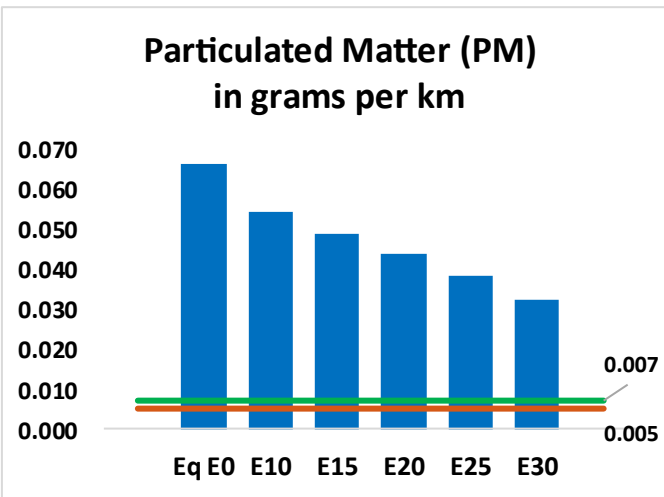
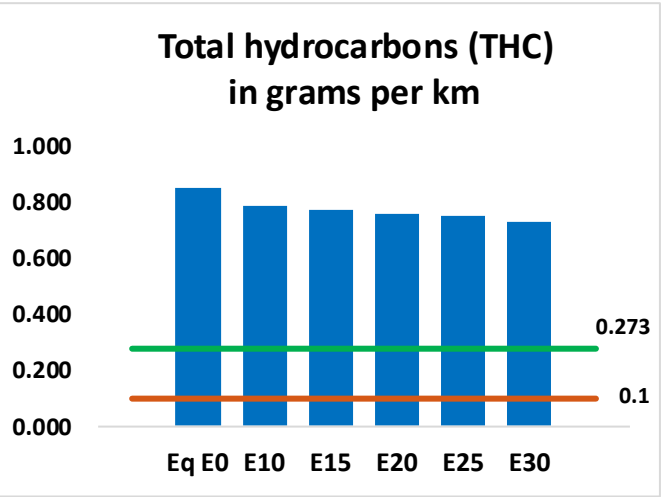
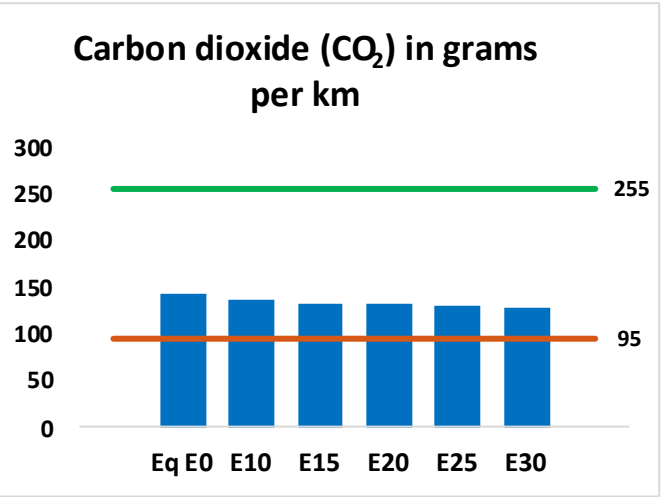
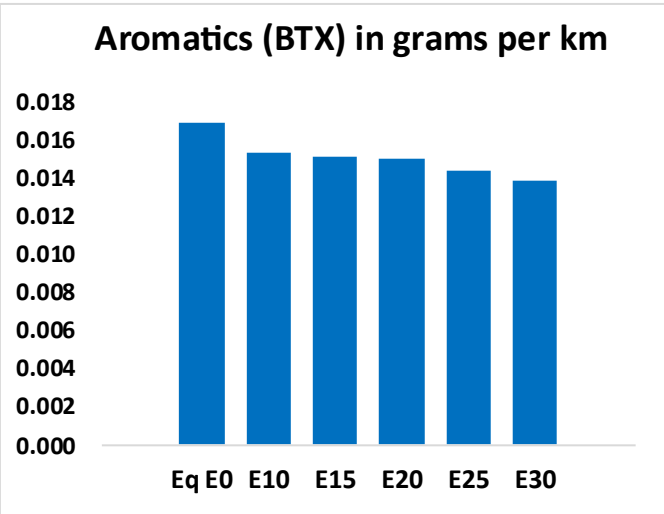
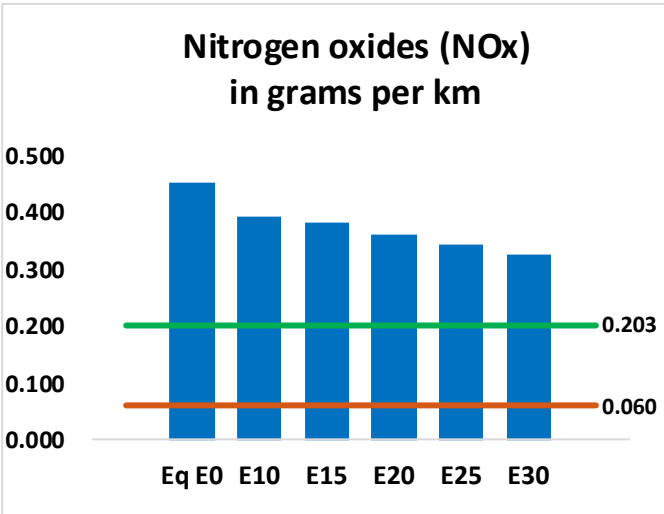
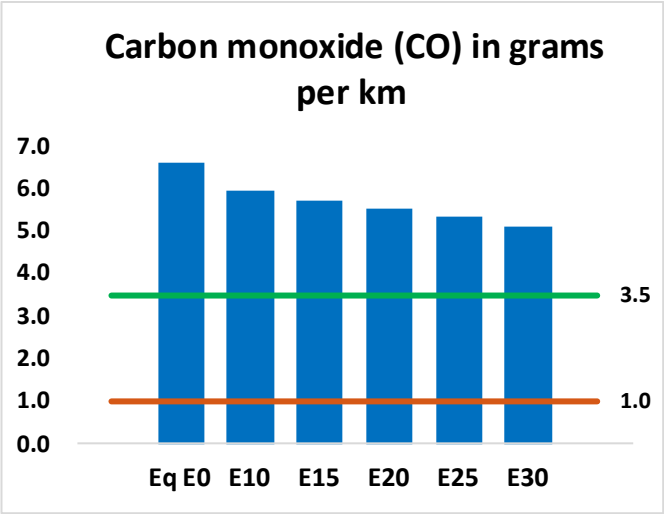
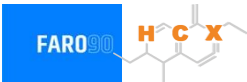
Gasoline Vehicle Fleet - Colombia



Vehicle Type	Engine	Exhaust	Recovery	Age	Number of Vehicles
Light automobiles	Electric	-	-	Any	19,724
	Multi FI-Pt	Hybrid	PVC Tank	Any	67,899
	Multi FI-Pt	Euro VI		<4 years, < 80 tkm	396440
		Euro V		4-8 yeras, 80-160 tkm	845596
				> 8 years, > 160 tkm	472860
		Euro IV		> 8 years, > 160 tkm	892921
		Euro III		> 8 years, > 160 tkm	1037888
	multi	3-Ways		> 8 years, > 160 tkm	1460739
	single	3-Ways		> 8 years, > 160 tkm	573983
	carbon	3-Ways		> 8 years, > 160 tkm	516661
Medium automobiles	Multi FI-Pt	Euro VI	PCV Tank	<4 years, < 80 tkm	308621
		Euro V		4-8 years, 80-160 tkm	313792
				> 8 years, > 160 tkm	206075
		Euro IV		> 8 years, > 160 tkm	229333
	Euro III	> 8 years, > 160 tkm		206403	
	multi	3-Ways		> 8 years, > 160 tkm	61447
	single	3-Ways		> 8 years, > 160 tkm	537
	carbon	3-Ways		> 8 years, > 160 tkm	5
Small Engines	Electric	-	-	Any	3,139
	Multi FI-Pt	Hybrid	PCV	Any	0
	FI 4-ciclos	Euro V		<4 years, < 80 tkm	2507101
		Euro IV		4-8 years, 80-160 tkm	1382788
				> 8 years, > 160 tkm	1816667
		Euro III		> 8 years, > 160 tkm	626370
		Euro II		> 8 years, > 160 tkm	6119537
		none		> 8 years, > 160 tkm	701662
				> 8 years, > 160 tkm	385219
Heavy	Electric	-	-	Any	2,064
	Multi FI-Pt	Hybrid	PCV	Any	1,421
	FI	Euro VI		<4 years, < 80 tkm	55876
		Euro V		4-8 yeras, 80-160 tkm	68512
				> 8 years, > 160 tkm	39235
		Euro IV		> 8 years, > 160 tkm	38784
		Euro III		> 8 years, > 160 tkm	44605
	3-Ways	> 8 years, > 160 tkm	24477		

Colombia – Vehicular Emissions

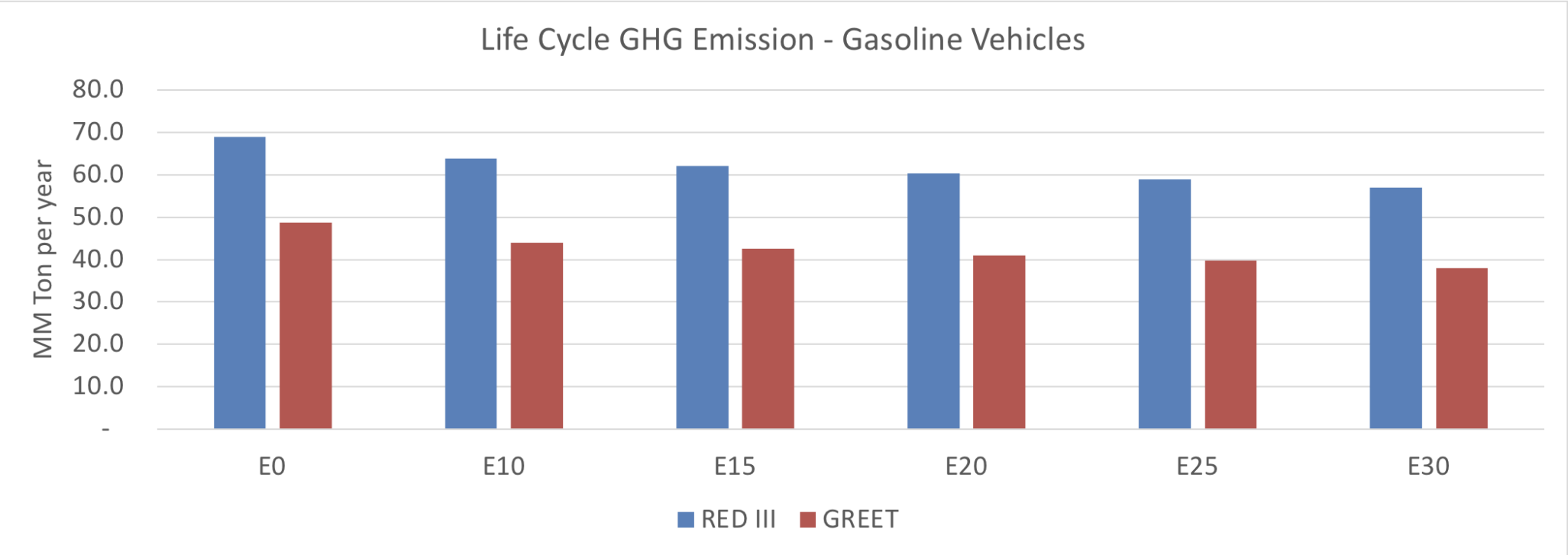
United States
Euro 6



Colombia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,805,179	2,668,777	2,532,376	2,436,063	2,346,260	2,279,353	2,182,488	-10%	-16%
CO2	60,851,816	59,287,055	57,809,226	56,646,956	56,071,819	55,804,909	54,776,264	-5%	-8%
VOC	361,584	349,020	336,456	329,133	323,132	318,931	311,867	-7%	-11%
NOx	193,404	178,576	168,261	162,933	154,431	147,729	139,862	-13%	-20%
SOx	695	642	590	545	503	456	408	-15%	-28%
NH3	25,965	25,708	25,450	25,571	25,859	26,061	26,229	-2%	0%
N2O	1,547	1,539	1,533	1,540	1,572	1,589	1,607	-1%	2%
CH4	92,034	92,034	92,034	93,415	95,439	97,188	98,845	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	3,677	3,541	3,404	3,321	3,251	3,200	3,119	-7%	-12%
Acetaldehyde	9,309	10,684	15,676	23,873	32,525	37,166	43,915	68%	249%
Formaldehyde	37,018	35,990	41,954	49,263	51,610	57,095	62,154	13%	39%
Benzene	7,198	6,917	6,555	6,451	6,397	6,118	5,921	-9%	-11%
PM2.5	4,333	3,939	3,545	3,199	2,860	2,496	2,115	-18%	-34%
PM10	24,028	21,843	19,659	17,737	15,856	13,840	11,730	-18%	-34%

Colombia – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	217.6
2035 Target (MMT/year)	125.5
Delta	92.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.6%	12.5%	15.8%	18.4%	21.9%
RED II/III	0.0%	7.5%	9.9%	12.5%	14.7%	17.4%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	5.1%	6.6%	8.3%	9.7%	11.6%
RED II/III	0.0%	5.6%	7.4%	9.4%	11.0%	13.0%

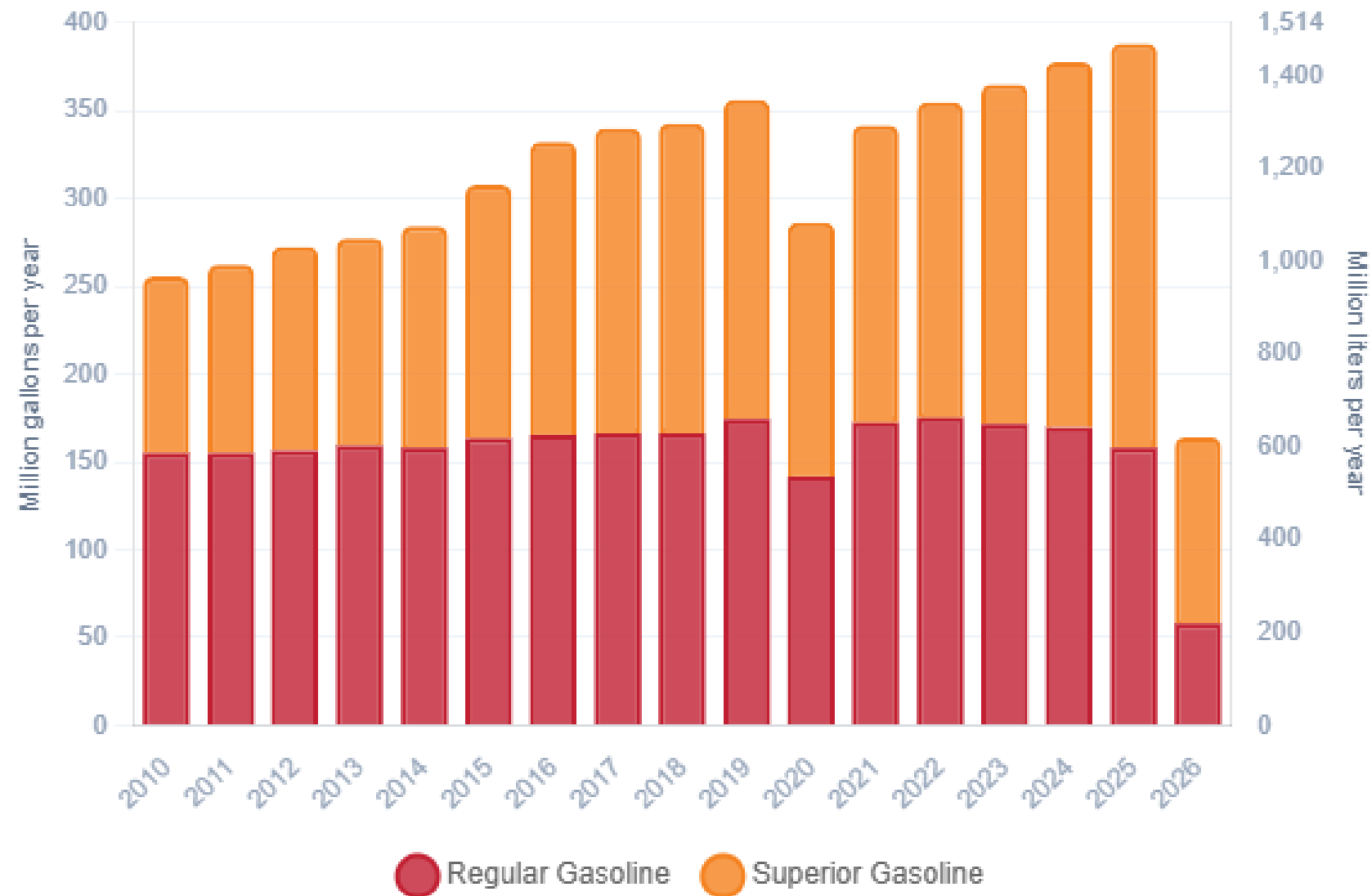
Ethanol Blending in Gasoline - Costa Rica



In 2025, gasoline demand was 387 million gallons (1468 million liters). Costa Rica has two grades of gasoline RON 95 and RON 91, with a market share of 41% for RON 91 and 59% for RON 95. The Limon refinery stopped production in 2012. Gasoline supply is done with imports mainly from the United States and some from Europe.

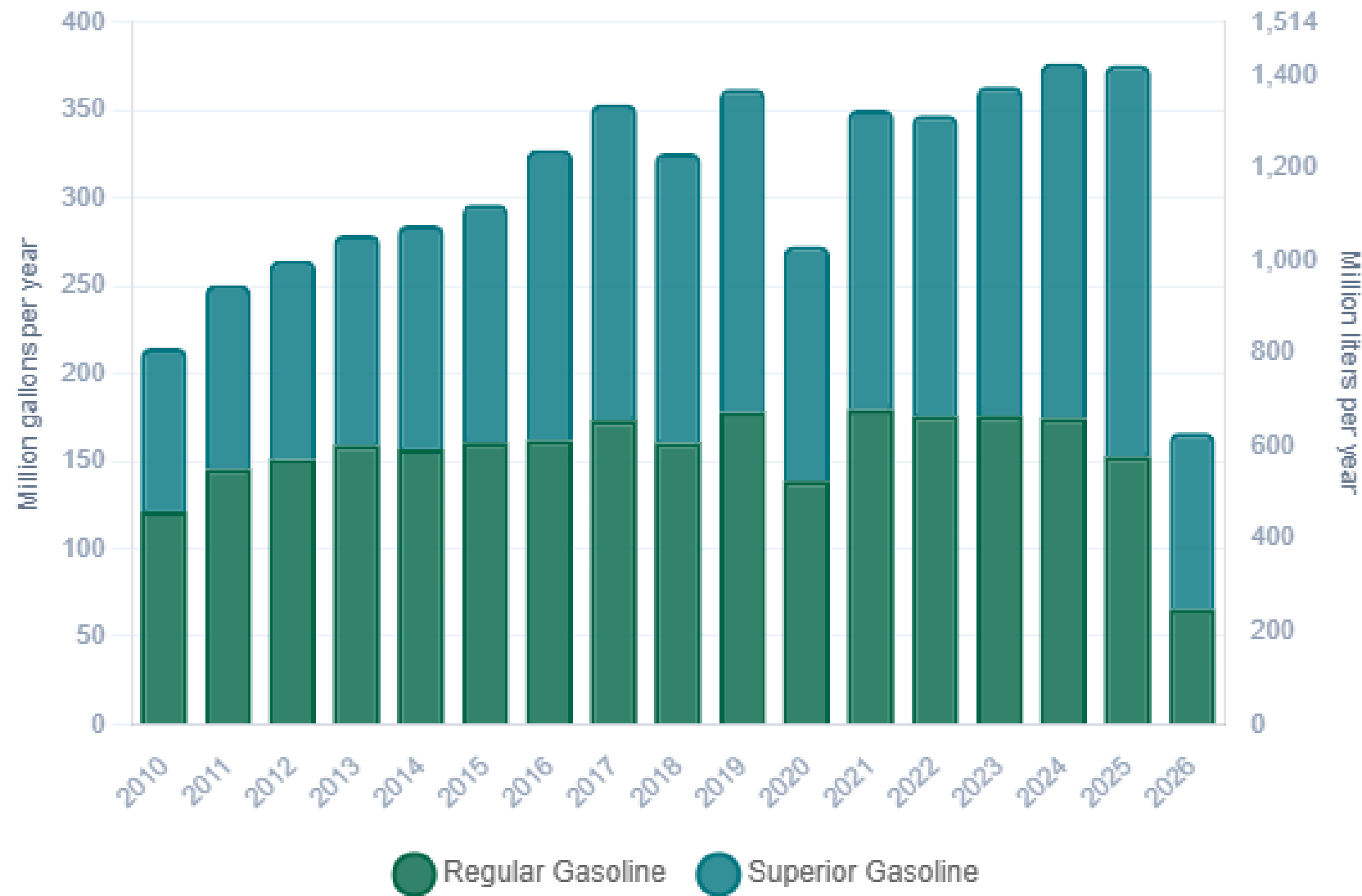
Gasoline blends with up to 10% v/v of ethanol are allowed (INTE E1:2019). However, this mandate is not yet reflected on the gasoline market. Current ethanol consumption is less than 0.2% because there is no stakeholder's consensus for implementation. In 2025, ethanol production was 7.5 million gallons (28.2 million liters) and ethanol exports mainly to Europe was almost 6 million gallons (22.5 million liters). Introduction of E10 is expected by the Government in 2026.

Gasoline Demand in Costa Rica



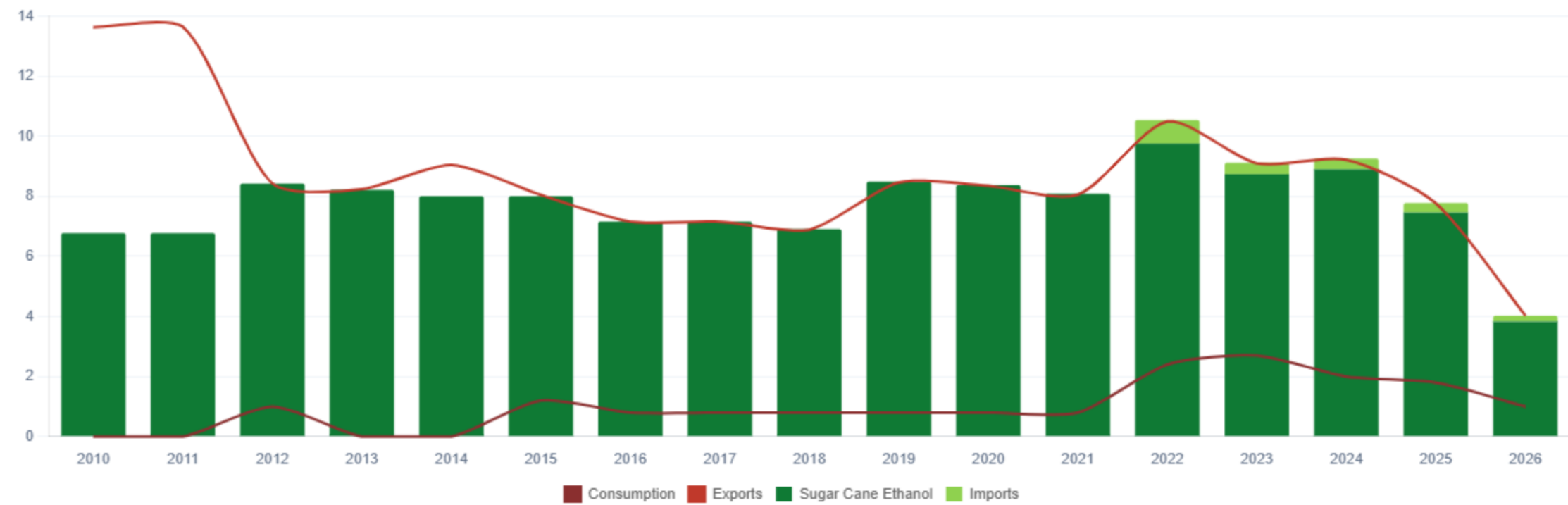
Source: Sepse

Gasoline Imports in Costa Rica



Source: Sepse

Ethanol Balance in Costa Rica



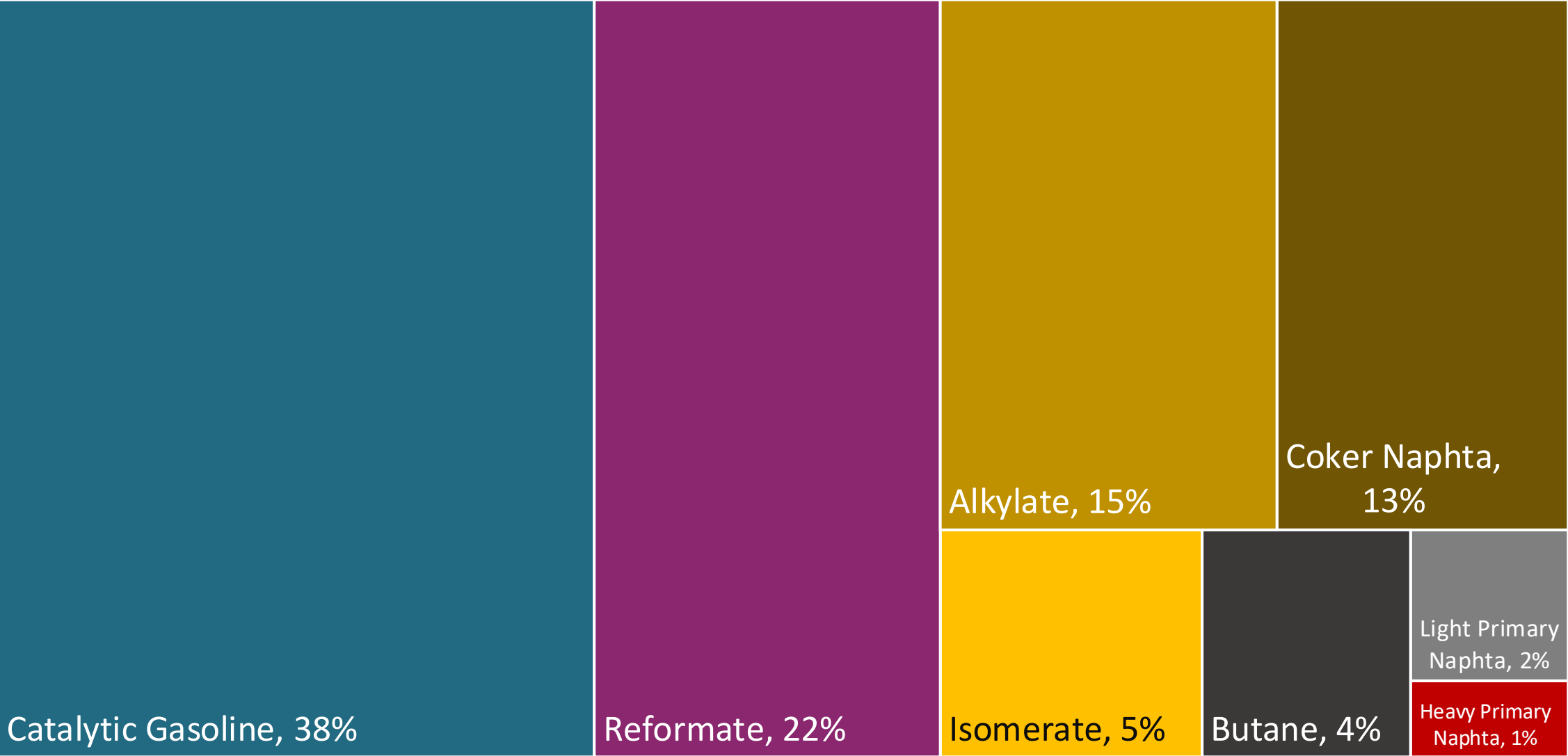
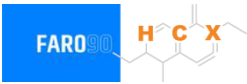
Source: Sepse

Gasoline Quality in Costa Rica

Name	INTE E1:2019		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	RON 91	RON 95	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1,5% v/v	< 1,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 35% v/v	< 35% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18% v/v	< 18% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	> 79	> 83	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% m/m (3,7% m/m if ethanol is added)	2,7% m/m (3,7% m/m if ethanol is added)	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10% v/v	< 10% v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa (< 76 kPa if ethanol is added)	< 69 kPa (< 76 kPa if ethanol is added)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-		-	-	-	-
Ethers 5 or more C Atoms	-		Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

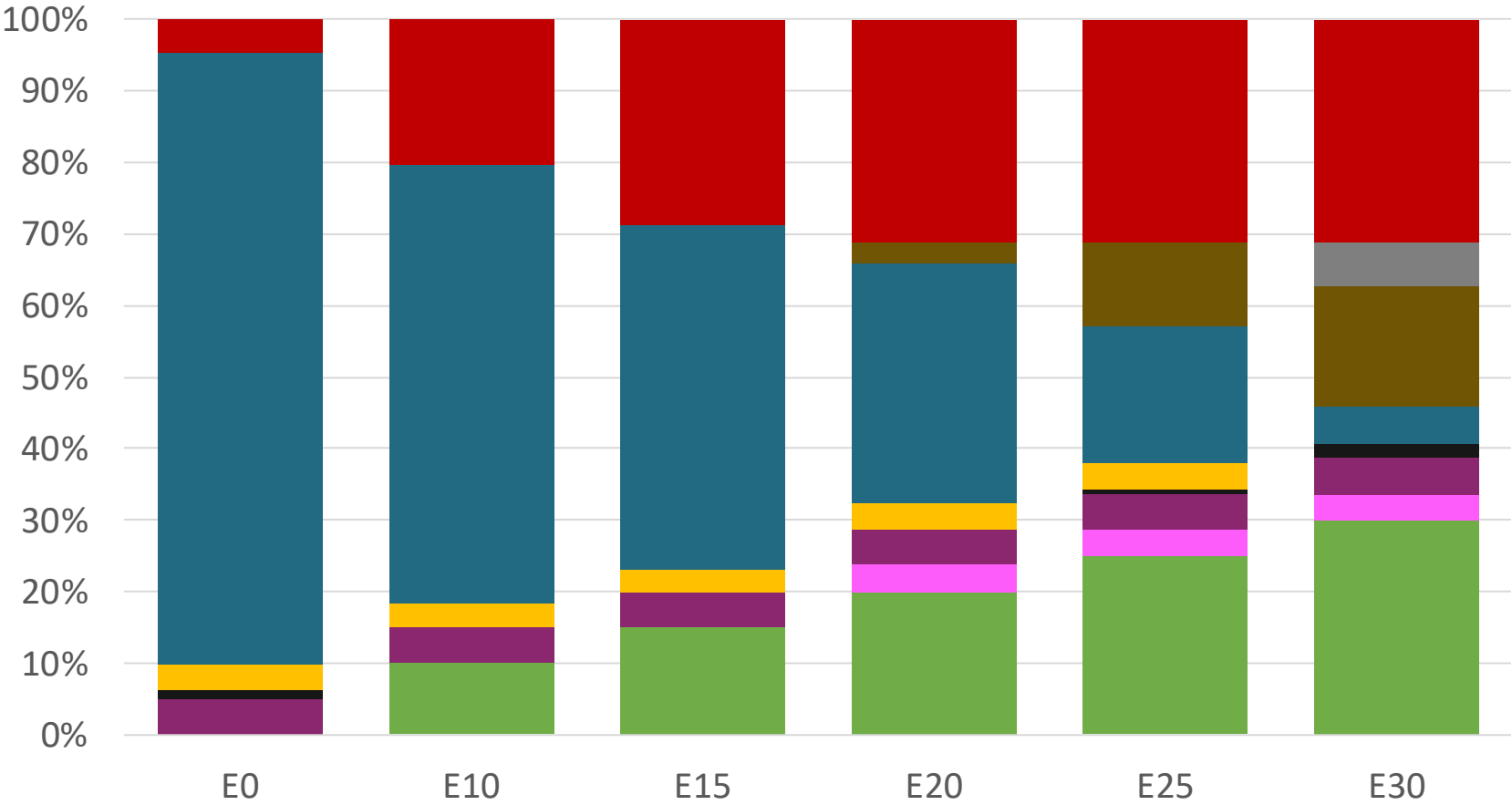
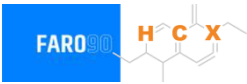
Source: INTECO

Blending Components - Costa Rica



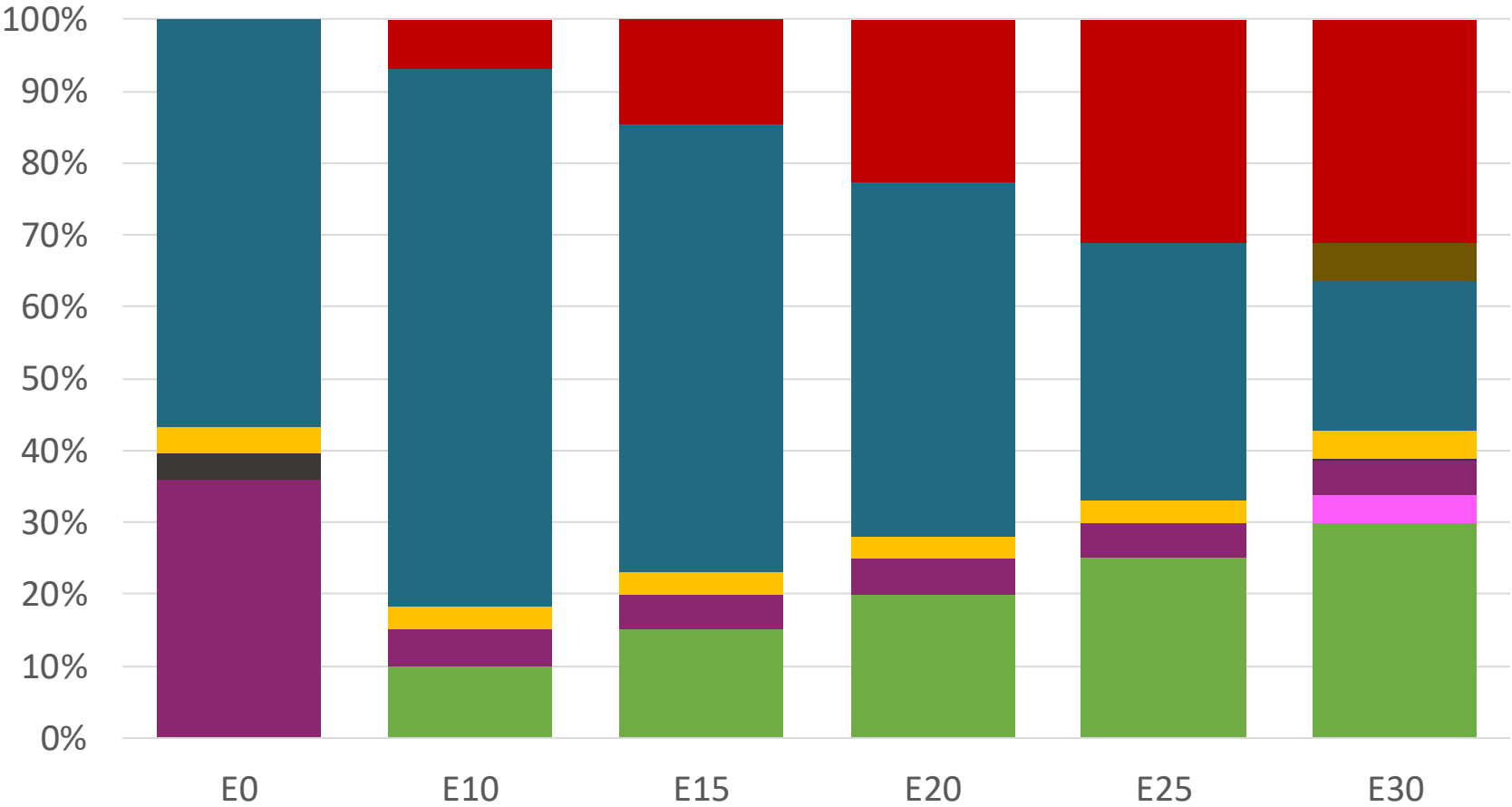
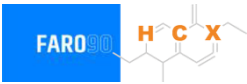
Source: Faro90

Costa Rica – Regular – Constant Octane



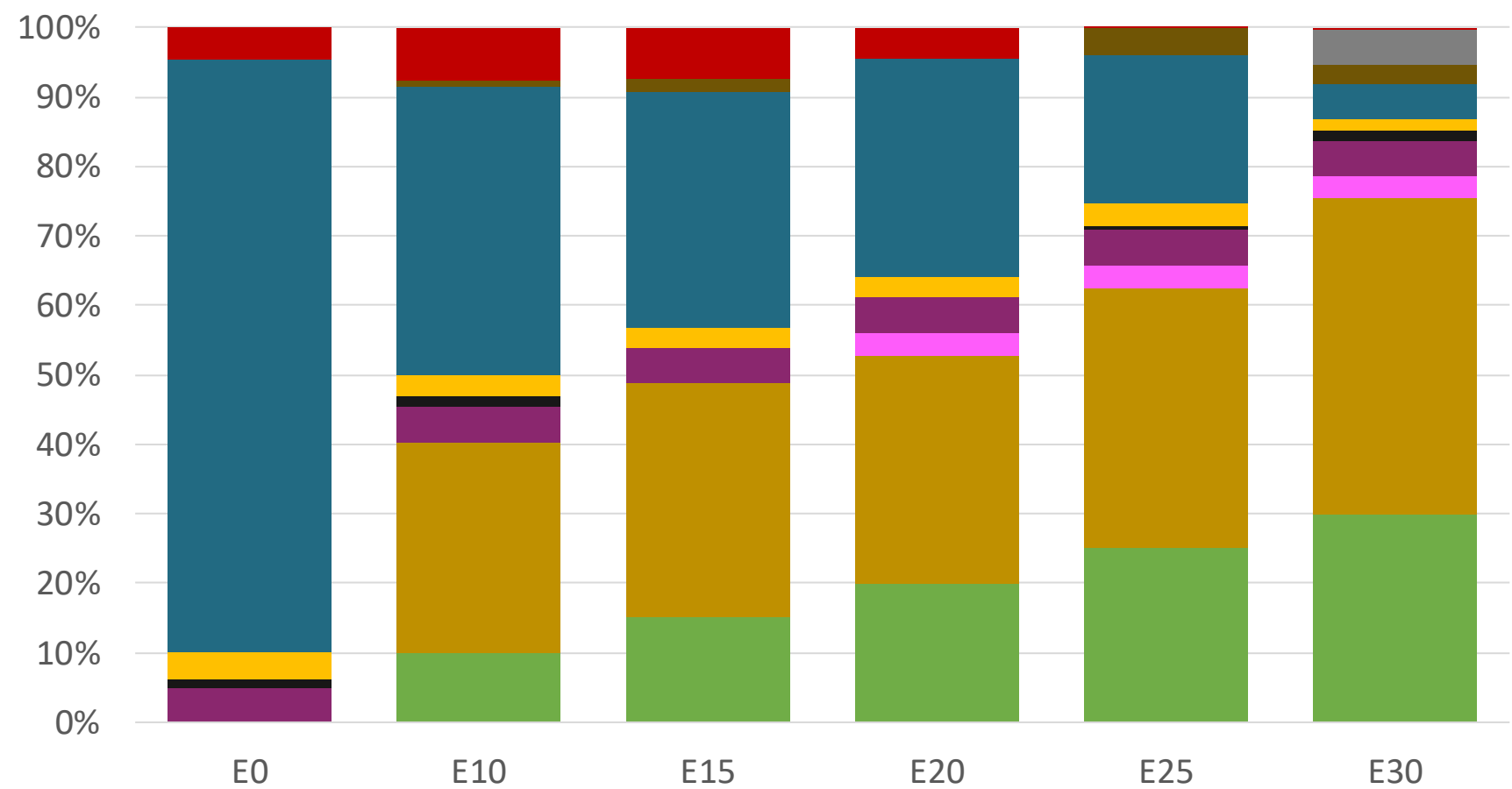
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.076	\$ 1.817	\$ 1.673	\$ 1.526	\$ 1.375	\$ 1.257

Costa Rica – Premium – Constant Octane



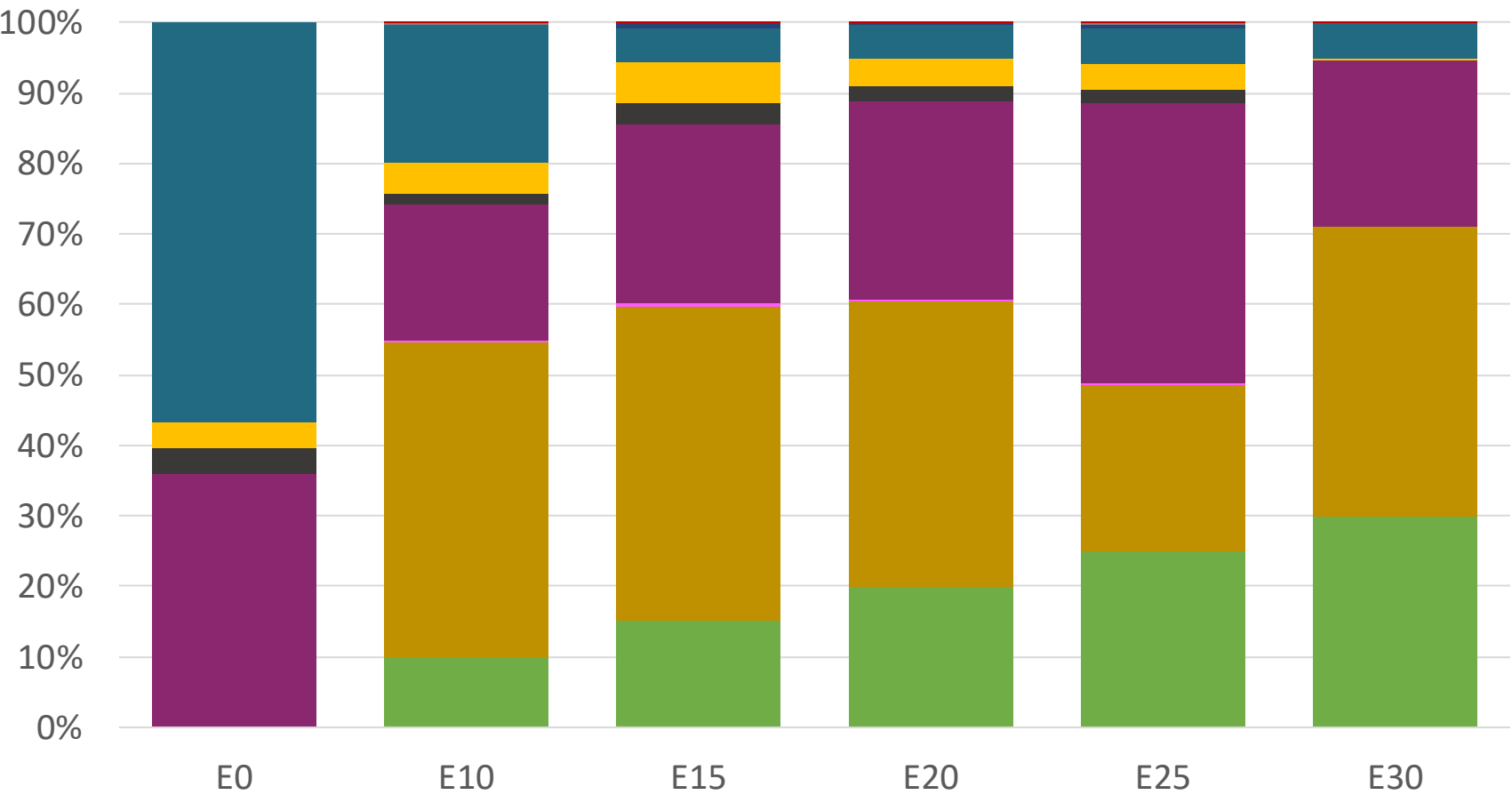
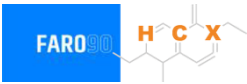
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.326	\$ 2.023	\$ 1.888	\$ 1.746	\$ 1.597	\$ 1.448

Costa Rica – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076

Costa Rica – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326

Gasoline Vehicular Fleet in Costa Rica

Type of Vehicle	Engine	Exhaust	Recovery	Age	Number of vehicles
Small automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	27,697
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	97,126
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	683,789
		Euro III		> 8 years, > 160 tkm	452,075
		Euro IV		> 8 years, > 160 tkm	302,621
		Euro V / Euro 6		> 8 years, > 160 tkm	184,548
				4-8 years, 80-160 mkm	99,330
		< 4años, < 80 mkm		51,317	
	Electric			Any	25,675
	Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm
Single FI-Pt		3-Ways	> 8 years, > 160 tkm		38,418
Multi FI-Pt		3-Ways	> 8 years, > 160 tkm		84,705
		Euro III	> 8 years, > 160 tkm		36,200
		Euro IV	> 8 years, > 160 tkm		20,968
		Euro V / Euro 6	> 8 years, > 160 tkm		15,637
			4-8 years, 80-160 mkm		15,172
		< 4años, < 80 mkm	21,595		
Electric			Any	3,206	
Small Engines		FI 4-cycles	None	PCV	> 8 years, > 160 tkm
	Euro II		> 8 years, > 160 tkm		23,591
	Euro III		> 8 years, > 160 tkm		70,066
	Euro IV		> 8 years, > 160 tkm		79,996
			> 8 years, > 160 tkm		100,035
	Euro V		4-8 years, 80-160 mkm		210,088
		< 4años, < 80 mkm	442,940		
	Electric			Any	2,954
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	67,455
	FI	3-Ways		> 8 years, > 160 tkm	13,399
		Euro III		> 8 years, > 160 tkm	33,310
		Euro IV		> 8 years, > 160 tkm	2,265
		Euro V / Euro 6		> 8 years, > 160 tkm	6,267
				4-8 years, 80-160 mkm	2,087
		< 4años, < 80 mkm		15,991	
	Electric			Any	439

Vehicular Fleet: **2,276,347**

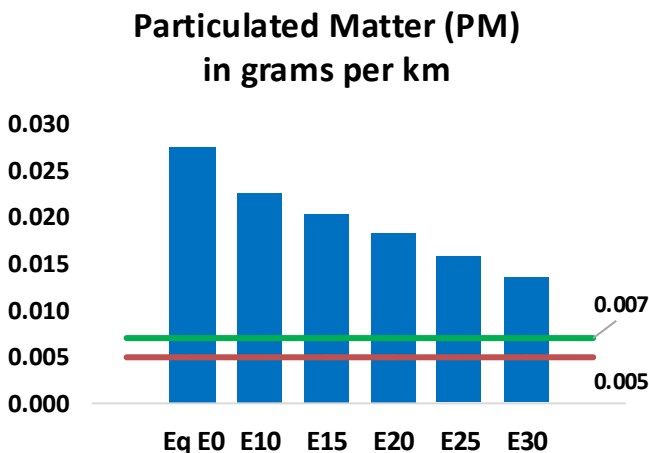
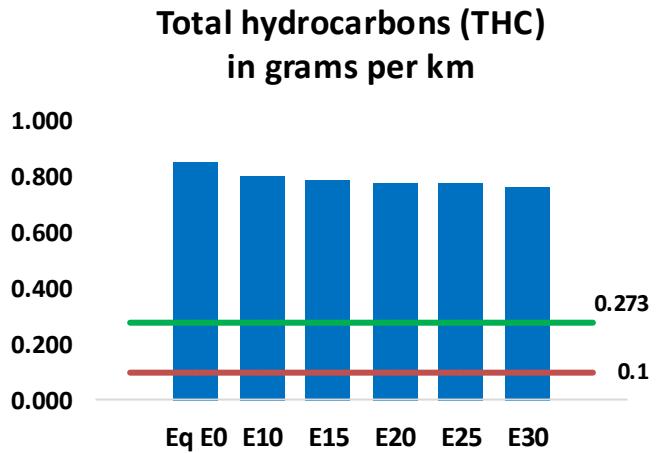
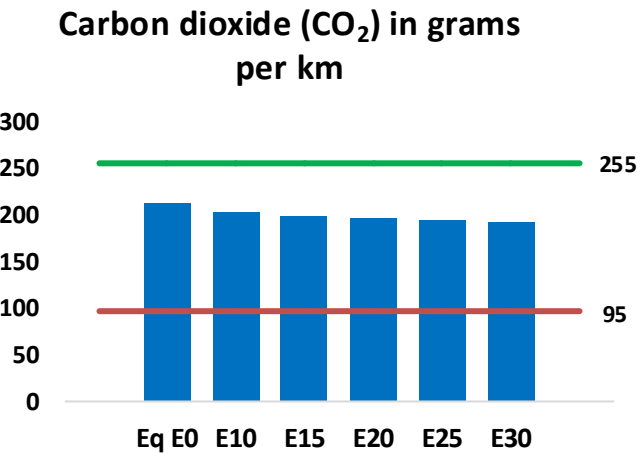
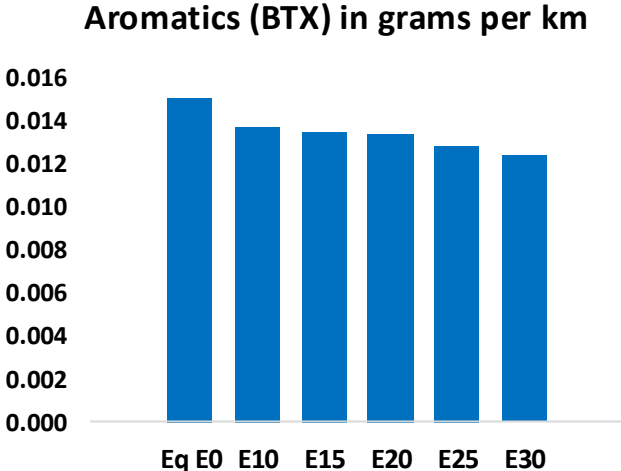
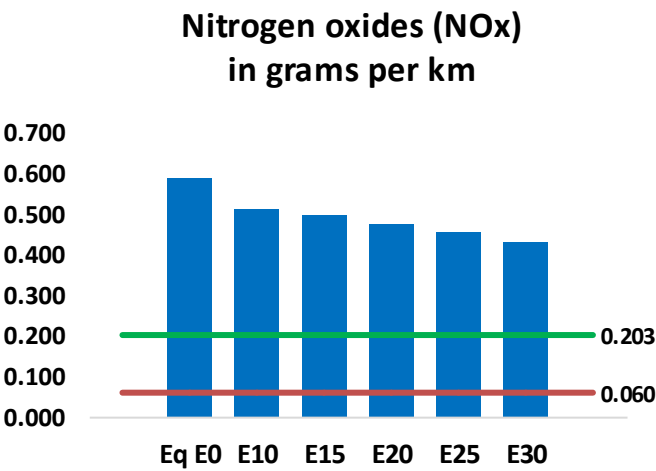
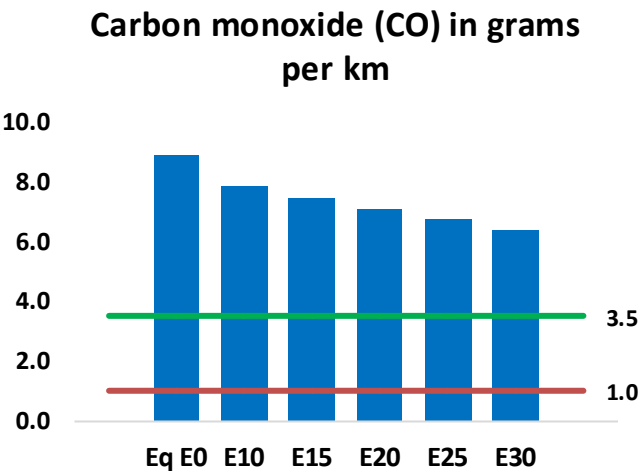
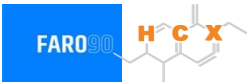
Average Age: **11 years**

Motorcycles: **19.6%**

Source: INEC , Faro 90

Costa Rica – Vehicular Emissions

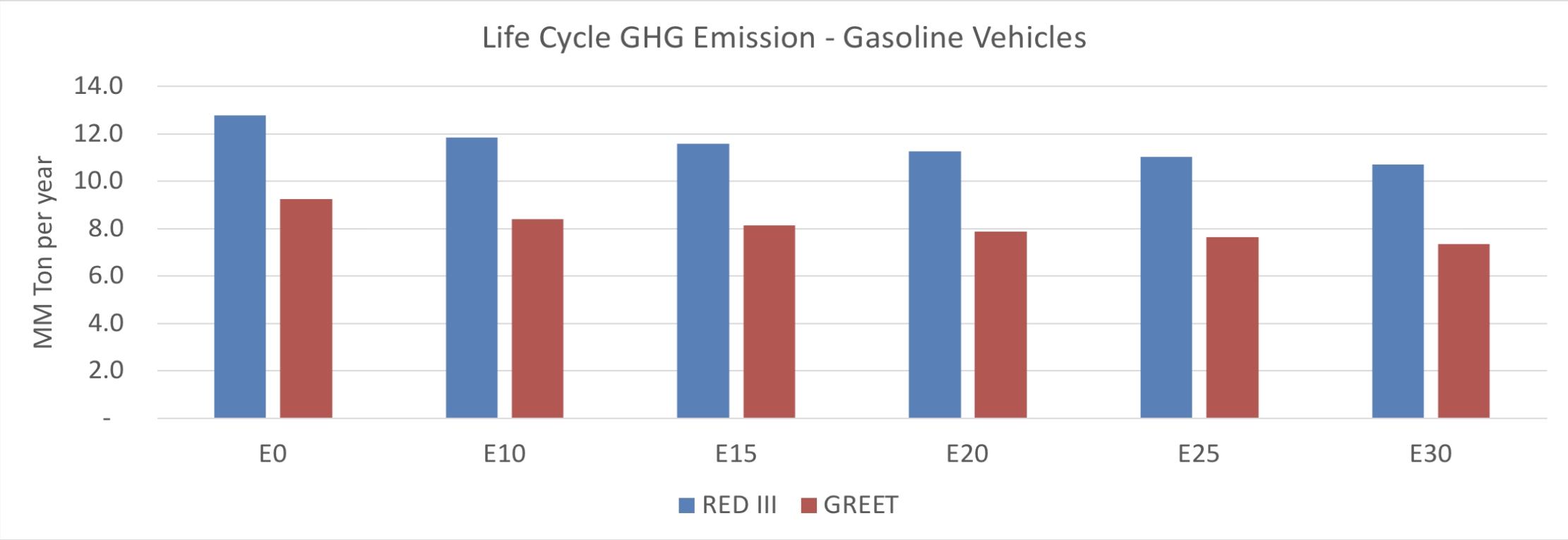
United States
Euro 6



Costa Rica – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	518,660	488,252	457,844	434,773	412,406	395,463	374,023	-12%	-20%
CO2	12,391,357	12,072,722	11,771,789	11,535,114	11,417,998	11,351,652	11,142,408	-5%	-8%
VOC	49,572	48,103	46,633	45,886	45,344	44,991	44,296	-6%	-9%
NOx	34,366	31,731	29,965	29,088	27,650	26,534	25,212	-13%	-20%
SOx	142	131	120	111	103	93	83	-15%	-28%
NH3	4,369	4,326	4,282	4,303	4,351	4,385	4,413	-2%	0%
N2O	172	171	171	172	175	177	179	-1%	2%
CH4	10,464	10,464	10,464	10,621	10,851	11,050	11,239	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	397	385	374	368	364	362	356	-6%	-8%
Acetaldehyde	833	956	1,402	2,135	2,909	3,324	3,928	68%	249%
Formaldehyde	2,557	2,486	2,898	3,403	3,565	3,944	4,293	13%	39%
Benzene	881	847	802	790	783	749	725	-9%	-11%
PM2.5	728	662	596	538	481	420	356	-18%	-34%
PM10	875	796	716	646	578	504	427	-18%	-34%

Costa Rica – Gasoline Vehicle Fleet GEI Emissions

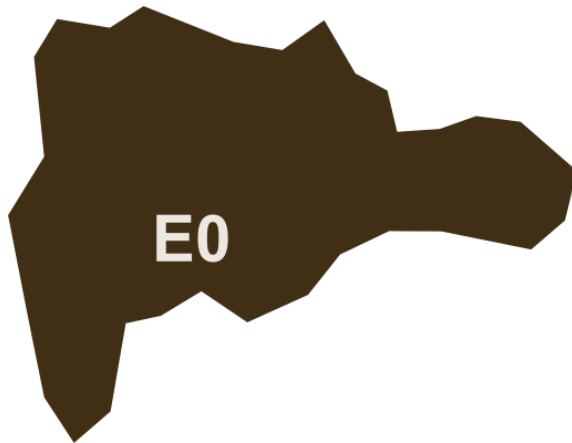


Country Emissions 2025 (MMT/year)	17.1
2035 Target (MMT/year)	8.5
Delta	8.6

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.1%	11.9%	14.9%	17.3%	20.5%
RED II/III	0.0%	7.2%	9.4%	11.8%	13.7%	16.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.8%	12.8%	16.0%	18.6%	22.1%
RED II/III	0.0%	10.7%	14.0%	17.5%	20.4%	24.2%

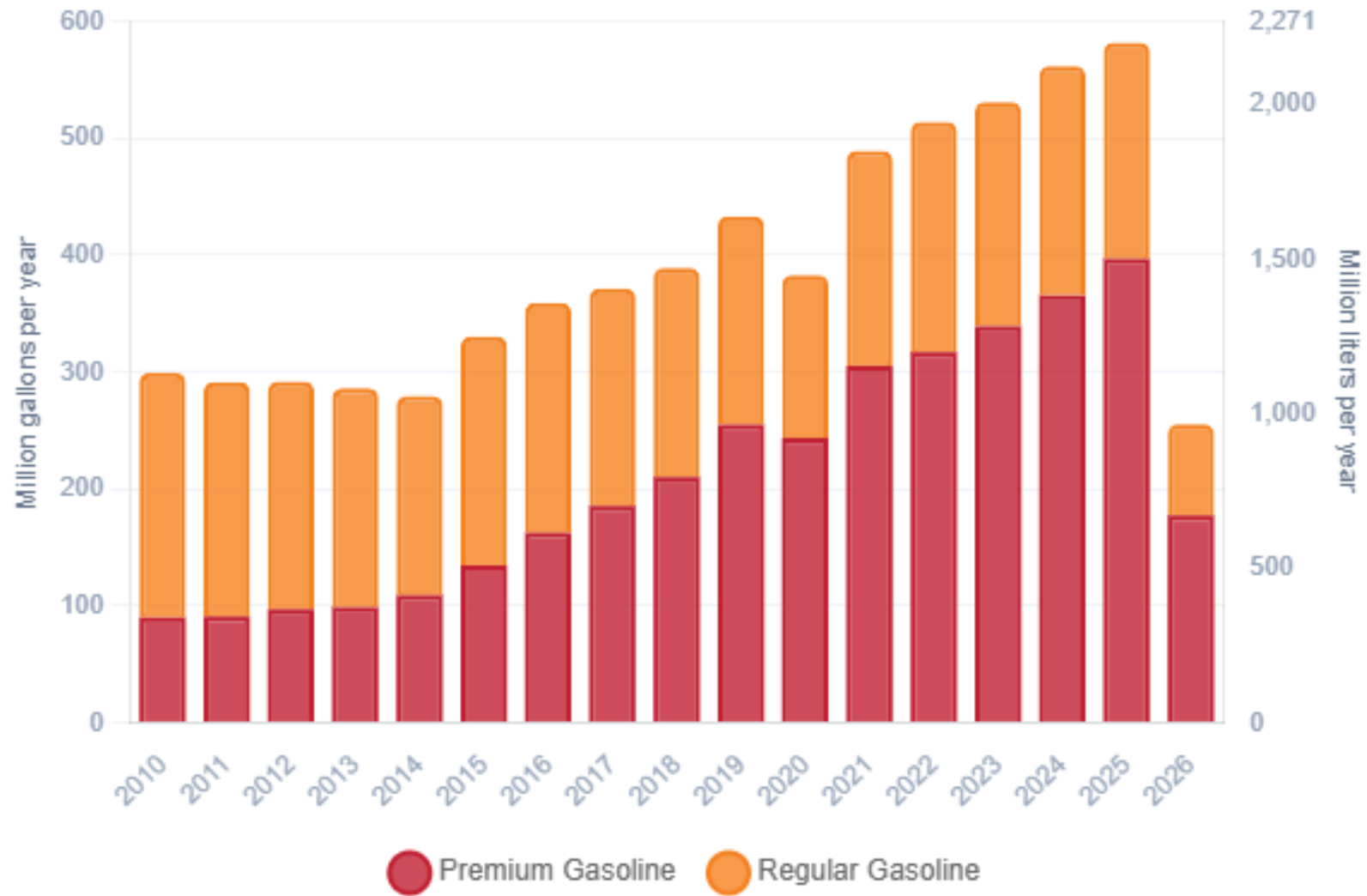
Ethanol Blending in Gasoline – Dominican Republic



National demand is covered up by local production representing 12%, and imports from United States, Trinidad and Tobago, Caribbean, among other countries. In 2025, gasoline consumption was 581 million gallons (2,200 million liters). Regular gasoline consumption was 32% of total demand and premium 68%. The Dominican Government applies subsidies to gasoline prices.

Dominican Republic does not currently have an ethanol mandate.

Gasoline Demand in Dominican Republic



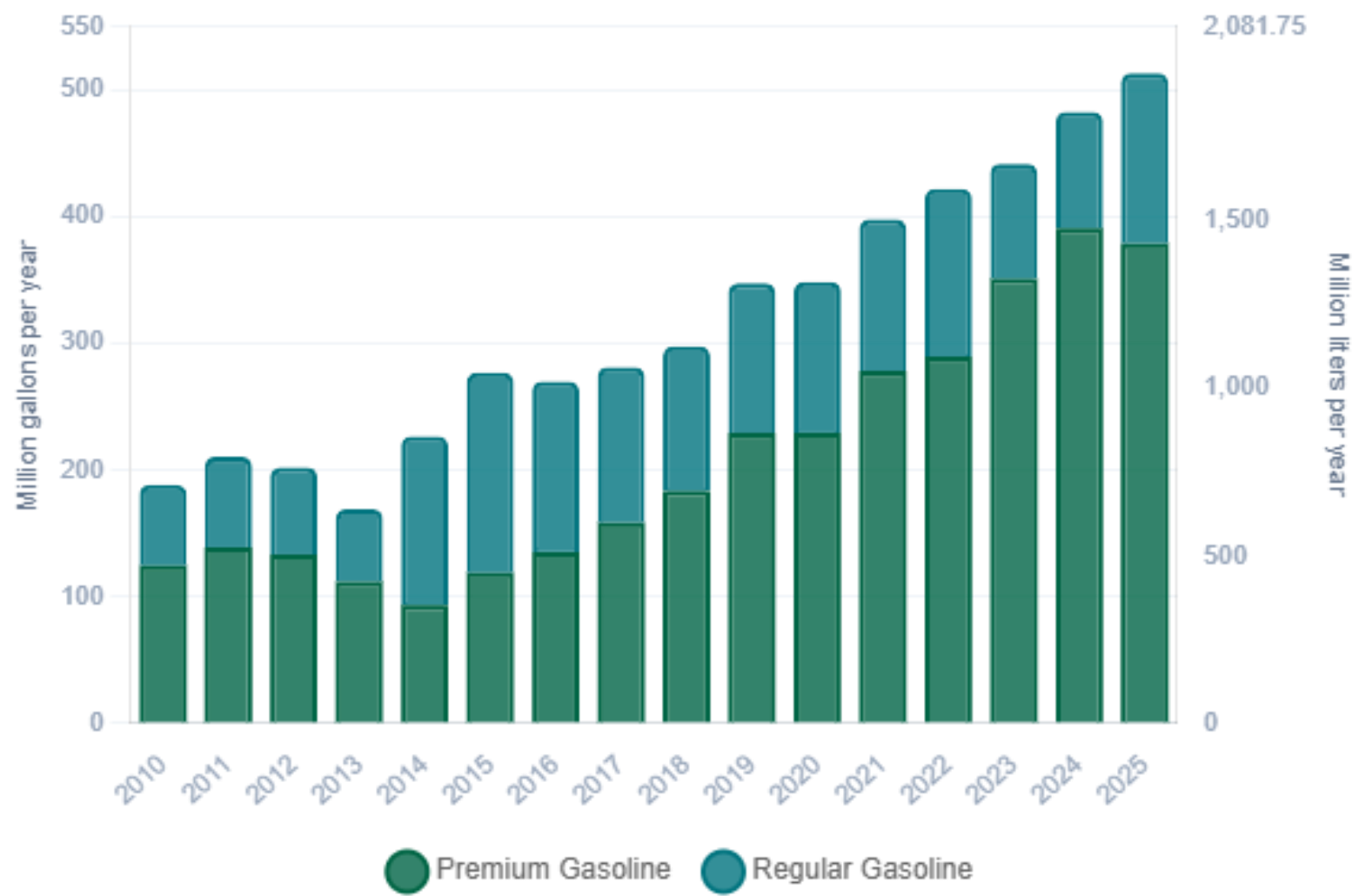
Source:
DGII

Gasoline Production in Dominican Republic



Source:
DGII

Gasoline Imports in Dominican Republic



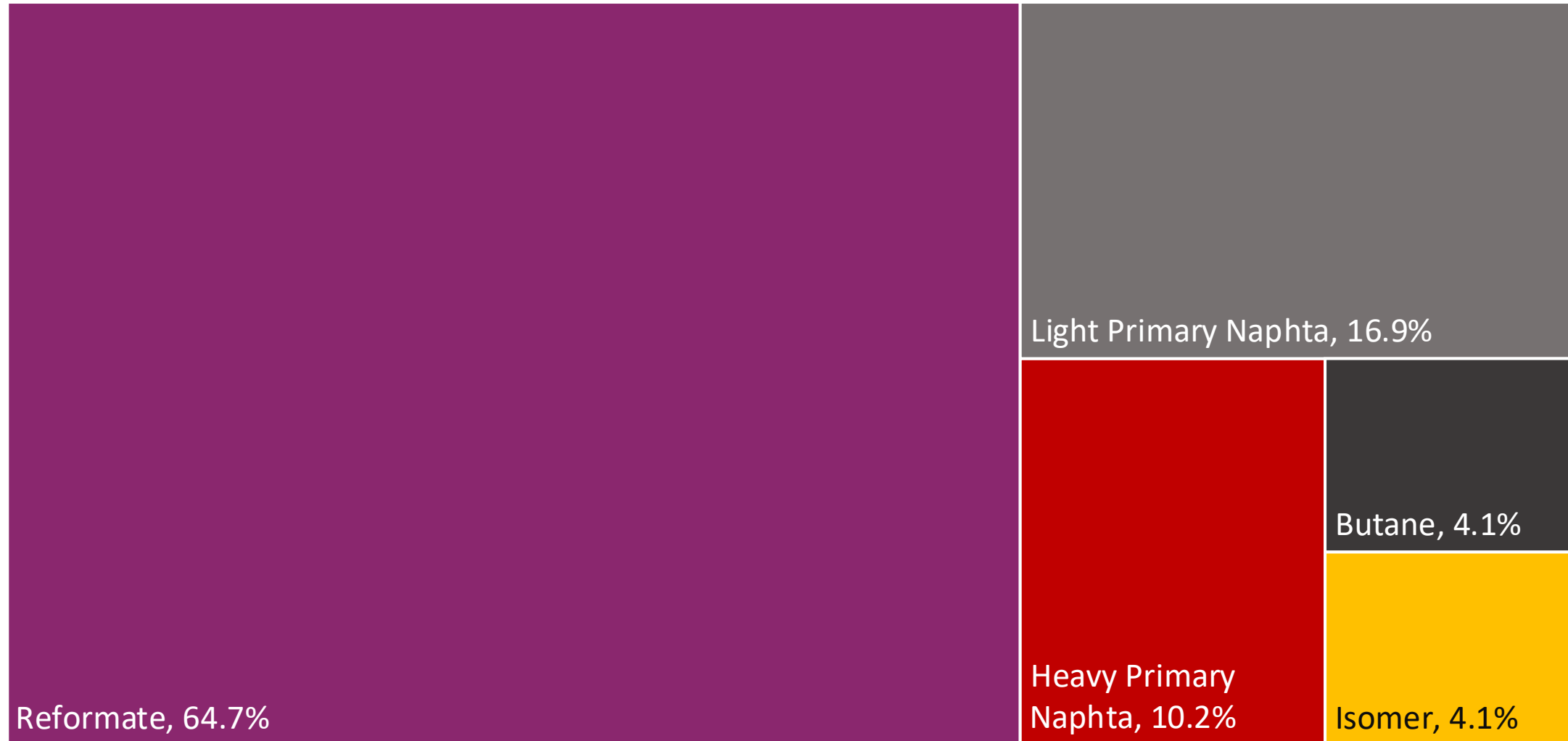
Source:
DGII

Gasoline Quality in Dominican Republic

Name	NORDOM 476 3rd Revision				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2015				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Regular Gasoline	Premium Gasoline	Oxygenated Regular Gasoline	Oxygenated Premium Gasoline	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	-	-	-	-	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	-	-	-	-	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 8,3 mg/l	< 8,3 mg/l	< 8,3 mg/l	< 8,3 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 89	> 95	> 90	> 96	> 95	> 95	> 98	> 98
MON	> 76	> 82	> 77	> 83	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 1.500 mg/kg	< 1.500 mg/kg	< 1.500 mg/kg	< 1.500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content			< 3.5 %m/m	< 3.5 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)			< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 61 kPa	< 61 kPa	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

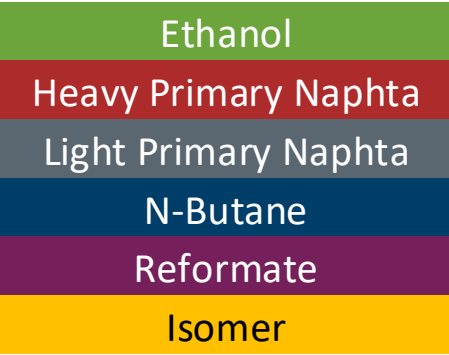
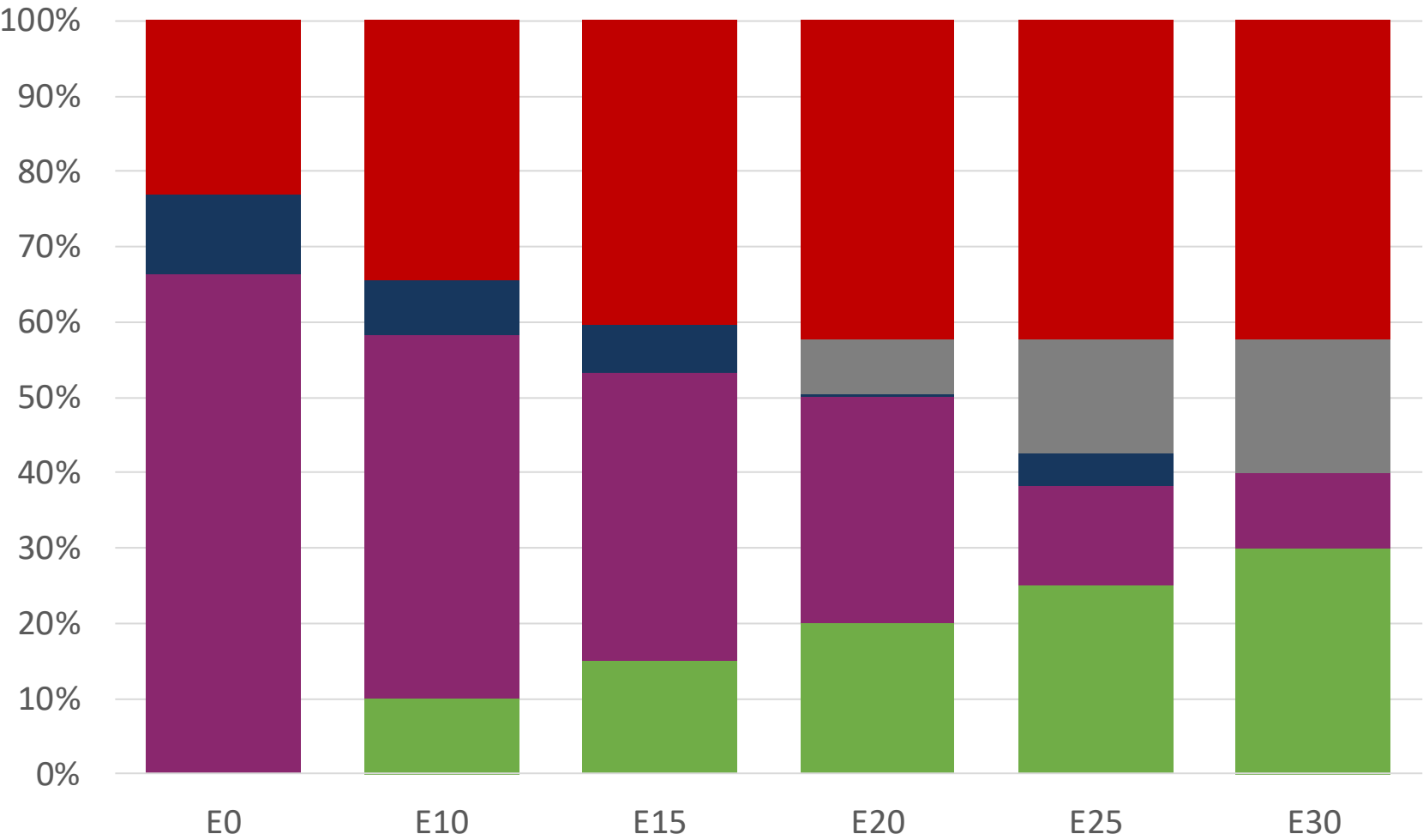
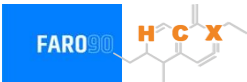
Source: NORDOM

Available Blending Components



Source: Faro90

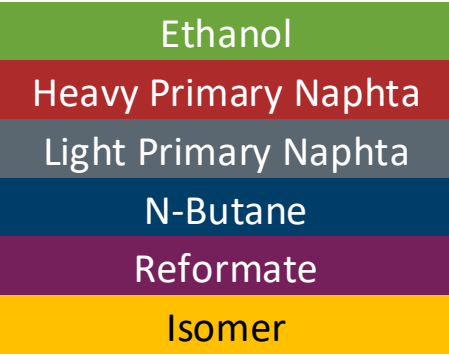
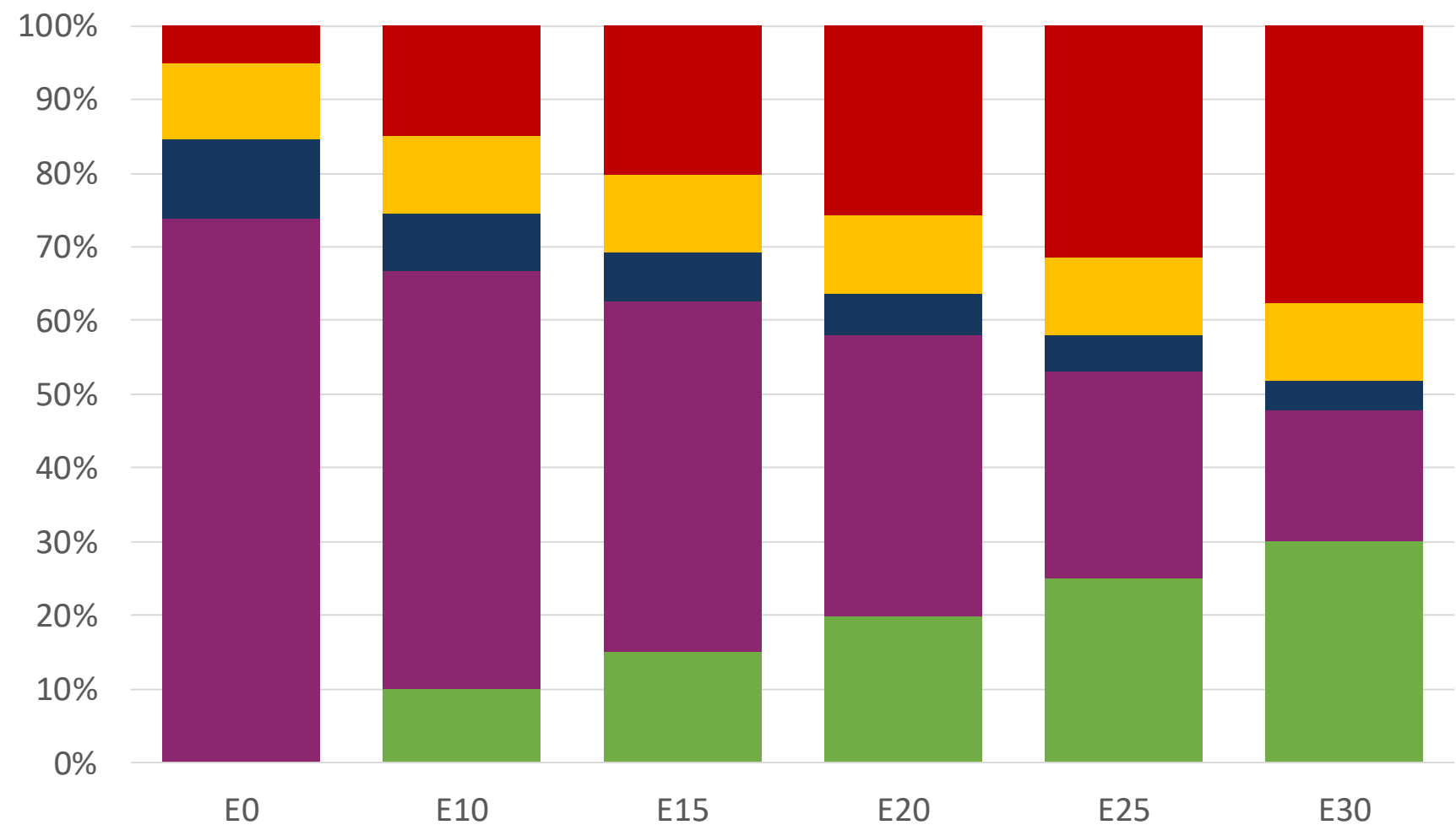
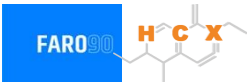
Dominican Republic – Regular – Constante Octane



Octane (RON)	90.0	90.0	90.0	90.0	90.0	91.8
Price (US\$/gal)	\$ 2.146	\$ 1.849	\$ 1.684	\$ 1.589	\$ 1.338	\$ 1.328

Source: Faro90

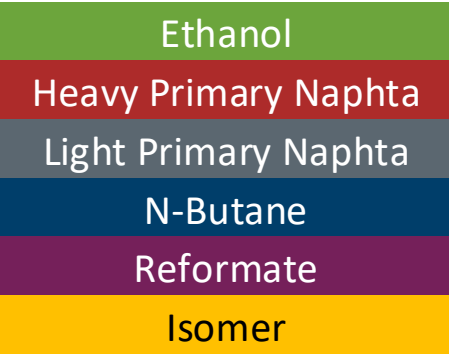
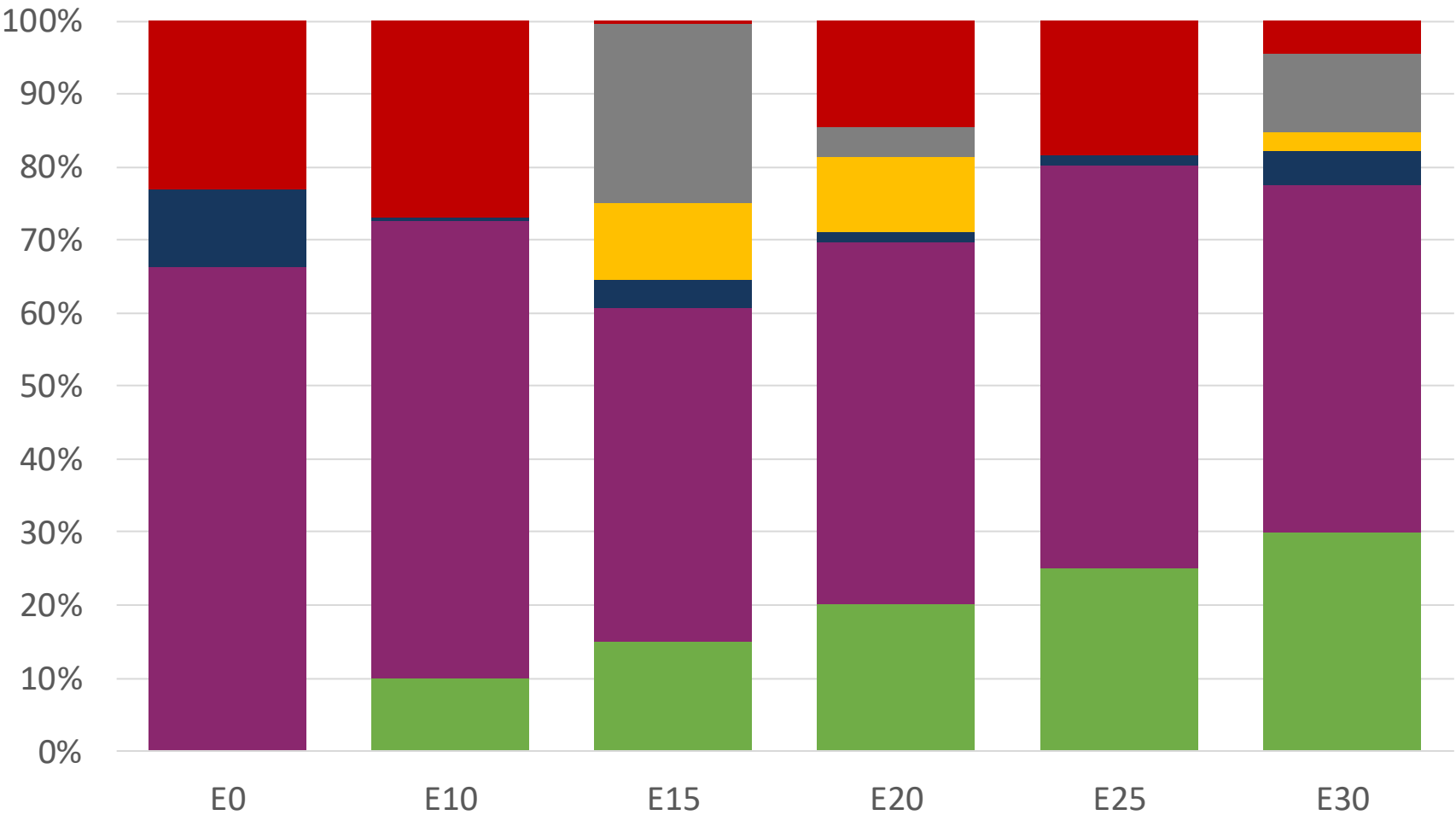
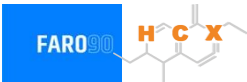
Dominican Republic– Premium – Constant Octane



Octane (RON)	96.0	96.0	96.0	96.0	96.0	96.0
Price (US\$/gal)	\$ 2.454	\$ 2.184	\$ 2.035	\$ 1.877	\$ 1.712	\$ 1.538

Source: Faro90

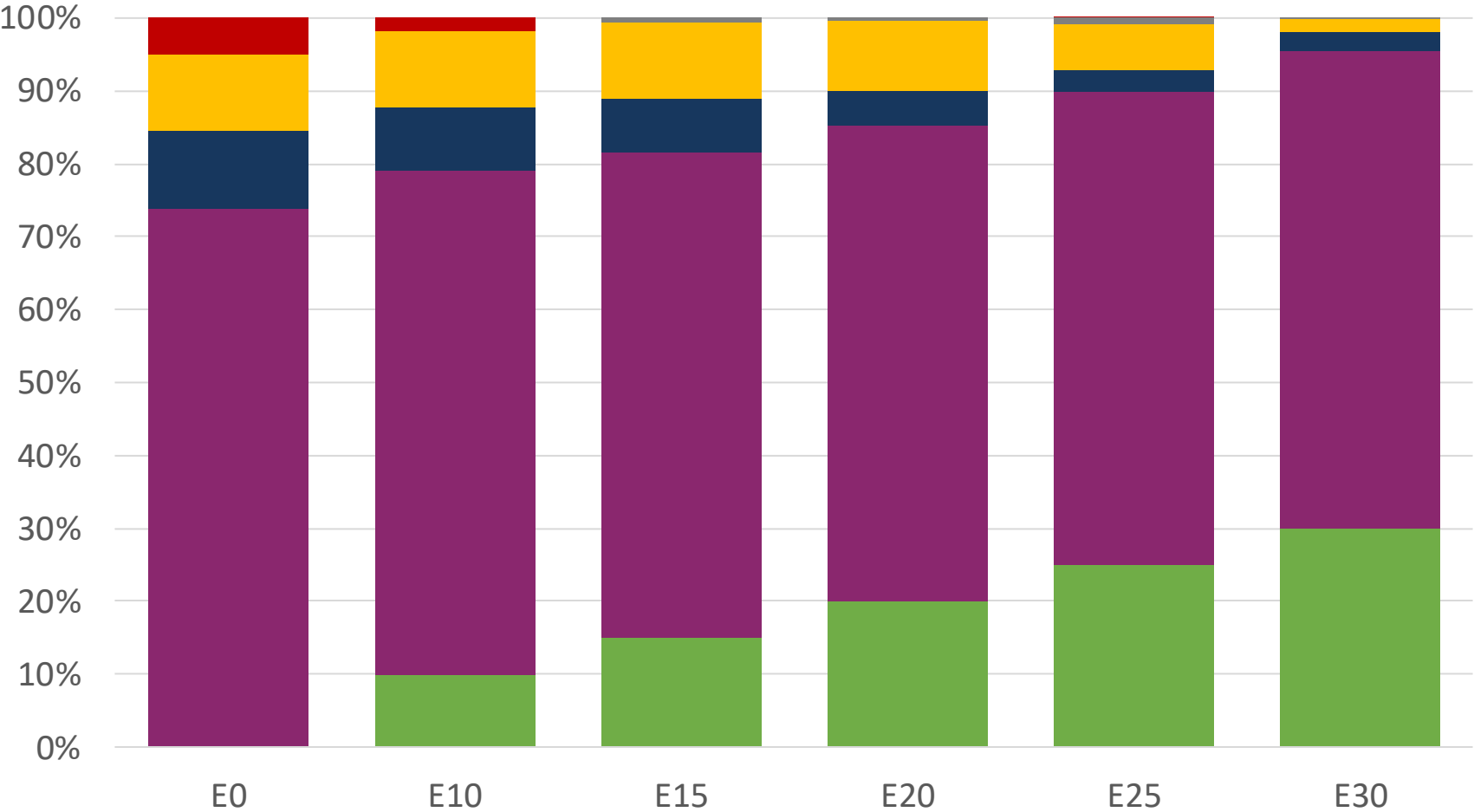
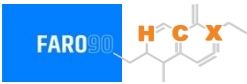
Dominican Republic – Regular – Increased Octane



Octane (RON)	90.0	93.2	96.8	99.0	101.3	104.3
Price (US\$/gal)	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146

Source: Faro90

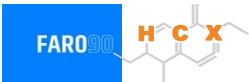
Dominican Republic – Premium – Increased Octane



Octane (RON)	96.0	100.8	102.9	104.6	106.3	108.2
Price (US\$/gal)	\$ 2.454	\$ 2.454	\$ 2.454	\$ 2.454	\$ 2.454	\$ 2.454

Source: Faro90

Gasoline Vehicular Fleet in Dominican Republic



Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	102,256				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	153,384				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	291,768				
		Euro III		> 8 years, > 160 tkm	102,402				
		Euro IV		> 8 years, > 160 tkm	75,866				
		Euro V / Euro 6		> 8 years, > 160 tkm	107,925				
				4-8 years, 80-160 tkm	178,225				
		< 4 years, < 80 tkm		148,308					
	Hybrid			Any	7,337				
Electric			Any	5,510					
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	27,660				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	41,490				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	119,696				
		Euro III		> 8 years, > 160 tkm	195,723				
		Euro IV		> 8 years, > 160 tkm	133,309				
		Euro V / Euro 6		> 8 years, > 160 tkm	126,253				
				4-8 years, 80-160 tkm	213,698				
		< 4 years, < 80 tkm		330,755					
	Hybrid			Any	7,400				
Electric			Any	5,670					
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	839,636				
		Euro II		> 8 years, > 160 tkm	86,144				
		Euro III		> 8 years, > 160 tkm	1,182,587				
		Euro IV		> 8 years, > 160 tkm	164,270				
		Euro V		4-8 years, 80-160 tkm	519,609				
				4-8 years, 80-160 tkm	133,556				
				< 4 years, < 80 tkm	900,975				
	Electric			Any	19,916				
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	36,882				
	FI	3-Ways		> 8 years, > 160 tkm	172,522				
		Euro III		> 8 years, > 160 tkm	50,357				
		Euro IV		> 8 years, > 160 tkm	34,363				
		Euro V / Euro 6		> 8 years, > 160 tkm	28,094				
				4-8 years, 80-160 tkm	48,287				
		< 4 years, < 80 tkm		73,979					
	Hybrid			Any	282				
	Electric			Any	976				

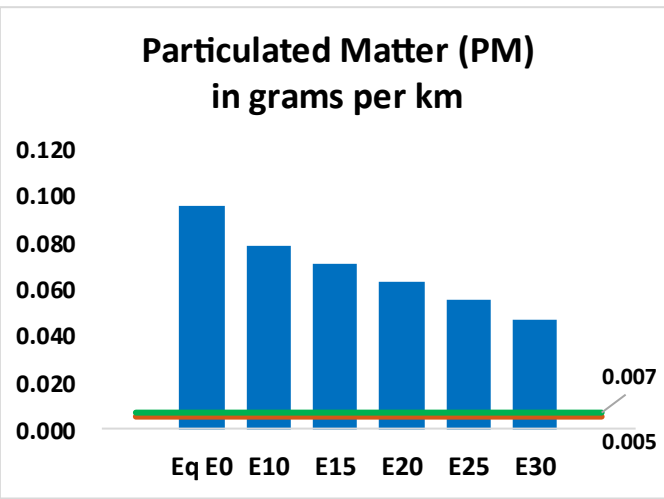
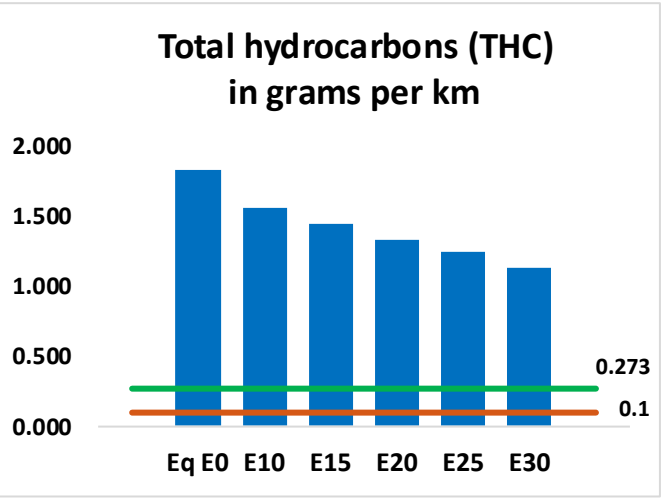
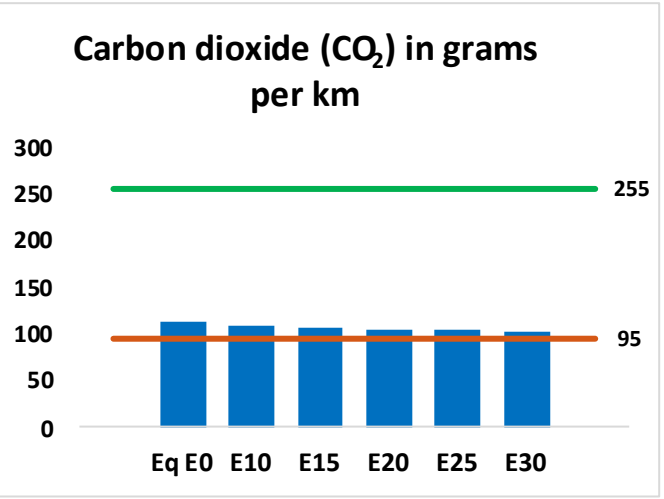
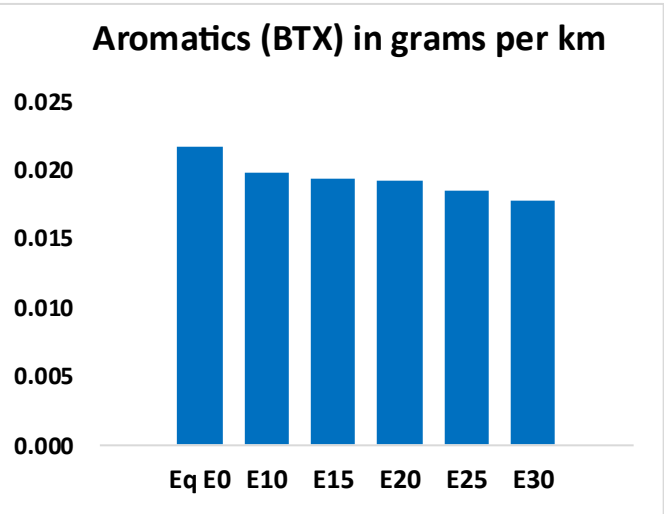
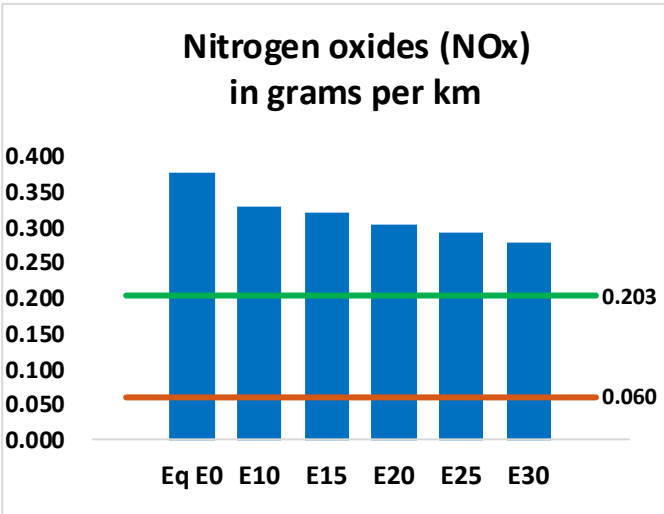
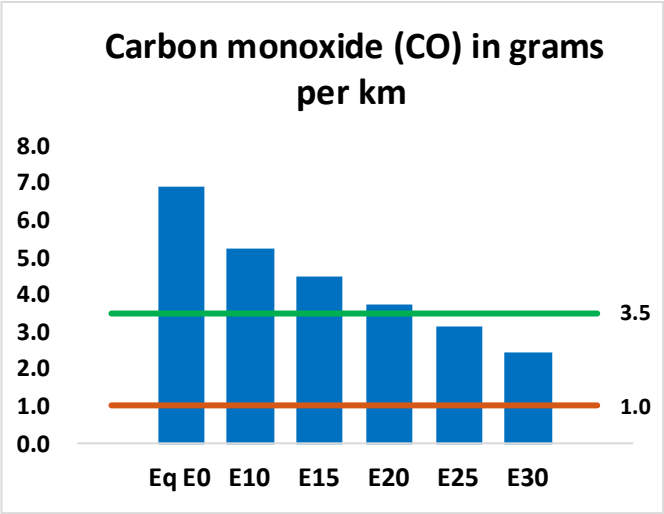
Gasoline Vehicular Fleet: 5,455,680

Average Age: 10 years

Motorcycles: 62%

Source: Dirección General de Impuestos Internos (DGII), 2024

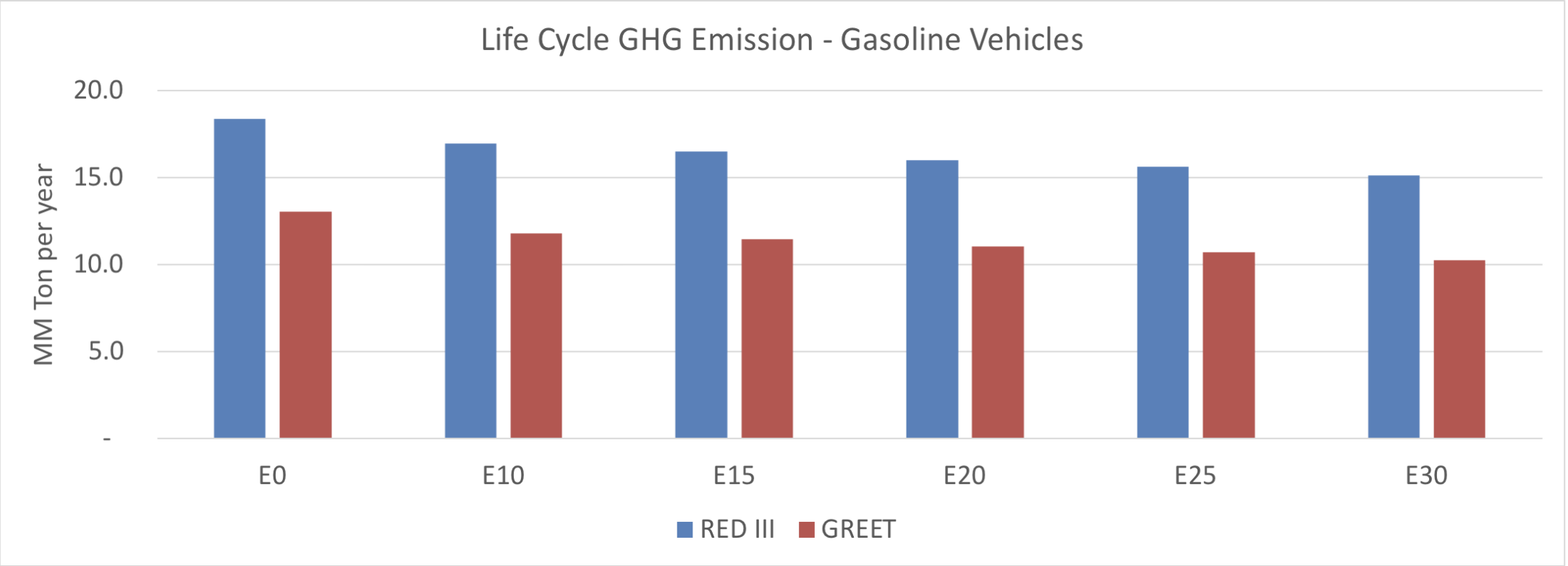
Vehicular Emissions – Dominican Republic



Dominican Republic – Vehicular Fleet

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	941,919	828,510	715,101	614,167	508,913	426,867	336,644	-24%	-46%
CO2	15,484,691	15,086,513	14,710,456	14,414,699	14,268,346	14,209,005	13,947,092	-5%	-8%
VOC	248,823	230,217	211,611	196,467	181,273	169,604	153,982	-15%	-27%
NOx	51,415	47,473	44,844	43,545	41,407	39,752	37,790	-13%	-19%
SOx	171	158	145	134	124	112	100	-15%	-28%
NH3	8,851	8,763	8,676	8,717	8,815	8,884	8,941	-2%	0%
N2O	195	194	193	194	198	200	202	-1%	2%
CH4	55,203	55,203	55,203	56,031	57,245	58,294	59,288	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,204	1,102	1,000	915	829	762	674	-17%	-31%
Acetaldehyde	4,414	5,066	7,433	11,320	15,422	17,623	20,823	68%	249%
Formaldehyde	18,492	17,978	20,957	24,608	25,781	28,520	31,048	13%	39%
Benzene	2,959	2,843	2,694	2,651	2,629	2,515	2,433	-9%	-11%
PM2.5	1,101	1,000	900	812	726	634	537	-18%	-34%
PM10	12,004	10,913	9,822	8,861	7,922	6,914	5,860	-18%	-34%

Dominican Republic – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	51.6
2035 Target (MMT/year)	36.1
Delta	15.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.5%	12.4%	15.5%	18.1%	21.5%
RED II/III	0.0%	7.7%	10.2%	12.8%	15.0%	17.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	8.0%	10.4%	13.1%	15.2%	18.1%
RED II/III	0.0%	9.2%	12.1%	15.2%	17.8%	21.1%

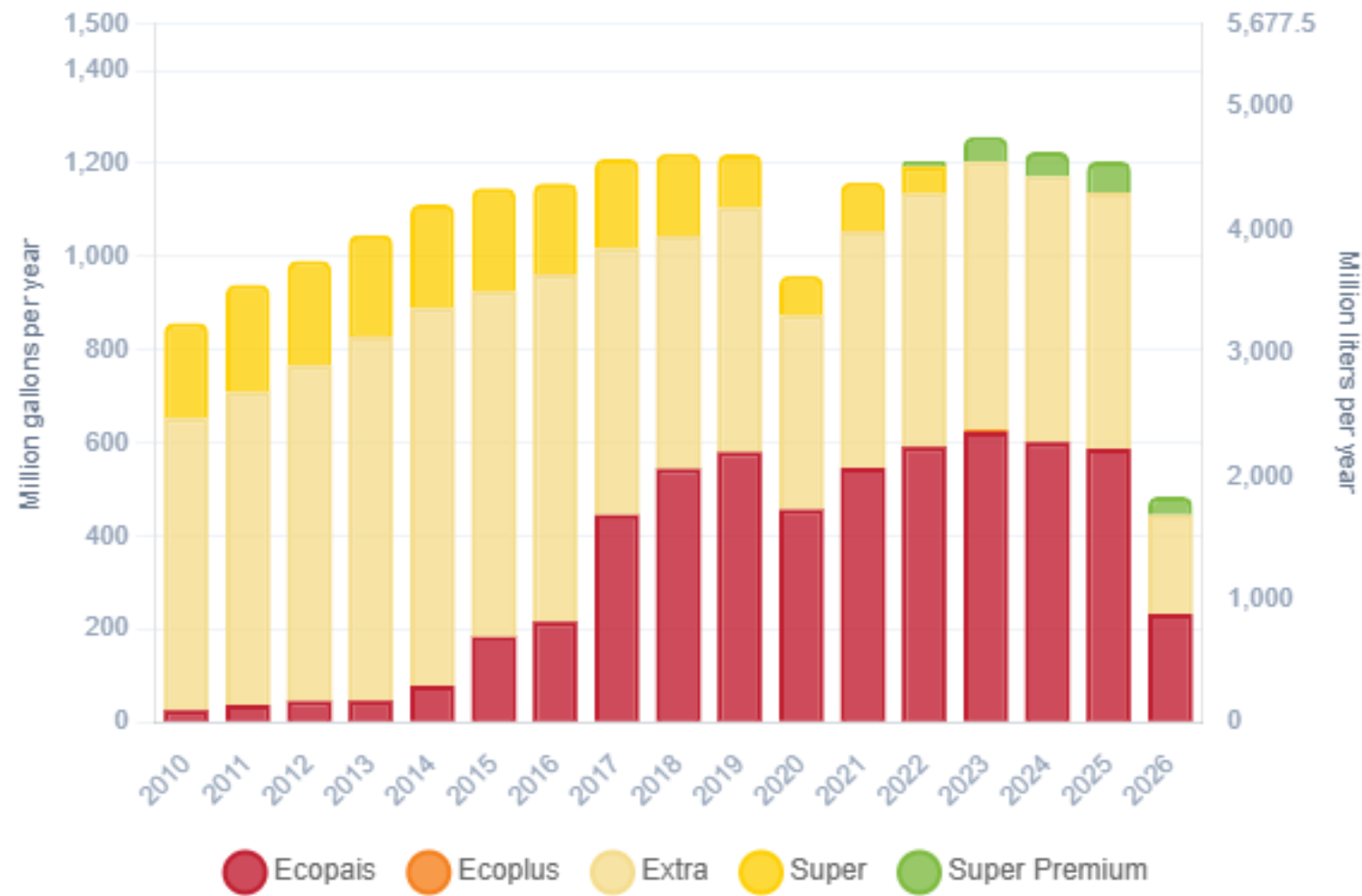
Ethanol Blending in Gasoline - Ecuador



In 2025, gasoline consumption in Ecuador was 1,205 million gallons (4,561 million liters). A new plan was established to implement new grades of gasoline: four grades with 5 different types according to the current quality norm (NTE INEN 935:2021): Extra (RON 85), Ecopais E10 (RON 85), Ecoplus (RON 89), Súper (RON 92) and Super Premium (RON 95). Super Premium (RON 95) will gradually substitute Super (RON 92). Gasoline Ecoplus (RON 89) was recently introduced in the market. Ecuador has local production and imports naphtha (unfinished gasolines) to supply national demand.

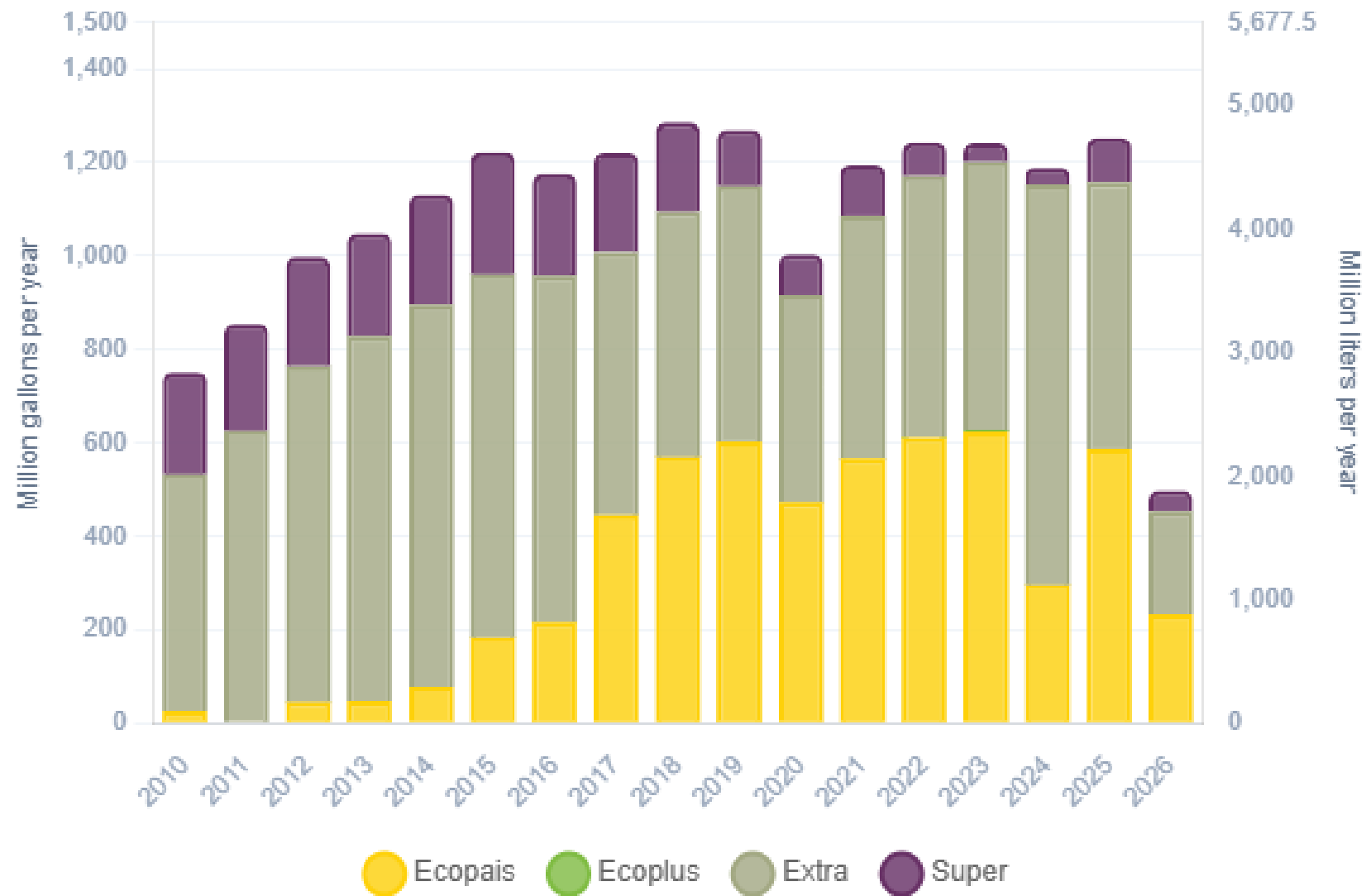
There is no specific mandate for ethanol blending, however there is a permit to blend it with gasoline. Ethanol has been blending in gasoline since 2009. Today the usual blend is E10 for Ecoplus and Super. Ecopais gasoline is supplied in some provinces with an ethanol content between 3-4%. Gasoline Ethanol is produced, consumed and exported to neighboring countries for fuel and industrial use.

Gasoline Demand in Ecuador



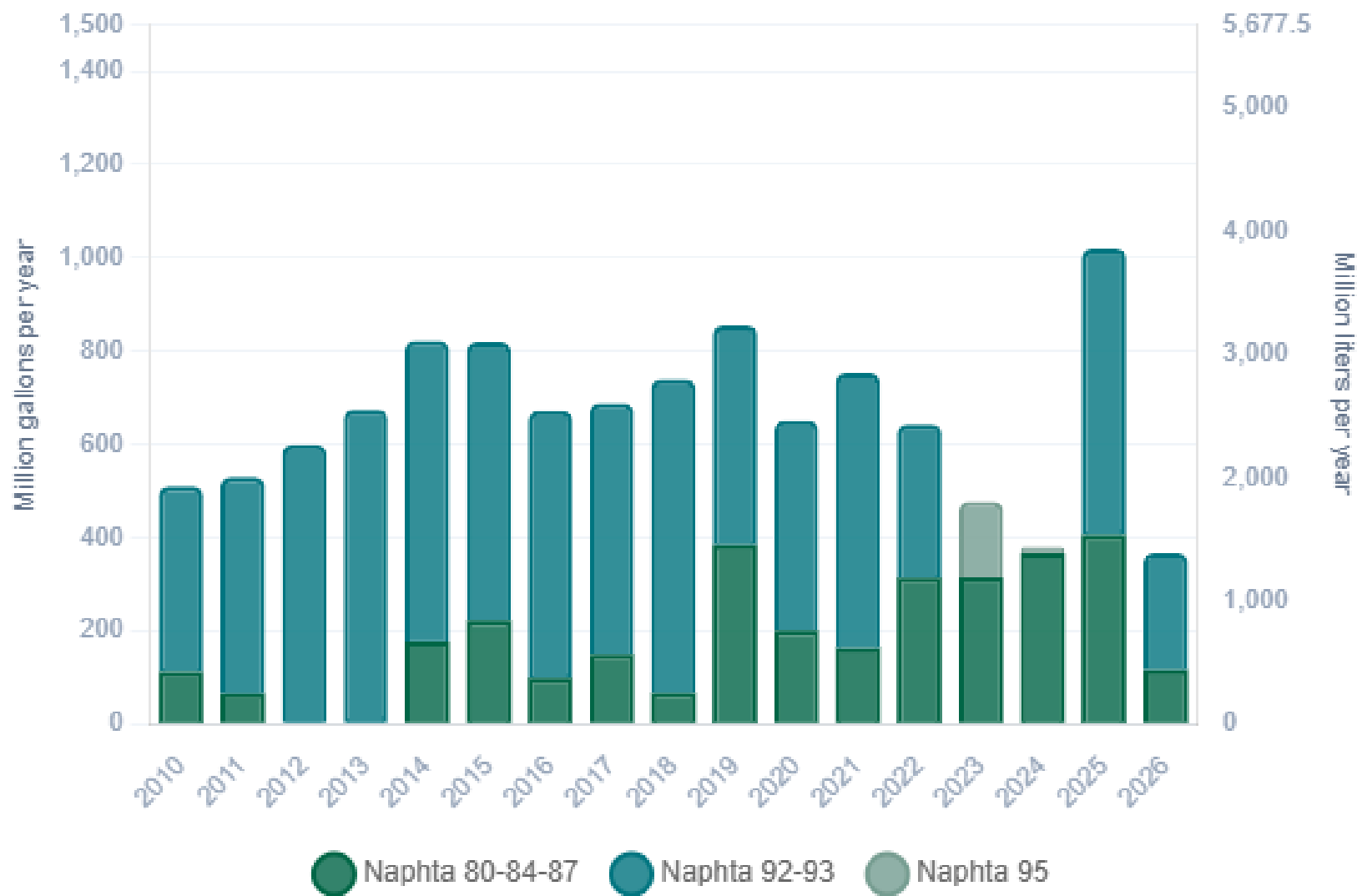
Source: EP PETROEQUADOR, 2024

Gasoline Production in Ecuador



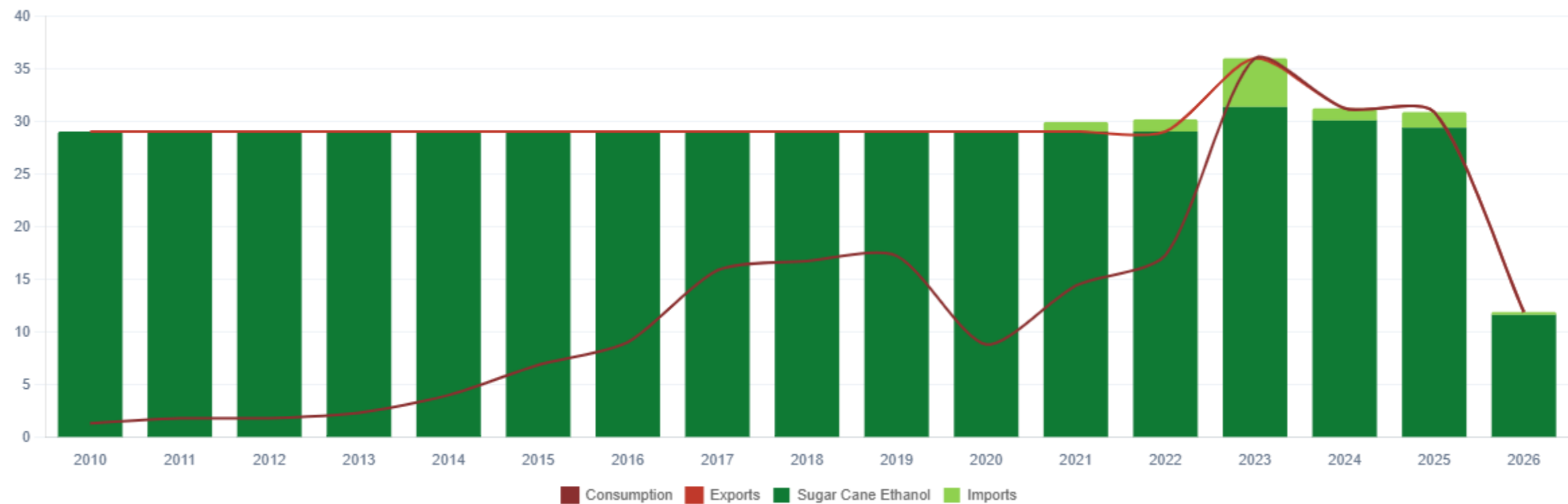
Source: EP PETROEQUADOR, 2024

Gasoline Imports in Ecuador



Source: EP PETROEQUADOR, 2024

Ethanol Balance in Ecuador

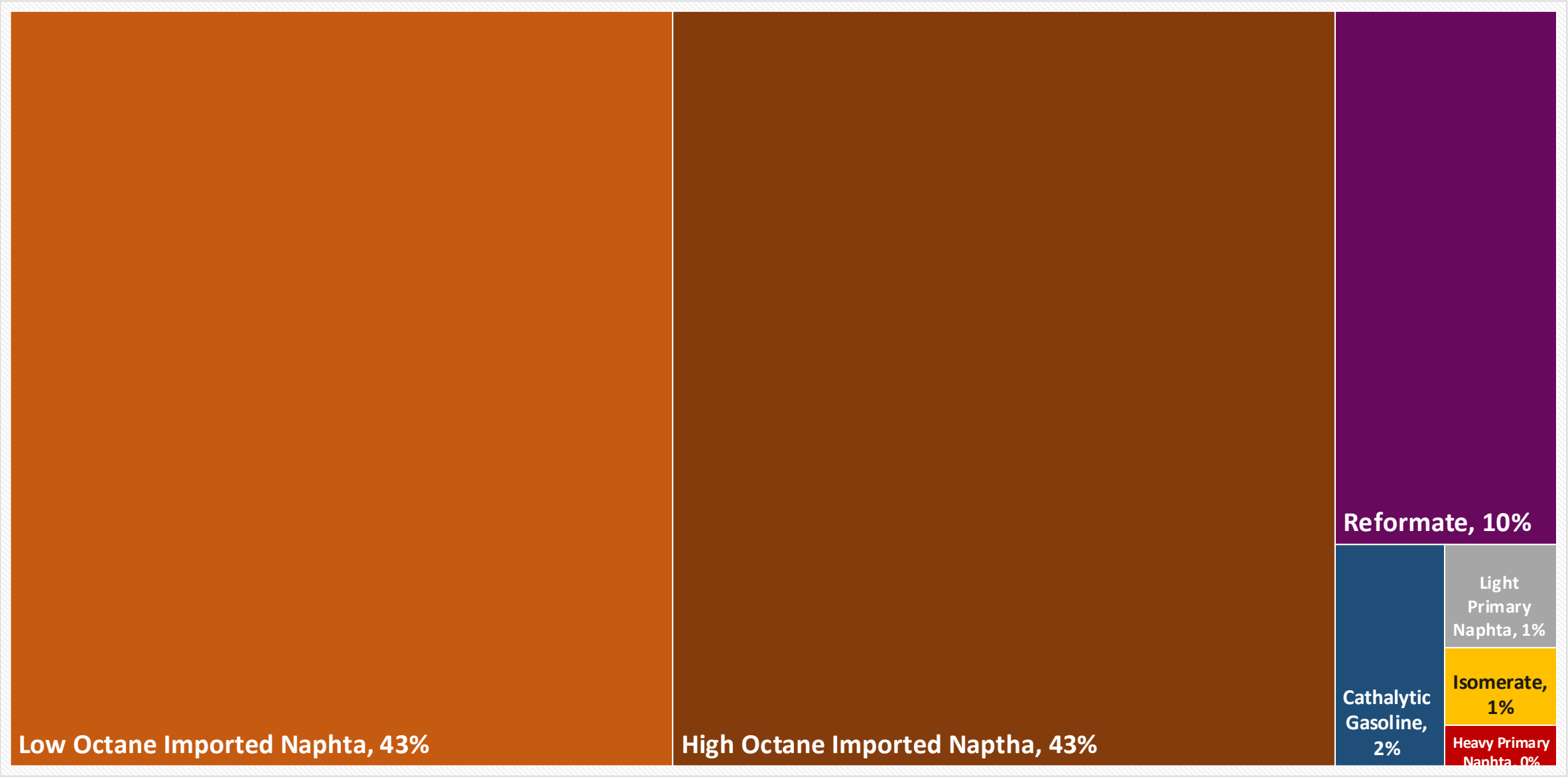
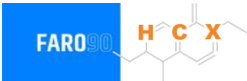


Source: EP PETROECUADOR, 2024; CEDRSSA, 2020

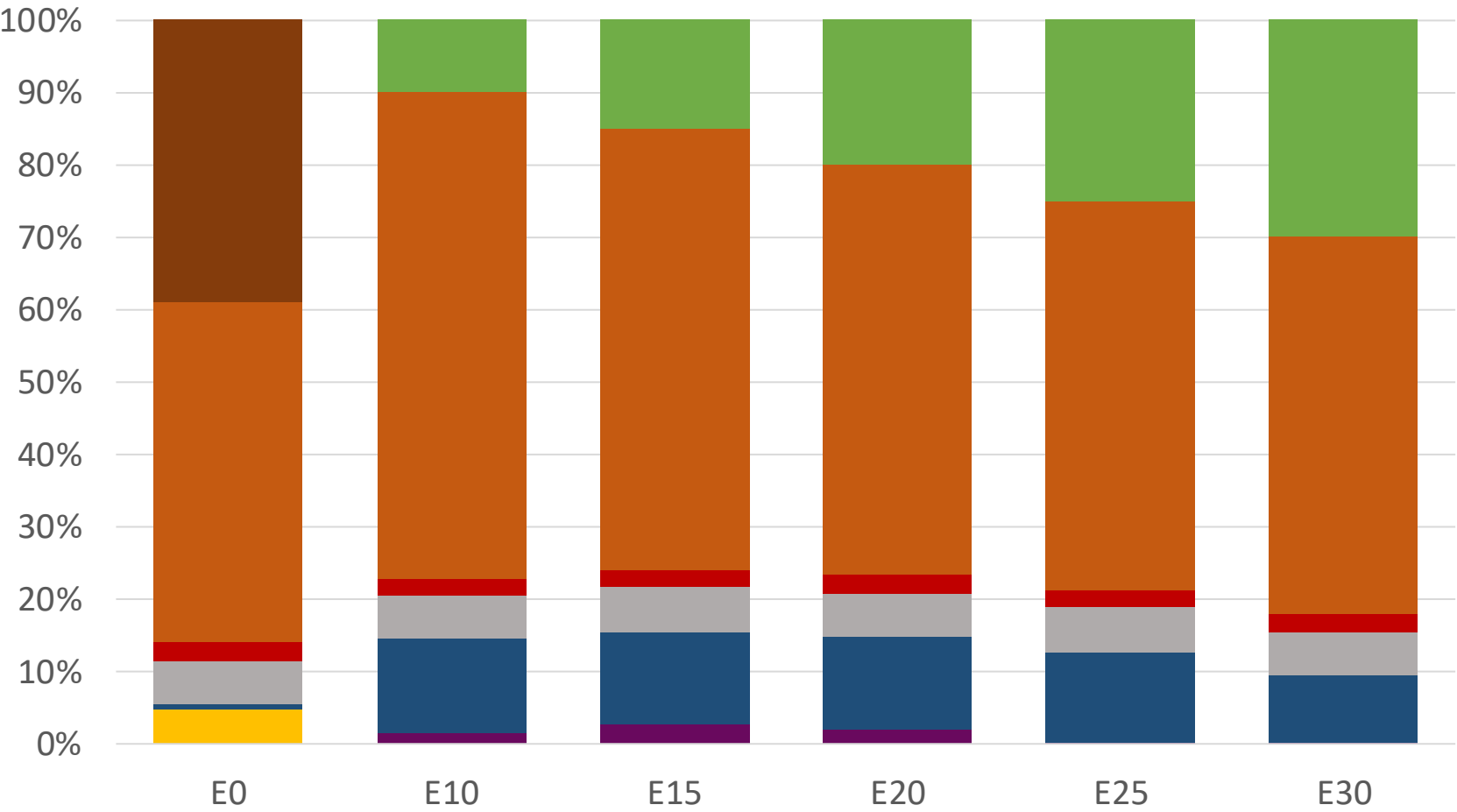
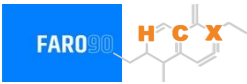
Gasoline Quality in Ecuador

Name	NTE INEN 935:2021				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021				2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	RON 85	RON 89	RON 92	RON 93	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 2 %v/v	<2 %v/v	<1,3 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 30 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 18 %v/v	< 25 %v/v	< 25 %v/v	< 25 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0 mg/l	< 0 mg/l	< 0 mg/l	< 0 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	85	89	92	95	> 95	> 95	> 98	> 98
MON					> 85	> 88	> 85	> 88
AKI								
Sulfur Content	650 mg/kg	650 mg/kg	450 mg/kg	<300 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)					<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<60 kPa	<60 kPa	<60 kPa	<62 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Available Blending Components

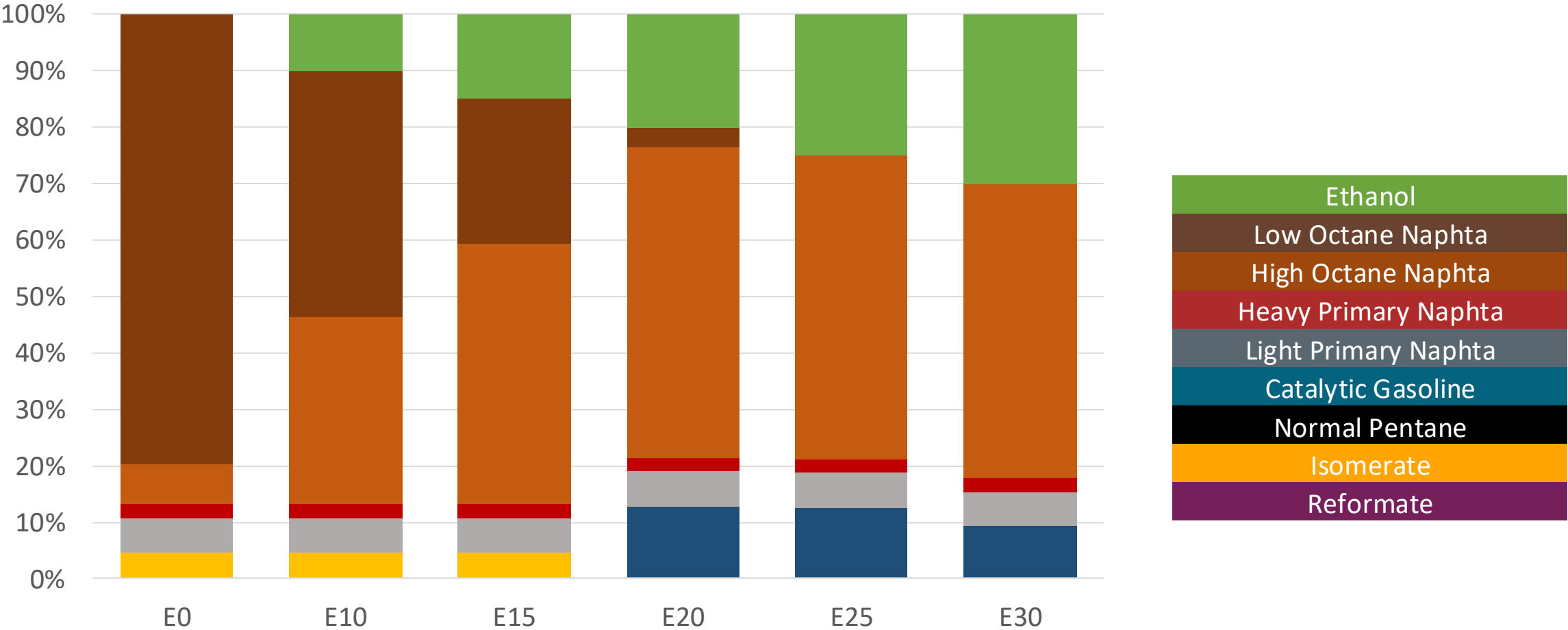


Ecuador –RON 85 Gasoline – Constant Octane



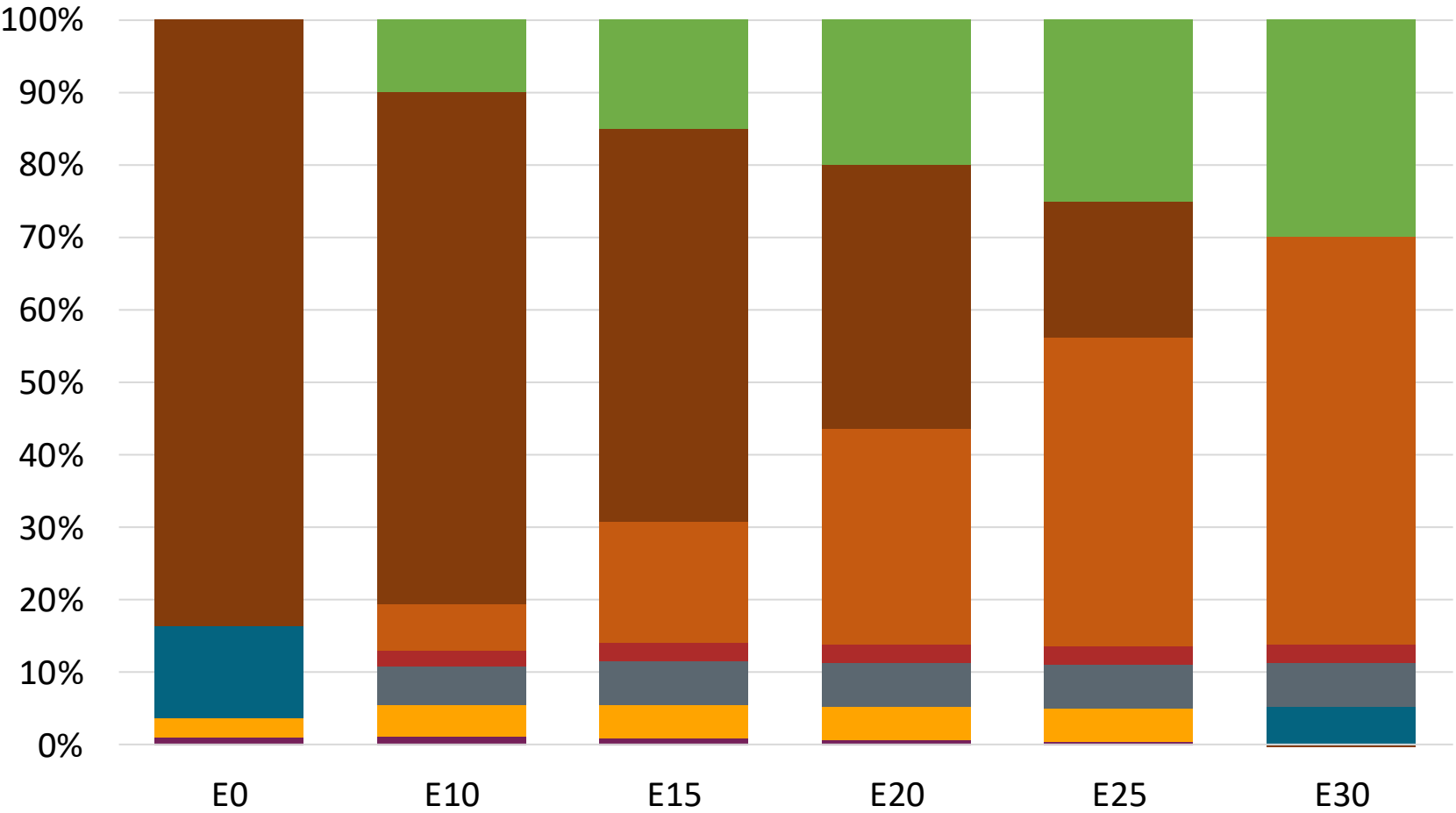
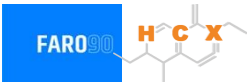
Octane (RON)	85.0	85.9	87.9	89.6	91.0	92.5
Price (US\$/gal)	\$ 1.925	\$ 1.860	\$ 1.849	\$ 1.826	\$ 1.797	\$ 1.773

Ecuador – RON 89 Gasoline– Constant Octane



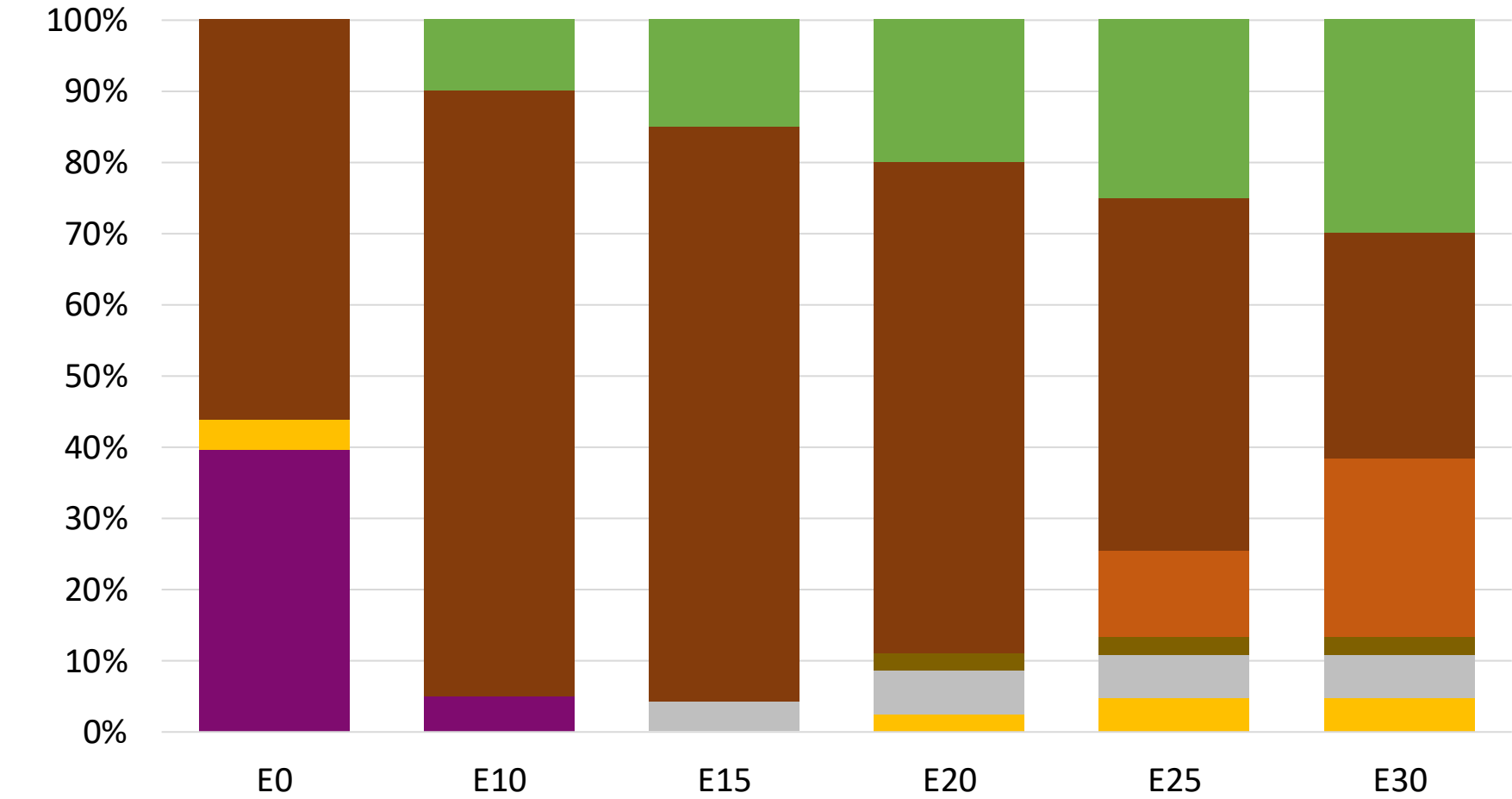
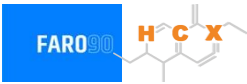
Octane (RON)	89.0	89.0	89.0	89.6	91.0	92.5
Price (US\$/gal)	\$ 1.992	\$ 1.896	\$ 1.849	\$ 1.821	\$ 1.797	\$ 1.773

Ecuador – RON 92 Gasoline – Constant Octane



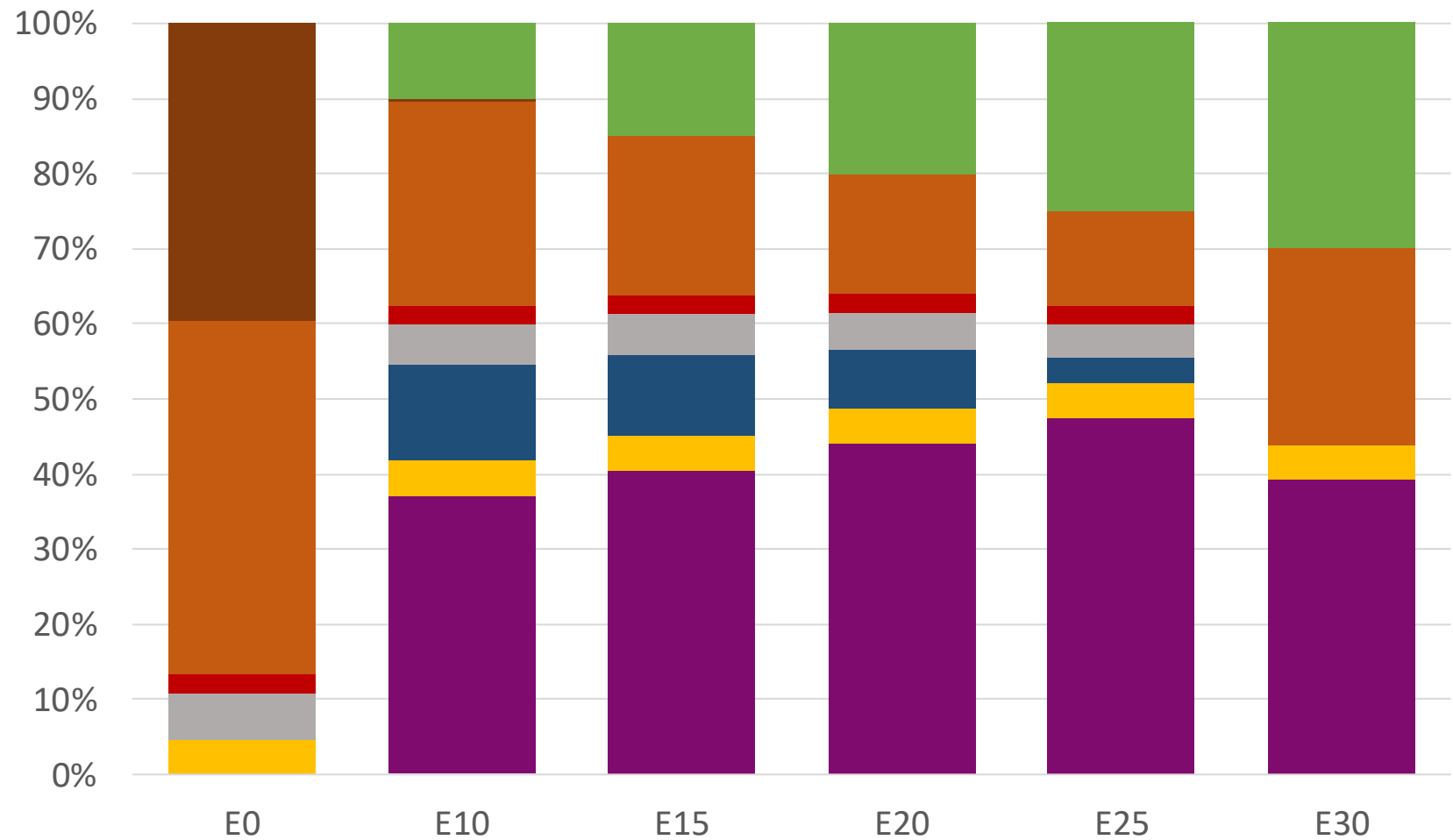
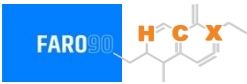
Octane (RON)	92.0	92.0	92.0	92.0	92.0	92.1
Price (US\$/gal)	\$ 2.123	\$ 1.958	\$ 1.901	\$ 1.853	\$ 1.805	\$ 1.764

Ecuador – RON 95 Gasoline – Constant Octane



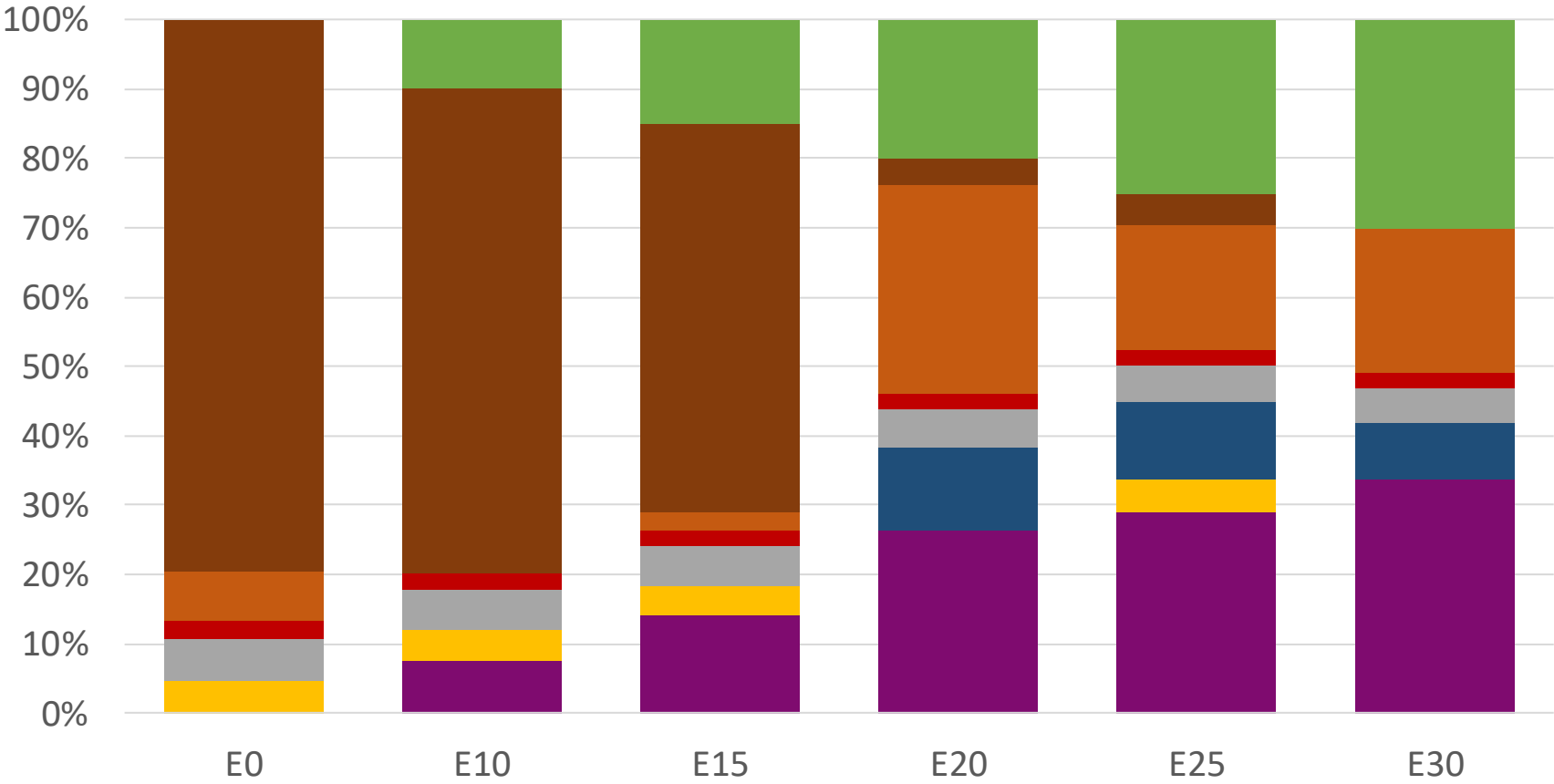
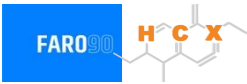
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$ 2.287	\$ 2.087	\$ 1.993	\$ 1.903	\$ 1.853	\$ 1.806

Ecuador – RON 85 Gasoline – Increased Octane



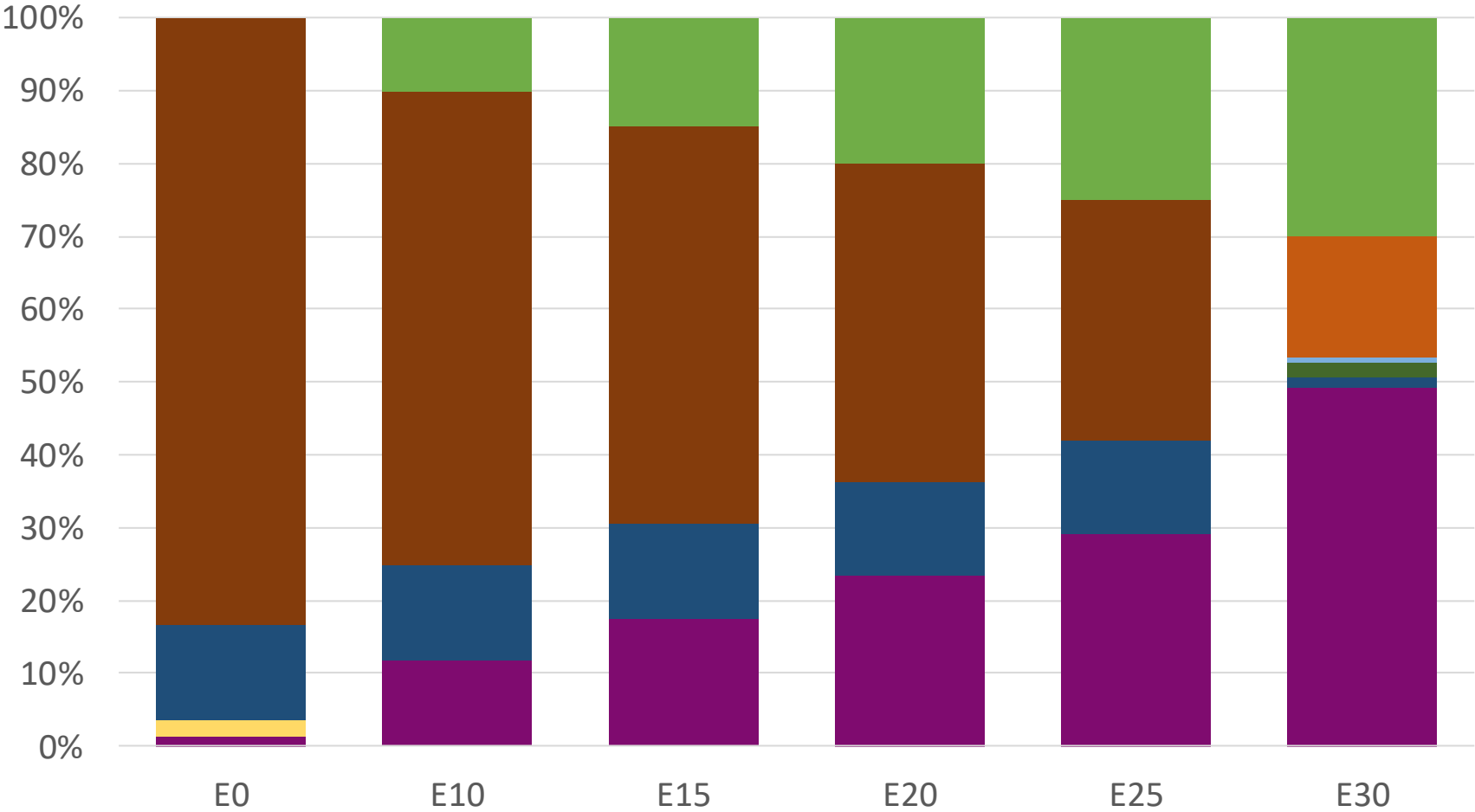
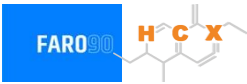
Octane (RON)	85.0	92.7	94.9	97.1	99.1	100.1
Price (US\$/gal)	\$ 1.925	\$ 2.086	\$ 2.086	\$ 2.086	\$ 2.086	\$ 2.086

Ecuador – Gasoline RON 89 – Increased Octane



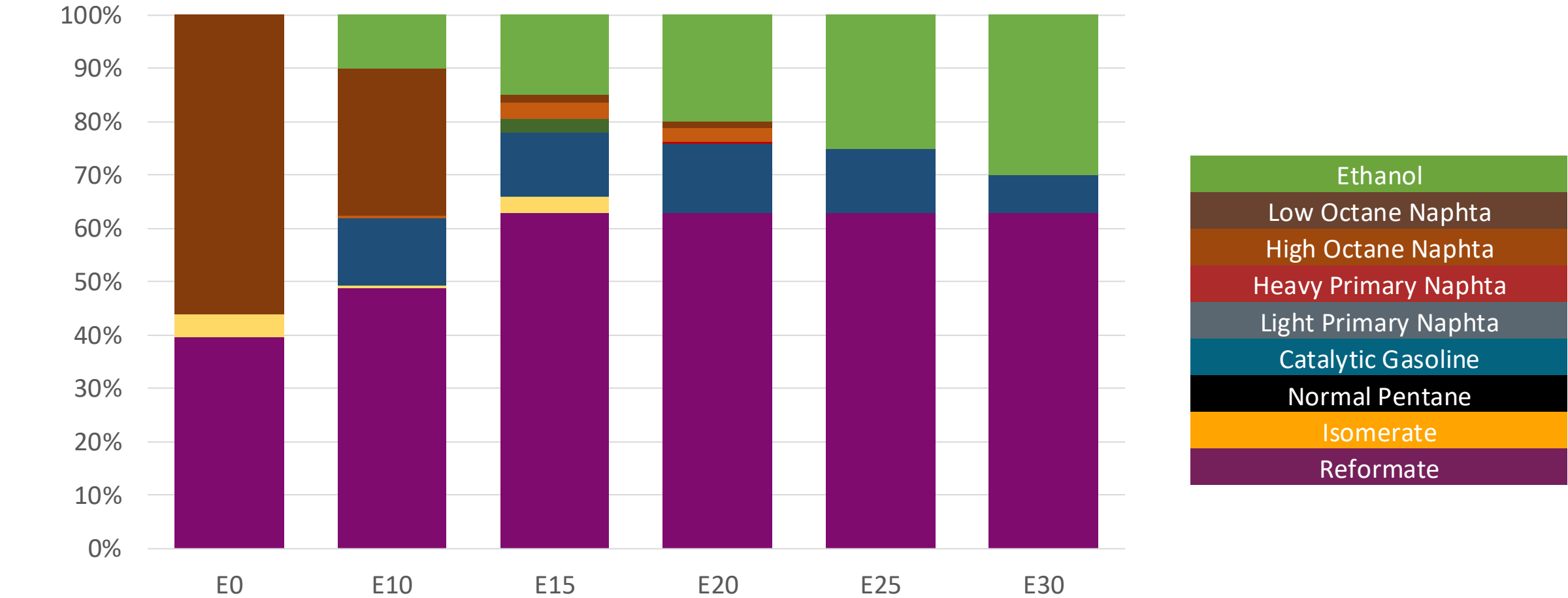
Octane (RON)	89.0	93.0	94.6	94.4	97.0	98.6
Price (US\$/gal)	\$ 1.992	\$ 1.992	\$ 1.992	\$ 1.992	\$ 1.992	\$ 1.992

Ecuador – RON 92 Gasoline – Increased Octane



Octane (RON)	92.0	95.5	97.3	99.1	100.8	101.4
Price (US\$/gal)	\$ 2.116	\$ 2.123	\$ 2.123	\$ 2.123	\$ 2.123	\$ 2.123

Ecuador – RON 95 Gasoline – increased Octane



Octane (RON)	95.0	98.4	99.9	101.8	103.5	104.8
Price (US\$/gal)	\$ 2.287	\$ 2.287	\$ 2.287	\$ 2.287	\$ 2.275	\$ 2.246

Gasoline Vehicle Fleet - Ecuador

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles
Light automobiles	Electric	-	-	Any	7,904
	Multi FI-Pt	Hybrid	PVC Tank	Any	74,475
	Multi FI-Pt	Euro VI		< 4 years, < 80 tkm	560,856
		Euro V		4-8 years, 80-160 tkm	35054
				> 8 years, > 160 tkm	75949
				Euro IV	> 8 years, > 160 tkm
		Euro III		> 8 years, > 160 tkm	245375
	multi	3-Ways		> 8 years, > 160 tkm	72444
	single	3-Ways		> 8 years, > 160 tkm	32717
	carbon	3-Ways		> 8 years, > 160 tkm	74781
Medium automobiles	Multi FI-Pt	Euro VI	PCV Tank	< 4 years, < 80 tkm	766222
		Euro V		4-8 years, 80-160 tkm	47889
				> 8 years, > 160 tkm	103759
				Euro IV	> 8 years, > 160 tkm
	Euro III	> 8 years, > 160 tkm		335222	
	multi	3-Ways		> 8 years, > 160 tkm	98970
	single	3-Ways		> 8 years, > 160 tkm	44696
	carbon	3-Ways		> 8 years, > 160 tkm	102163
Small Engines	Electric	-	-	Any	777
	Multi FI-Pt	Hybrid	PCV	Any	6
	FI 4-cycles	Euro V		< 4 years, < 80 tkm	1126726
		Euro IV		4-8 years, 80-160 tkm	180656
				> 8 years, > 160 tkm	85574
				> 8 years, > 160 tkm	0
		Euro III		> 8 years, > 160 tkm	71312
		Euro II		> 8 years, > 160 tkm	115684
		none		> 8 years, > 160 tkm	797107
Heavy	Electric	-	-	Any	2,253
	Multi FI-Pt	Hybrid	PCV	Any	1,551
	FI	Euro VI		< 4 years, < 80 tkm	70366
		Euro V		4-8 years, 80-160 tkm	86280
				> 8 years, > 160 tkm	49410
				Euro IV	> 8 years, > 160 tkm
		Euro III		> 8 years, > 160 tkm	30745
		3-Ways		> 8 years, > 160 tkm	15402

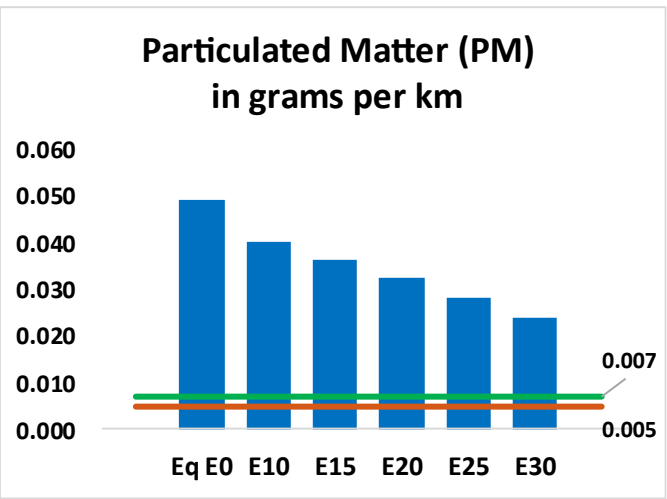
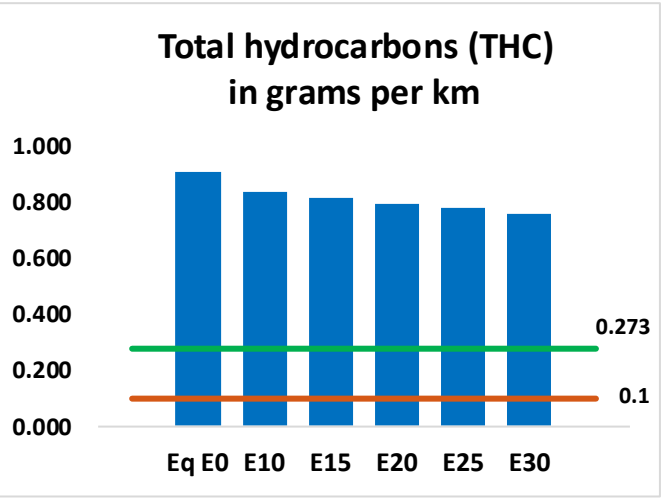
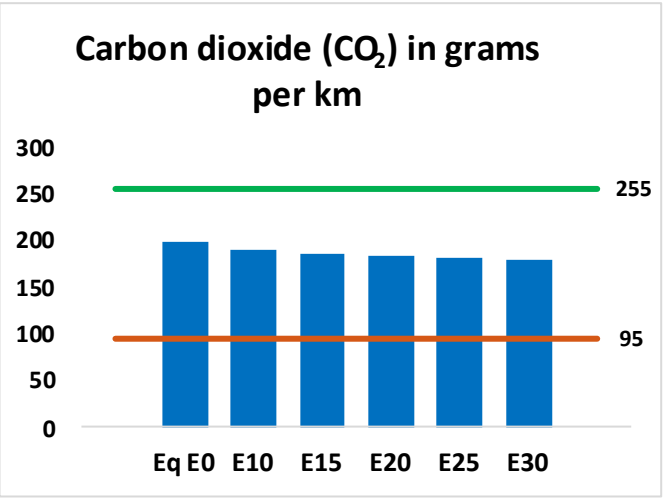
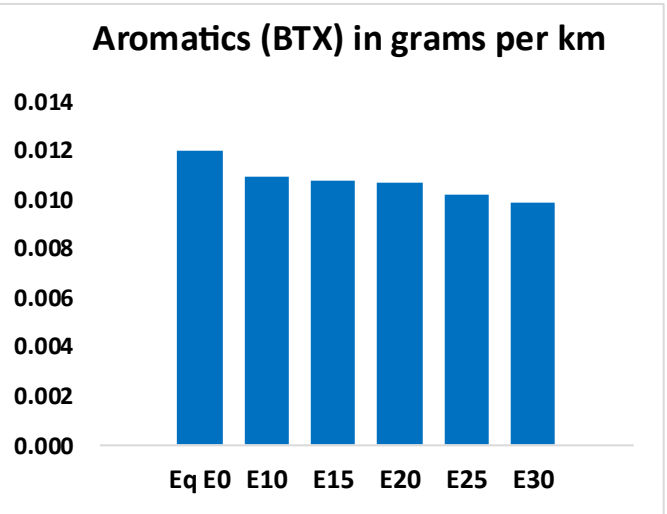
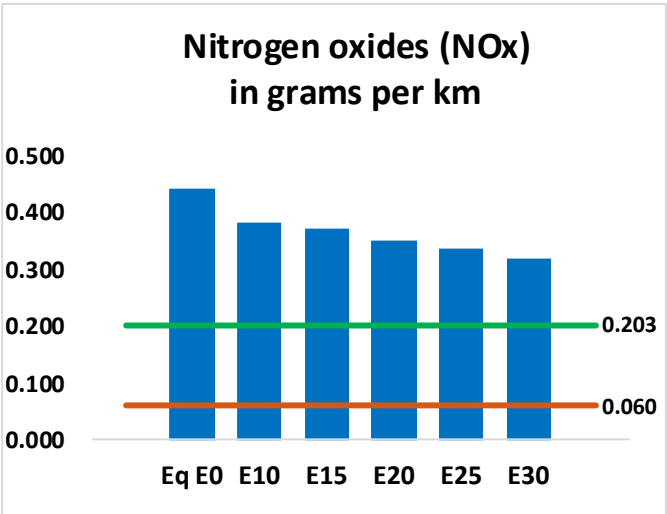
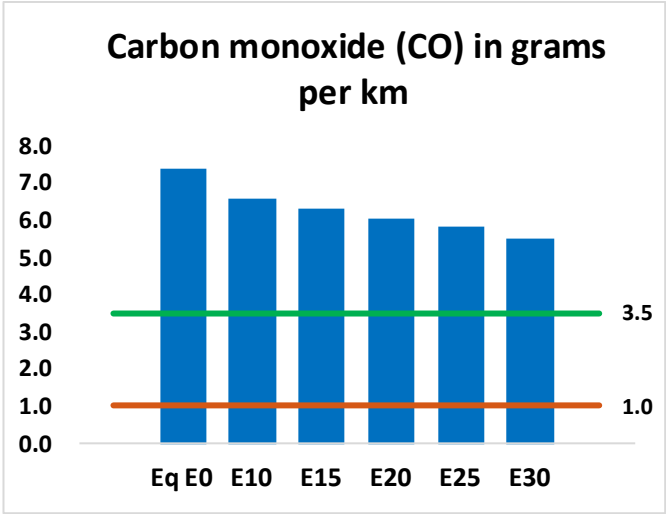
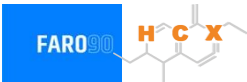
Vehicle Fleet: **4,986,666**

Age: 13 years

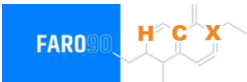
Source: CINAIE, 2023; analysis de Faro90

Ecuador – Vehicular Emissions

United States
Euro 6

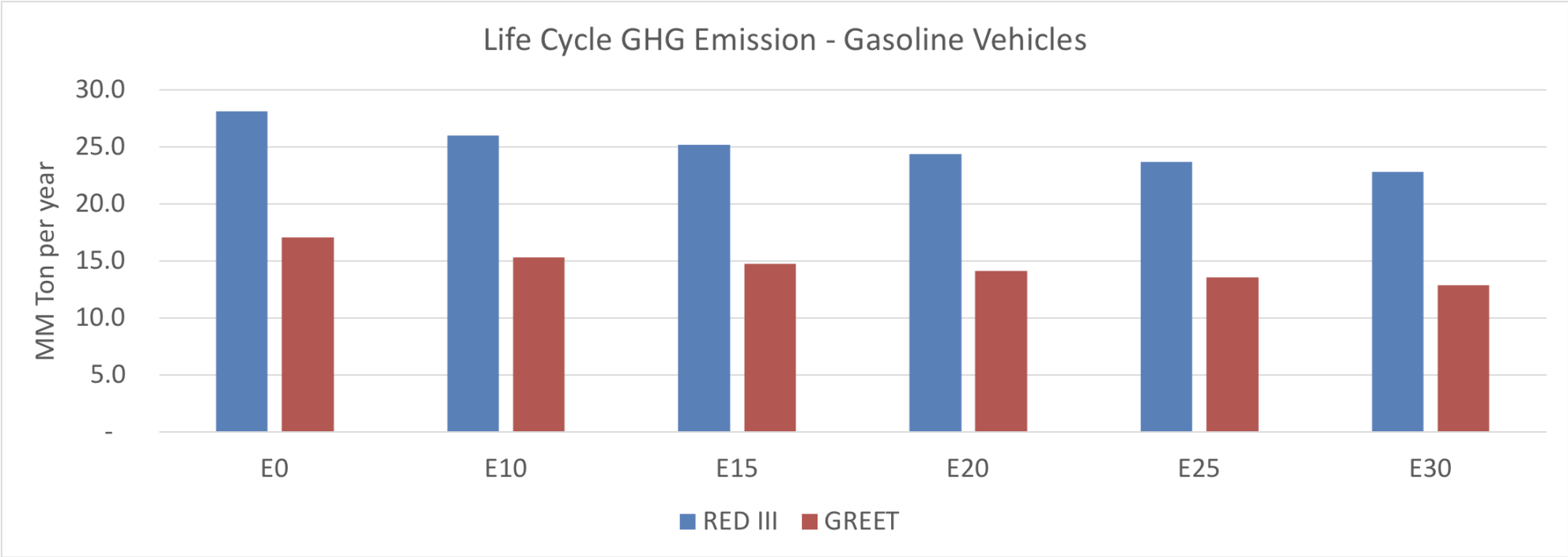


Ecuador – Gasoline Vehicle Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	866,631	820,176	773,721	739,588	707,051	682,578	647,991	-11%	-18%
CO2	23,349,725	22,749,304	22,182,239	21,736,259	21,515,571	21,390,678	20,996,386	-5%	-8%
VOC	106,789	102,613	98,437	95,802	93,512	91,861	89,272	-8%	-12%
NOx	51,967	47,982	45,211	43,779	41,495	39,694	37,580	-13%	-20%
SOx	266	246	226	209	192	175	156	-15%	-28%
NH3	8,890	8,802	8,714	8,755	8,853	8,923	8,980	-2%	0%
N2O	468	466	464	466	476	481	486	-1%	2%
CH4	23,868	23,868	23,868	24,226	24,751	25,205	25,634	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	920	892	863	848	837	830	816	-6%	-9%
Acetaldehyde	2,386	2,739	4,019	6,120	8,337	9,527	11,257	68%	249%
Formaldehyde	9,034	8,783	10,239	12,022	12,595	13,934	15,169	13%	39%
Benzene	1,408	1,353	1,282	1,262	1,252	1,197	1,158	-9%	-11%
PM2.5	1,325	1,205	1,084	978	875	763	647	-18%	-34%
PM10	4,452	4,047	3,642	3,286	2,938	2,564	2,173	-18%	-34%

Ecuador – Gasoline Vehicle Fleet GEI Emissions

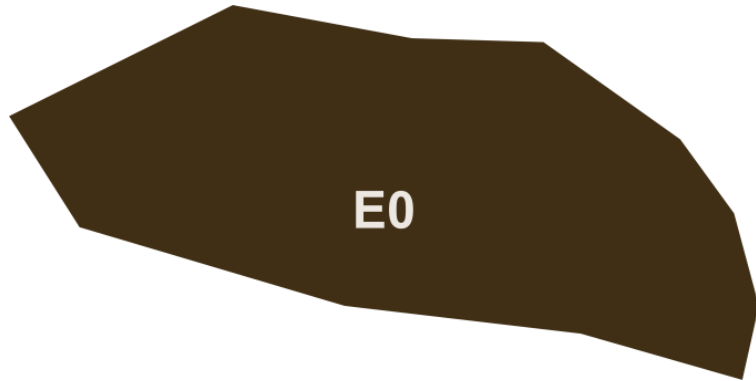


Country Emissions 2025 (MMT/year)	101.4
2035 Target (MMT/year)	71.0
Delta	30.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	10.0%	13.5%	17.1%	20.3%	24.4%
RED II/III	0.0%	7.5%	10.3%	13.2%	15.7%	18.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	5.6%	7.6%	9.6%	11.4%	13.6%
RED II/III	0.0%	7.0%	9.6%	12.2%	14.5%	17.3%

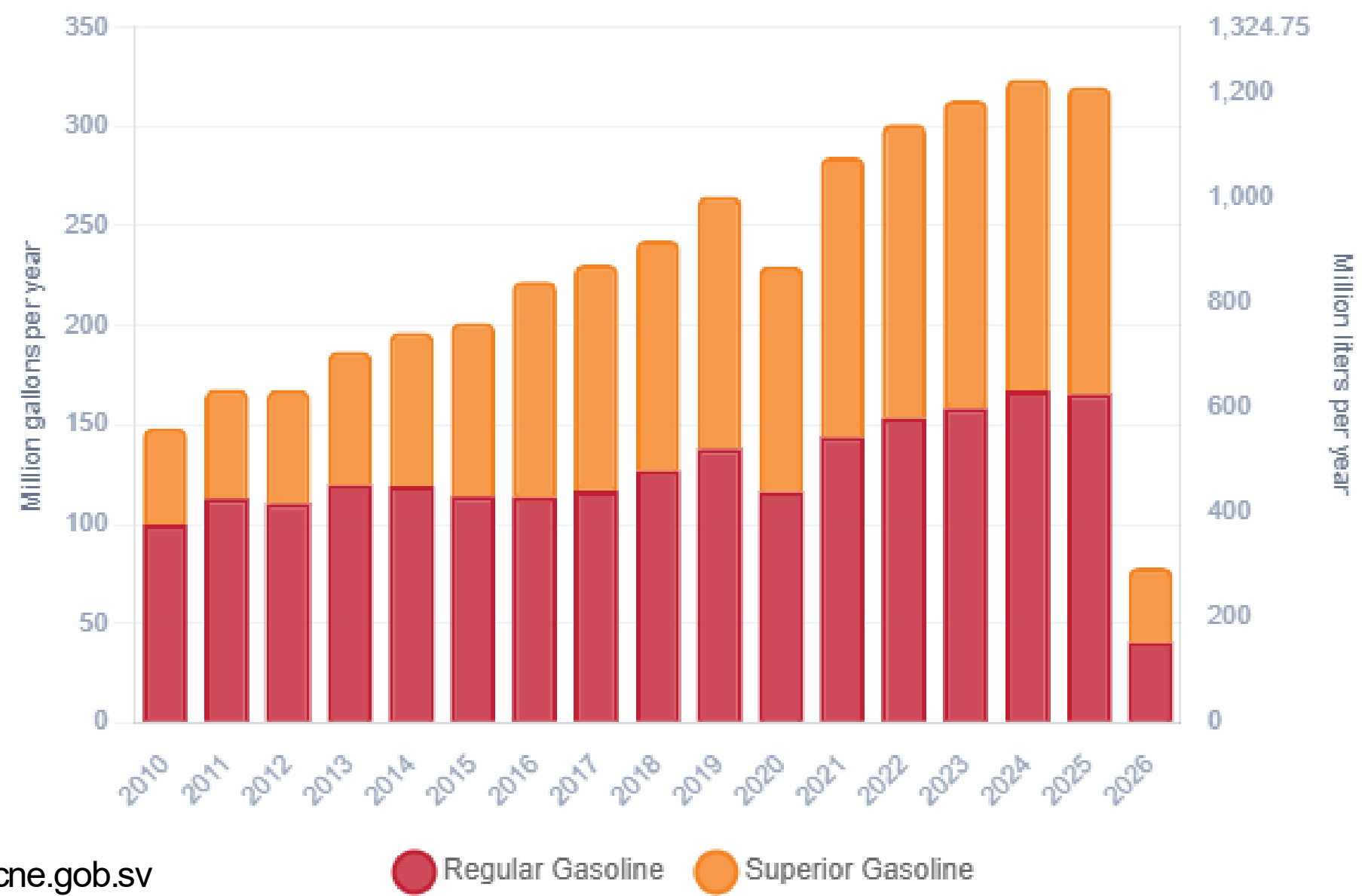
Ethanol Blending in Gasoline - El Salvador



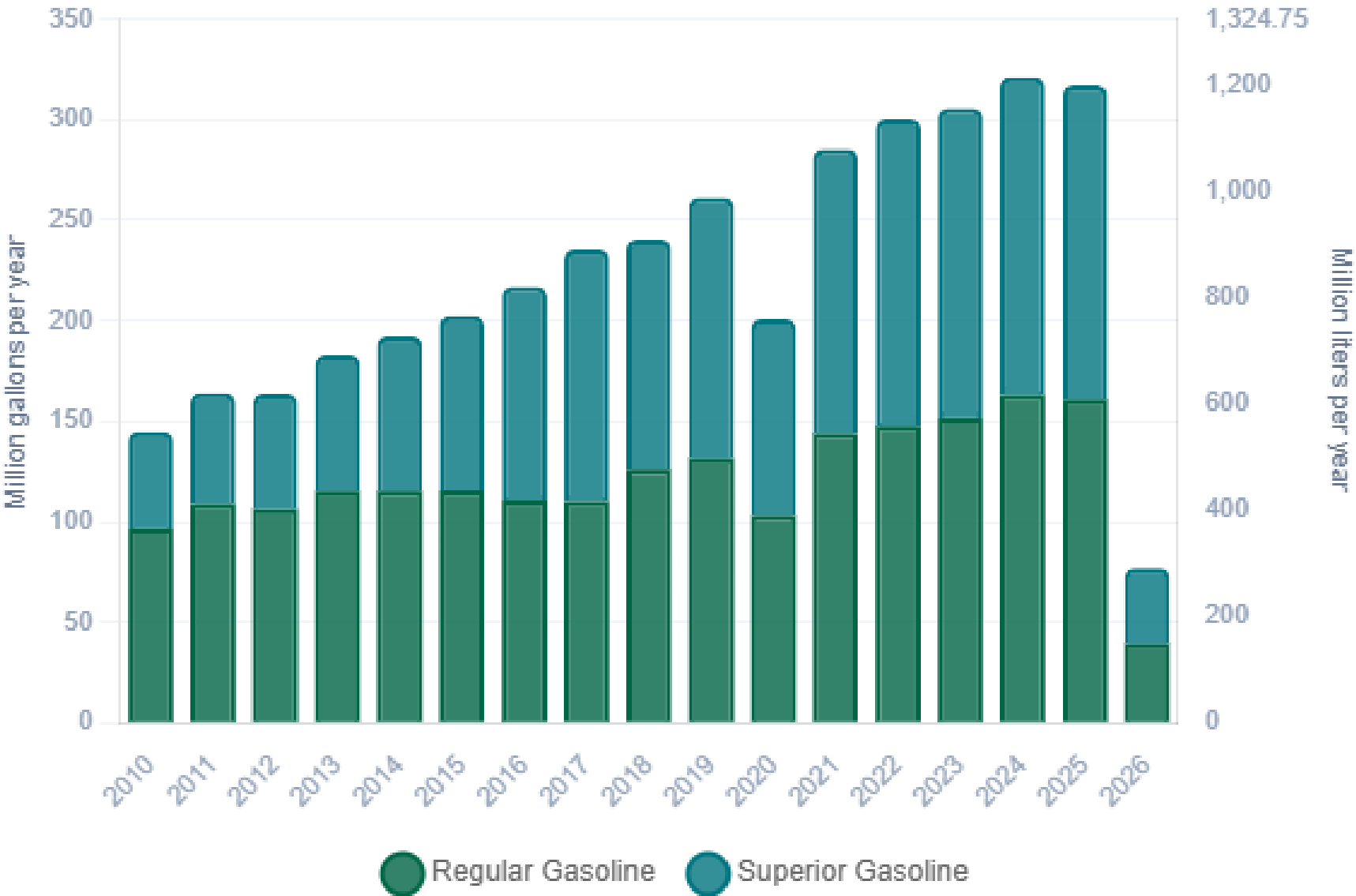
In 2025, gasoline demand was 328 million gallons (1,241 million liters). Regular gasoline RON 91 represented 51% of the volume consumed while Superior gasoline RON 95 reached 49%. Gasoline is supplied only by imports, distributed mainly by Chevron, Puma and Uno companies and imported mainly from the United States.

El Salvador has no national mandate for ethanol. It has the potential to produce ethanol from maize and sugar cane. In the last year there has been an interest for its use.

Gasoline Demand in El Salvador



Gasoline Imports in El Salvador



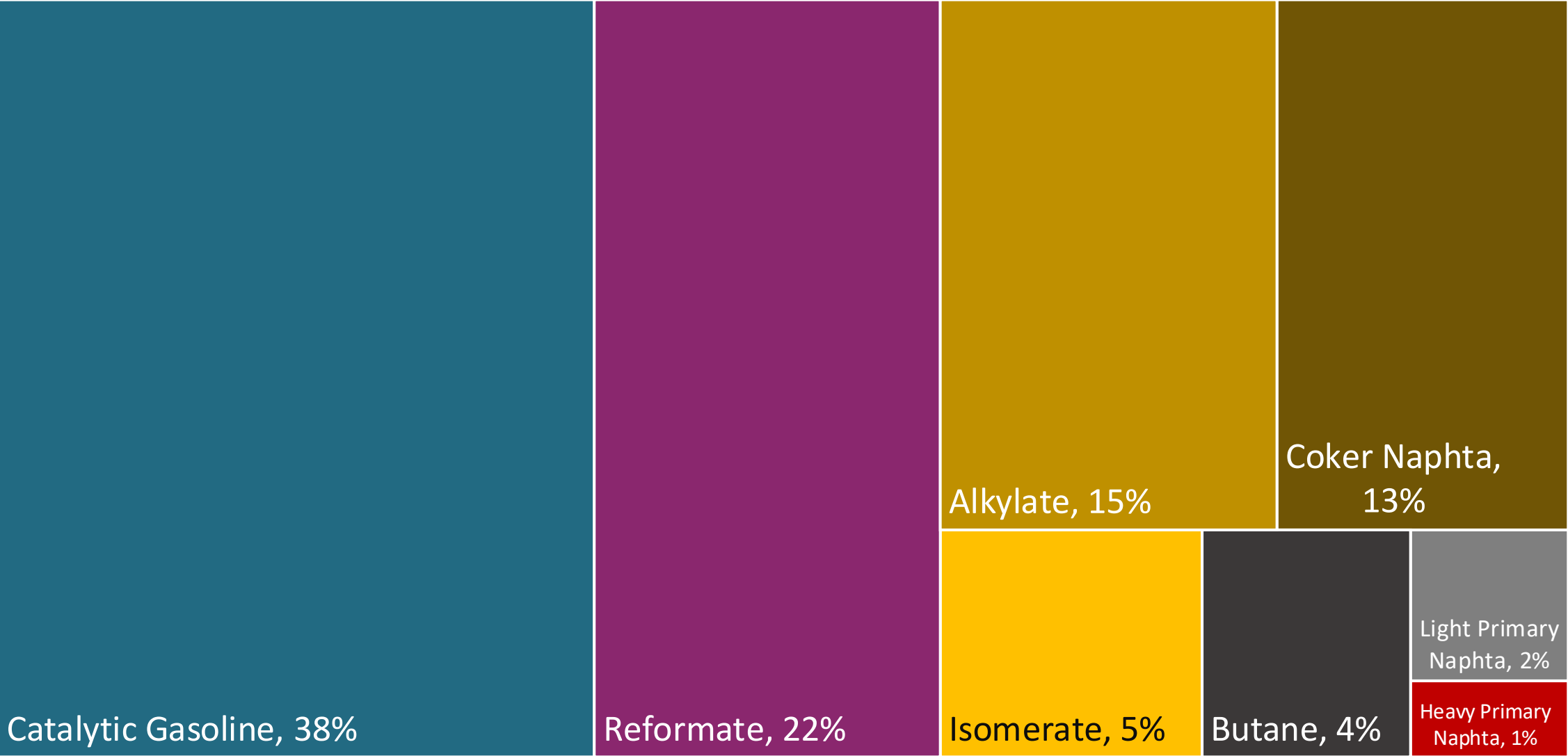
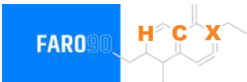
Source: cne.gob.sv

Gasoline Quality in El Salvador

Name	RTCA 75.01.19:19	RTCA 75.01.20:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	5% v/v	5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	2,0% v/v	2,0% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	0,7% v/v	0,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

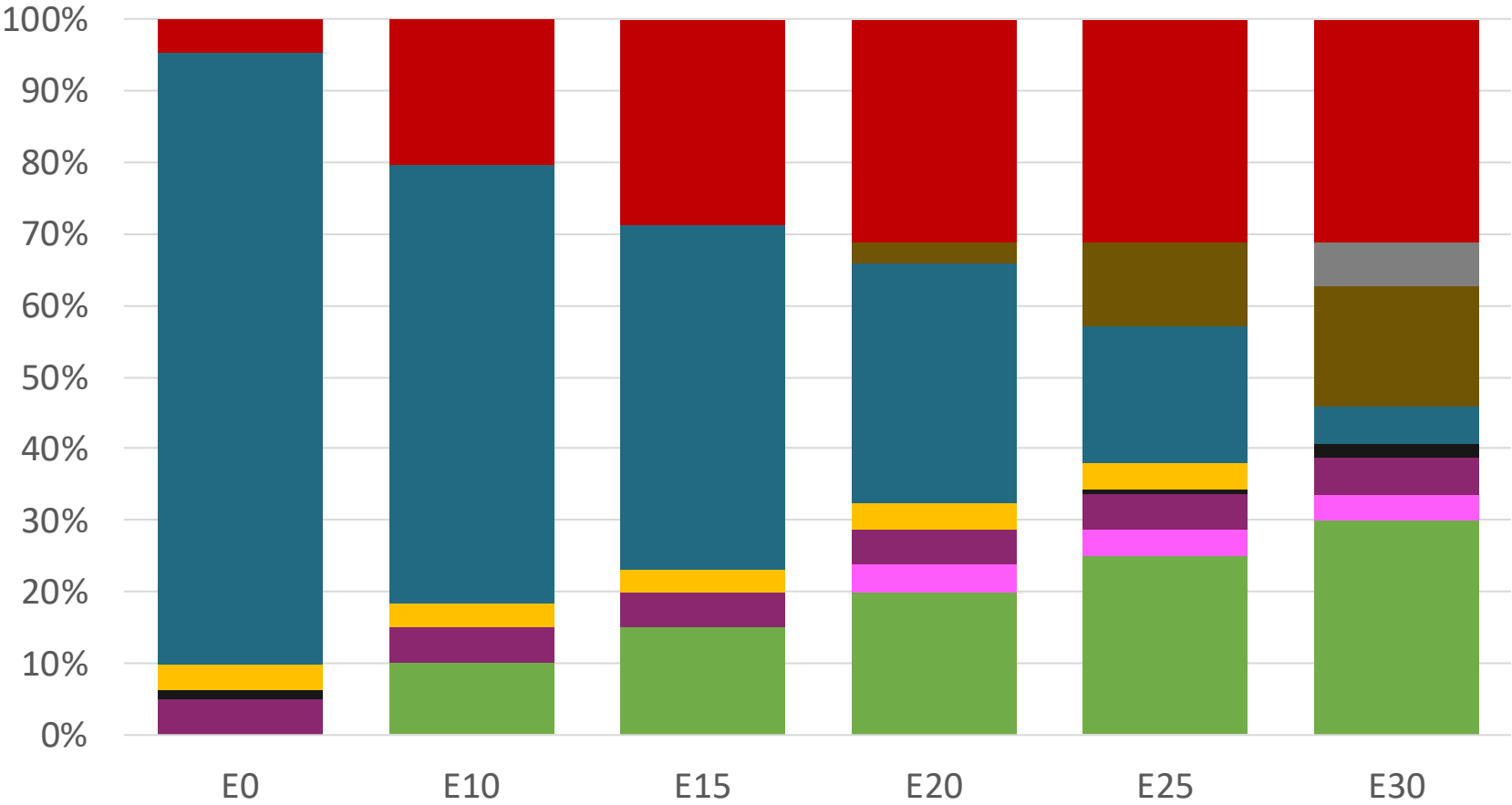
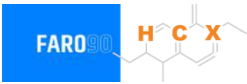
Source: RICA

Blending Components- El Salvador



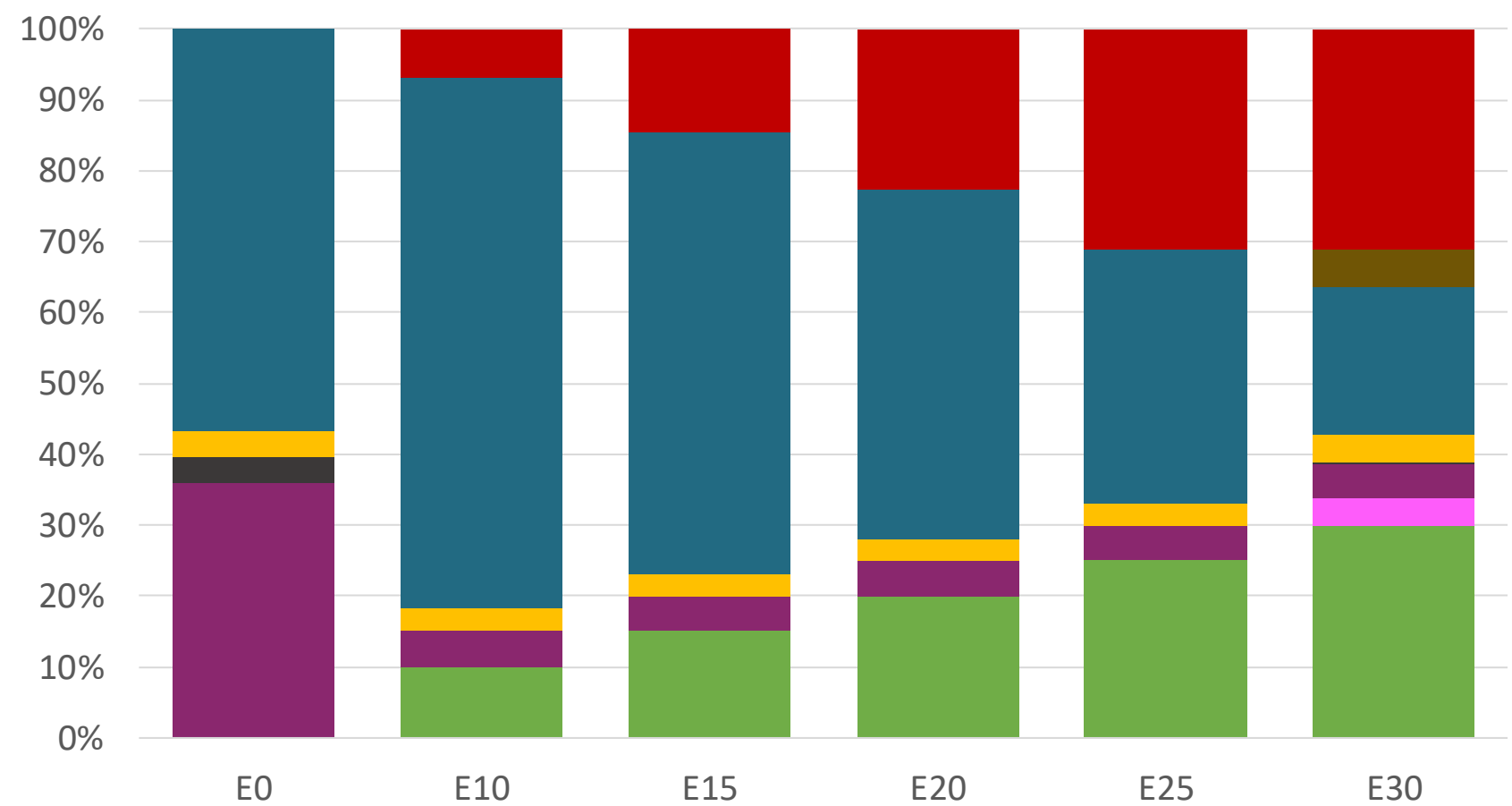
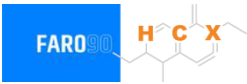
Source: Faro90

EI Salvador – Regular – Constant Octane



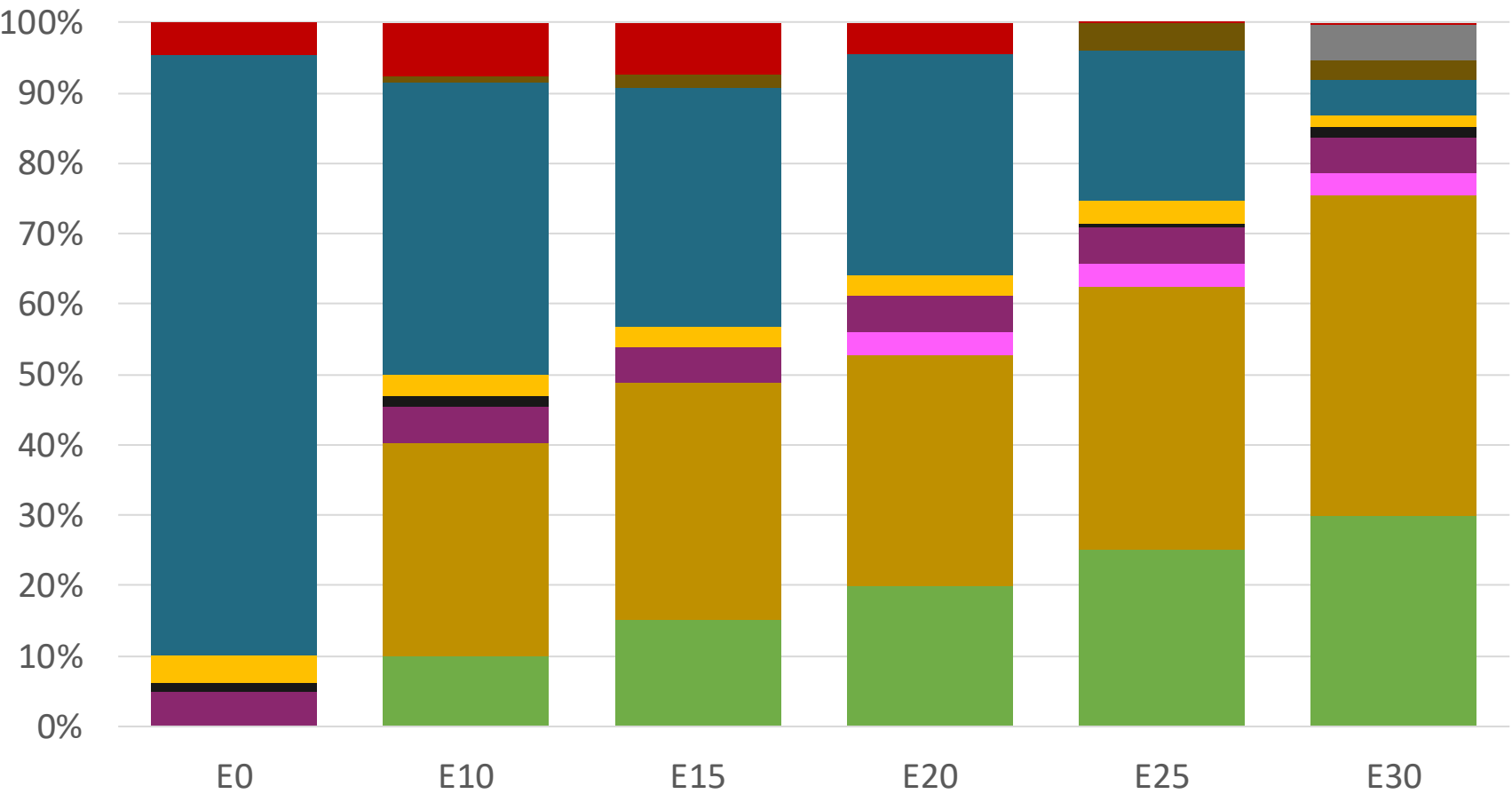
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.126	\$. 1.867	\$. 1.723	\$. 1.576	\$ 1.425	\$. 1.307

El Salvador – Premium – Constant Octane



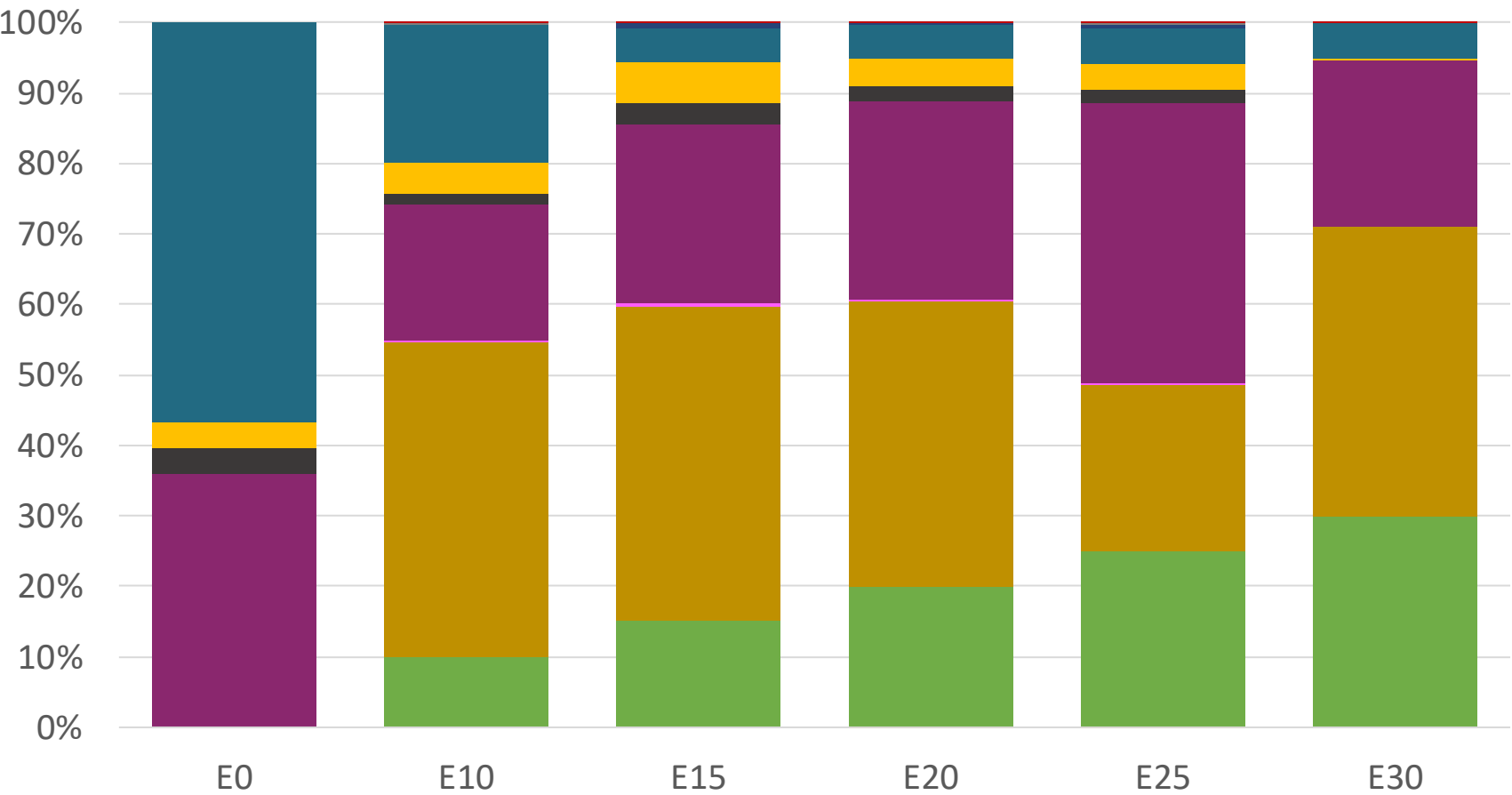
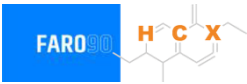
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.376	\$ 2.073	\$ 1.938	\$ 1.796	\$ 1.647	\$ 1.498

El Salvador – Regular – Increased Octane



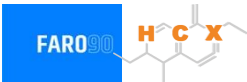
Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126

EI Salvador – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376

Gasoline Vehicular Fleet in El Salvador



Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles			
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	5			
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	739			
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	15,239			
		Euro III		> 8 years, > 160 tkm	37,987			
		Euro IV		> 8 years, > 160 tkm	95,925			
		Euro V / Euro 6		> 8 years, > 160 tkm	143,401			
				> 8 years, > 160 tkm	178,359			
				4-8 years, 80-160 tkm	167,508			
				< 4 years, < 80 tkm				
	Hybrid			Any	152			
Electric			Any	119				
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	12			
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	1,239			
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	17,096			
		Euro III		> 8 years, > 160 tkm	32,828			
		Euro IV		> 8 years, > 160 tkm	64,441			
		Euro V / Euro 6		> 8 years, > 160 tkm	73,656			
				> 8 years, > 160 tkm	67,493			
				4-8 years, 80-160 tkm	54,893			
				< 4 years, < 80 tkm				
	Hybrid			Any	74			
Electric			Any	58				
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	3,788			
		Euro II		> 8 years, > 160 tkm	16,390			
		Euro III		> 8 years, > 160 tkm	42,231			
		Euro IV		> 8 years, > 160 tkm	26,009			
		Euro V		4-8 years, 80-160 tkm	122,044			
				4-8 years, 80-160 tkm	250,554			
			< 4 years, < 80 tkm	241,102				
	Hybrid			Any	154			
	Electric			Any	120			
	Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	1,093		
FI		3-Ways	> 8 years, > 160 tkm		24,117			
		Euro III	> 8 years, > 160 tkm		21,929			
		Euro IV	> 8 years, > 160 tkm		22,379			
		Euro V / Euro 6	> 8 years, > 160 tkm		16,188			
			> 8 years, > 160 tkm		26,181			
		4-8 years, 80-160 tkm	114,901					
		< 4 years, < 80 tkm						
Hybrid			Any	20				
Electric			Any	16				

Vehicular Fleet: 1,652,936

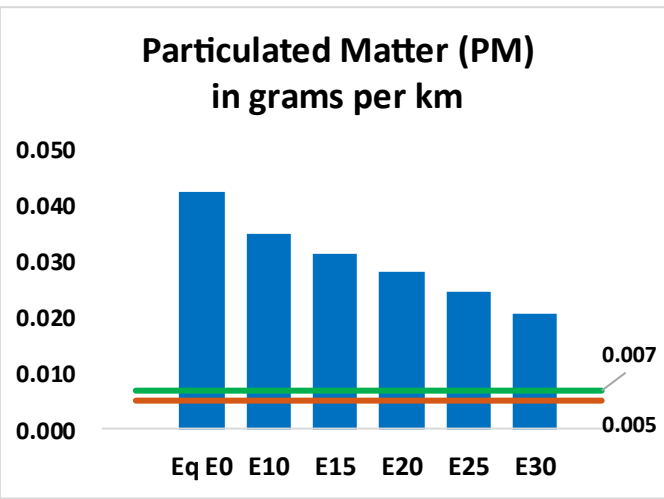
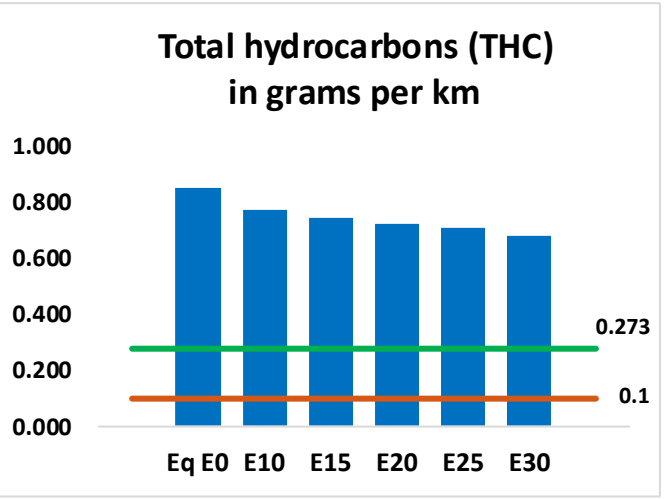
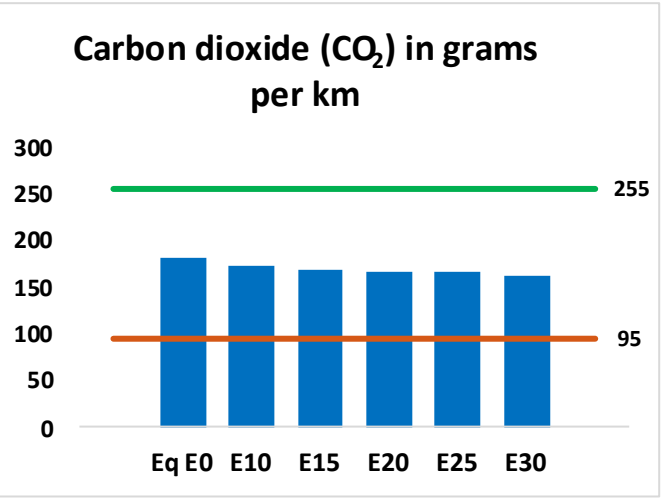
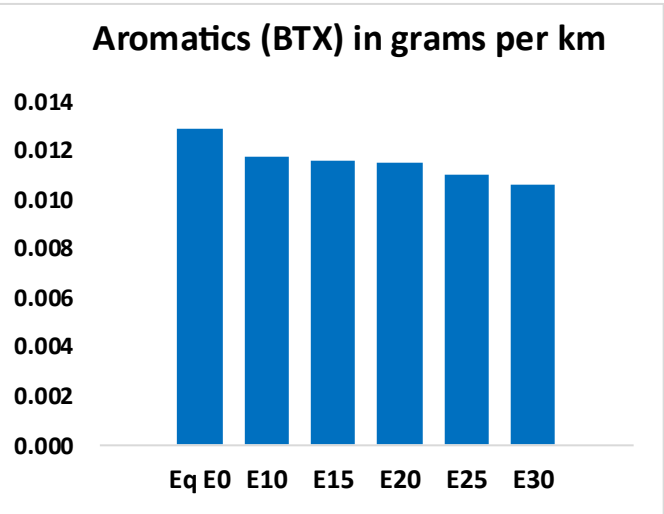
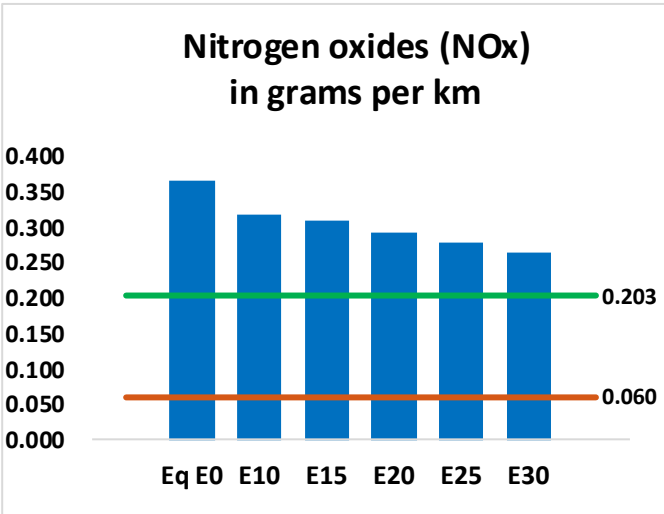
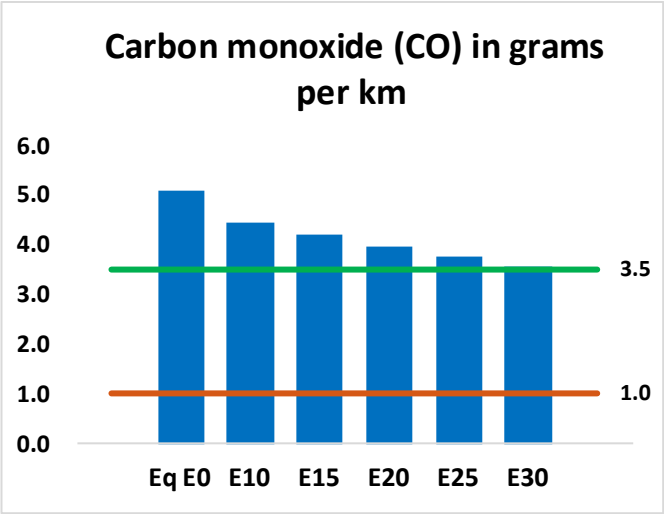
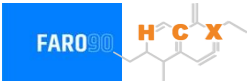
Average Age: 9 years

Motorcycles: 33%

Source: Registro Publico, Faro 90

El Salvador - Vehicular Emissions

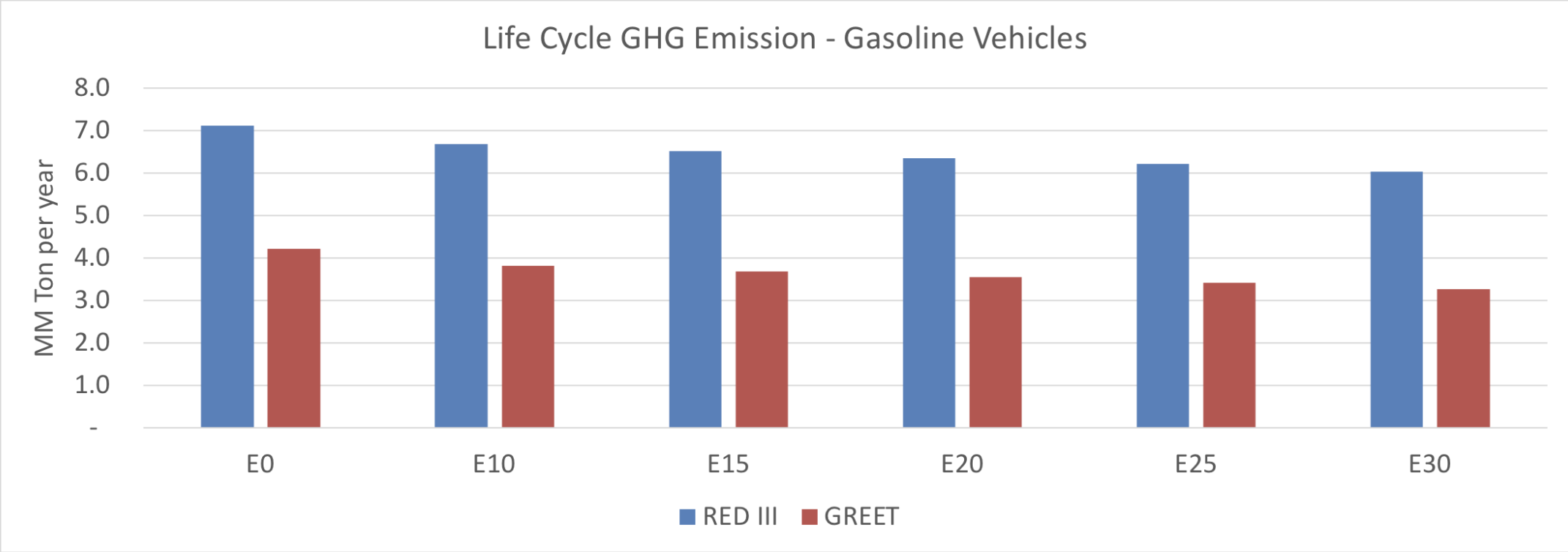
United States
Euro 6



El Salvador - Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	164,291	154,054	143,818	135,896	128,140	122,240	115,143	-12%	-22%
CO2	5,847,536	5,697,171	5,555,160	5,443,472	5,388,204	5,364,902	5,266,011	-5%	-8%
VOC	27,515	26,272	25,029	24,181	23,406	22,833	21,985	-9%	-15%
NOx	11,842	10,934	10,304	9,979	9,460	9,051	8,570	-13%	-20%
SOx	67	62	57	52	48	44	39	-15%	-28%
NH3	2,277	2,254	2,232	2,242	2,267	2,285	2,300	-2%	0%
N2O	71	71	70	71	72	73	74	-1%	2%
CH4	6,419	6,419	6,419	6,515	6,656	6,778	6,894	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	139	130	121	115	108	103	96	-12%	-22%
Acetaldehyde	385	442	649	988	1,346	1,538	1,817	68%	249%
Formaldehyde	1,520	1,478	1,723	2,023	2,120	2,345	2,553	13%	39%
Benzene	418	402	381	375	372	356	344	-9%	-11%
PM2.5	359	326	293	265	237	207	175	-18%	-34%
PM10	1,018	925	833	751	672	586	497	-18%	-34%

El Salvador – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	14.8
2035 Target (MMT/year)	10.3
Delta	4.4

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.4%	12.7%	16.0%	18.8%	22.6%
RED II/III	0.0%	6.3%	8.6%	10.8%	12.8%	15.2%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	8.9%	12.0%	15.2%	17.9%	21.5%
RED II/III	0.0%	10.1%	13.8%	17.4%	20.6%	24.5%

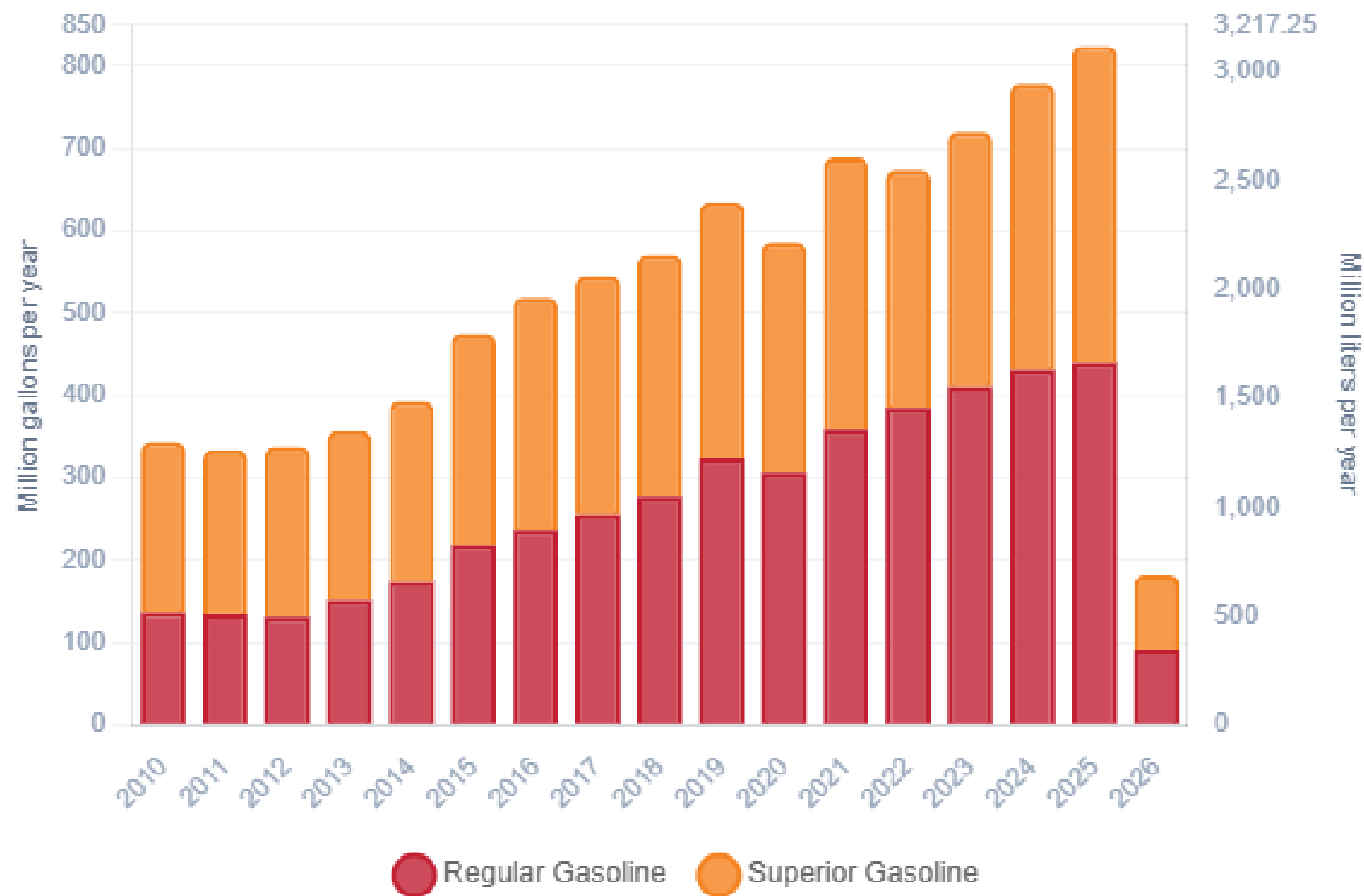
Ethanol Blending in Gasoline - Guatemala



In 2025, gasoline consumption was 824 million gallons (3,119 million liters). Gasoline is supplied only by imports, mainly from United States. There are two grades of gasoline: (RON 91 - AKI 85) and Superior (RON 95 - AKI 89). 53% of gasoline is gasoline Regular.

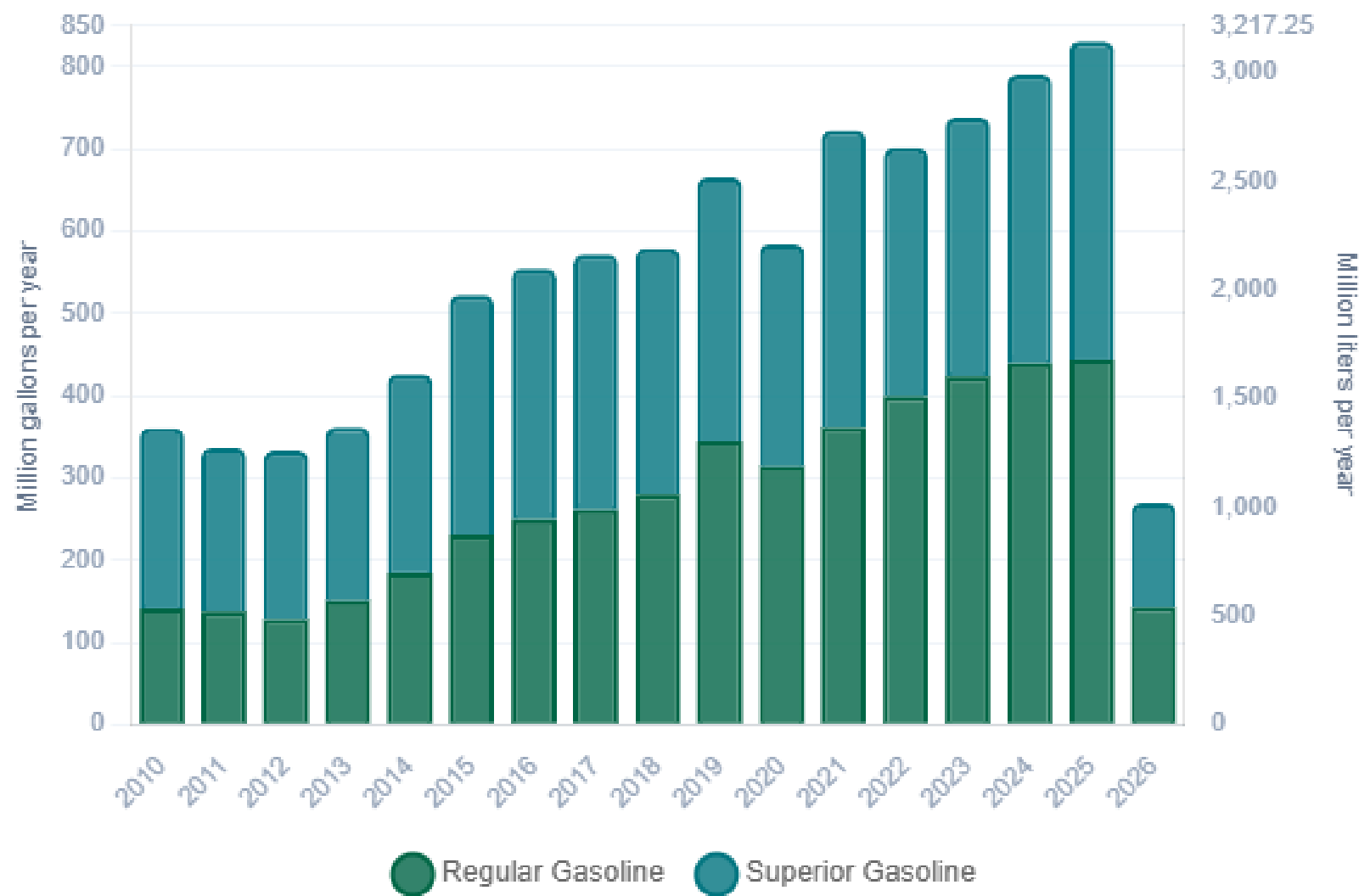
Guatemala approved the mandate for E10 blends throughout the territory in 2025. It has local ethanol advanced ethanol production that is exported to Europe. Guatemala is the main ethanol producer in Central America.

Gasoline Demand in Guatemala



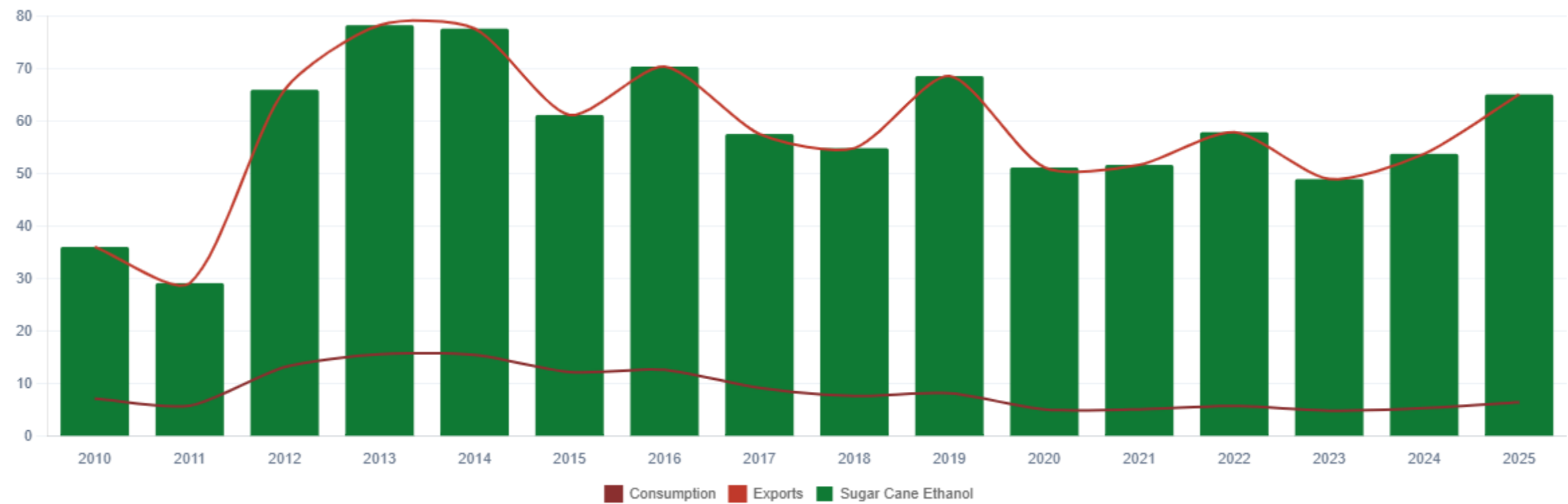
Source: MEM, 2025

Gasoline Imports in Guatemala



Source: MEM, 2025

Ethanol Balance in Guatemala



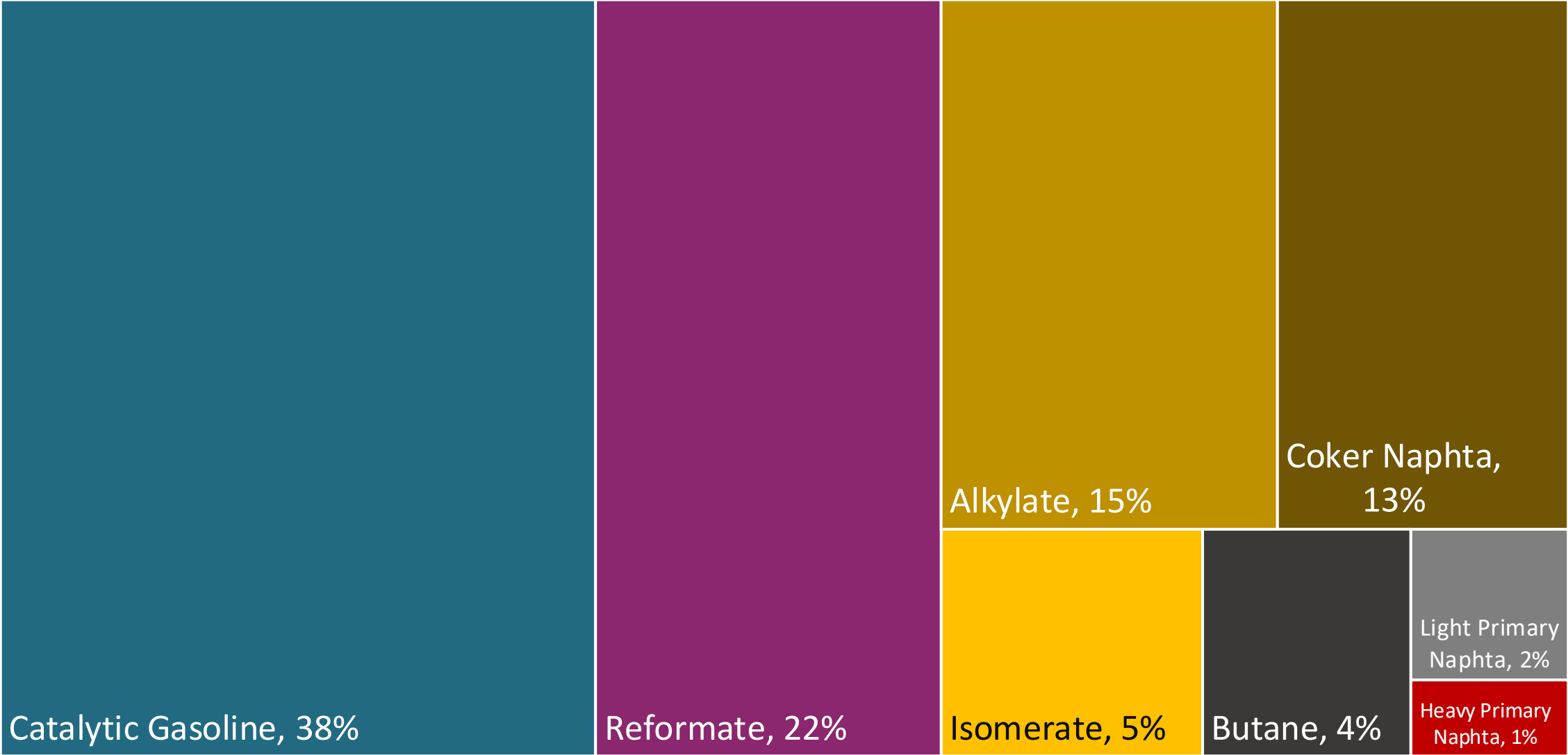
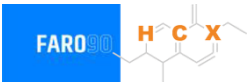
Source: Asociación de productores de etanol de Guatemala

Gasoline Quality in Guatemala

Name	Ministerial Accord Number 320-2022		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2022		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Regular	Gasoline Superior	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 2,5% v/v	< 2,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	- / < 50% v/v	- / < 50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	- / < 30% v/v	- / < 30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,5 mg/l	< 2,5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	-	-	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	10% v/v	10% v/v	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

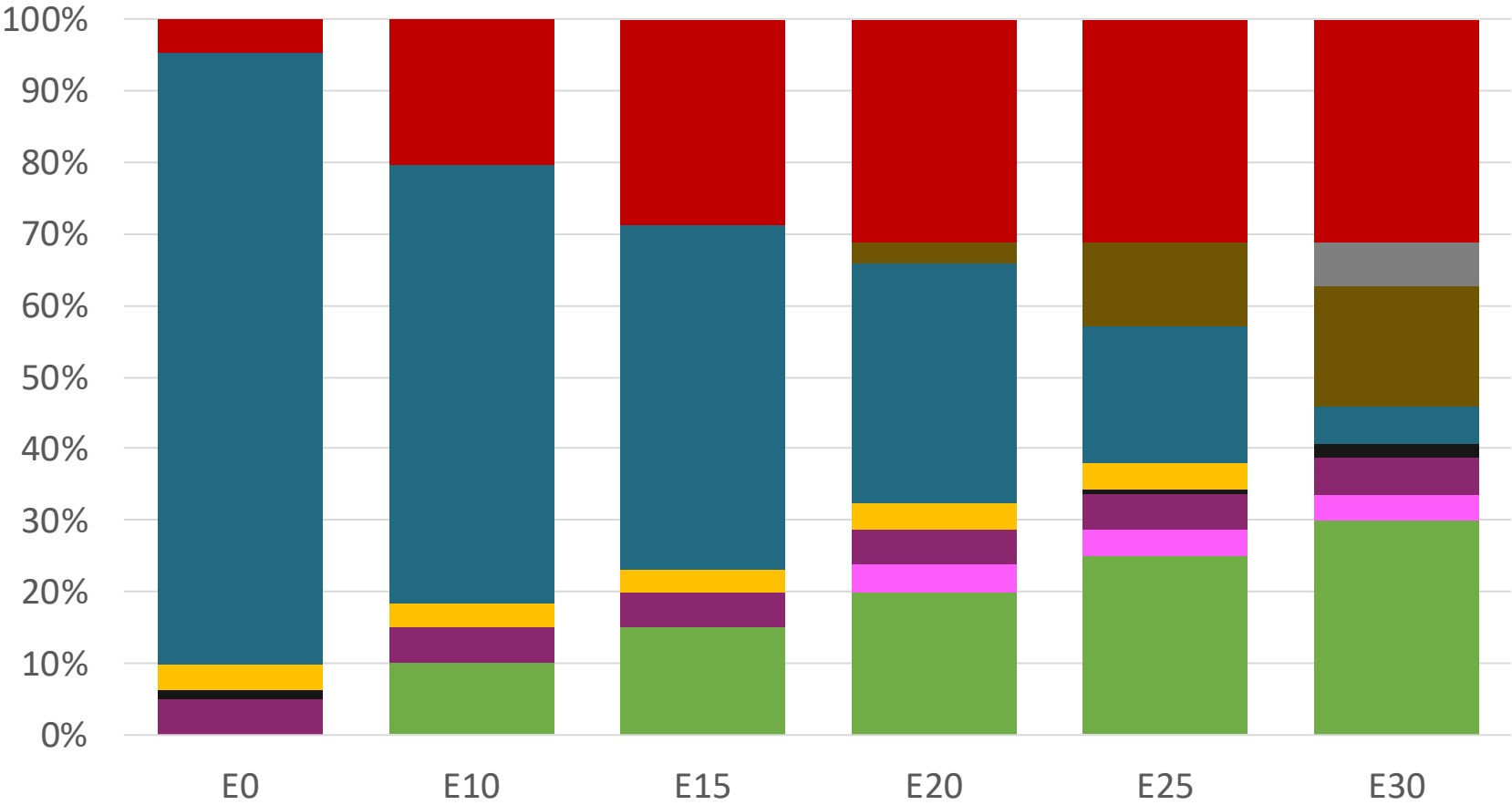
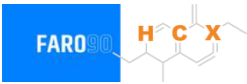
Source: MEM, 2022

Blending Components - Guatemala



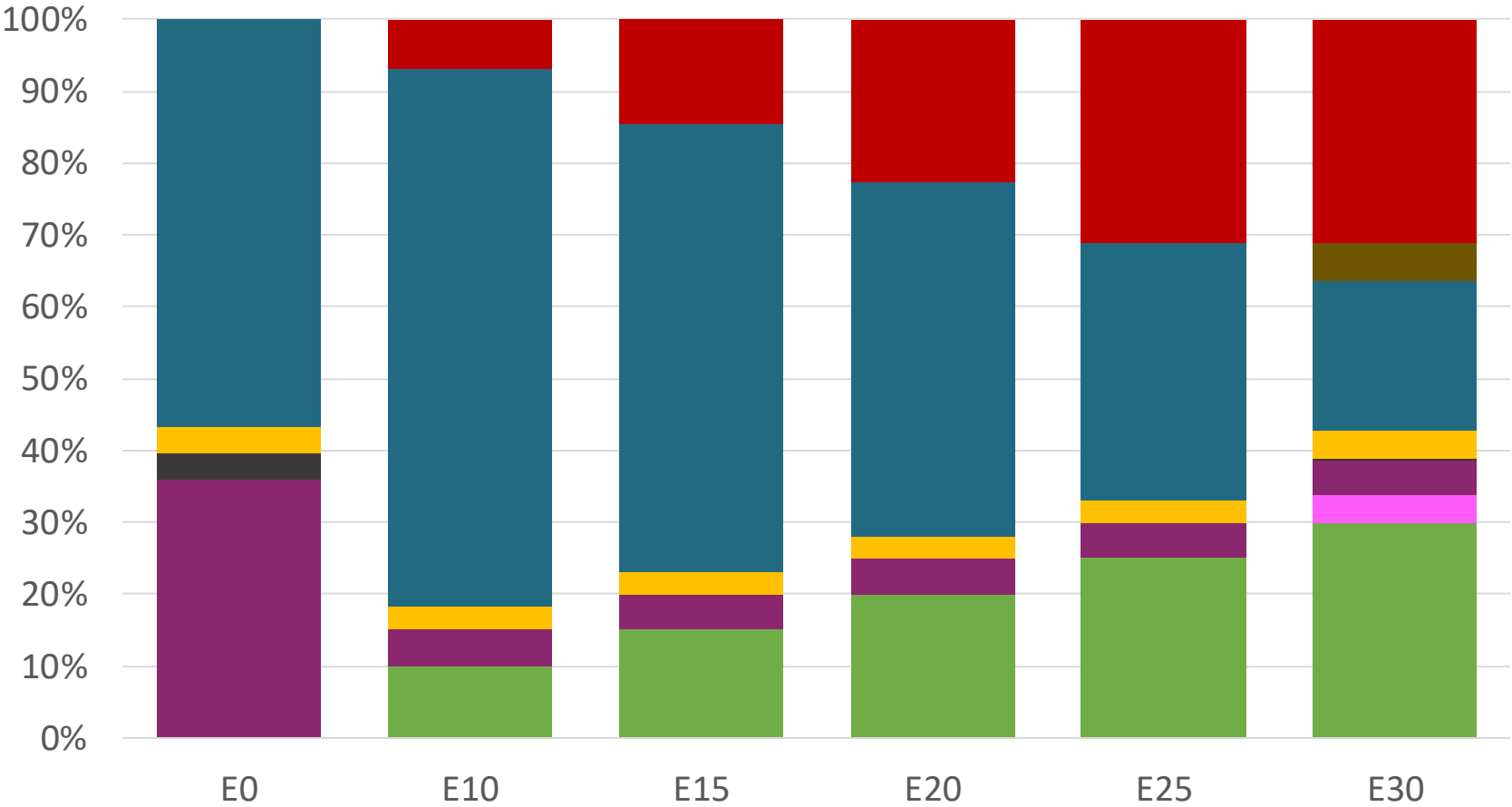
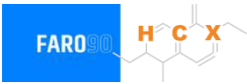
Source: Faro90

Guatemala – Regular – Constant Octane



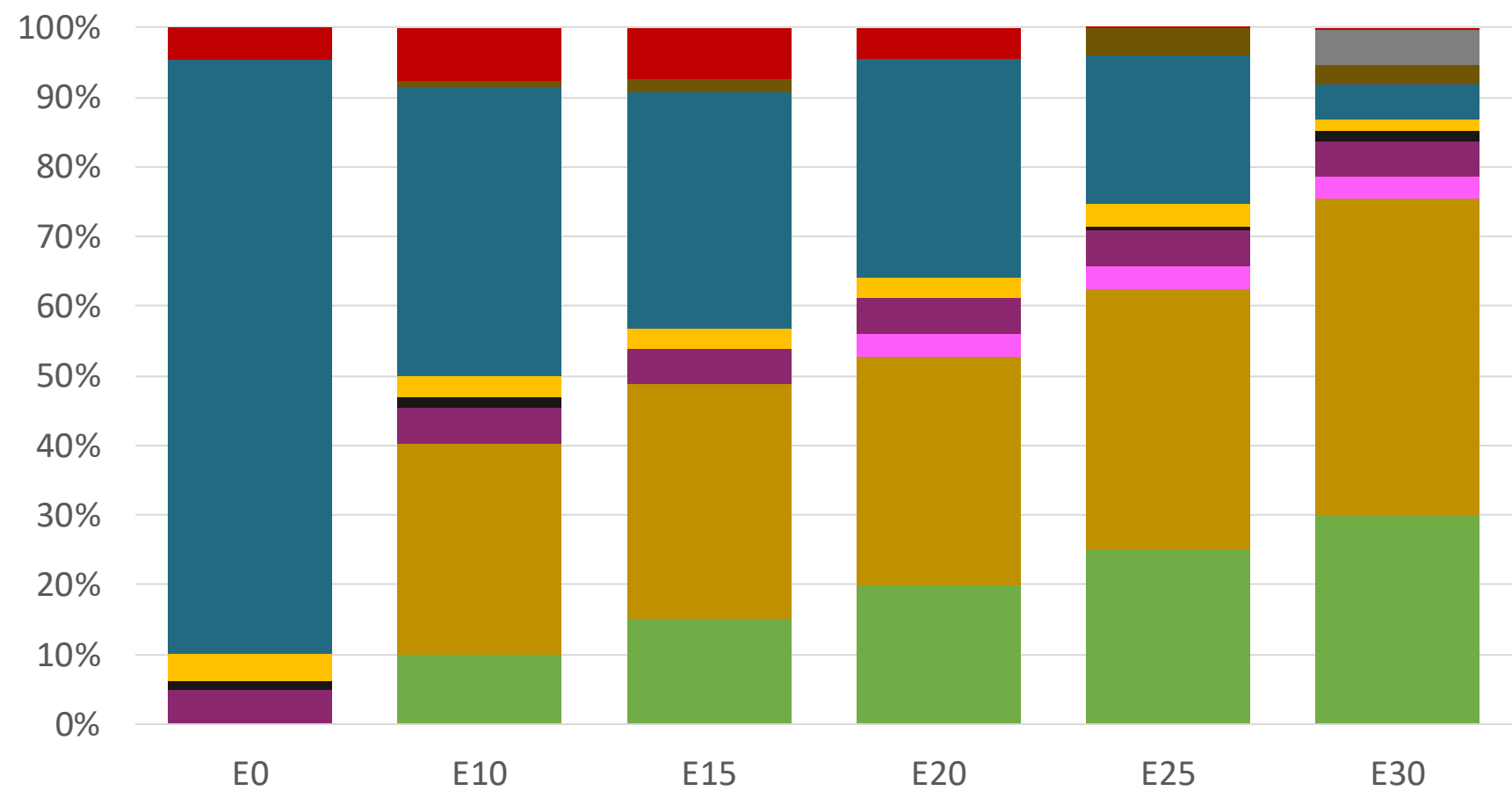
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.126	\$. 1.867	\$. 1.723	\$. 1.576	\$. 1.425	\$. 1.307

Guatemala – Premium – Constant Octane



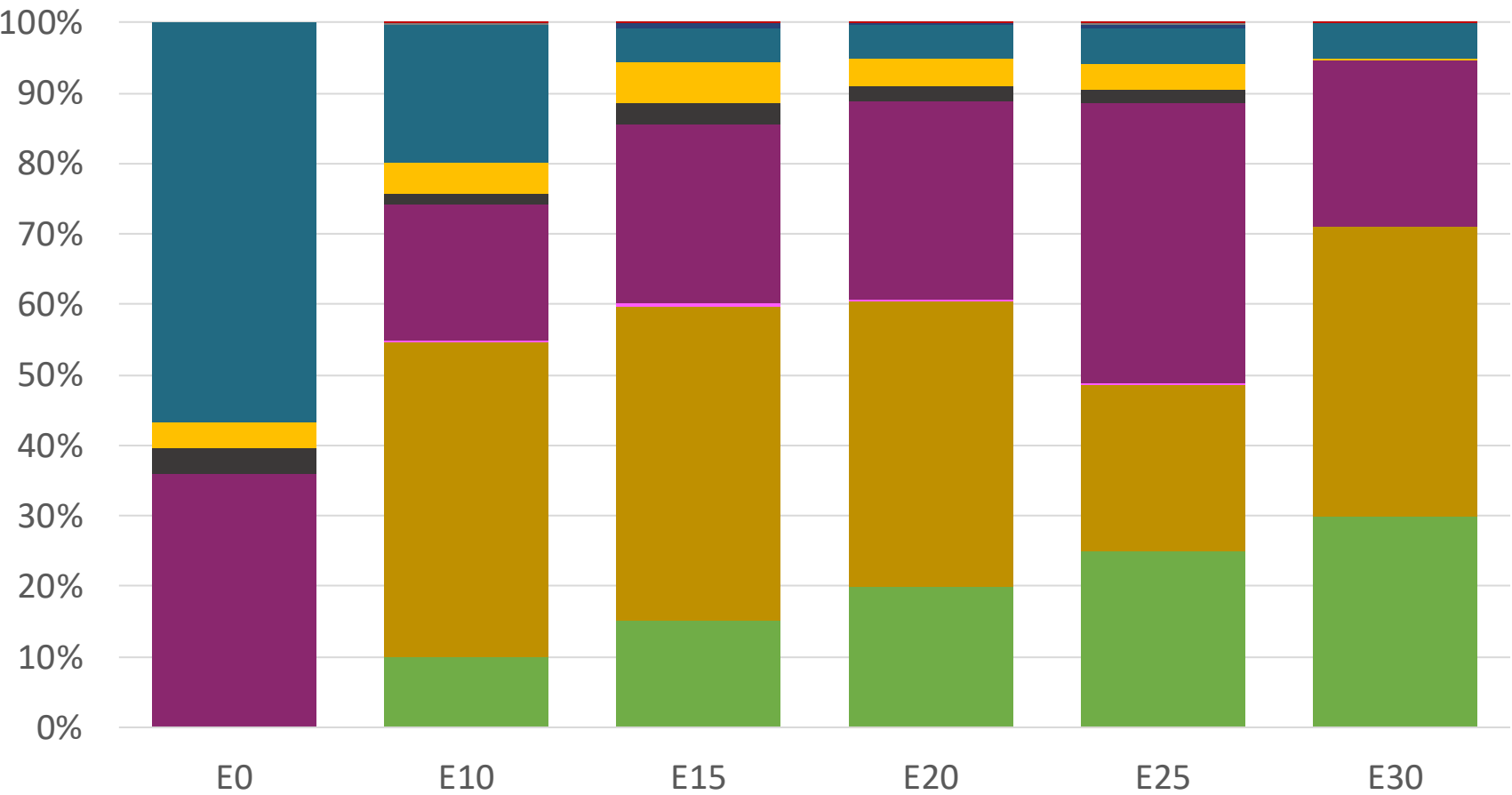
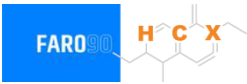
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.376	\$ 2.073	\$ 1.938	\$ 1.796	\$ 1.647	\$ 1.498

Guatemala – Regular – Increased Octane



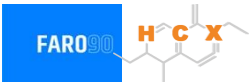
Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126

Guatemala – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376	\$ 2.376

Gasoline Vehicular Fleet - Guatemala



Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	14,668				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	51,436				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	362,121				
		Euro III		> 8 years, > 160 tkm	239,410				
		Euro IV		> 8 years, > 160 tkm	160,262				
		Euro V / Euro 6		> 8 years, > 160 tkm	97,733				
				4-8 years, 80-160 tkm	52,603				
				< 4 years, < 80 tkm	27,177				
		Hybrid			Any	6,278			
	Electric			Any	785				
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	33,594				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	294,887				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	650,178				
		Euro III		> 8 years, > 160 tkm	277,862				
		Euro IV		> 8 years, > 160 tkm	160,947				
		Euro V / Euro 6		> 8 years, > 160 tkm	120,025				
				4-8 years, 80-160 tkm	116,456				
				< 4 years, < 80 tkm	165,761				
		Hybrid			Any	11,362			
	Electric			Any	1,420				
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	49,306				
		Euro II		> 8 years, > 160 tkm	59,976				
		Euro III		> 8 years, > 160 tkm	645,317				
		Euro IV		> 8 years, > 160 tkm	141,330				
		Euro V		4-8 years, 80-160 tkm	483,674				
				4-8 years, 80-160 tkm	200,576				
				< 4 years, < 80 tkm	1,442,641				
		Electric			Any	974			
	Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	16,289			
FI		3-Ways	> 8 years, > 160 tkm		232,495				
		Euro III	> 8 years, > 160 tkm		74,845				
		Euro IV	> 8 years, > 160 tkm		36,066				
		Euro V / Euro 6	> 8 years, > 160 tkm		20,853				
			4-8 years, 80-160 tkm		27,307				
			< 4 years, < 80 tkm		52,267				
		Hybrid			Any	9			
		Electric			Any	215			

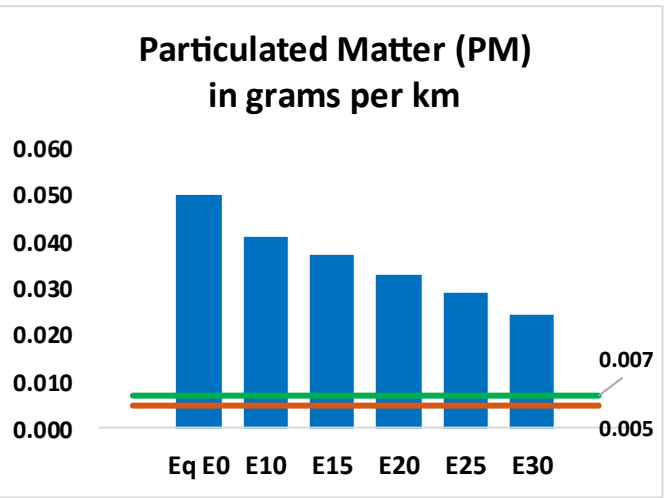
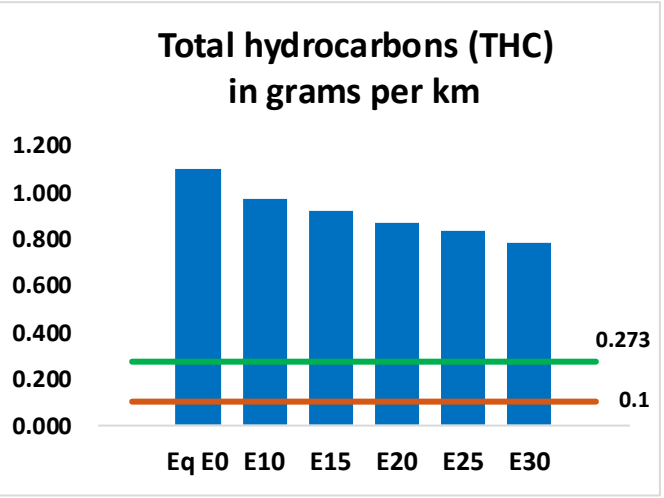
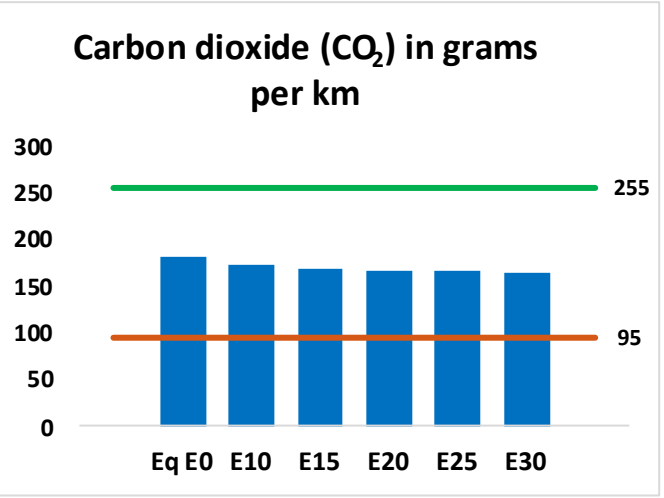
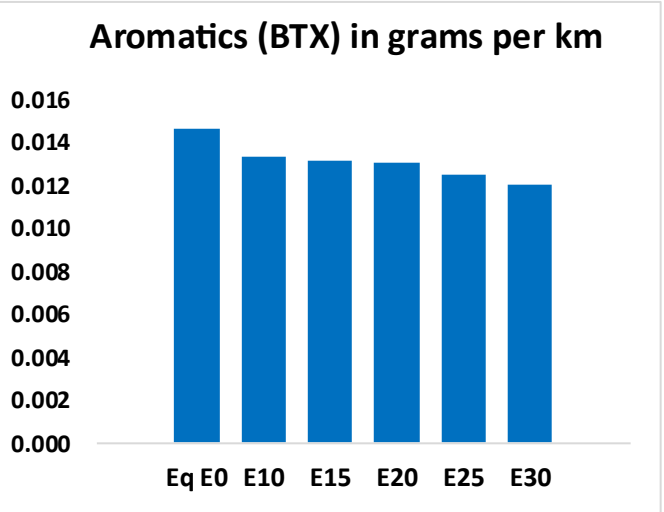
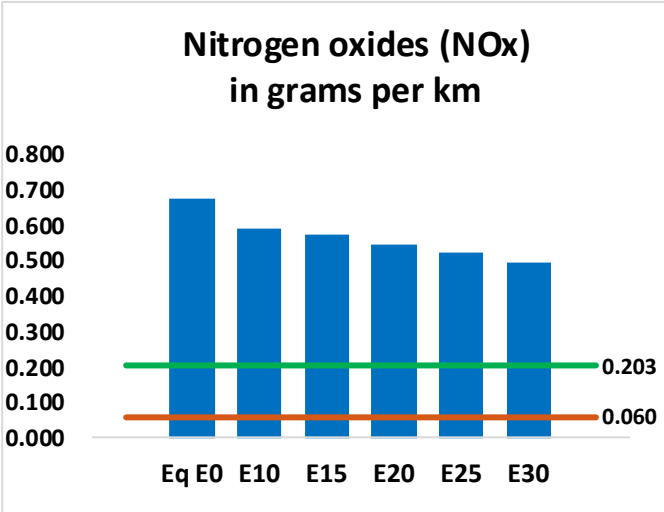
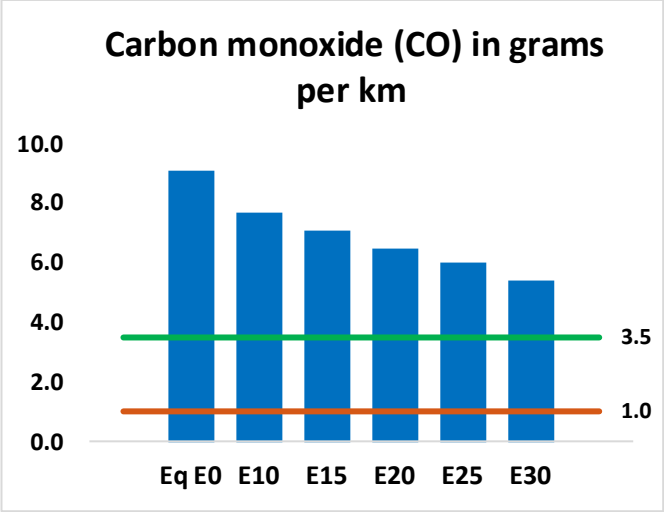
Gasoline Vehicular Fleet: 5,915,613

Average age: 13 years

Motorcycles: 48%

Source: INE, analysis Faro 90

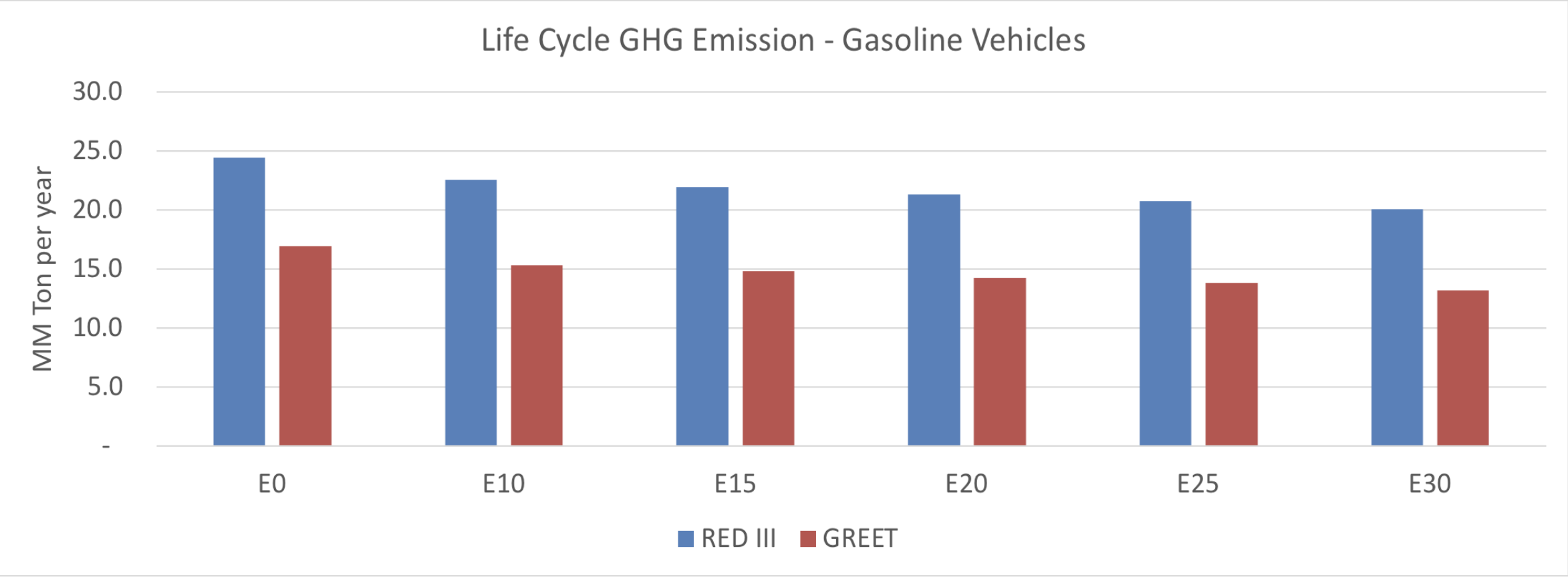
Guatemala – Vehicular Emissions



Guatemala – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	964,717	887,827	810,937	747,397	683,210	633,780	572,436	-16%	-29%
CO2	19,217,409	18,723,247	18,256,539	17,889,486	17,707,854	17,630,789	17,305,803	-5%	-8%
VOC	115,959	109,110	102,260	97,050	91,993	88,160	82,847	-12%	-21%
NOx	71,550	66,064	62,410	60,606	57,636	55,337	52,612	-13%	-19%
SOx	220	203	186	172	159	144	129	-15%	-28%
NH3	7,058	6,988	6,918	6,951	7,029	7,084	7,130	-2%	0%
N2O	524	522	520	522	533	539	545	-1%	2%
CH4	26,581	26,581	26,581	26,980	27,565	28,070	28,548	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	951	905	859	827	797	774	742	-10%	-16%
Acetaldehyde	2,182	2,504	3,675	5,596	7,624	8,712	10,294	68%	249%
Formaldehyde	7,888	7,669	8,940	10,497	10,998	12,166	13,244	13%	39%
Benzene	1,554	1,493	1,415	1,392	1,381	1,320	1,278	-9%	-11%
PM2.5	1,102	1,002	902	813	727	635	538	-18%	-34%
PM10	4,202	3,820	3,438	3,102	2,773	2,420	2,051	-18%	-34%

Guatemala – Gasoline Vehicle Fleet GEI Emissions

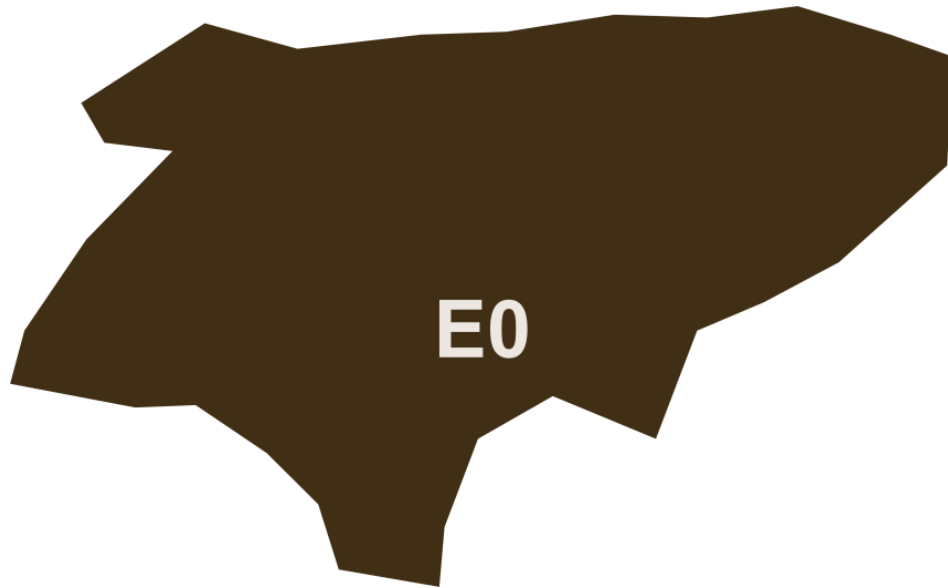


Country Emissions 2025 (MMT/year)	43.5
2035 Target (MMT/year)	21.7
Delta	21.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.8%	12.7%	15.9%	18.6%	22.1%
RED II/III	0.0%	7.7%	10.1%	12.8%	15.0%	17.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	7.6%	9.9%	12.4%	14.5%	17.2%
RED II/III	0.0%	8.6%	11.4%	14.4%	16.9%	20.0%

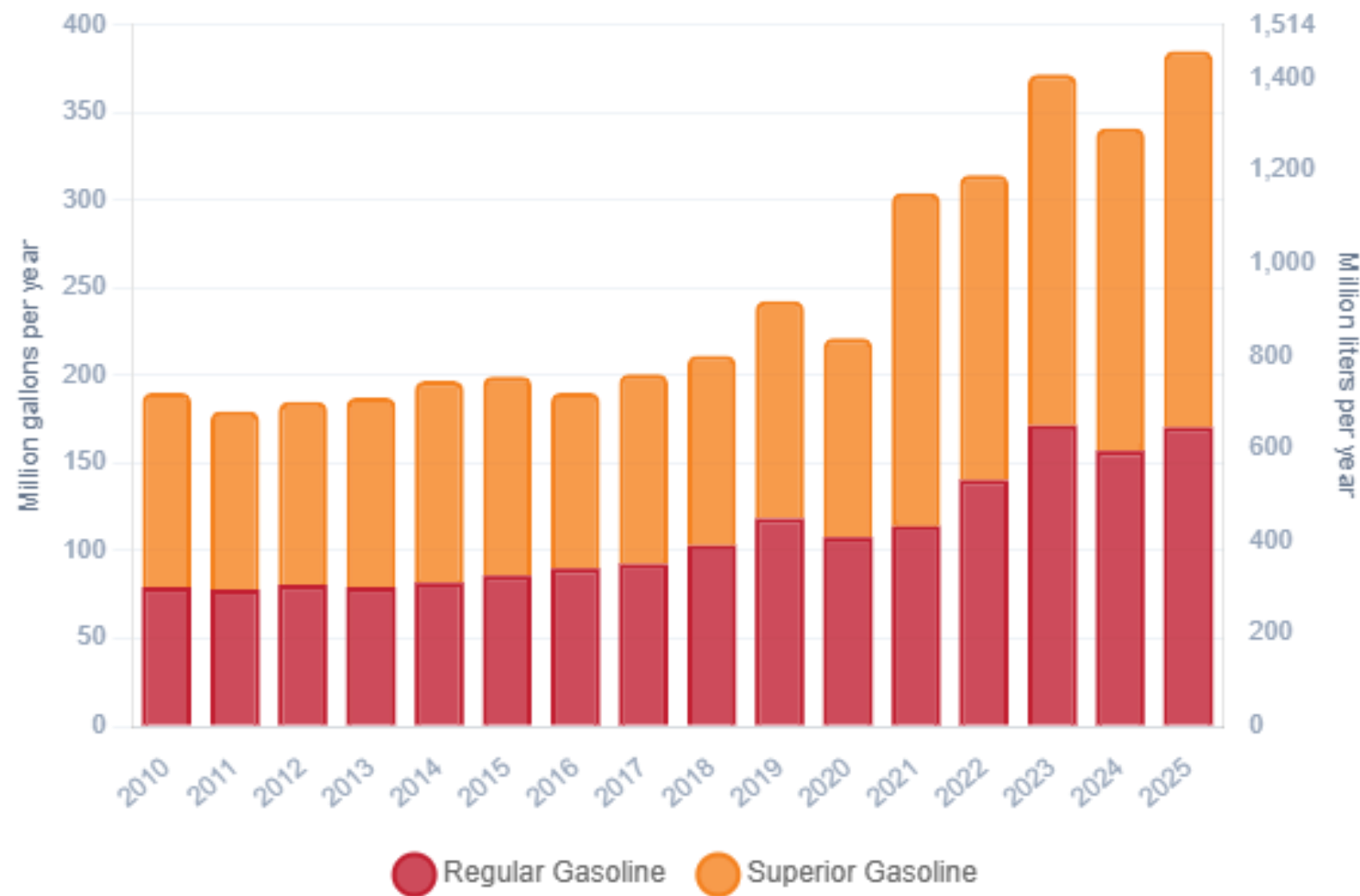
Ethanol Blending in Gasoline - Honduras



In 2025, gasoline demand was 385 million gallons (1,457 million liters). Regular gasoline RON 91 represented 44% while Superior gasoline RON 95 reached 56% of national demand. Honduras does not have gasoline production capacity and supplies its market with imports mainly from the United States.

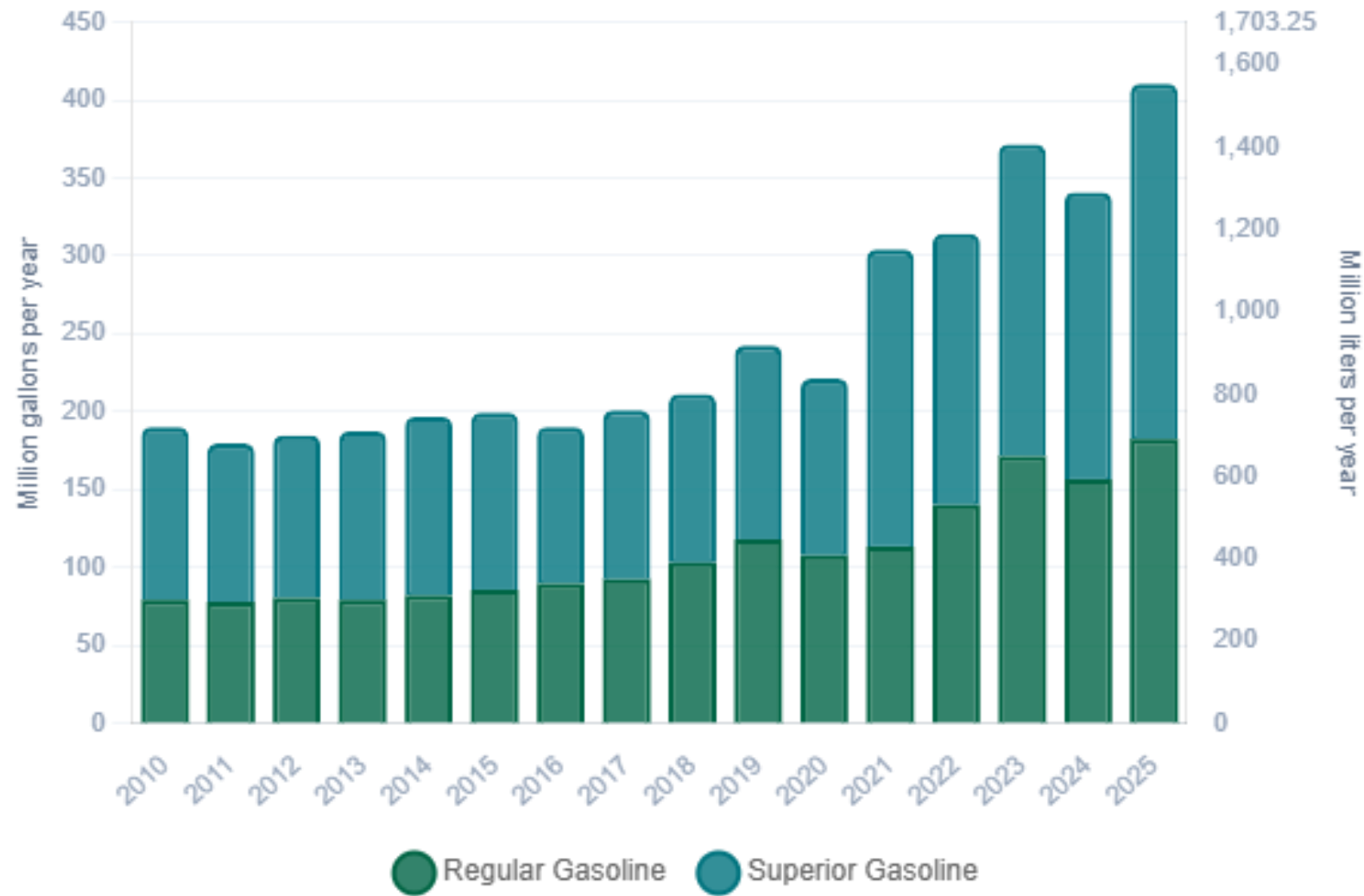
Honduras has no national mandate for ethanol blends. A new biofuel's law is under discussion.

Gasoline Demand in Honduras



Source: CEPAL, Secretaría de Energía Honduras

Gasoline Imports in Honduras

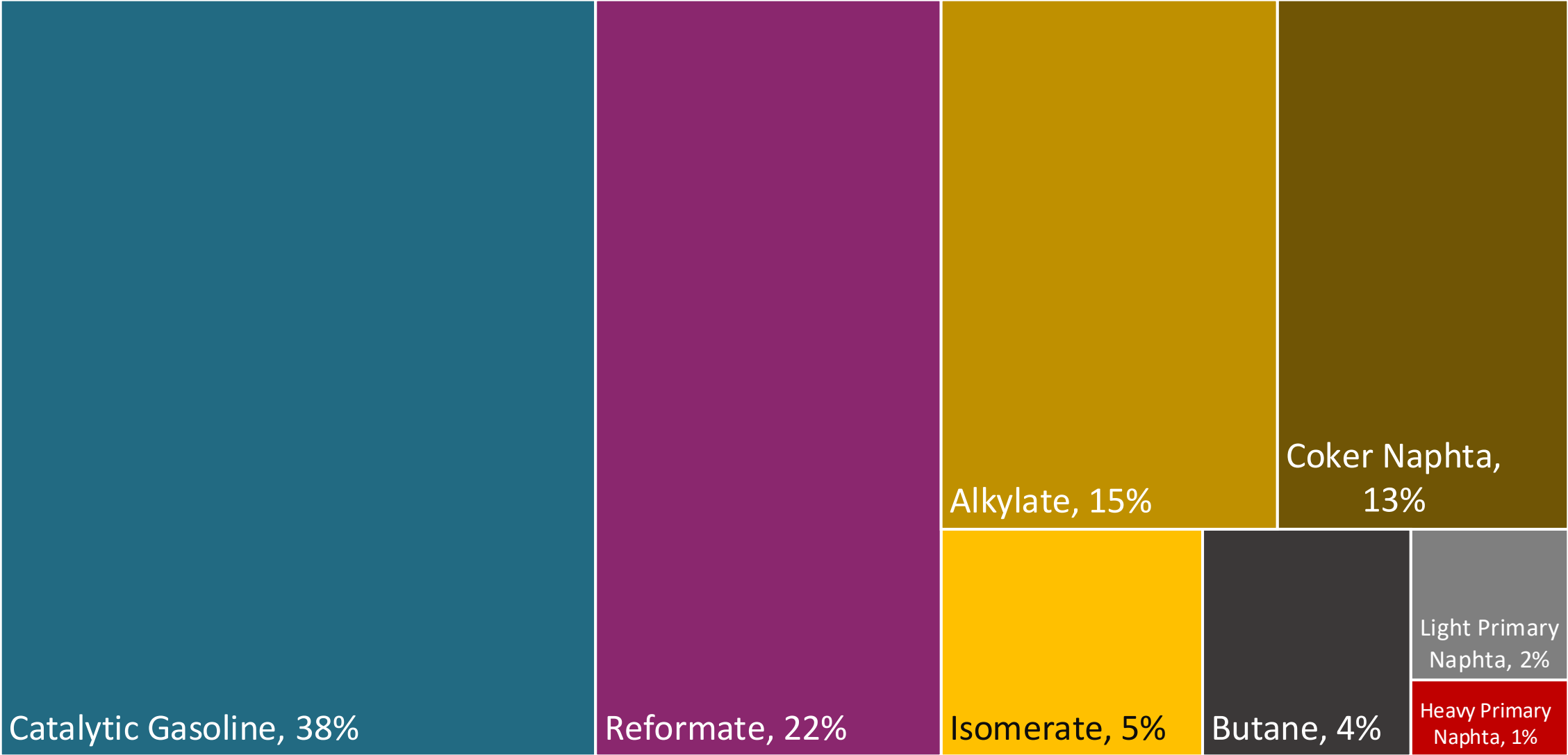
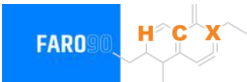


Source: CEPAL, Secretaría de Energía Honduras

Gasoline Quality in Honduras

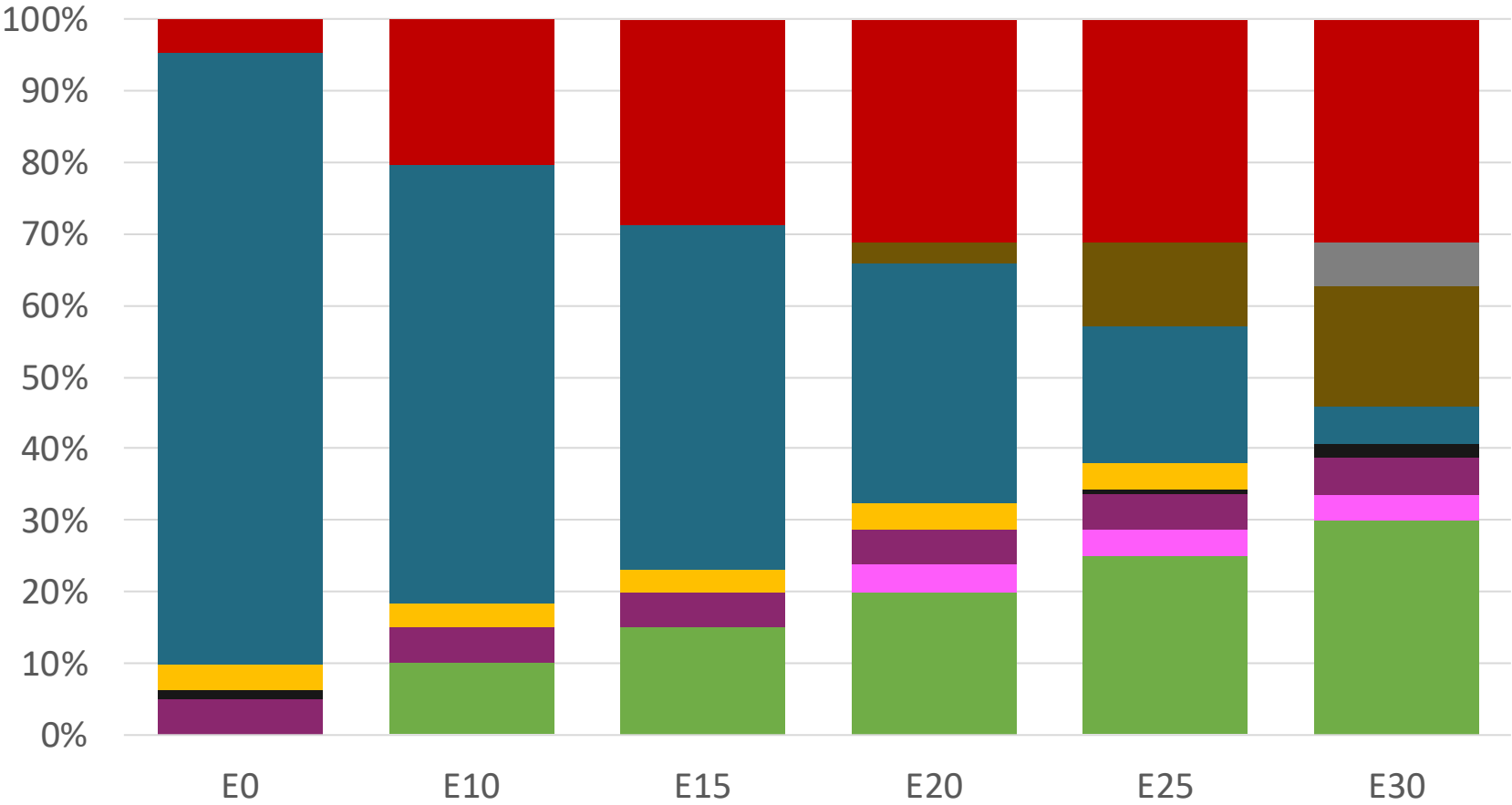
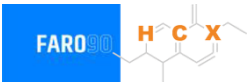
Name	RTCA 75.01.08:19 p.m.	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	2,5% v/v	2,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0,25% v/v	0,25% v/v	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Blending Components - Honduras



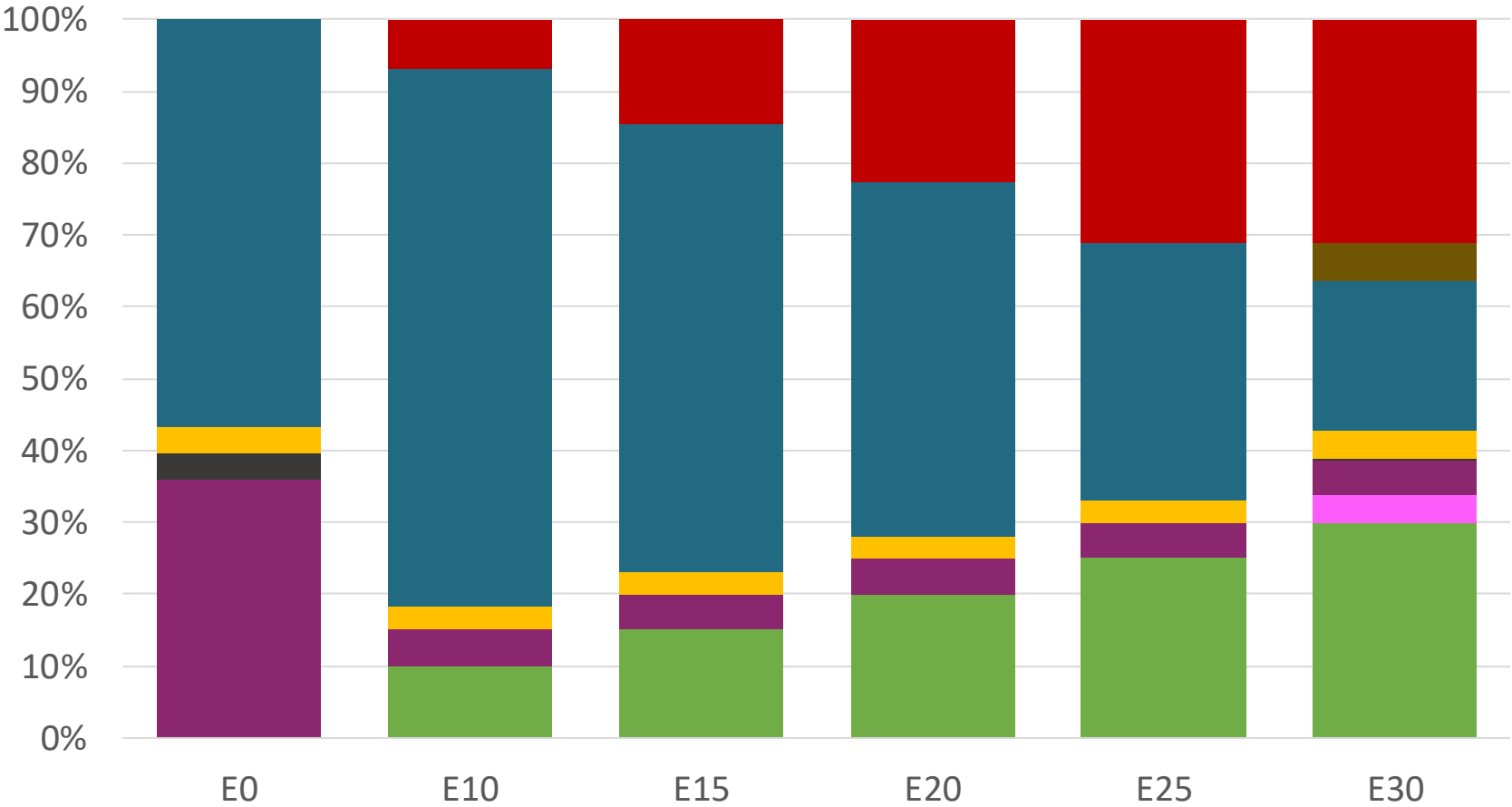
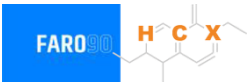
Source: Faro90

Honduras – Regular – Constant Octane



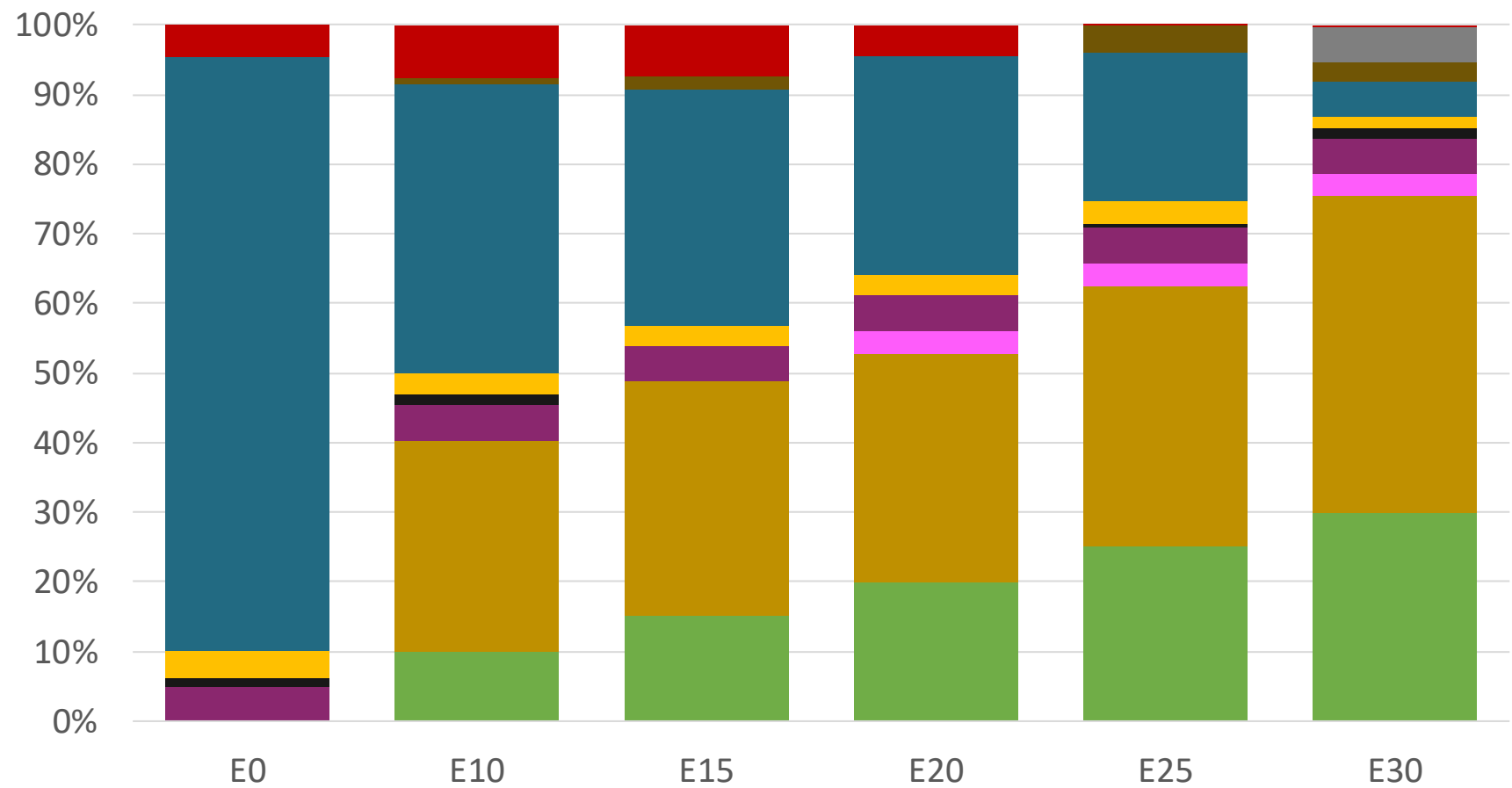
Octane (RON)	88.0	88.0	88.0	88.0	88.0	90.7
Price (USD/gal)	\$ 1.946	\$ 1.674	\$ 1.523	\$ 1.381	\$ 1.237	\$ 1.266

Honduras – Premium – Constant Octane



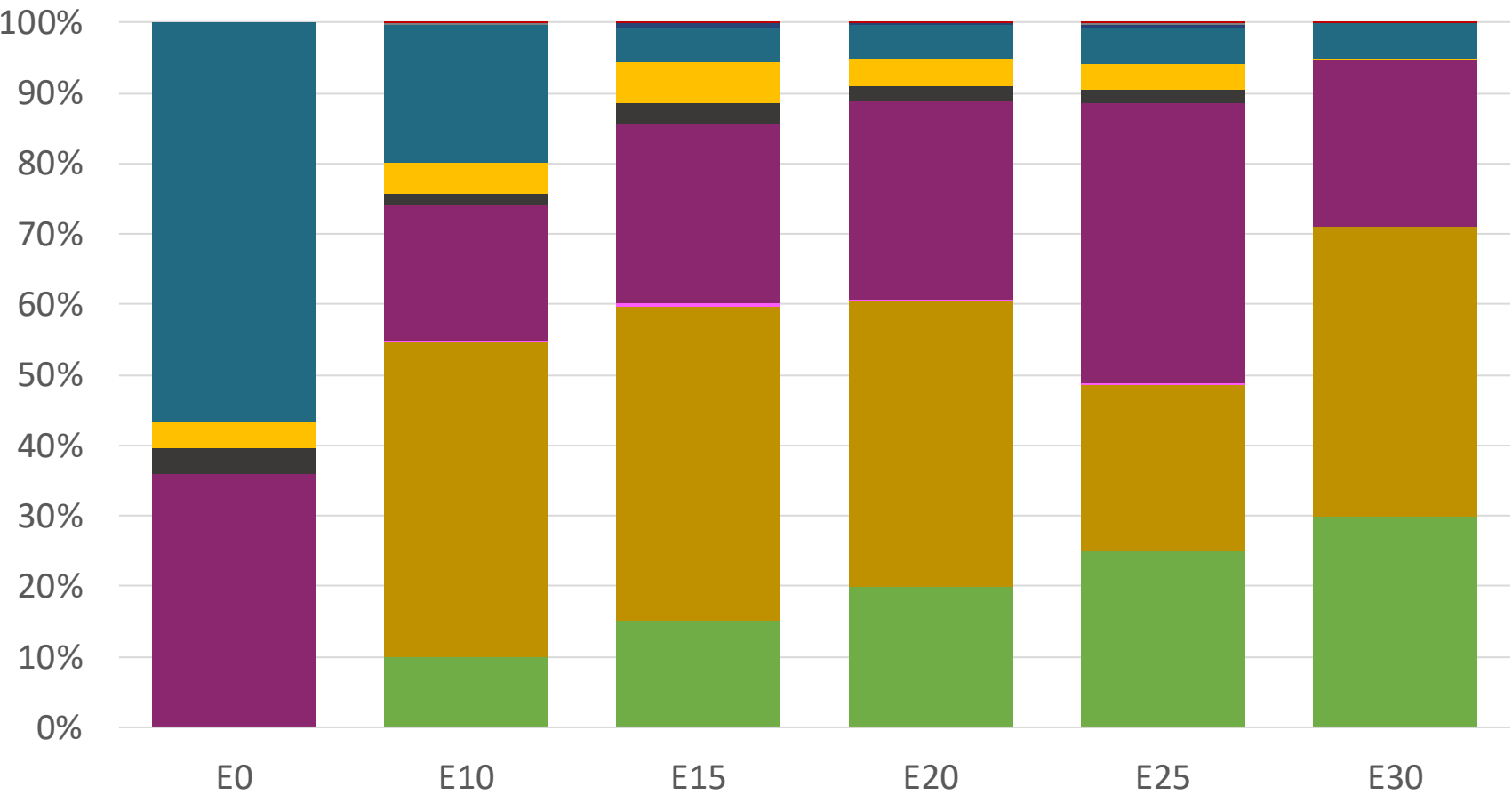
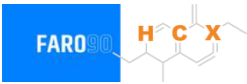
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.326	\$ 2.023	\$ 1.888	\$ 1.746	\$ 1.597	\$ 1.448

Honduras – Regular – Increased Octane



Octane (RON)	88.0	91.6	95.3	97.2	97.9	100.1
Price (USD/gal)	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946

Honduras – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326

Gasoline Vehicular Fleet in Honduras

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles						
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	6,004						
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	20,322						
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	147,410						
		Euro III		> 8 years, > 160 tkm	102,548						
		Euro IV		> 8 years, > 160 tkm	73,989						
		Euro V / Euro 6		> 8 years, > 160 tkm	55,318						
				4-8 years, 80-160 tkm	16,440						
				< 4 years, < 80 tkm	12,198						
	Carburator	3-Ways		PCV Tank	> 8 years, > 160 tkm	15,239					
	Single FI-Pt	3-Ways			> 8 years, > 160 tkm	138,486					
Medium automobiles	Multi FI-Pt	3-Ways	PCV Tank	> 8 years, > 160 tkm	309,468						
		Euro III		> 8 years, > 160 tkm	137,159						
		Euro IV		> 8 years, > 160 tkm	83,924						
		Euro V / Euro 6		> 8 years, > 160 tkm	82,046						
				4-8 years, 80-160 tkm	54,136						
				< 4 years, < 80 tkm	91,300						
				Small Engines	FI 4-cycles	PCV	> 8 years, > 160 tkm	23,510			
		> 8 years, > 160 tkm					29,000				
> 8 years, > 160 tkm	298,892										
> 8 years, > 160 tkm	136,978										
4-8 years, 80-160 tkm	181,607										
4-8 years, 80-160 tkm	239,066										
< 4 years, < 80 tkm	791,830										
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	7,503						
	FI	3-Ways		> 8 years, > 160 tkm	109,909						
		Euro III		> 8 years, > 160 tkm	36,890						
		Euro IV		> 8 years, > 160 tkm	18,371						
		Euro V / Euro 6		> 8 years, > 160 tkm	14,349						
				4-8 years, 80-160 tkm	13,125						
				< 4 years, < 80 tkm	29,889						

Vehicular Fleet: **3,046,872**

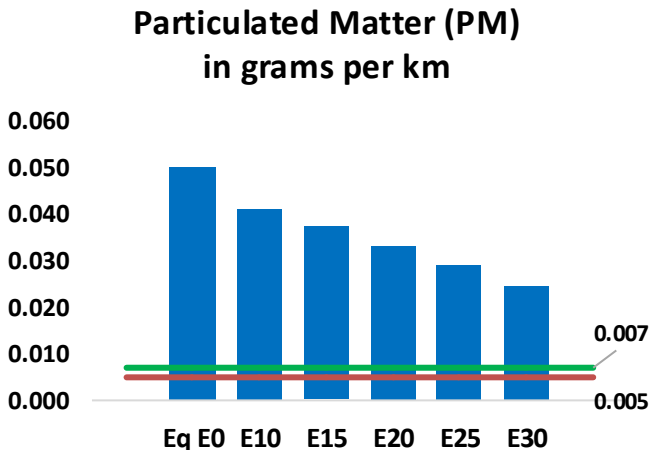
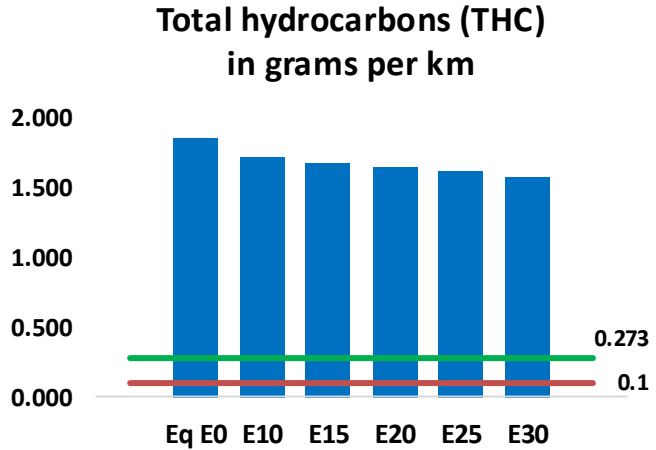
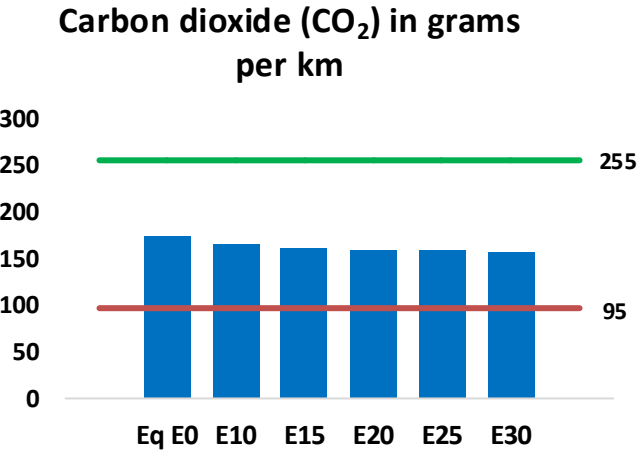
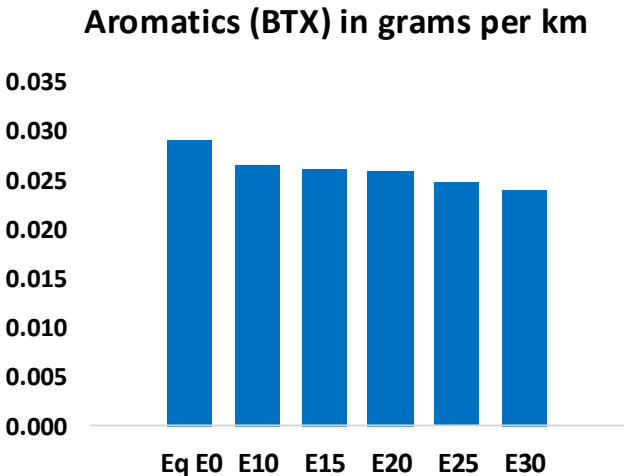
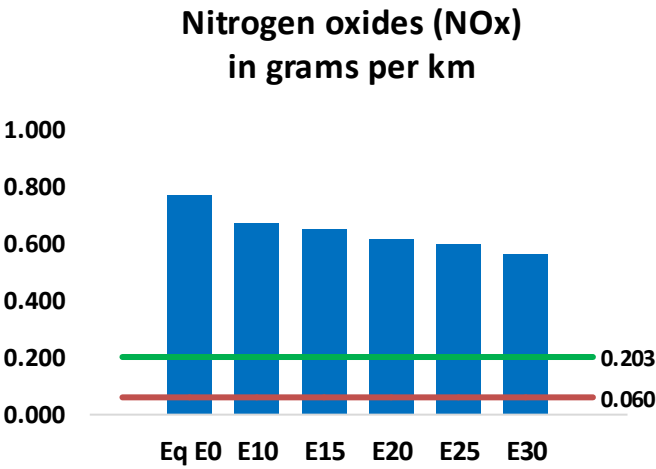
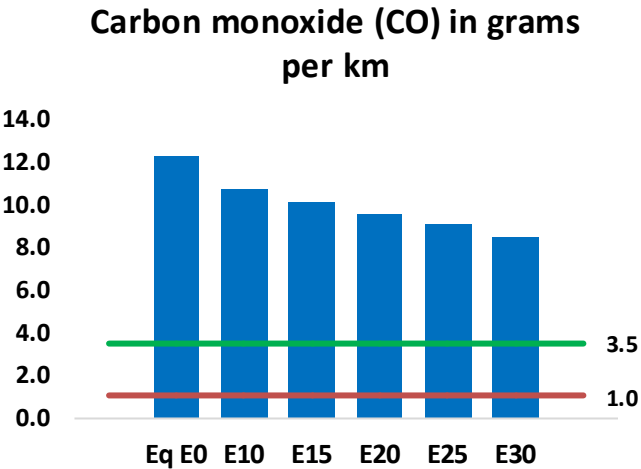
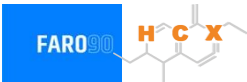
Average Age: **13 years**

Motorcycles: **50%**

Source: INE, analysis Faro 90

Honduras – Vehicular Emissions

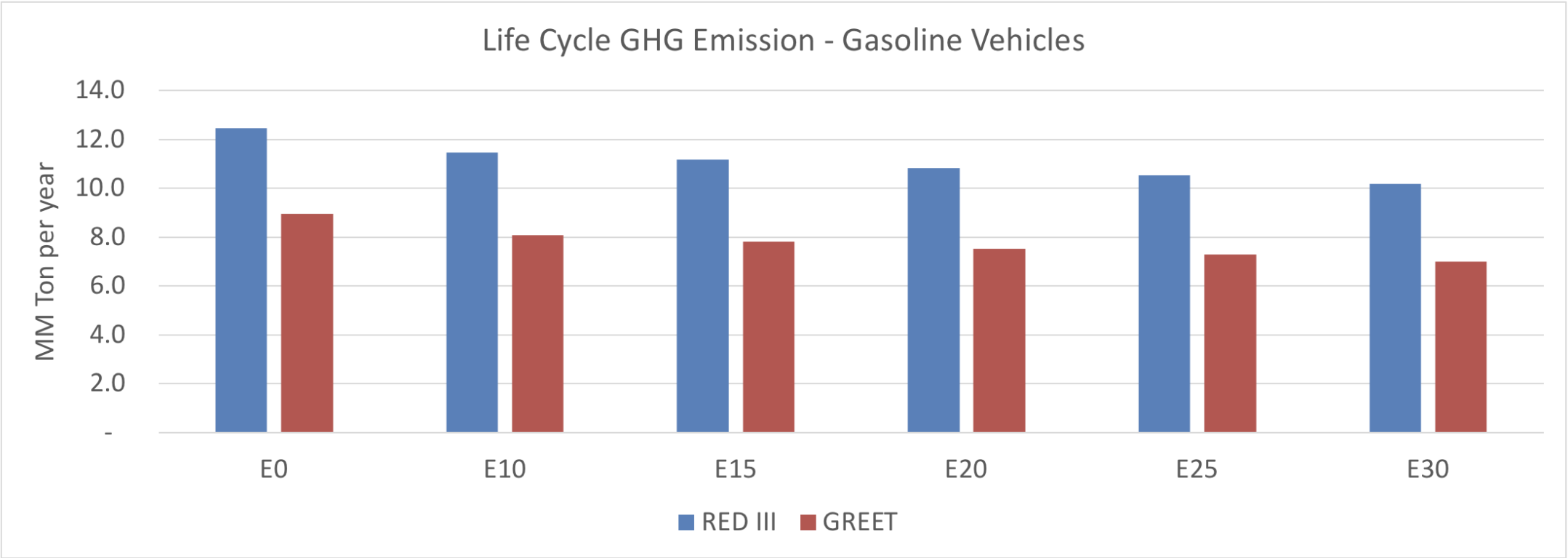
United States
Euro 6



Honduras – Gasoline Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	657,884	616,144	574,405	541,926	510,034	485,751	455,533	-13%	-22%
CO2	9,243,575	9,005,883	8,781,396	8,604,844	8,517,479	8,496,255	8,339,644	-5%	-8%
VOC	98,921	95,343	91,764	89,617	87,818	86,544	84,459	-7%	-11%
NOx	41,297	38,131	36,011	34,958	33,232	31,893	30,308	-13%	-20%
SOx	108	100	92	85	78	71	64	-15%	-28%
NH3	3,656	3,620	3,584	3,601	3,641	3,670	3,693	-2%	0%
N2O	169	168	167	168	171	173	175	-1%	2%
CH4	22,145	22,145	22,145	22,477	22,964	23,385	23,784	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	366	343	319	301	283	270	251	-13%	-23%
Acetaldehyde	925	1,062	1,558	2,373	3,232	3,694	4,364	68%	249%
Formaldehyde	3,506	3,409	3,974	4,666	4,888	5,408	5,887	13%	39%
Benzene	1,559	1,498	1,420	1,397	1,386	1,325	1,282	-9%	-11%
PM2.5	551	501	451	407	364	317	269	-18%	-34%
PM10	2,132	1,938	1,745	1,574	1,407	1,228	1,041	-18%	-34%

Honduras – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	30.7
2035 Target (MMT/year)	15.3
Delta	15.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.9%	12.7%	15.9%	18.5%	21.9%
RED II/III	0.0%	8.0%	10.4%	13.1%	15.3%	18.2%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	5.8%	7.4%	9.3%	10.8%	12.8%
RED II/III	0.0%	6.5%	8.4%	10.7%	12.5%	14.7%

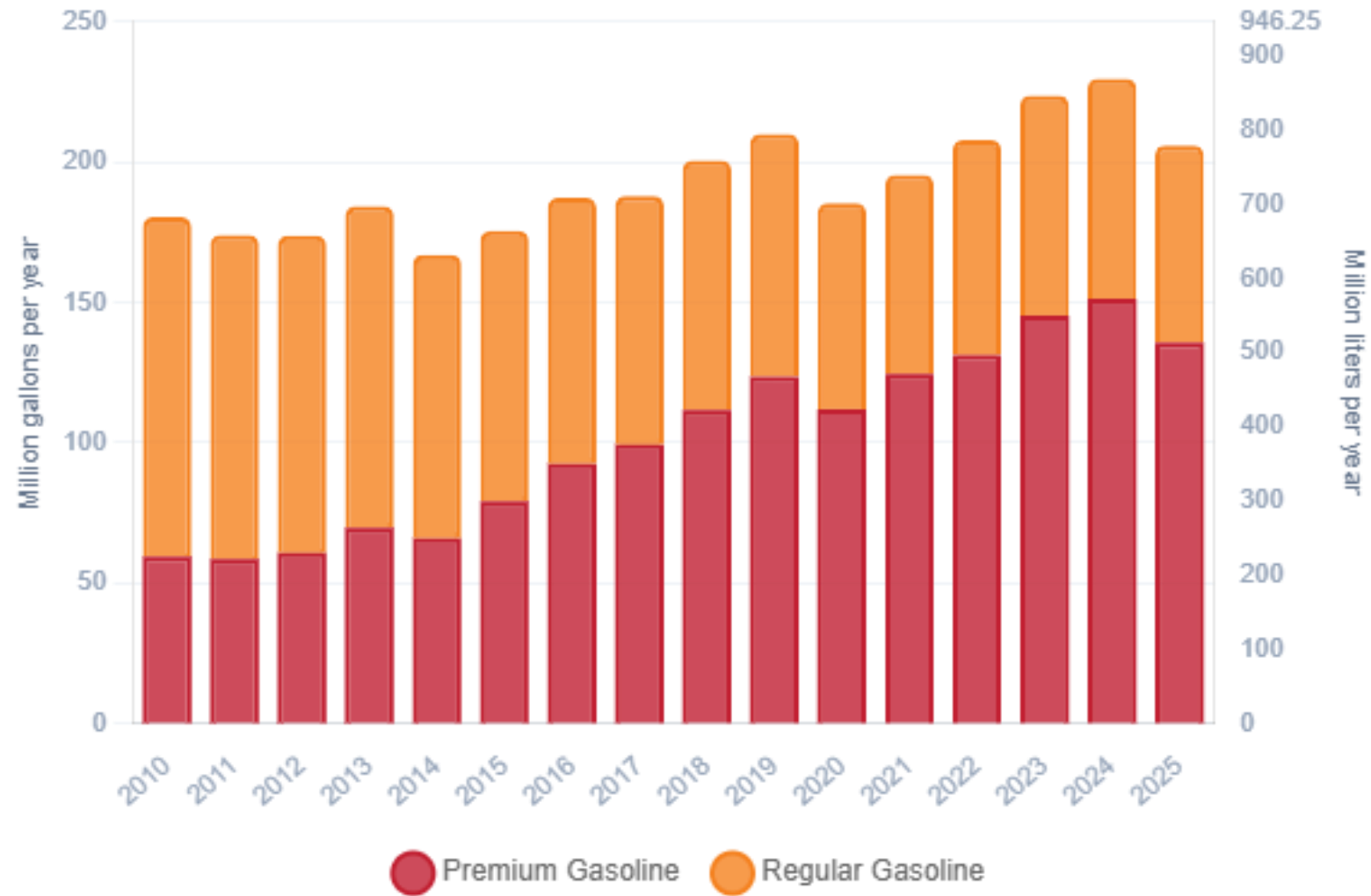
Ethanol Blending in Gasoline - Jamaica



In 2025, gasoline demand was 206 million gallons (779 million liters). Gasoline production was 34 million gallons (130 million liters) and imports 195 million gallons (739 million liters) mainly from United States. Regular grade (AKI 87) had a market share of 35% and Premium grade (AKI 90) 65%.

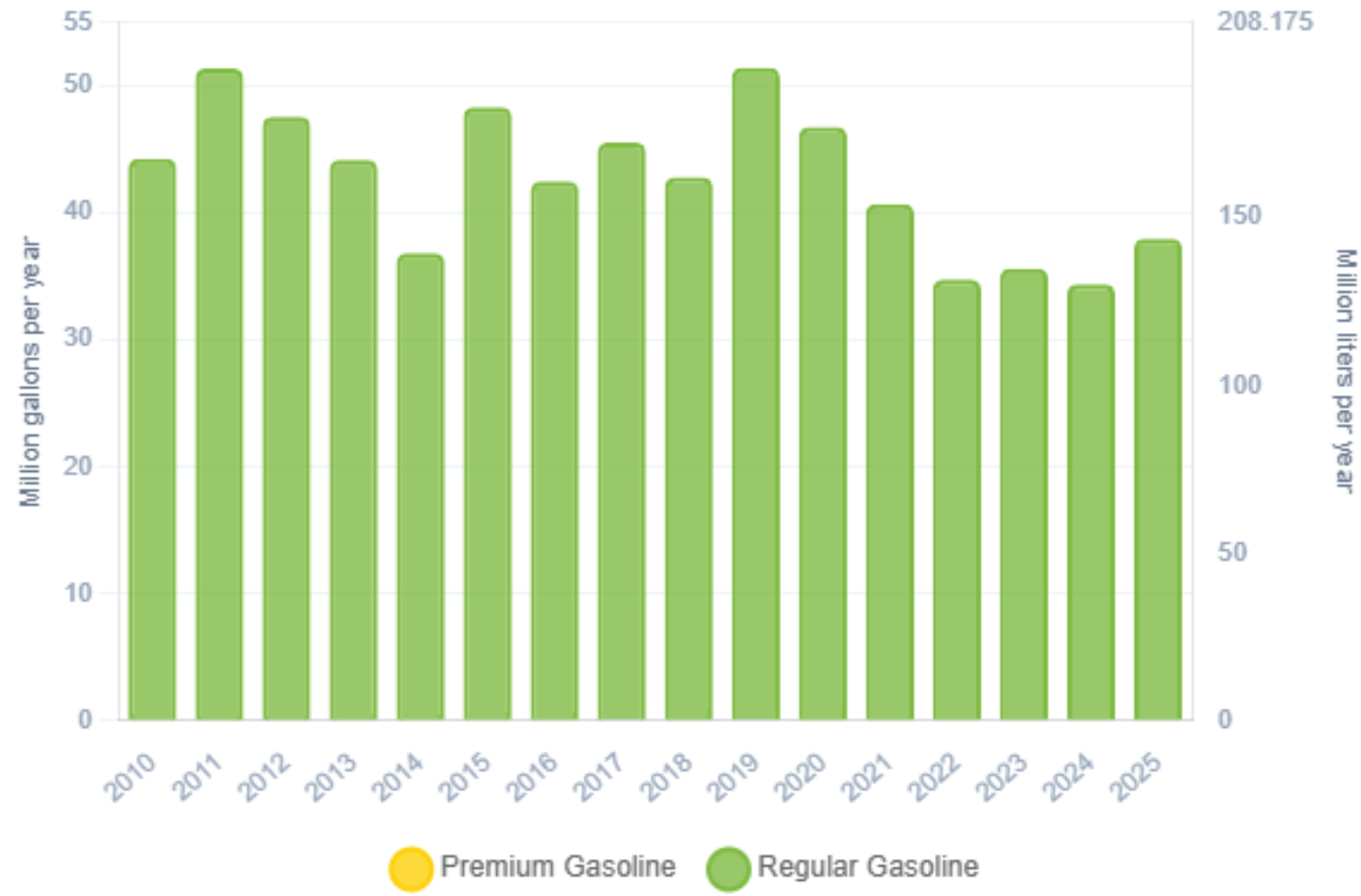
Current gasoline quality norm mandates E10 (JS 341:2017) . Previously, ethanol was supplied with local production to Petrojam, however, in recent years, ethanol is imported from the United States.

Gasoline Demand in Jamaica



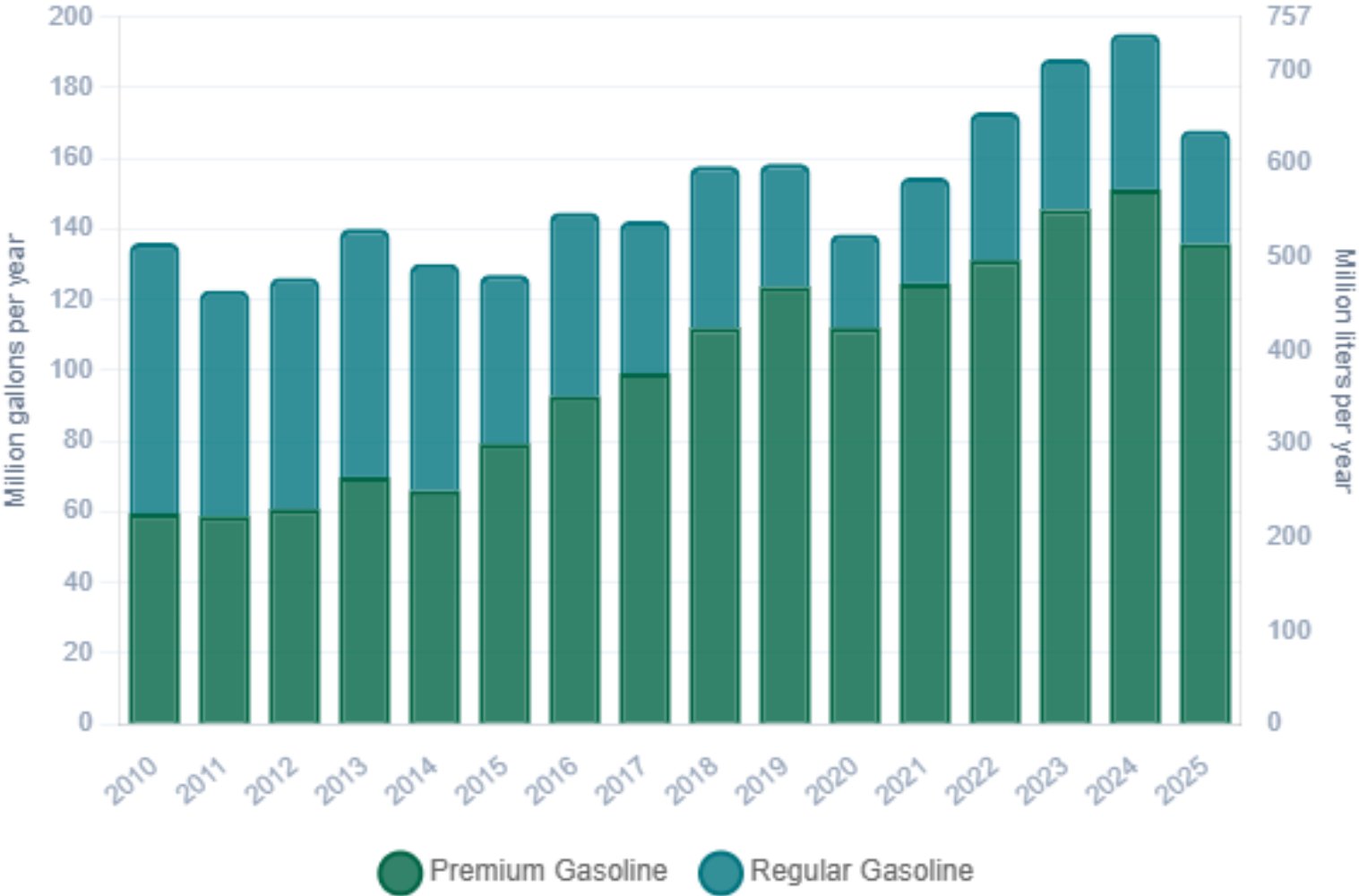
Source: Petrojam, MSET

Gasoline Production in Jamaica



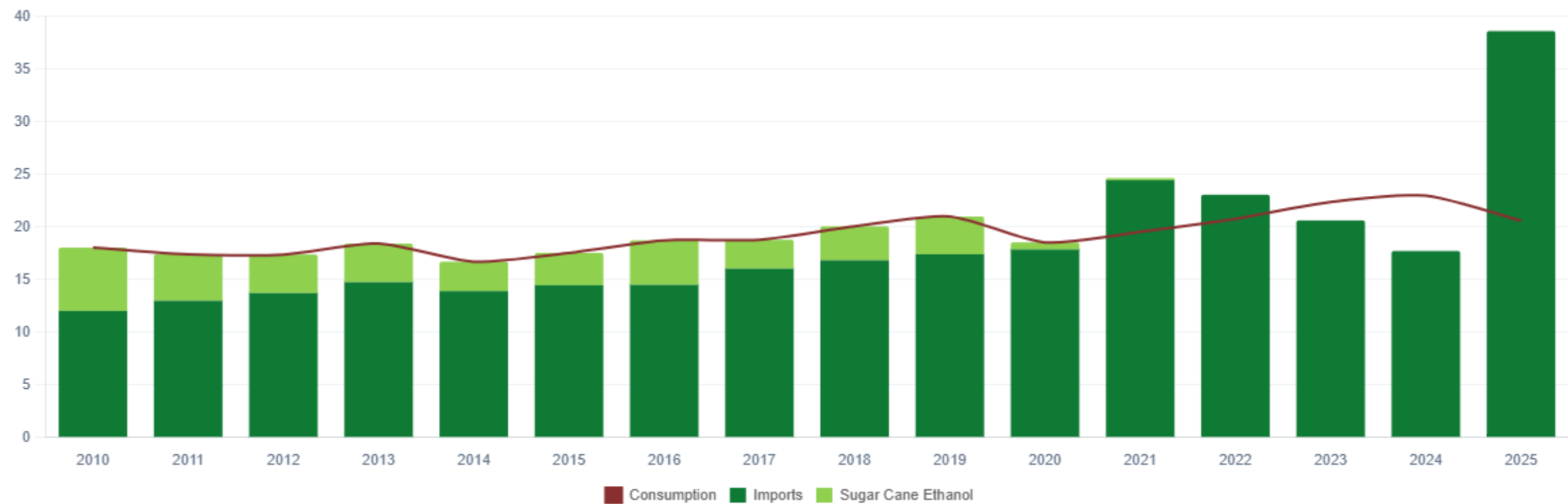
Source: Petrojam, MSET

Gasoline Imports in Jamaica



Source: Petrojam, MSET

Ethanol Balance in Jamaica

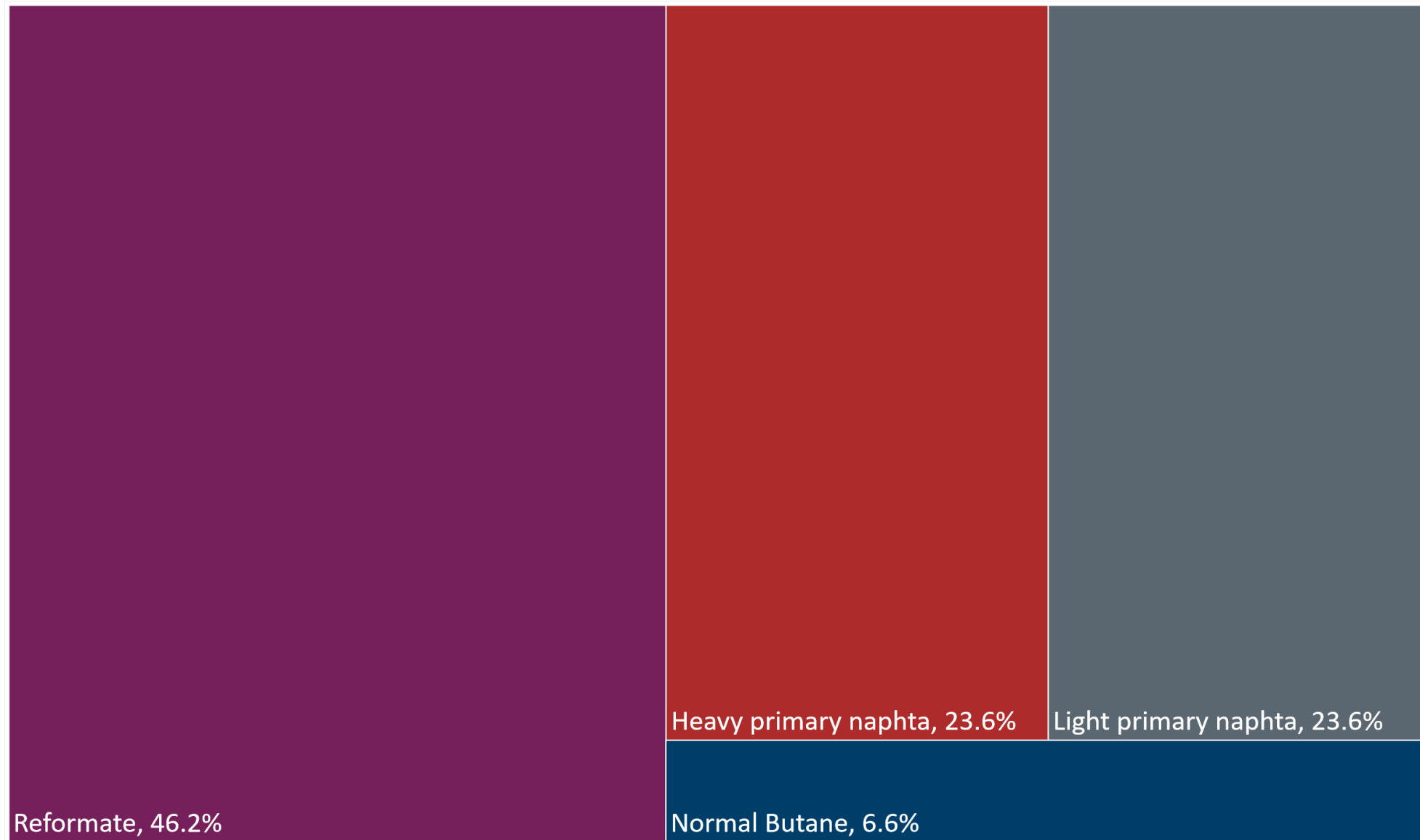


Source: MSET

Gasoline Quality in Jamaica

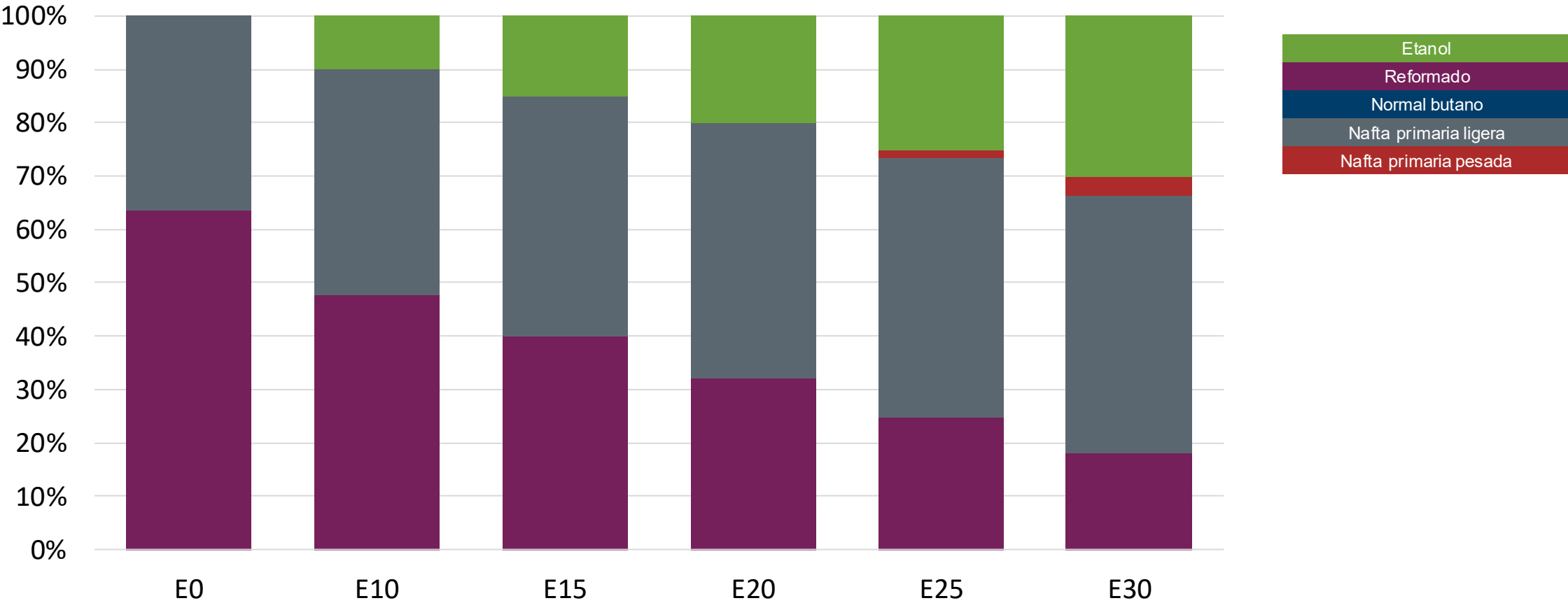
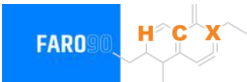
Name	JS 341:2017		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2017		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Unleaded Petrol AKI 87 E10	Unleaded Petrol AKI 90 E10	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 5 %v/v	< 5 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 45 %v/v	< 45 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	-	-	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 18 mg/l	< 18 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON			> 95	> 95	> 98	> 98
MON			> 85	> 88	> 85	> 88
AKI	90	87				
Sulfur Content	< 1.500 mg/kg	< 1.500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 4 %m/m (Gasoline with no ethanol)	< 4 %m/m (Gasoline with no ethanol)	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< 10 %v/v	< 10 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 61 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	< 15% v/v for Gasoline with no ethanol	< 15% v/v for Gasoline with no ethanol	-	-	-	-
Ethers 5 or more C Atoms	15% v/v (other oxygenates)	15% v/v (other oxygenates)	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Available Blending Components



Source: Faro90

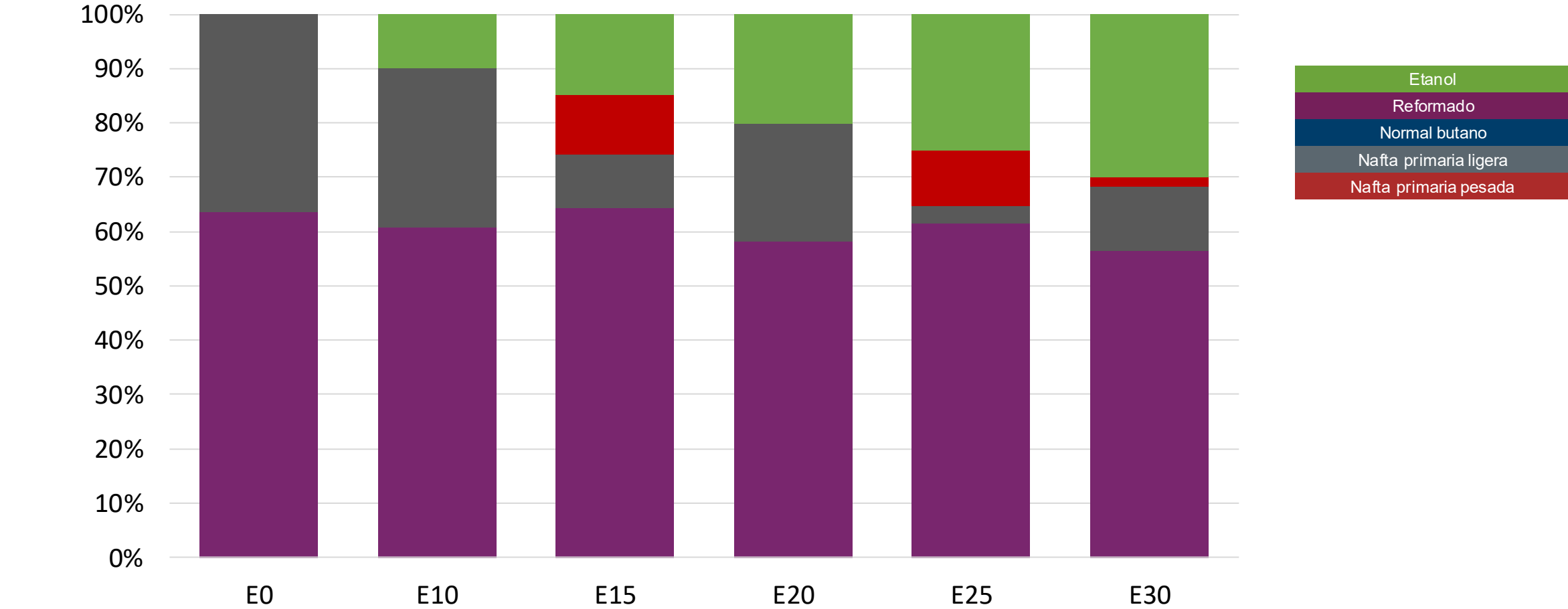
Regular Gasoline – Constant Octane



Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (USD/gal)	\$ 2.302	\$ 2.103	\$ 2.004	\$ 1.904	\$ 1.805	\$ 1.706

Source: Faro90

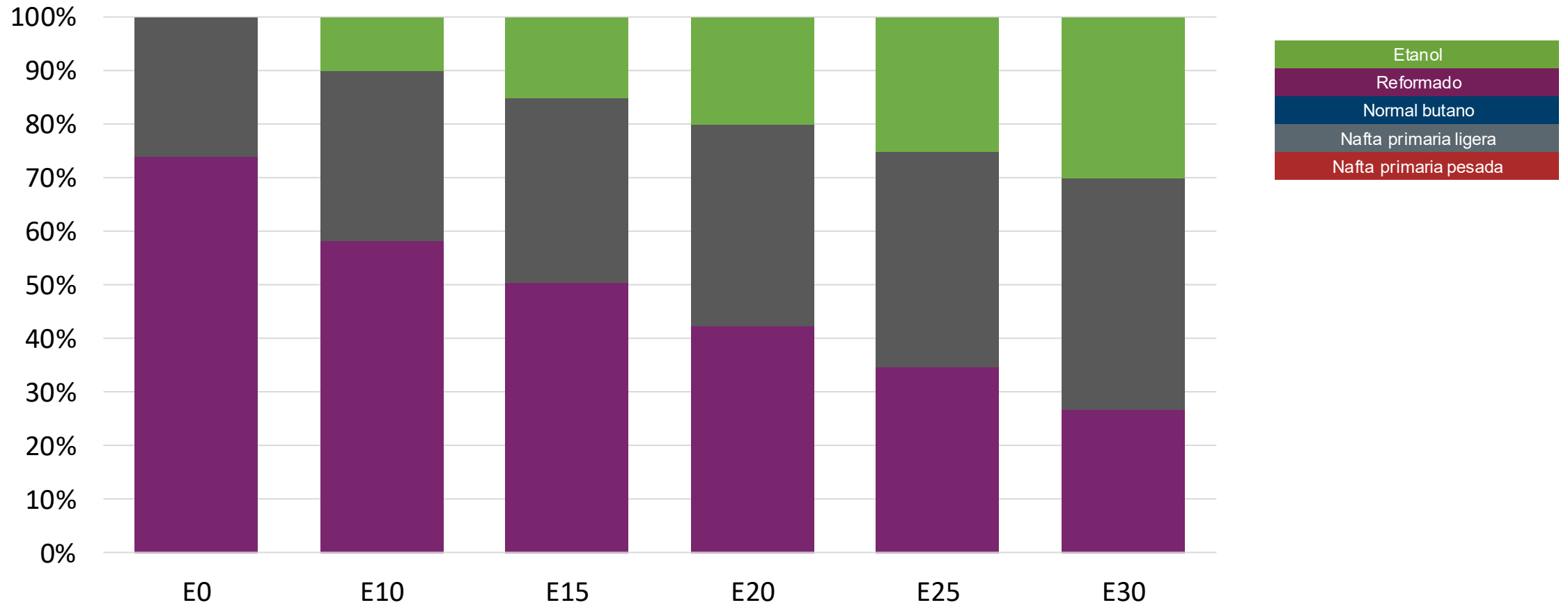
Regular Gasoline – Increased Octane



Octane (AKI)	87.0	90.8	92.6	94.5	96.4	98.3
Price (USD/gal)	\$ 2.302	\$ 2.302	\$ 2.302	\$ 2.302	\$ 2.302	\$ 2.302

Source: Faro90

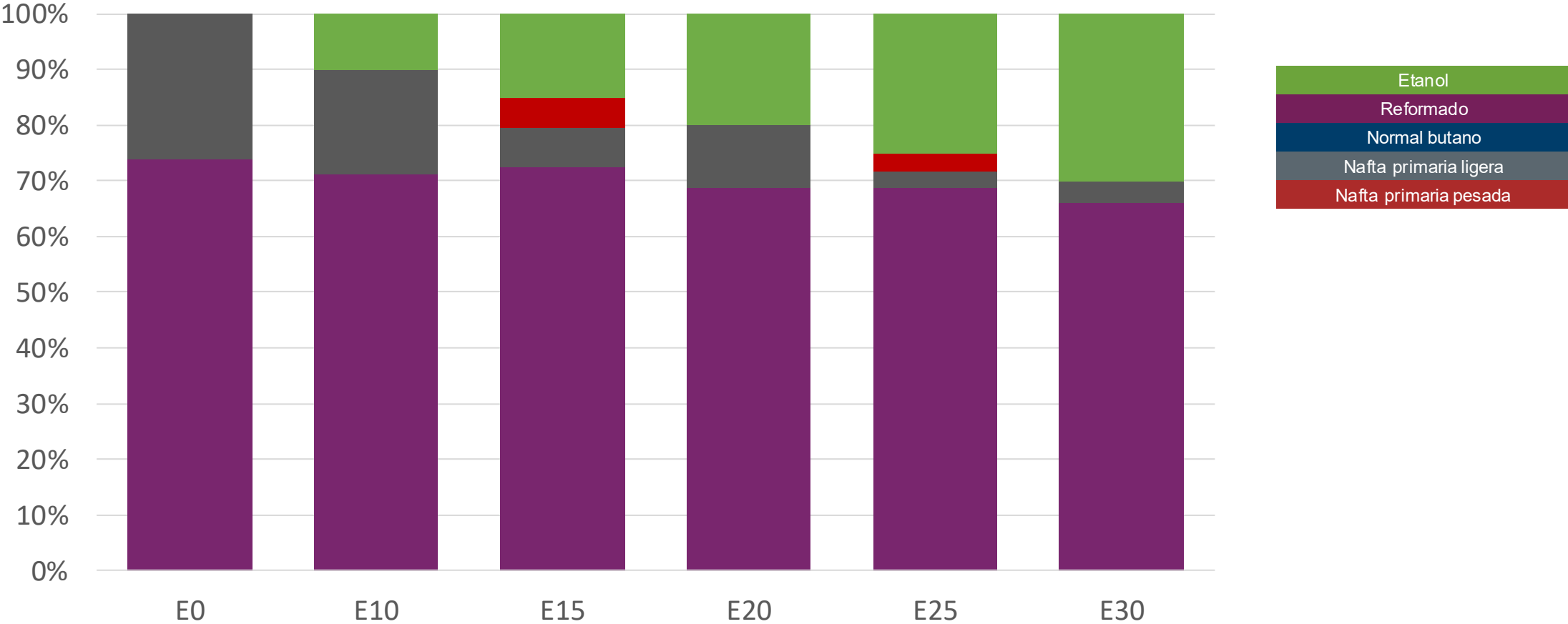
Premium Gasoline – Constant Octane



Octane (AKI)	90.0	90.0	90.0	90.0	90.0	90.0
Price (USD/gal)	\$ 2.460	\$ 2.261	\$ 2.162	\$ 2.062	\$ 1.963	\$ 1.864

Source: Faro90

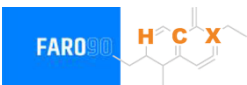
Premium Gasoline – Octane Increment



Octane (AKI)	90.0	93.8	95.6	97.5	99.4	101.3
Price (USD/gal)	\$ 2.460	\$ 2.460	\$ 2.460	\$ 2.460	\$ 2.460	\$ 2.460

Source: Faro90

Gasoline Vehicle Fleet - Jamaica

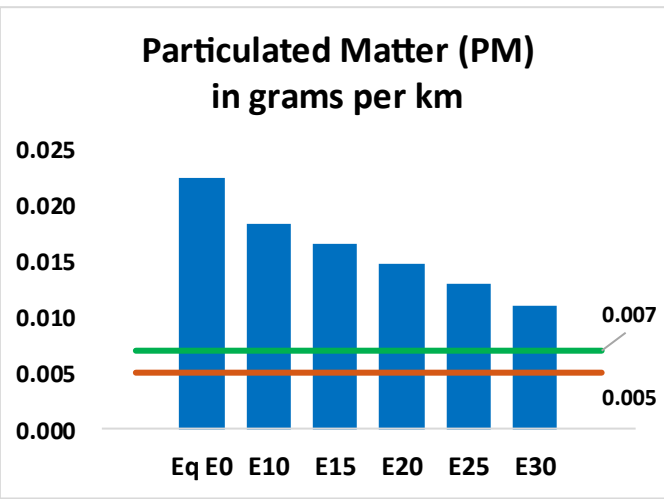
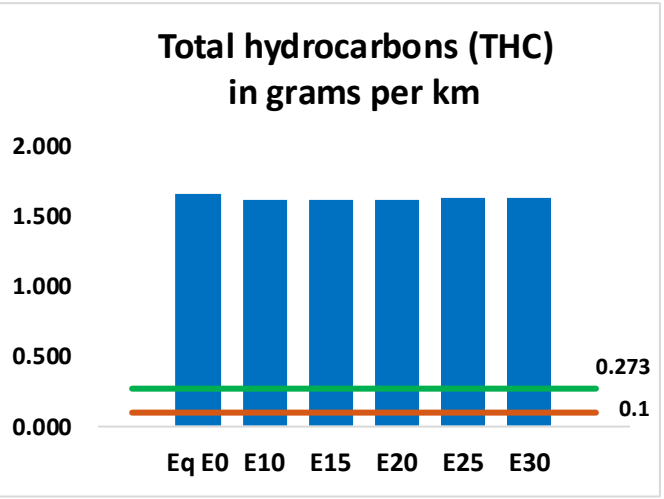
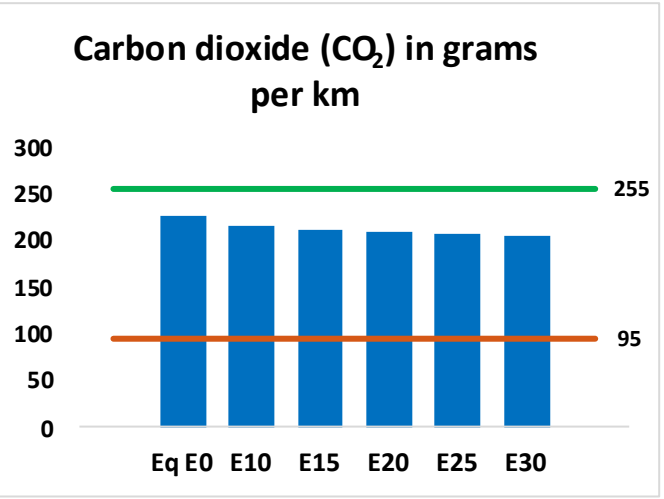
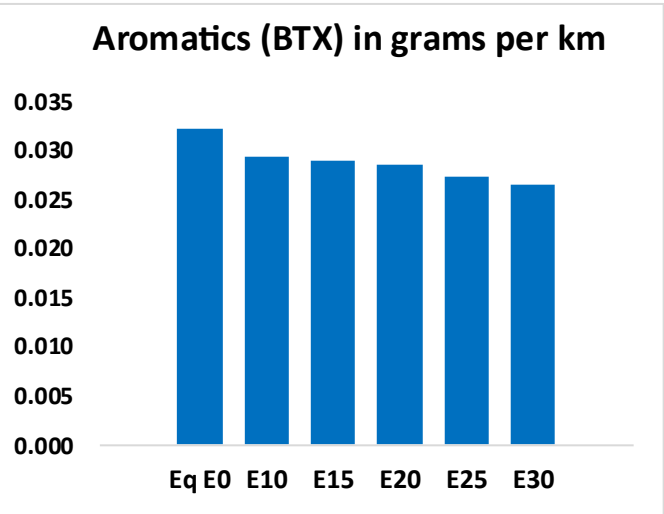
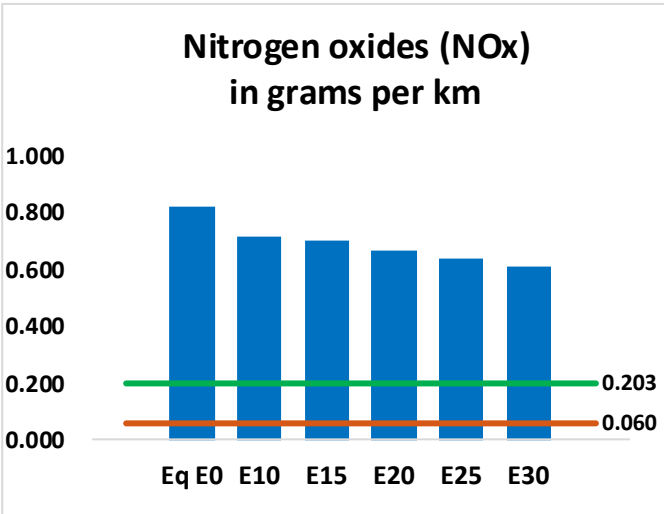
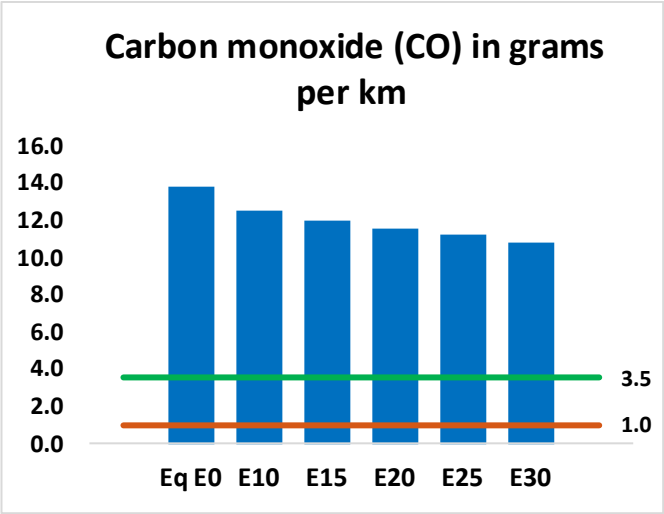
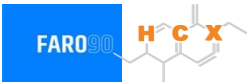


Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles					
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	4,644					
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	15,718					
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	114,014					
		Euro III		> 8 years, > 160 tkm	79,316					
		Euro IV		> 8 years, > 160 tkm	57,226					
		Euro V / Euro 6		> 8 years, > 160 tkm	42,786					
				4-8 years, 80-160 tkm	12,716					
				< 4 years, < 80 tkm	9,435					
	Carburator	3-Ways		PCV Tank	> 8 years, > 160 tkm	8,366				
	Single FI-Pt	3-Ways			> 8 years, > 160 tkm	76,028				
Medium automobiles	Multi FI-Pt	3-Ways	PCV Tank	> 8 years, > 160 tkm	169,895					
		Euro III		> 8 years, > 160 tkm	75,299					
		Euro IV		> 8 years, > 160 tkm	46,074					
		Euro V / Euro 6		> 8 years, > 160 tkm	45,043					
				4-8 years, 80-160 tkm	29,720					
				< 4 years, < 80 tkm	50,123					
		Small Engines		FI 4-ciclos	None	PCV	> 8 years, > 160 tkm	682		
Euro II	> 8 years, > 160 tkm		841							
Euro III	> 8 years, > 160 tkm		8,671							
Euro IV	> 8 years, > 160 tkm		3,974							
Euro V	4-8 years, 80-160 tkm		5,269							
	4-8 years, 80-160 tkm		6,936							
	< 4 years, < 80 tkm		22,972							
Carburator	3-Ways		PCV	> 8 years, > 160 tkm	1,135					
FI	3-Ways			> 8 years, > 160 tkm	16,622					
	Euro III			> 8 years, > 160 tkm	5,579					
	Euro IV	> 8 years, > 160 tkm		2,778						
	Euro V / Euro 6	> 8 years, > 160 tkm		2,170						
		4-8 years, 80-160 tkm		1,985						
		< 4 years, < 80 tkm		4,520						
		Hybrid		Any						
	Electric		Any							

Vehicular Fleet: **885,746**
Average Age: **20 years**
Motorcycles: **2.3%**

Jamaica – Vehicular Emissions

United States
Euro 6

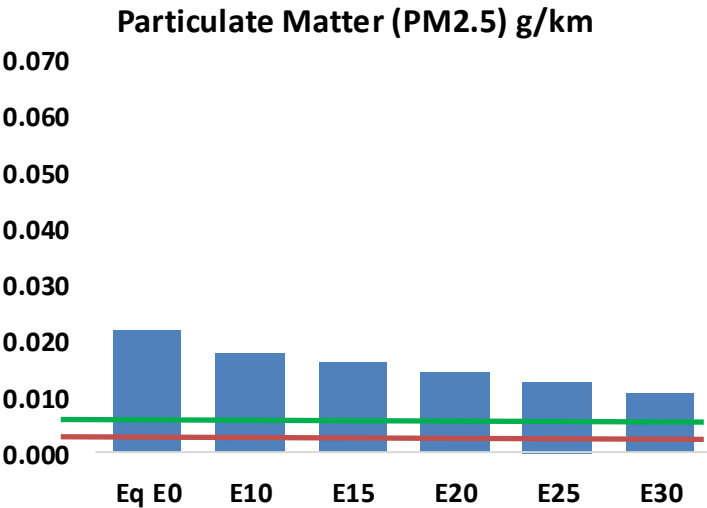
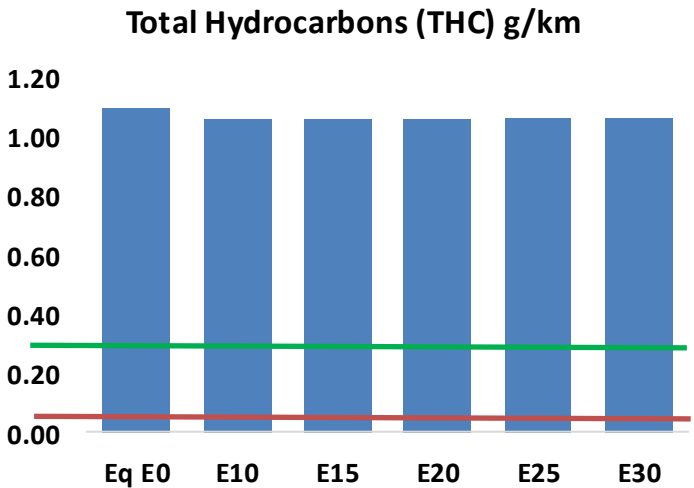
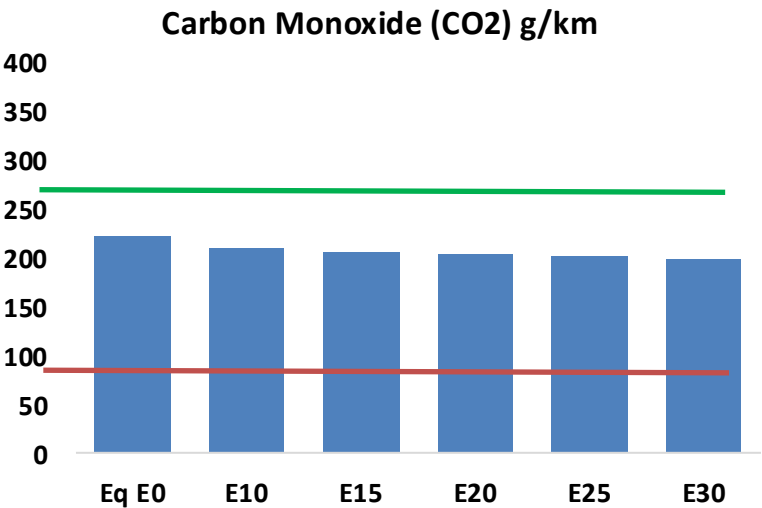
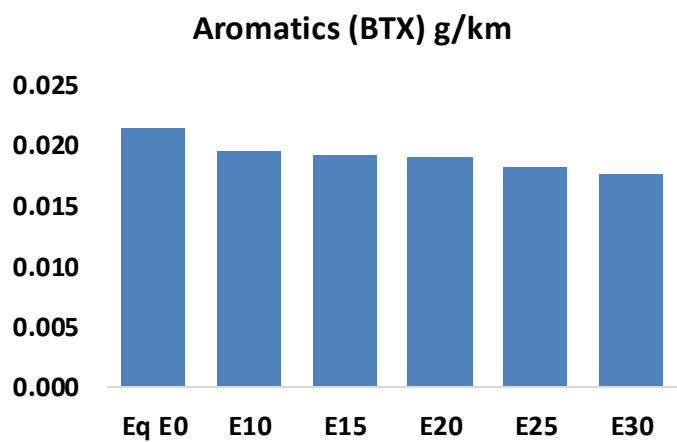
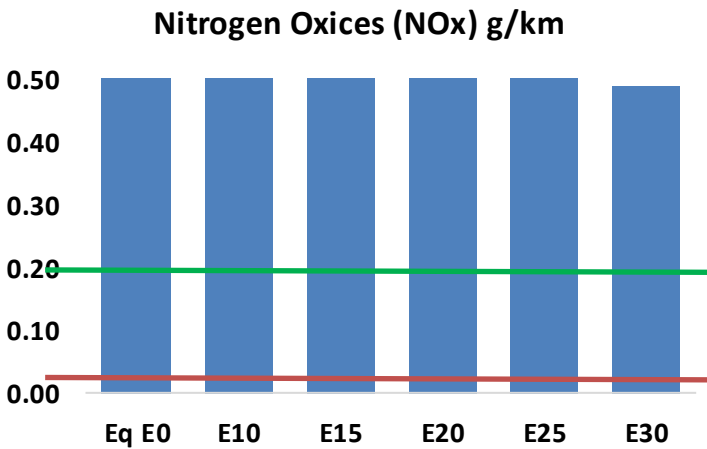
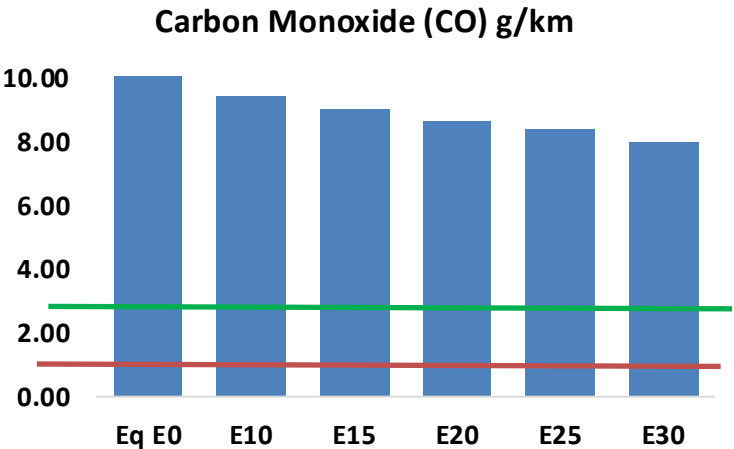
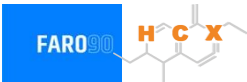


Jamaica – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	314,970	299,702	284,433	273,667	263,636	256,165	246,058	-10%	-16%
CO2	5,175,847	5,042,754	4,917,055	4,818,196	4,769,277	4,747,659	4,660,146	-5%	-8%
VOC	37,858	37,303	36,748	36,752	36,968	37,198	37,280	-3%	-2%
NOx	18,864	17,418	16,459	15,989	15,211	14,610	13,897	-13%	-19%
SOx	62	57	52	48	45	40	36	-15%	-28%
NH3	1,724	1,707	1,690	1,698	1,717	1,731	1,742	-2%	0%
N2O	93	93	92	93	95	96	97	-1%	2%
CH4	8,151	8,151	8,151	8,274	8,453	8,608	8,755	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	132	127	122	118	116	114	111	-8%	-12%
Acetaldehyde	231	265	388	592	806	921	1,088	68%	249%
Formaldehyde	668	649	757	889	931	1,030	1,122	13%	39%
Benzene	736	708	670	660	654	626	606	-9%	-11%
PM2.5	285	259	233	210	188	164	139	-18%	-34%
PM10	230	209	188	169	151	132	112	-18%	-34%

Jamaica – Vehicular Emissions

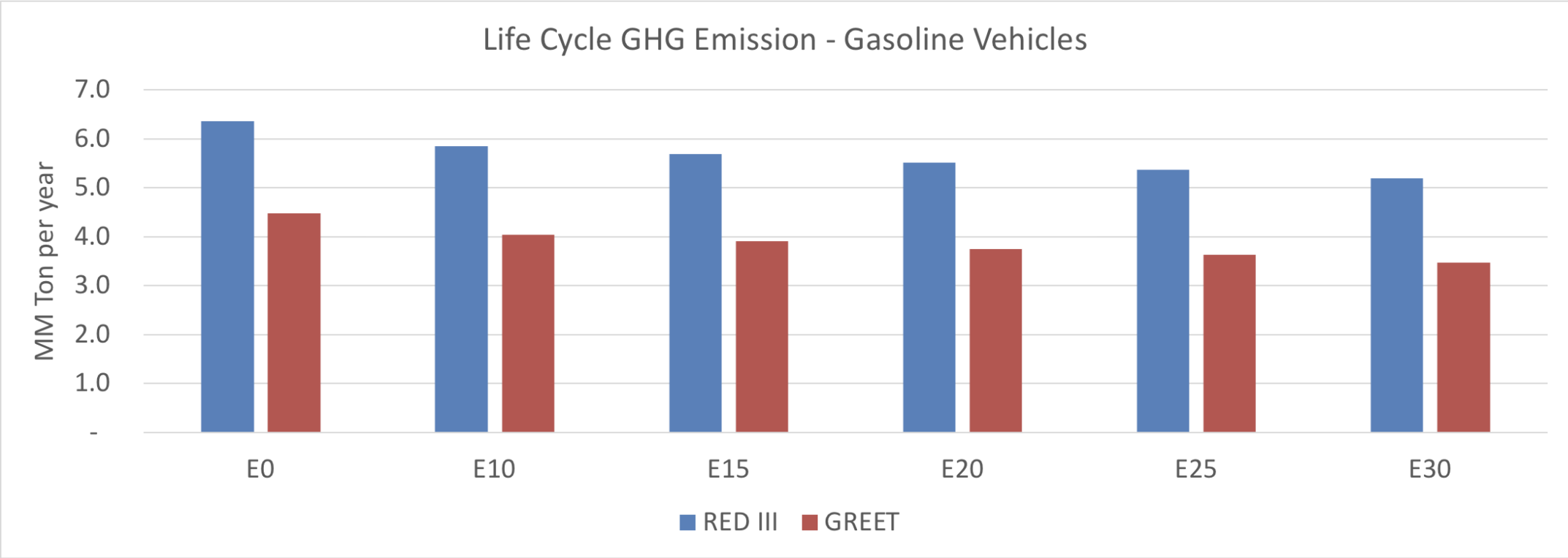
United States
Euro 6



Jamaica – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	166,859	158,208	149,556	143,282	137,342	132,889	127,019	-10%	-18%
CO2	3,490,486	3,400,730	3,315,961	3,249,293	3,216,303	3,199,172	3,140,202	-5%	-8%
VOC	17,388	17,100	16,813	16,782	16,845	16,922	16,924	-3%	-3%
NOx	10,587	9,775	9,234	8,968	8,528	8,188	7,785	-13%	-19%
SOx	41	38	35	32	29	27	24	-15%	-28%
NH3	1,181	1,169	1,158	1,163	1,176	1,186	1,193	-2%	0%
N2O	54	54	54	54	55	56	56	-1%	2%
CH4	3,735	3,735	3,735	3,791	3,873	3,944	4,011	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	95	92	89	87	86	86	84	-6%	-9%
Acetaldehyde	175	200	294	448	610	697	824	68%	249%
Formaldehyde	501	487	568	667	698	773	841	13%	39%
Benzene	339	326	308	304	301	288	279	-9%	-11%
PM2.5	200	181	163	147	132	115	97	-18%	-34%
PM10	147	134	120	109	97	85	72	-18%	-34%

Jamaica – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	9.4
2035 Target (MMT/year)	4.7
Delta	4.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.9%	12.9%	16.2%	18.9%	22.5%
RED II/III	0.0%	7.9%	10.5%	13.2%	15.5%	18.4%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.4%	12.2%	15.4%	17.9%	21.3%
RED II/III	0.0%	10.7%	14.1%	17.8%	20.9%	24.8%

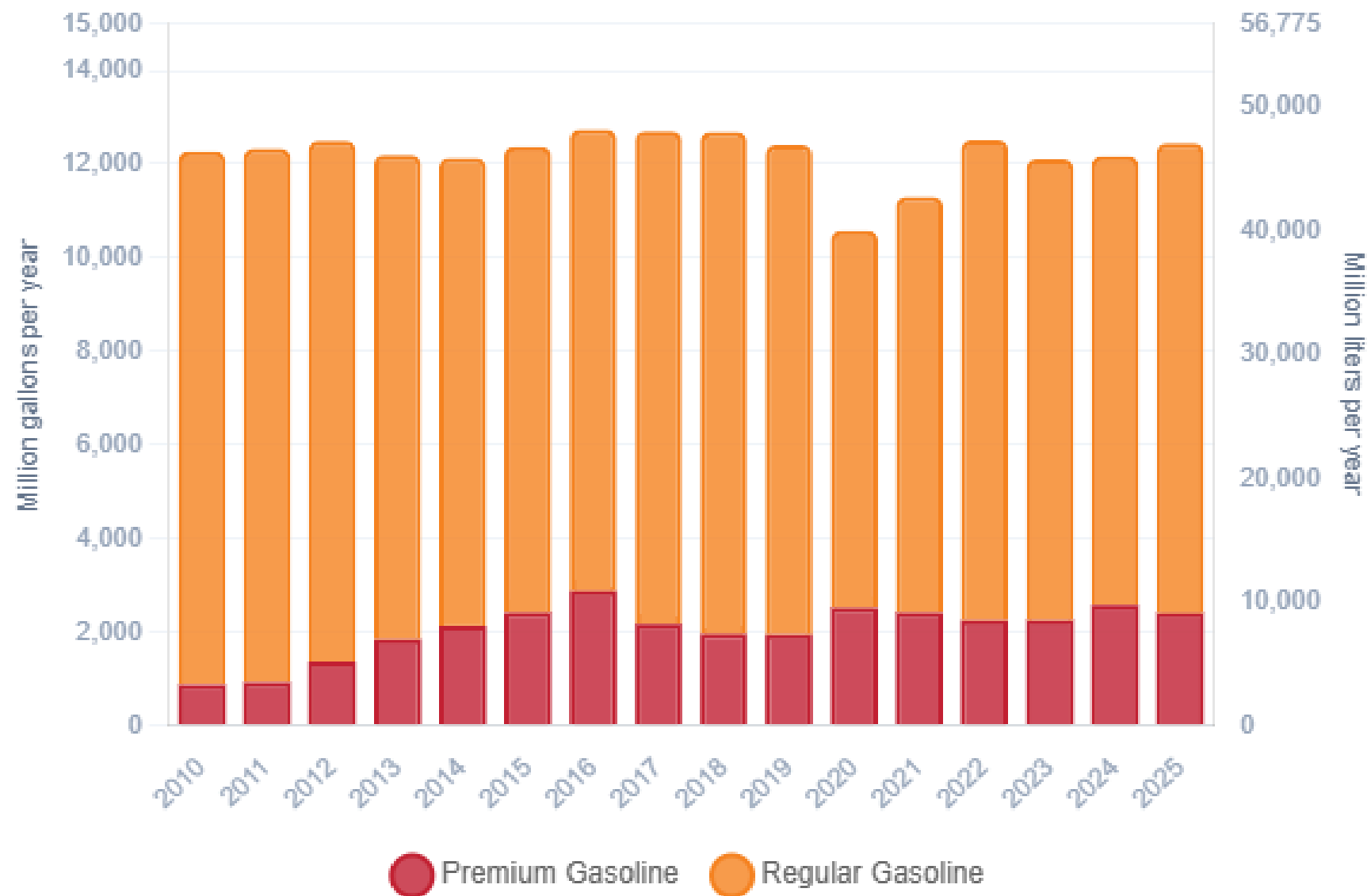
Ethanol Blending in Gasoline - Mexico



In 2025, gasoline consumption was 12,447 million gallons (47,116 million liters). There are two grades of gasoline, Regular (AKI 87) and Premium (AKI 91). Regular gasoline is the main grade in the market, representing 80% of the volume. There are stringent gasoline specifications for major metropolitan areas in Mexico City, Guadalajara and Monterrey. Demand is much higher than production, making necessary to import 40% of gasoline mainly from the United States.

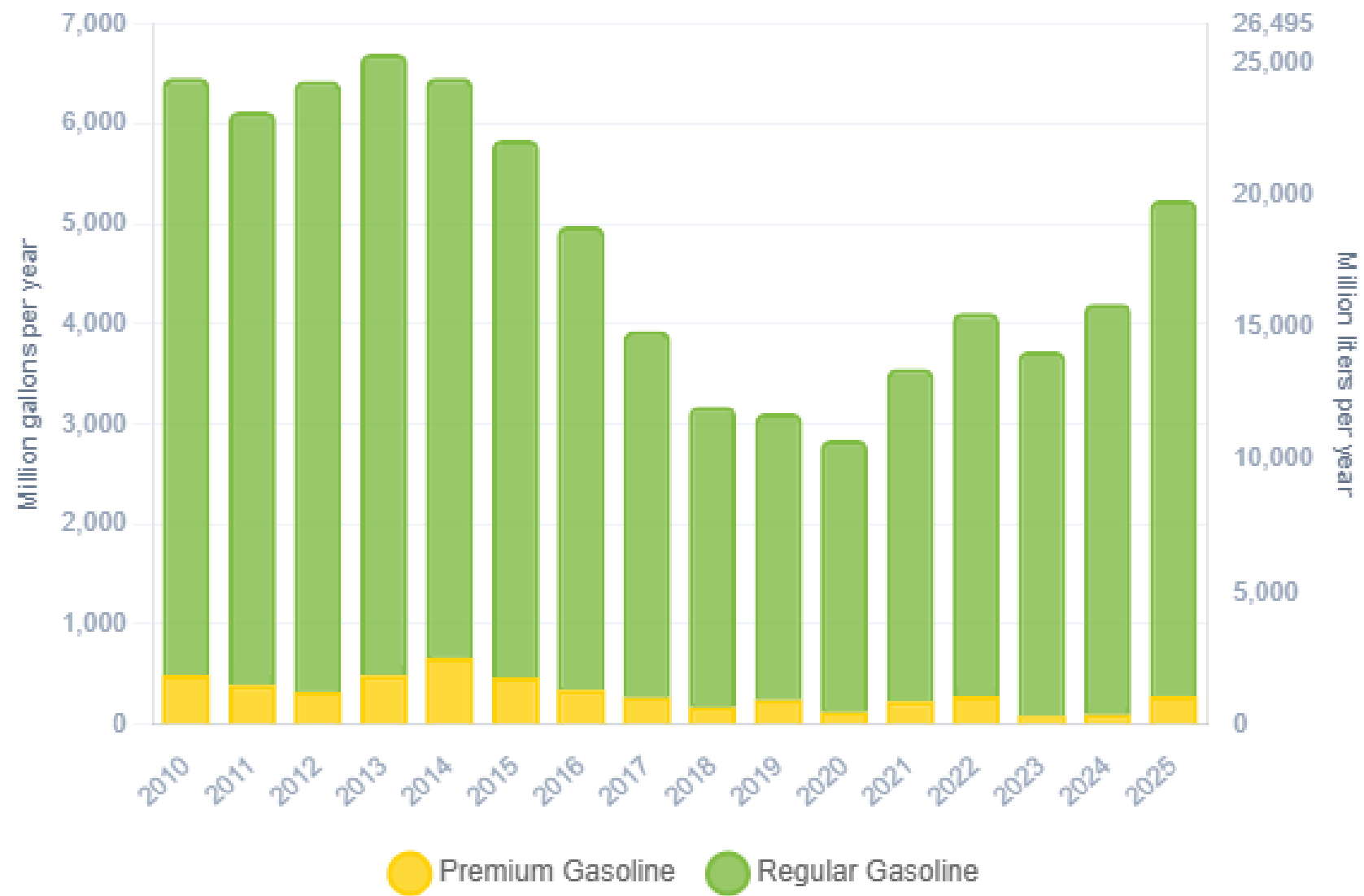
Gasoline quality norm (NOM-016-CRE de 2016) allows blends with a maximum of 5.8% v/v of ethanol with the exception of metropolitan areas of Valle de Mexico, Guadalajara and Monterrey. Current ethanol use is equivalent to an average of 1.06% in the Rest of the Country grade gasoline. In 2025, a new biofuels law was approved that fosters the production with agricultural and other organic residuals. Recently there are new state level efforts to increase ethanol consumption (e.g. Tamaulipas).

Gasoline Demand in Mexico



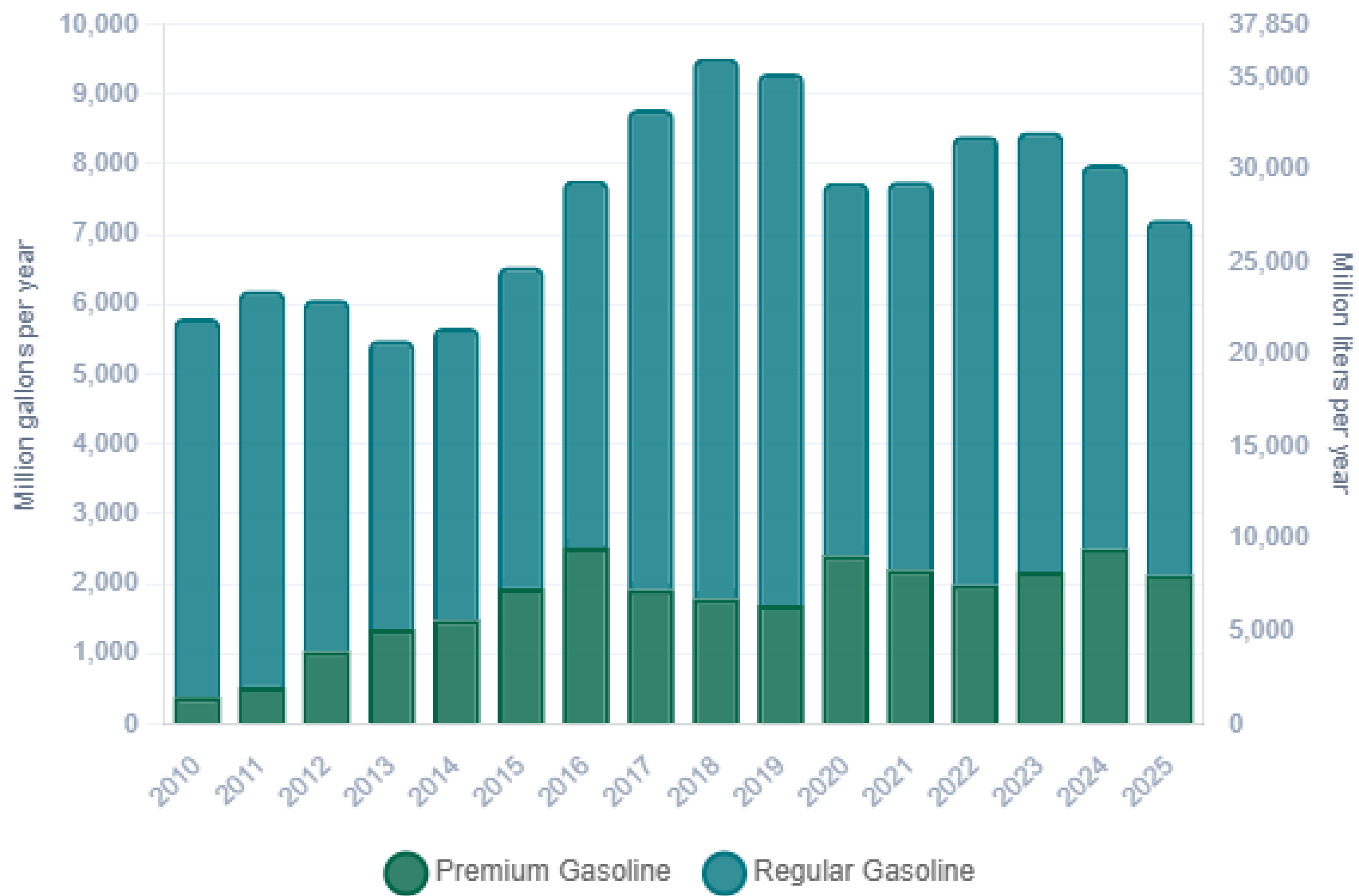
Source: SIE - SENER, 2025

Gasoline Production in Mexico



Source: SE - SENER, 2025

Gasoline Imports in Mexico

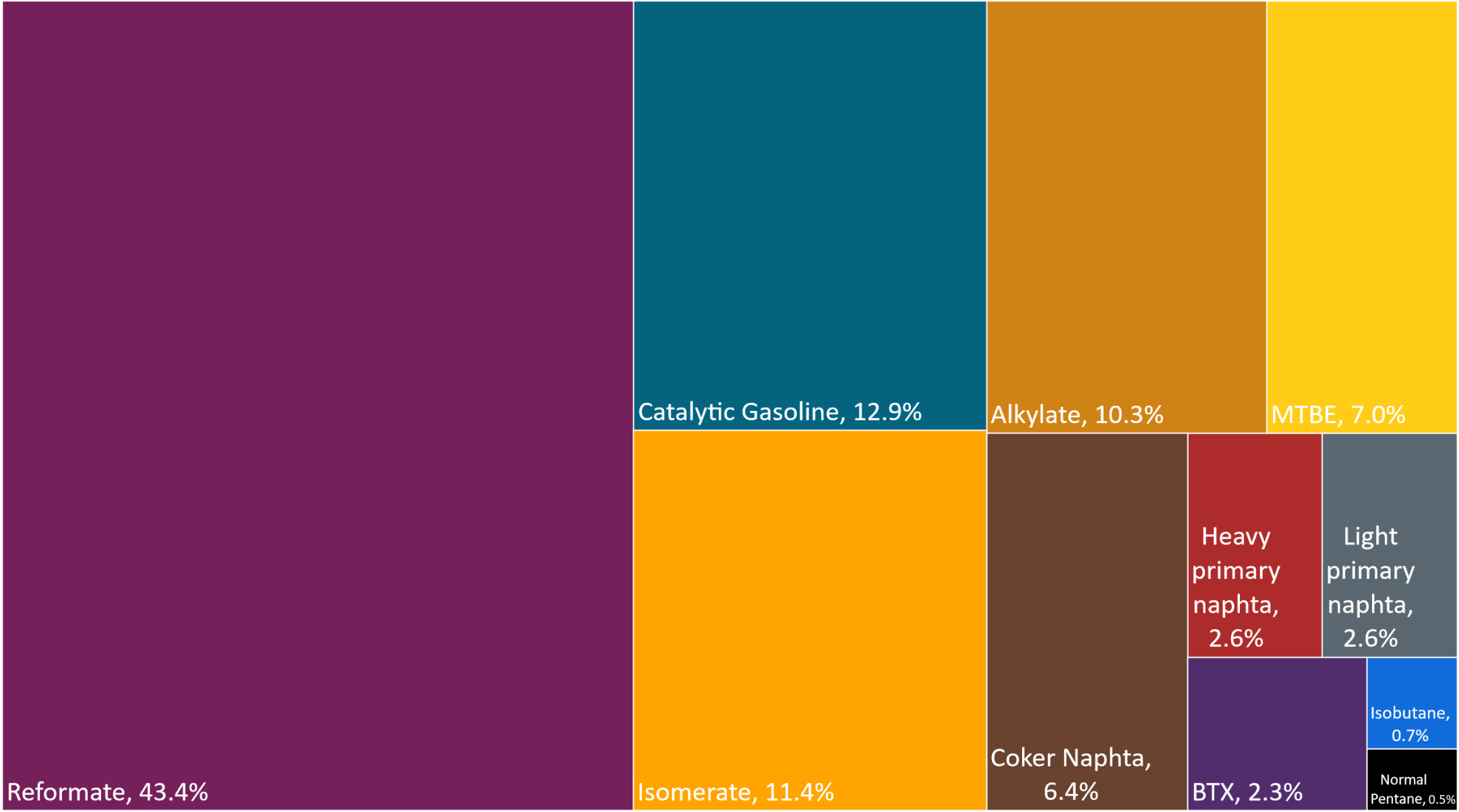


Source: SE - SENER, 2025

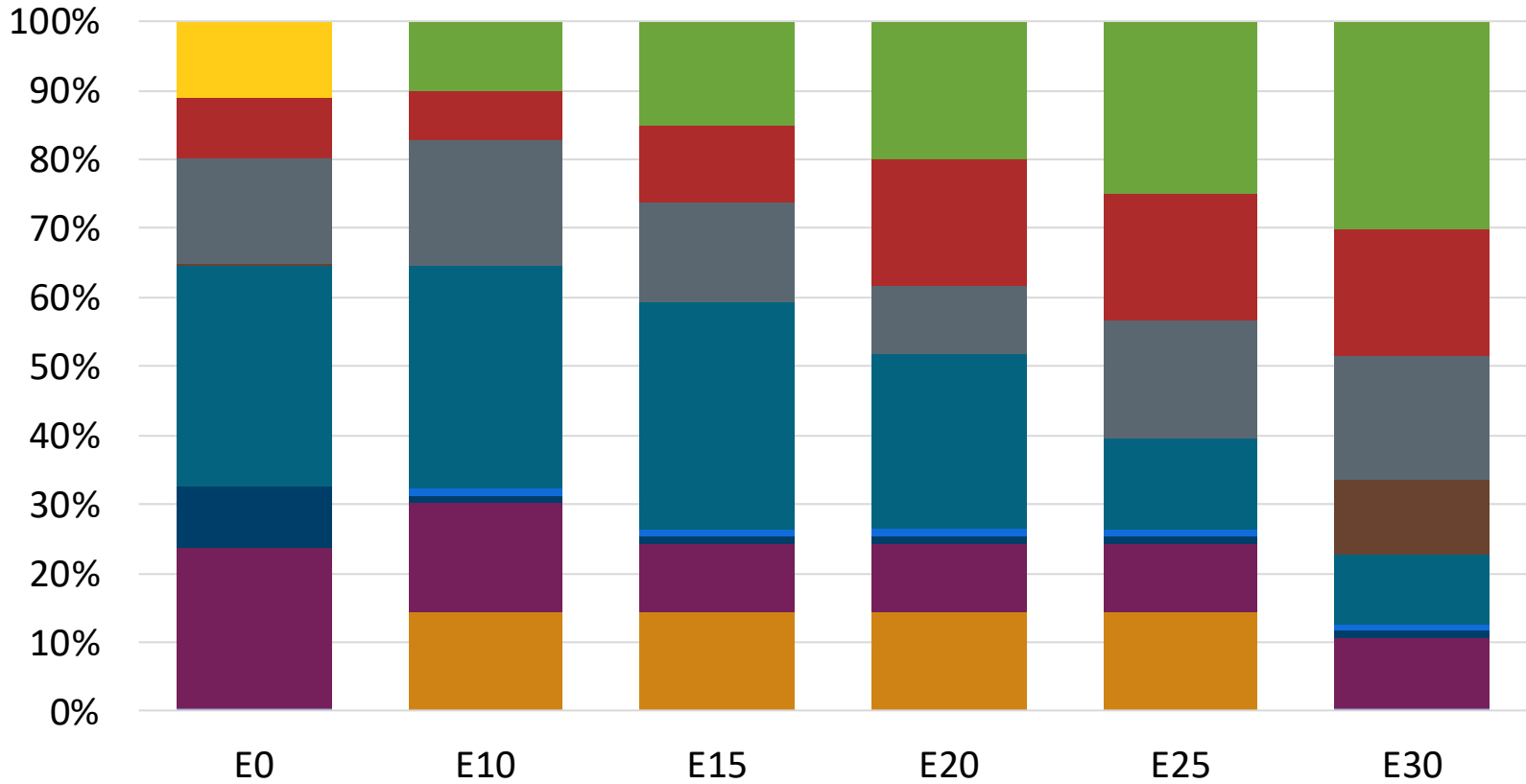
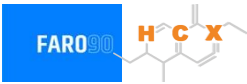
Gasoline Quality in Mexico

Name	NOM-016-CRE-2016						EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2016						2017			
Applicability	Mexico City	Mexico City	Guadalajara and Monterrey	Guadalajara and Monterrey	Rest of the country	Rest of the country	All countries			
Selected Grade	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	Gasoline Regular	Gasoline Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 2 %v/v	< 2 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 25 %v/v	< 25 %v/v	< 32 %v/v	22,6% v/v	Report	22,6% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 10 %v/v	< 10 %v/v	< 11,9 %v/v	< 11,9 %v/v	Report	< 2,5 mg/l	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	-	-	-	-	-	-	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON		94		94		94	> 95	> 95	> 98	> 98
MON	82		82		82		> 85	> 88	> 85	> 88
AKI										
Sulfur Content	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 80 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 3,7 %m/m	< 2,7 %m/m	< 3,7 %m/m
Ethanol (EtOH)	Not allowed	Not allowed	Not allowed	Not allowed	> 5,8 %v/v	> 5,8 %v/v	< 5 %v/v	< 10 %v/v	< 5 %v/v	< 10 %v/v
RVP 37.8°C (Summer)	< 62 kPa	< 62 kPa	< 62 kPa	< 62 kPa	< 69 kPa North, < 62 kPa Center, South, Pacific	< 69 kPa North, < 62 kPa Center, South, Pacific	< 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa	< 79 kPa				
RVP 37.8°C (Transition)	< 54 kPa	< 54 kPa	< 54 kPa Guadalajara	< 54 kPa Guadalajara						
MTBE							-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	-	-	Based on oxygen content	< 22 %v/v	Based on oxygen content	< 22 %v/v

Available Blending Components

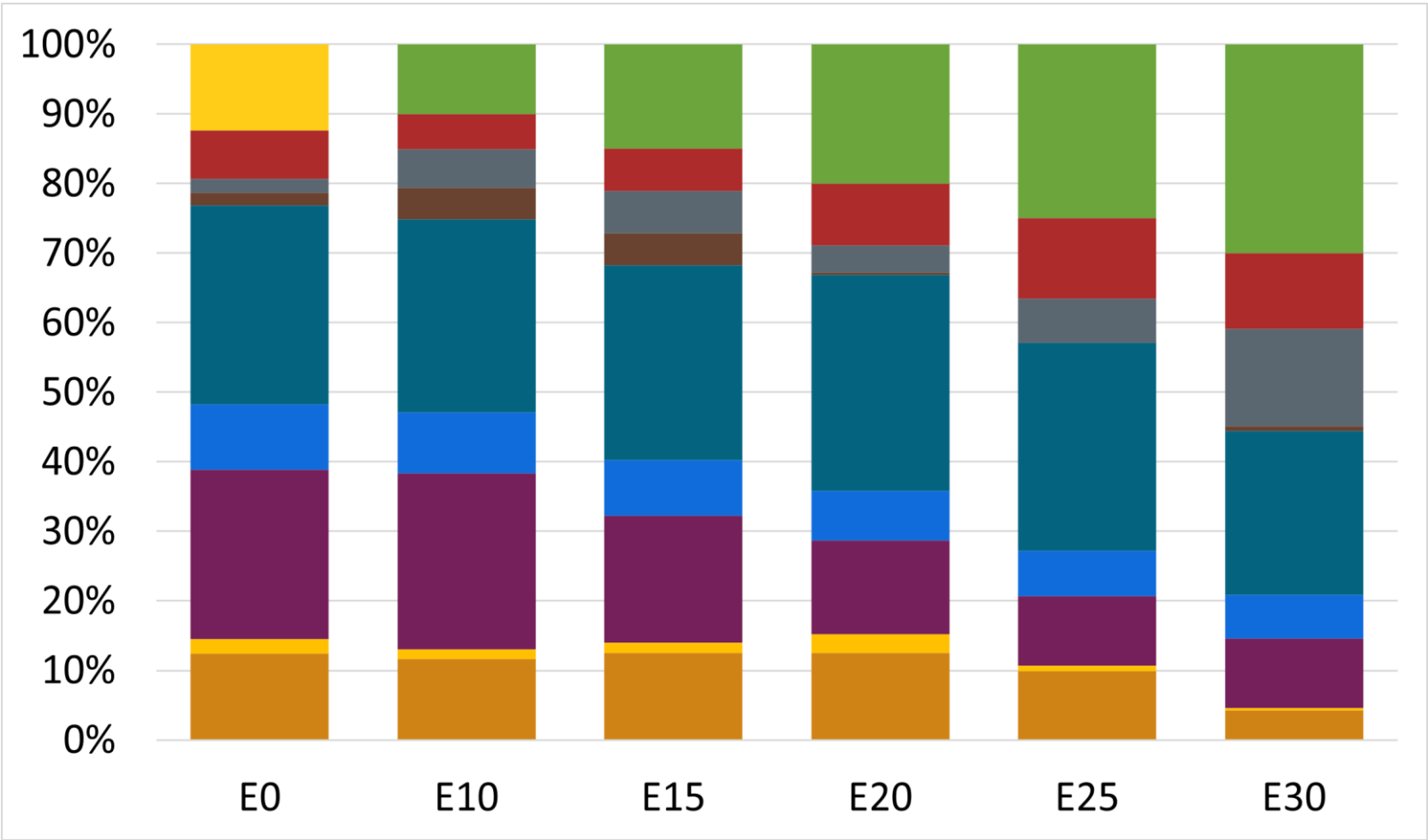
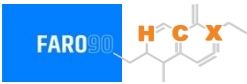


Regular Gasoline RP – Constant Octane



Octane (AKI)	87.1	87.0	87.0	87.0	87.0	87.0
Price (US\$/gal)	\$ 2.034	\$ 2.076	\$ 2.000	\$ 1.927	\$ 1.856	\$ 1.794

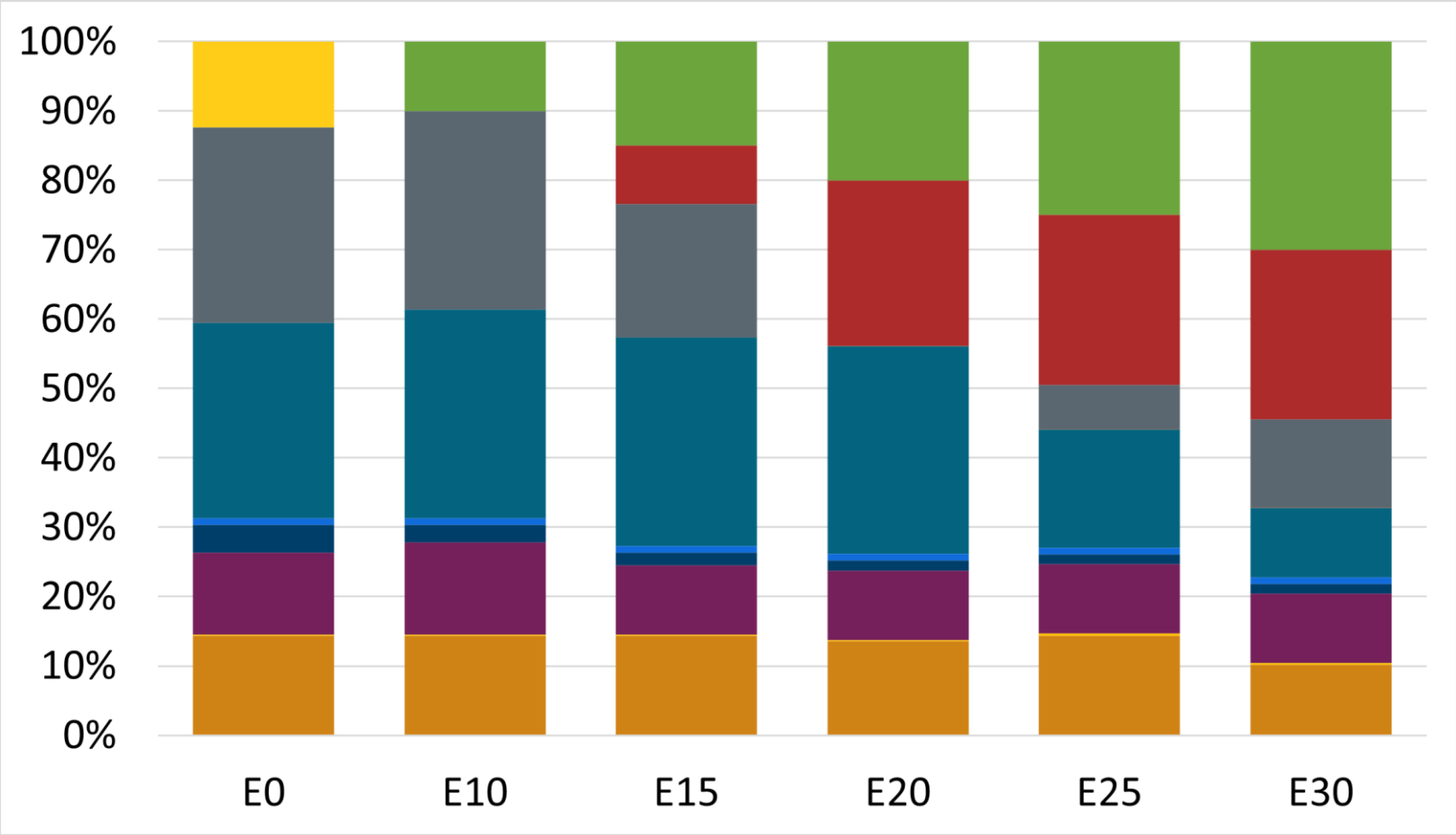
Premium Gasoline RP – Constant Octane



Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US\$/gal)	\$ 2.212	\$ 2.185	\$ 2.120	\$ 2.056	\$ 1.990	\$ 1.923

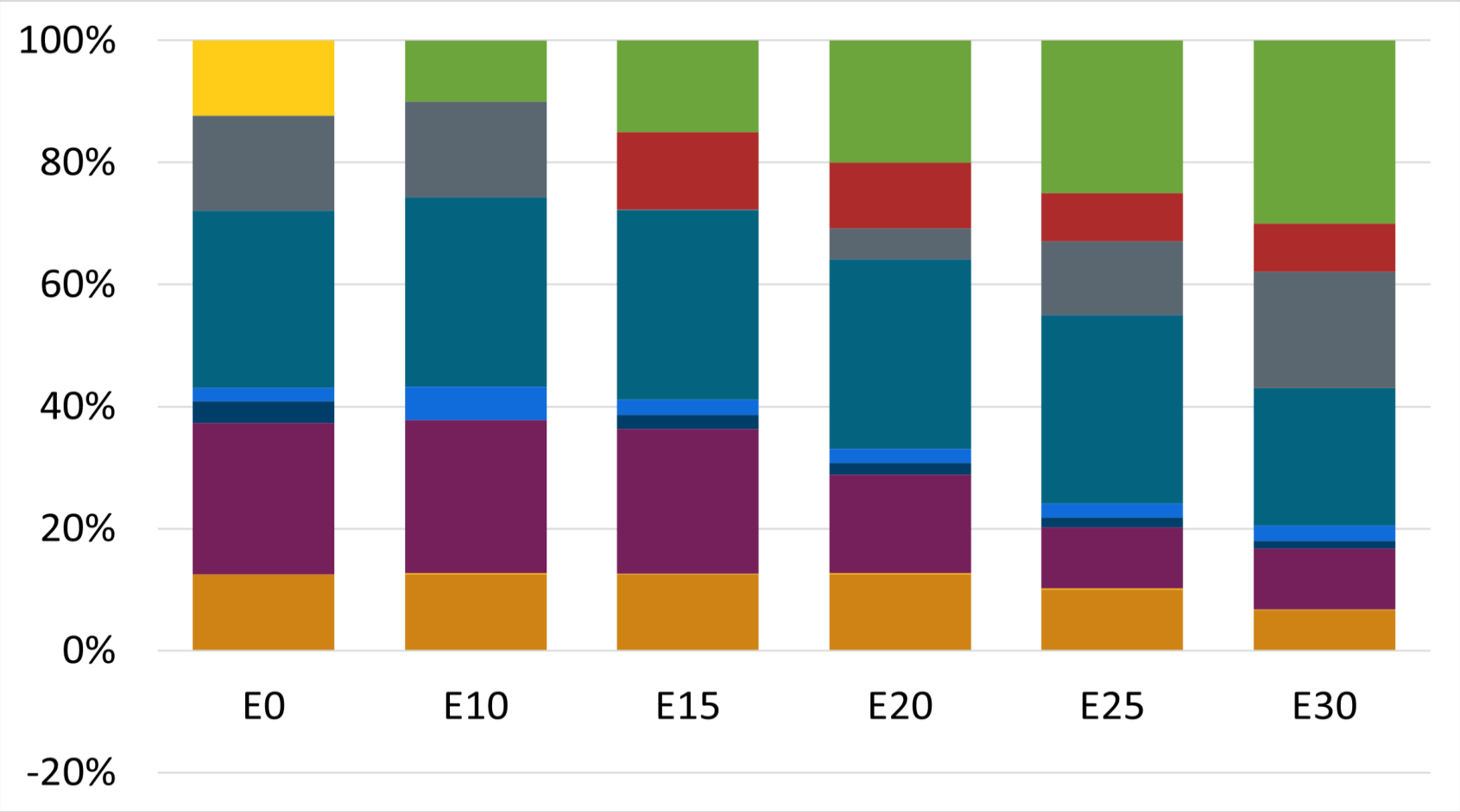


Regular Gasoline ZM – Constant Octane



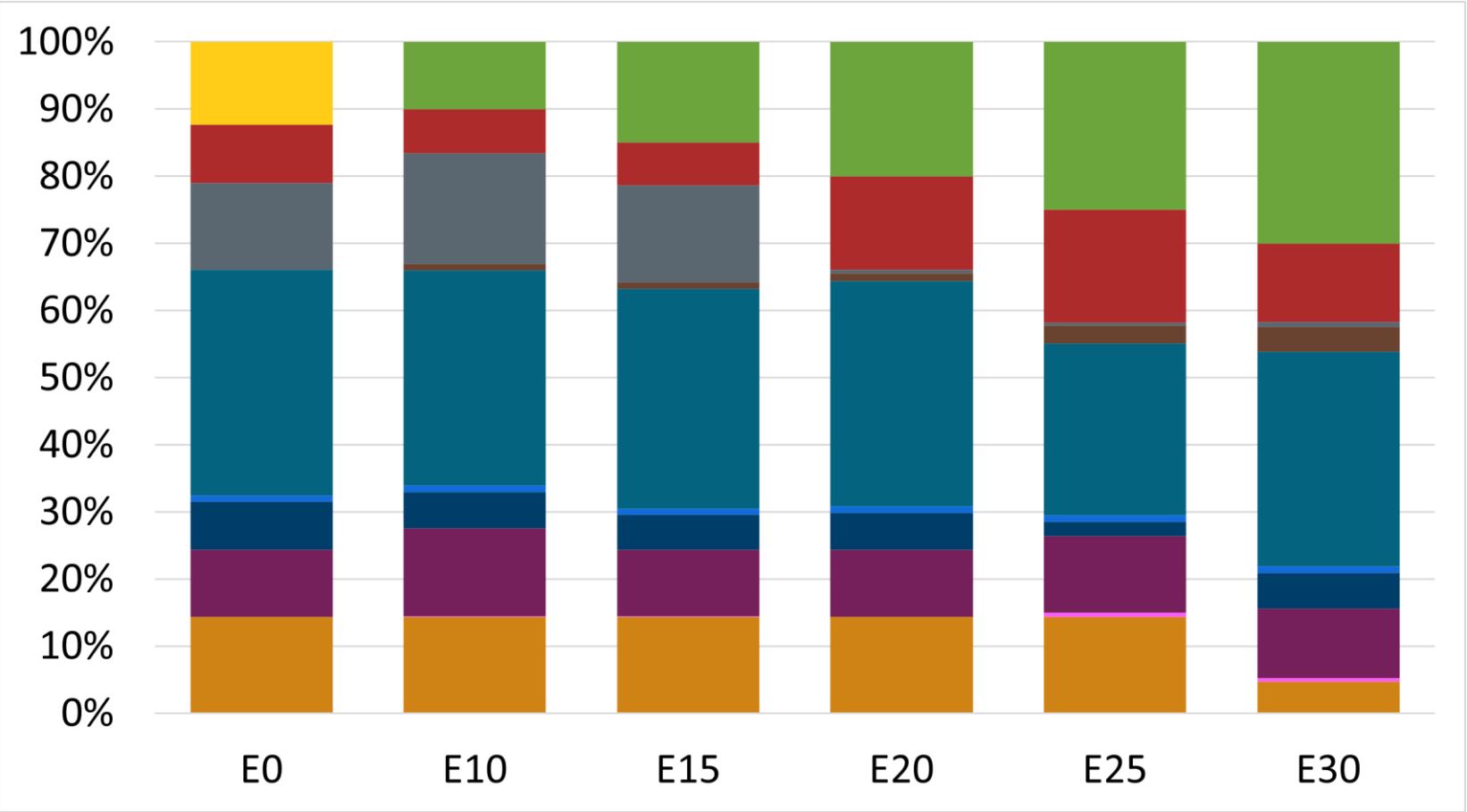
Octane (AKI)	87.0	87.0	87.0	87.0	87.0	87.0
Price (US\$/gal)	\$ 2.070	\$ 2.054	\$ 1.989	\$ 1.920	\$ 1.850	\$ 1.779

Premium Gasoline ZM – Constant Octane



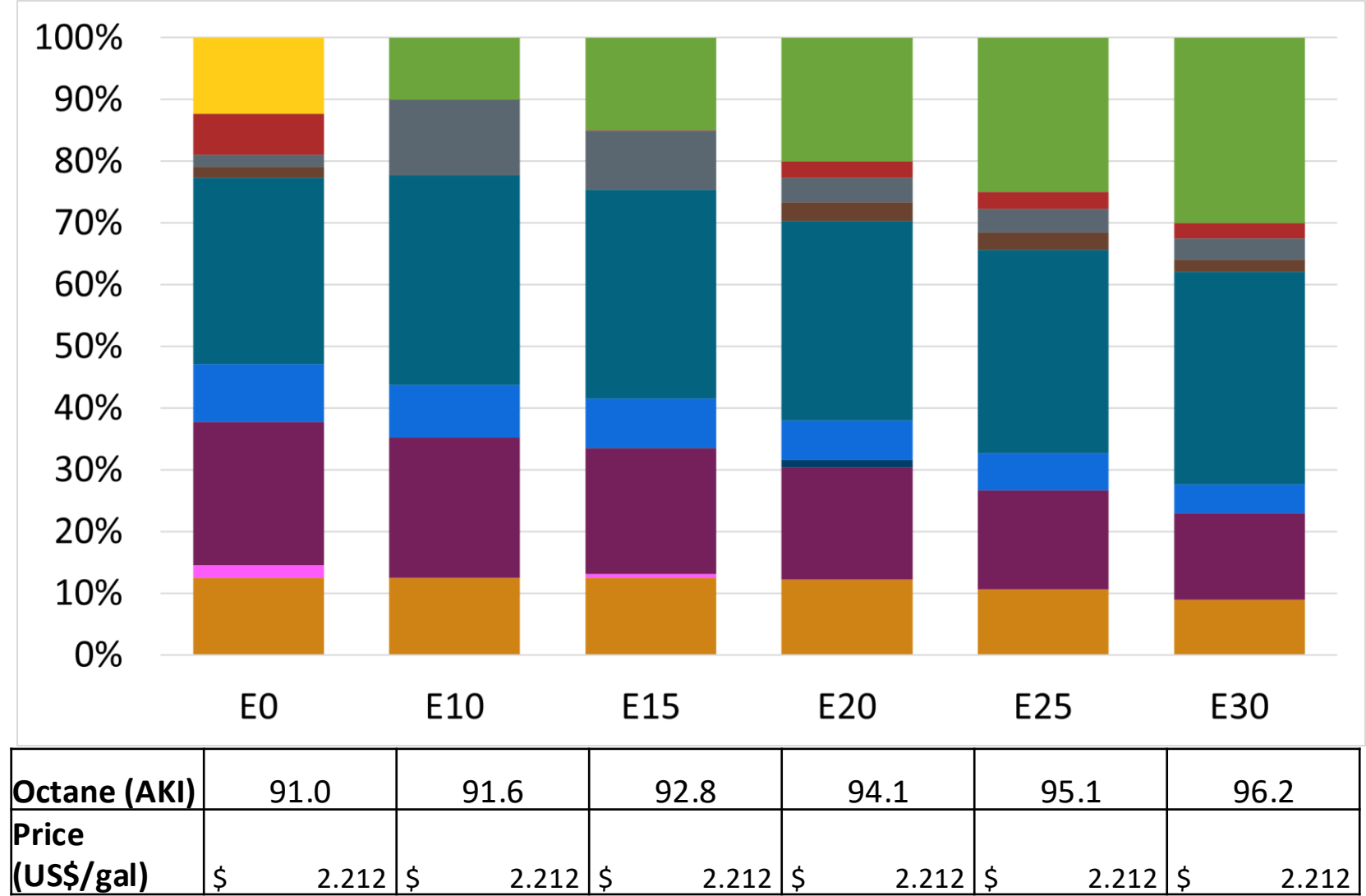
Octane (AKI)	91.0	91.0	91.0	91.0	91.0	91.0
Price (US\$/gal)	\$ 2.267	\$ 2.232	\$ 2.167	\$ 2.099	\$ 2.031	\$ 1.962

Regular Gasoline RP – Increased Octane

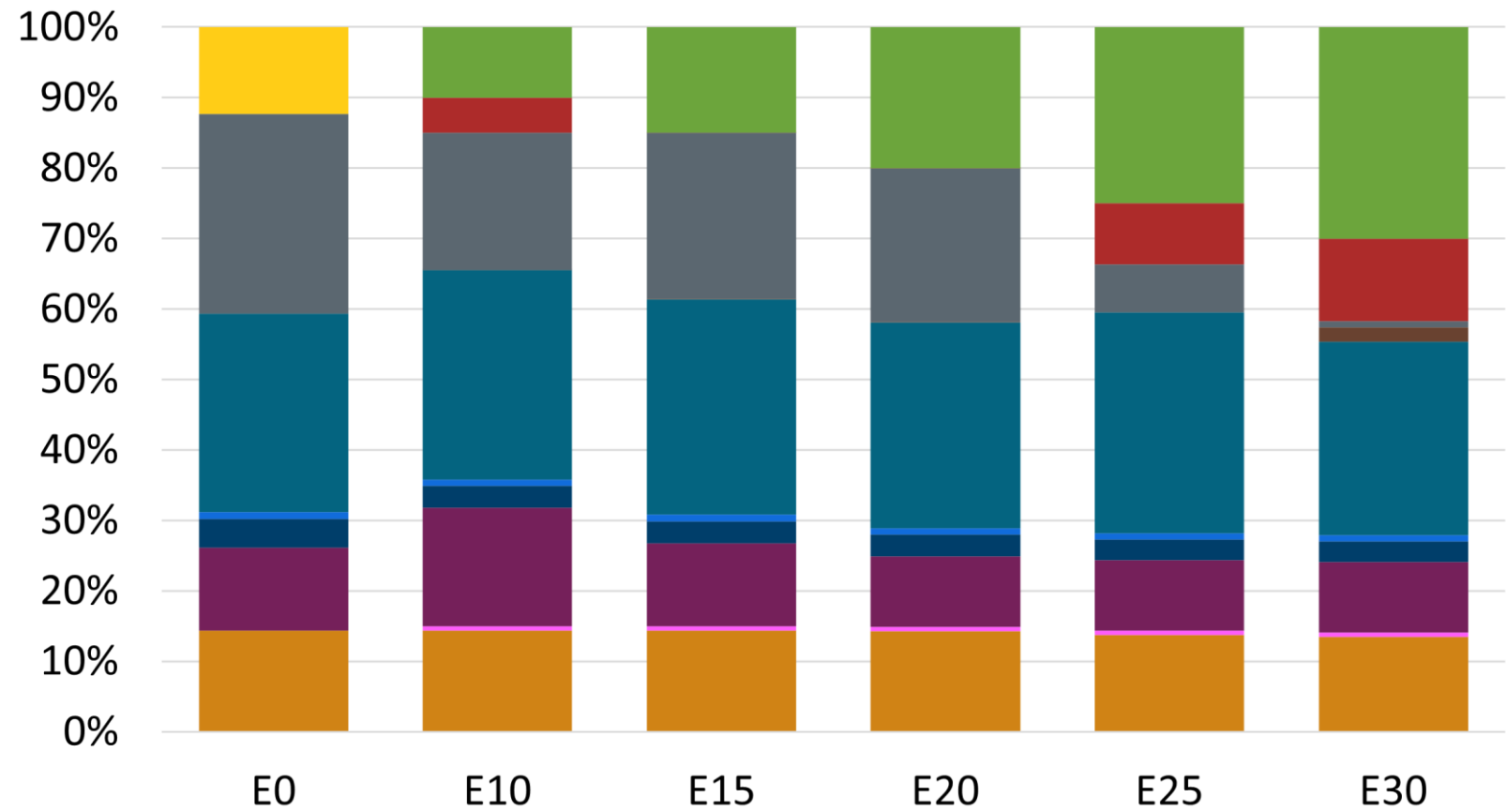


Octane (AKI)	87.0	87.2	88.6	90.1	90.5	92.9
Price (US\$/gal)	\$ 2.020	\$ 2.020	\$ 2.020	\$ 2.020	\$ 2.020	\$ 2.020

Premium Gasoline RP – Increased Octane

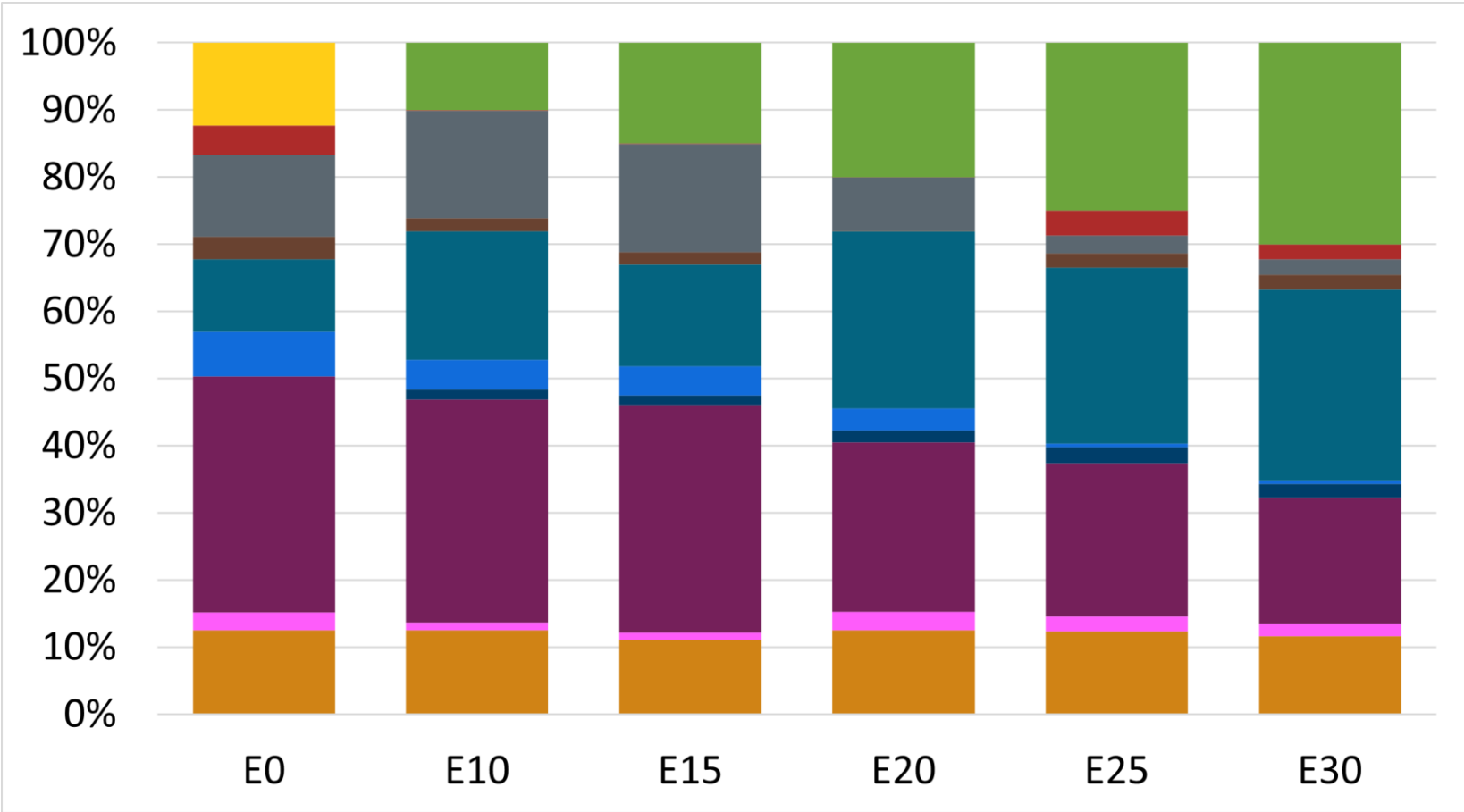


Regular Gasoline ZM – Increased Octane



Octane (AKI)	87.0	87.4	88.9	90.4	91.7	93.2
Price (US\$/gal)	\$ 2.070	\$ 2.070	\$ 2.070	\$ 2.070	\$ 2.070	\$ 2.070

Premium Gasoline ZM – Increased Octane



Octane (AKI)	91.0	91.6	93.0	94.3	95.1	96.5
Price (US\$/gal)	\$ 2.264	\$ 2.264	\$ 2.264	\$ 2.264	\$ 2.264	\$ 2.264

Vehicular Fleet in Mexico

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	3,465				
	Single FI	3-Ways		> 8 years, > 160 tkm	89,129				
	Multi FI	3-Ways		> 8 years, > 160 tkm	3,808,157				
		Euro III		> 8 years, > 160 tkm	5,537,051				
		Euro IV		> 8 years, > 160 tkm	4,183,161				
		Euro V		> 8 years, > 160 tkm	5,781,437				
				4-8 years, 80-160 tkm	4,462,625				
		< 4 years, < 80 tkm		4,281,459					
	Hybrid			Any	234,946				
Electric			Any	32,663					
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	343				
	Single FI	3-Ways		> 8 years, > 160 tkm	2,799				
	Multi FI	3-Ways		> 8 years, > 160 tkm	664,855				
		Euro III		> 8 years, > 160 tkm	1,967,943				
		Euro IV		> 8 years, > 160 tkm	1,334,250				
		Euro V		> 8 years, > 160 tkm	2,074,483				
				4-8 years, 80-160 tkm	2,261,569				
		< 4 years, < 80 tkm		3,723,434					
	Hybrid			Any	175,050				
	Electric			Any	28,023				
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	53,390				
		Euro II		> 8 years, > 160 tkm	85,136				
		Euro III		> 8 years, > 160 tkm	1,896,819				
		Euro IV		> 8 years, > 160 tkm	1,024,779				
		Euro V		4-8 years, 80-160 tkm	1,287,524				
				4-8 years, 80-160 tkm	1,223,855				
				< 4 years, < 80 tkm	2,499,594				
	Electric		Any	1,614,420					
	Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	18,254			
FI		3-Ways	> 8 years, > 160 tkm		2,630,270				
		Euro III	> 8 years, > 160 tkm		2,338,059				
		Euro IV	> 8 years, > 160 tkm		1,551,482				
		Euro V / Euro 6	> 8 years, > 160 tkm		2,170,211				
			4-8 years, 80-160 tkm		1,601,860				
		< 4 years, < 80 tkm	2,814,954						
Hybrid			Any		5,074				
Electric			Any	1,015					

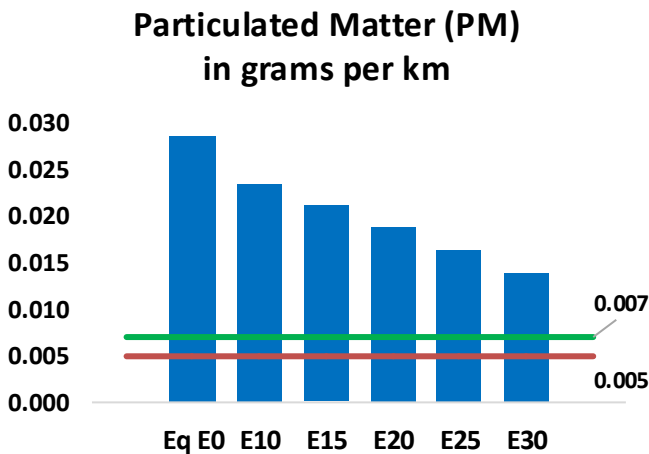
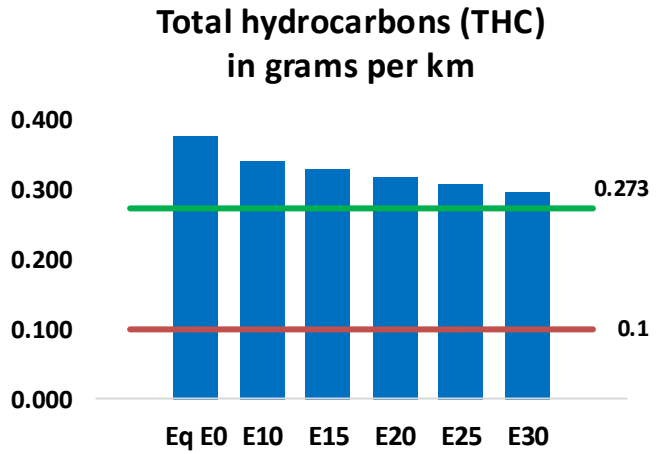
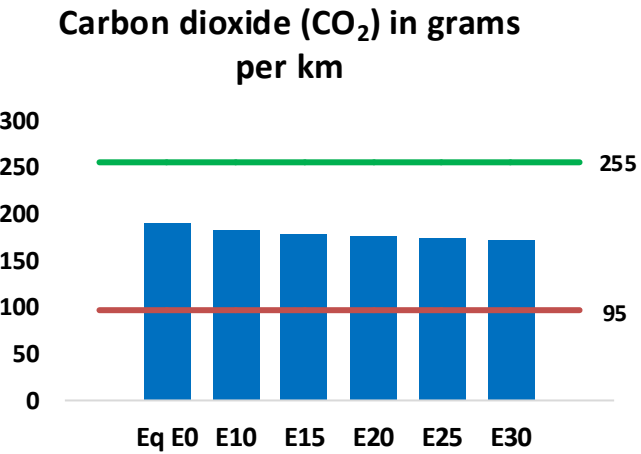
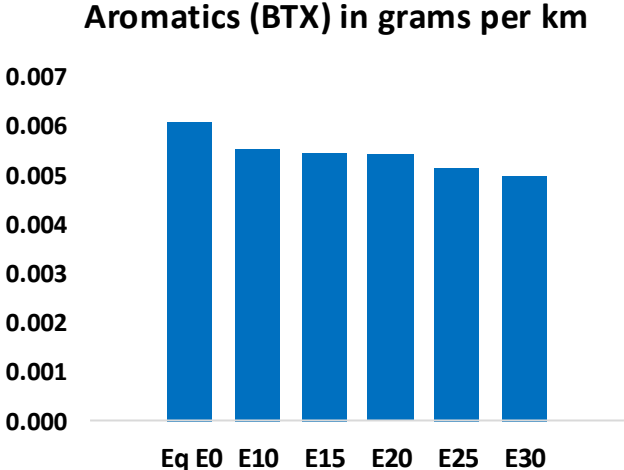
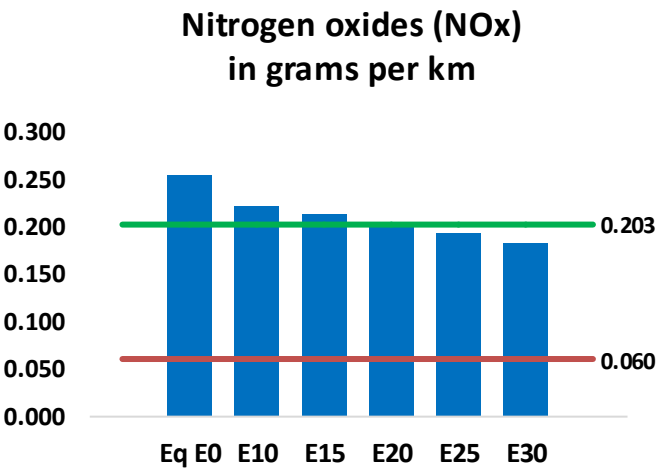
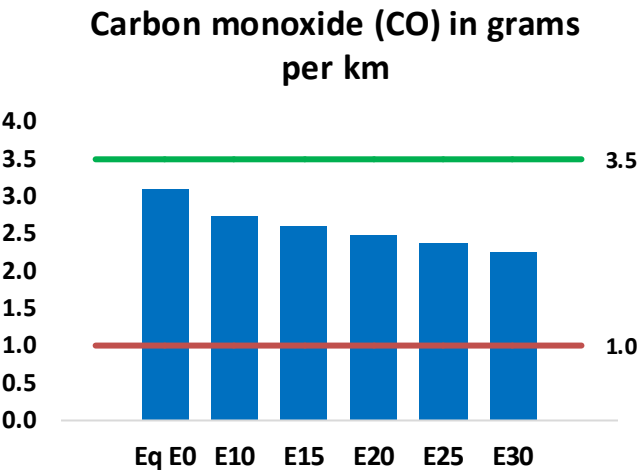
Gasoline Vehicular Fleet: **48,247,257**

Average Age: **10 years**

Total = 63,463,539

Vehicular Emissions in Mexico

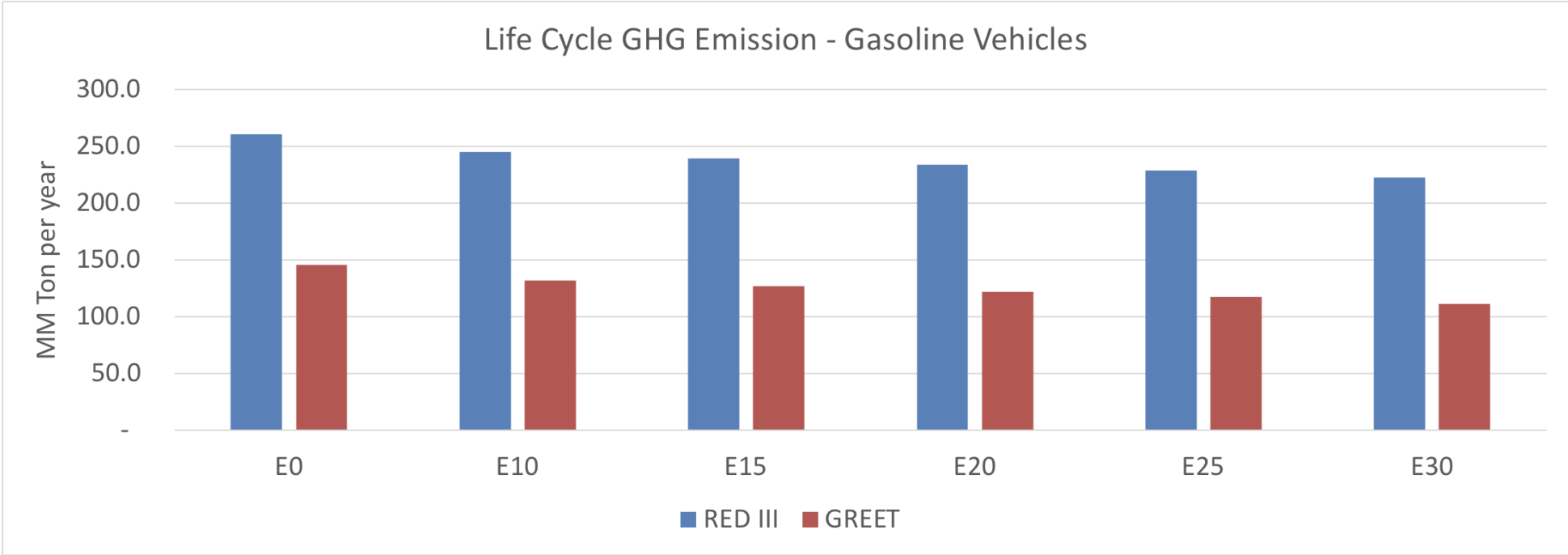
United States
Euro 6



Mexico – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	3,770,544	3,551,737	3,332,931	3,167,493	3,007,391	2,886,207	2,734,181	-12%	-20%
CO2	230,824,858	224,889,361	219,283,615	214,874,860	212,693,235	211,504,284	207,605,652	-5%	-8%
VOC	455,609	434,222	412,836	397,972	384,239	374,049	359,145	-9%	-16%
NOx	308,561	284,905	268,484	260,020	246,494	235,842	223,331	-13%	-20%
SOx	2,639	2,440	2,240	2,072	1,910	1,734	1,550	-15%	-28%
NH3	81,098	80,295	79,492	79,870	80,768	81,400	81,925	-2%	0%
N2O	3,065	3,049	3,037	3,051	3,114	3,149	3,185	-1%	2%
CH4	115,733	115,733	115,733	117,469	120,015	122,214	124,298	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	4,349	4,183	4,016	3,912	3,823	3,759	3,657	-8%	-12%
Acetaldehyde	10,181	11,685	17,146	26,111	35,573	40,649	48,032	68%	249%
Formaldehyde	35,972	34,973	40,769	47,871	50,152	55,482	60,398	13%	39%
Benzene	7,401	7,113	6,740	6,633	6,578	6,291	6,088	-9%	-11%
PM2.5	14,022	12,748	11,473	10,351	9,253	8,077	6,845	-18%	-34%
PM10	20,505	18,641	16,777	15,137	13,532	11,811	10,010	-18%	-34%

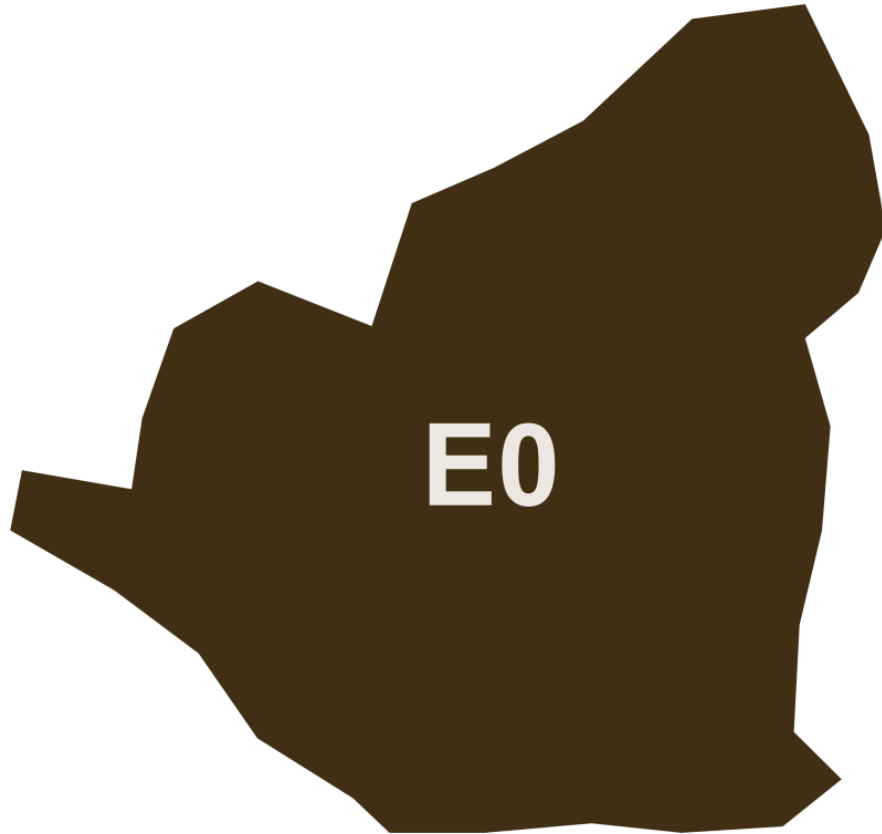
Mexico – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	751.4
2035 Target (MMT/year)	500.4
Delta	251.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.4%	12.9%	16.4%	19.4%	23.4%
RED II/III	0.0%	5.9%	8.2%	10.3%	12.2%	14.6%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	5.4%	7.5%	9.5%	11.2%	13.5%
RED II/III	0.0%	6.1%	8.5%	10.7%	12.7%	15.2%

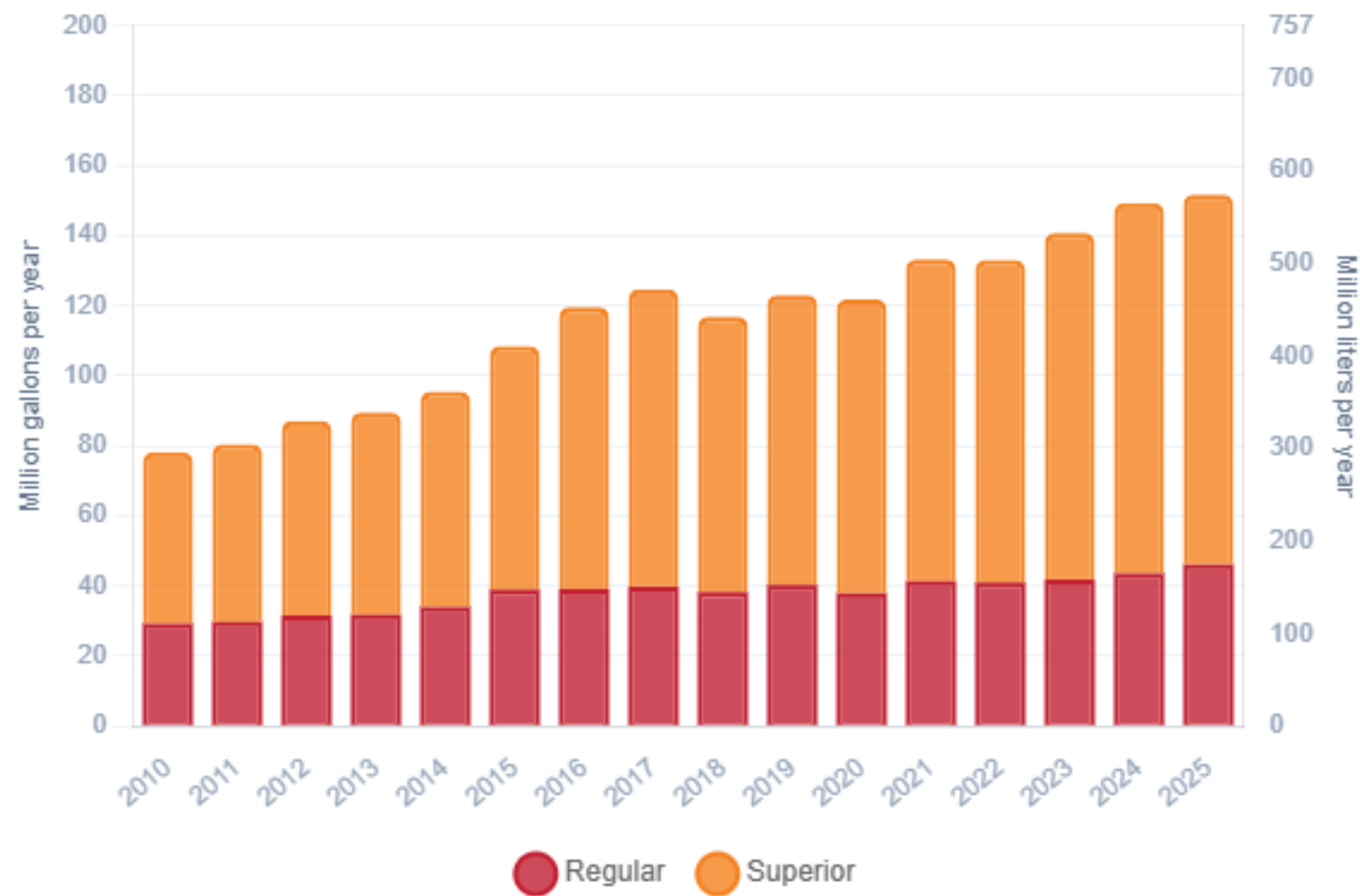
Ethanol Blending in Gasoline - Nicaragua



In 2025, gasoline demand was 151 million gallons (574 million liters). Nicaragua is the only country in Central America that has active refinery capacity, which produced 38% of gasoline national demand. The rest of demand is imported, previously from Venezuela and since 2016 mainly from United States. There are two gasoline grades in the country, regular with 30% and premium with 70% of market share.

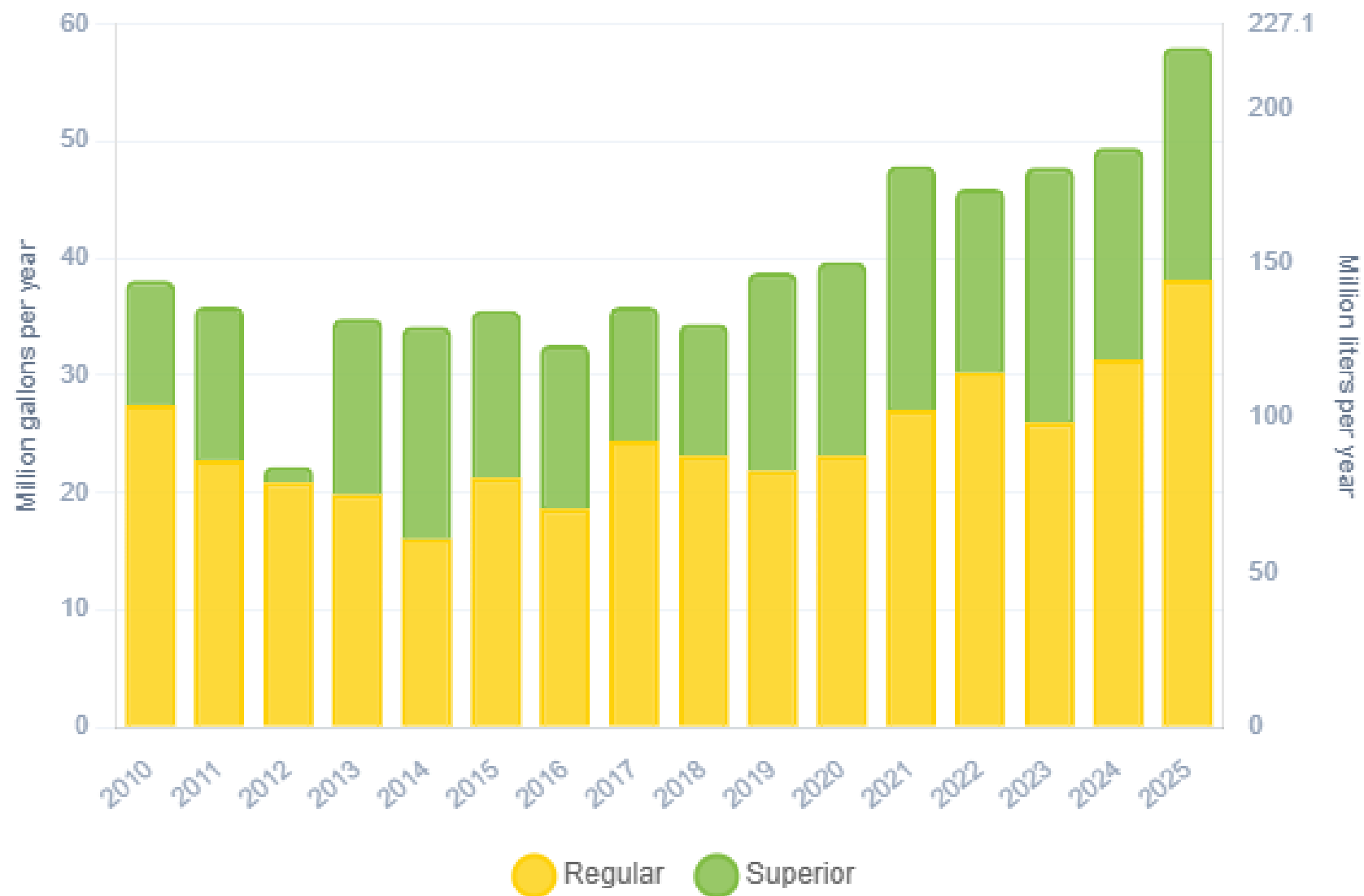
According to the Ministry of Energy and RTCA, Nicaragua has no ethanol mandate.

Gasoline Demand in Nicaragua



Source: MEM Nicaragua

Gasoline Production in Nicaragua



Source: MEM Nicaragua

Gasoline Imports in Nicaragua



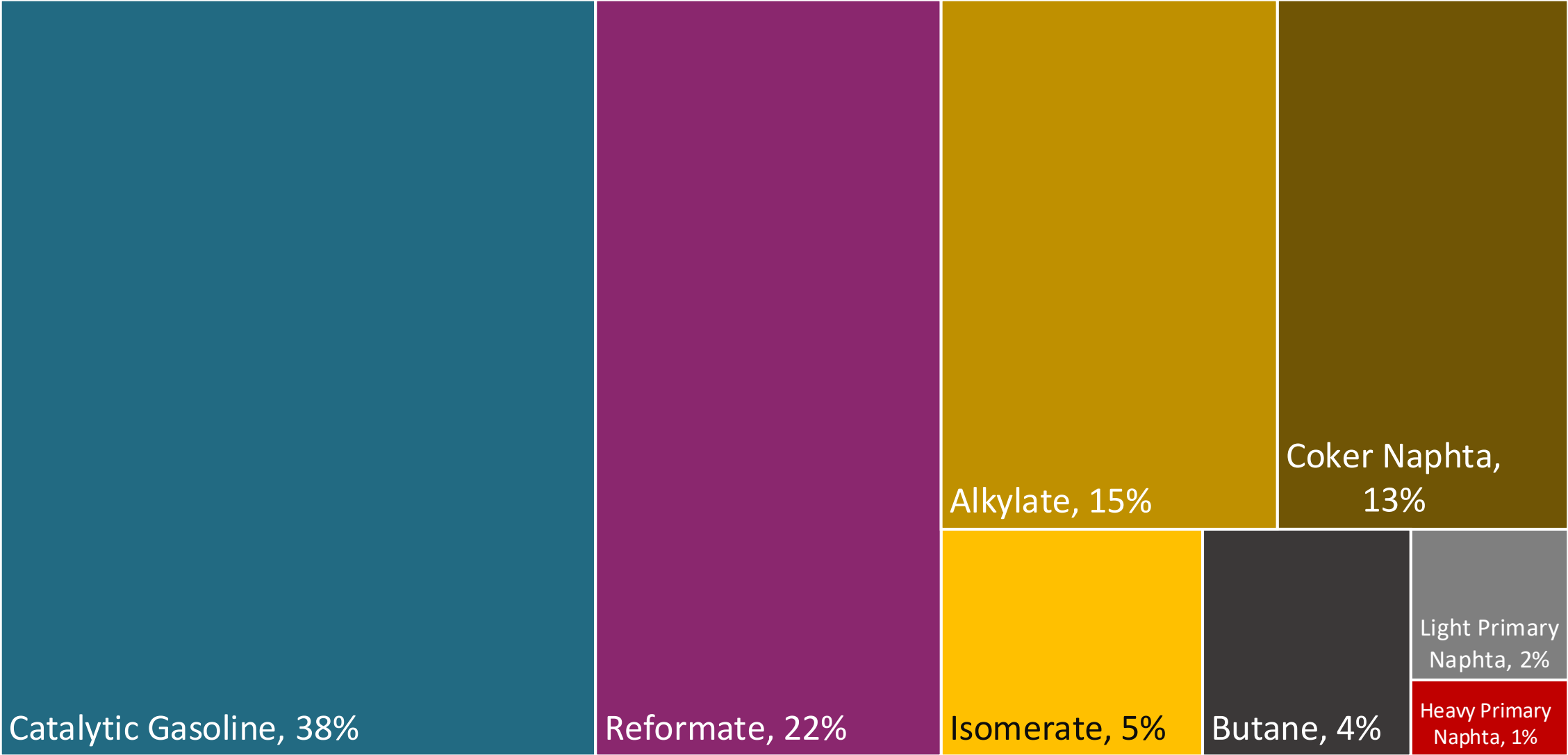
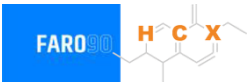
Source: MEM Nicaragua

Gasoline Quality in Nicaragua

Name	RTCA 75.01.20:19	RTCA 75.01.19:19	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021	2021	2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	Gasoline Premium	Gasoline Regular	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	5% v/v	5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	50% v/v	50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	30% v/v	30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	Report	Report	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 91	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg	< 500 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	2,7% v/v	2,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	-	-	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa	< 69 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	-	-	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

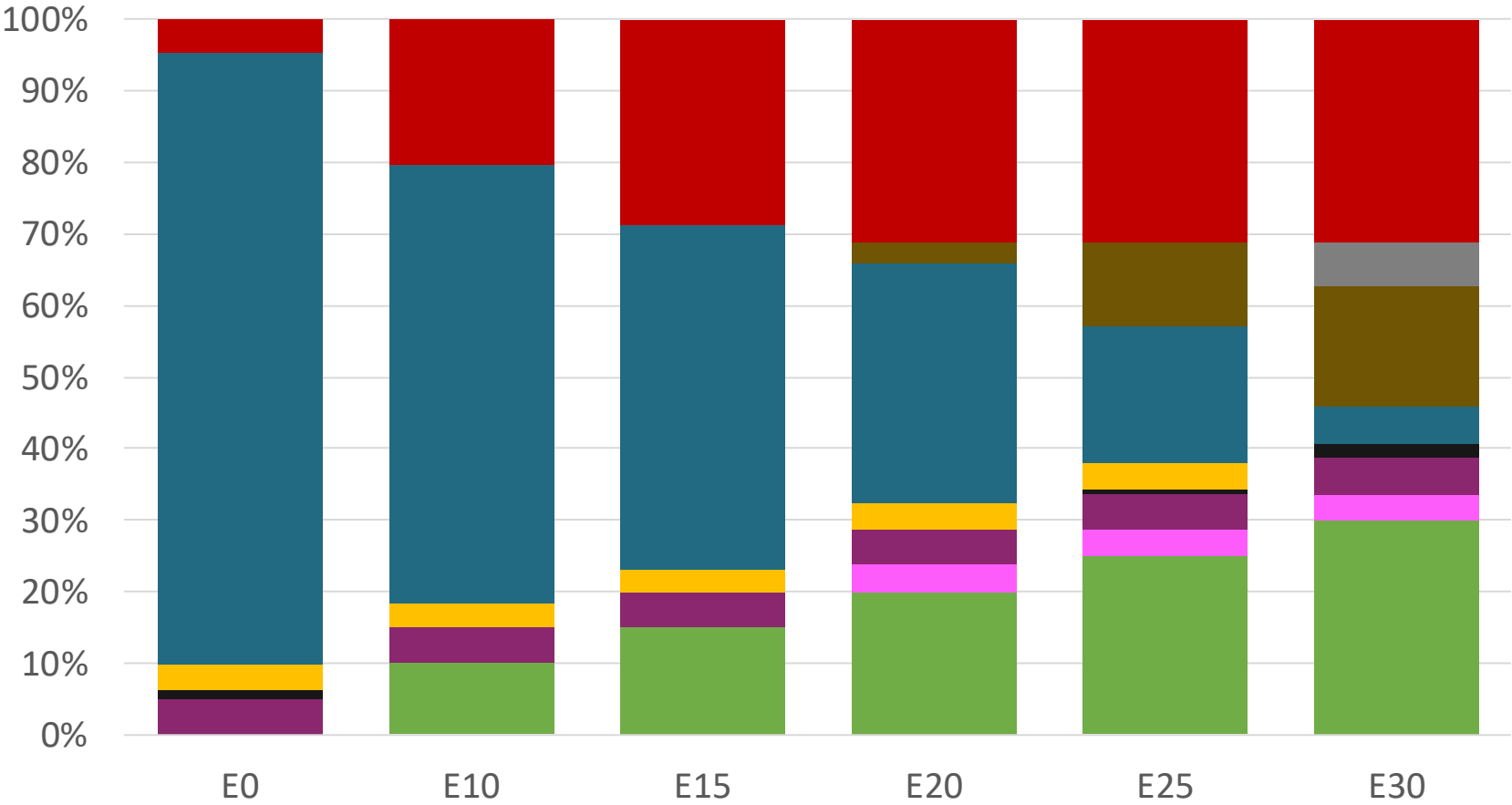
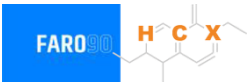
Source: RTCA

Blending Components - Nicaragua



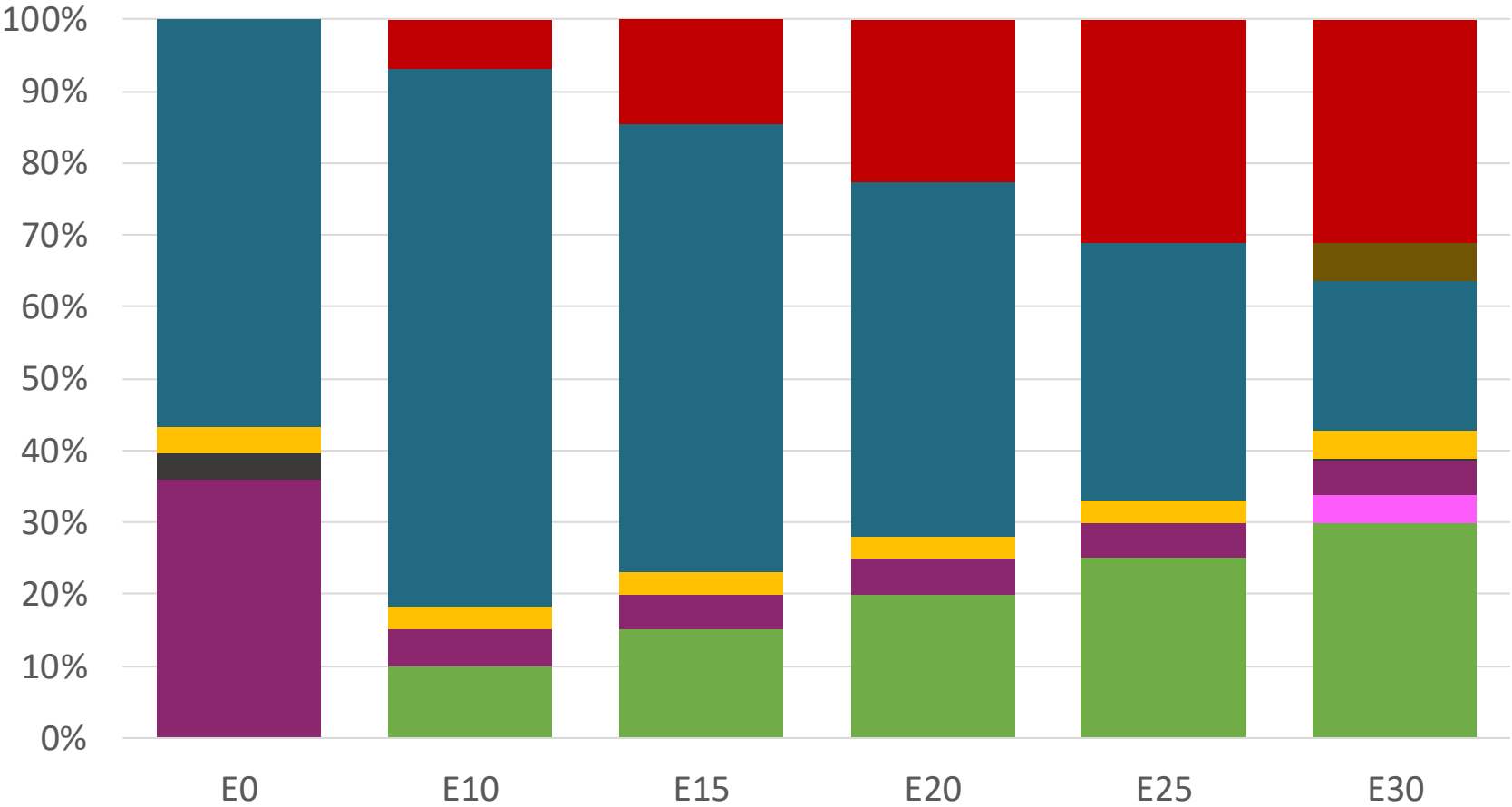
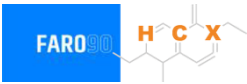
Source: Faro90

Nicaragua – Regular – Constant Octane



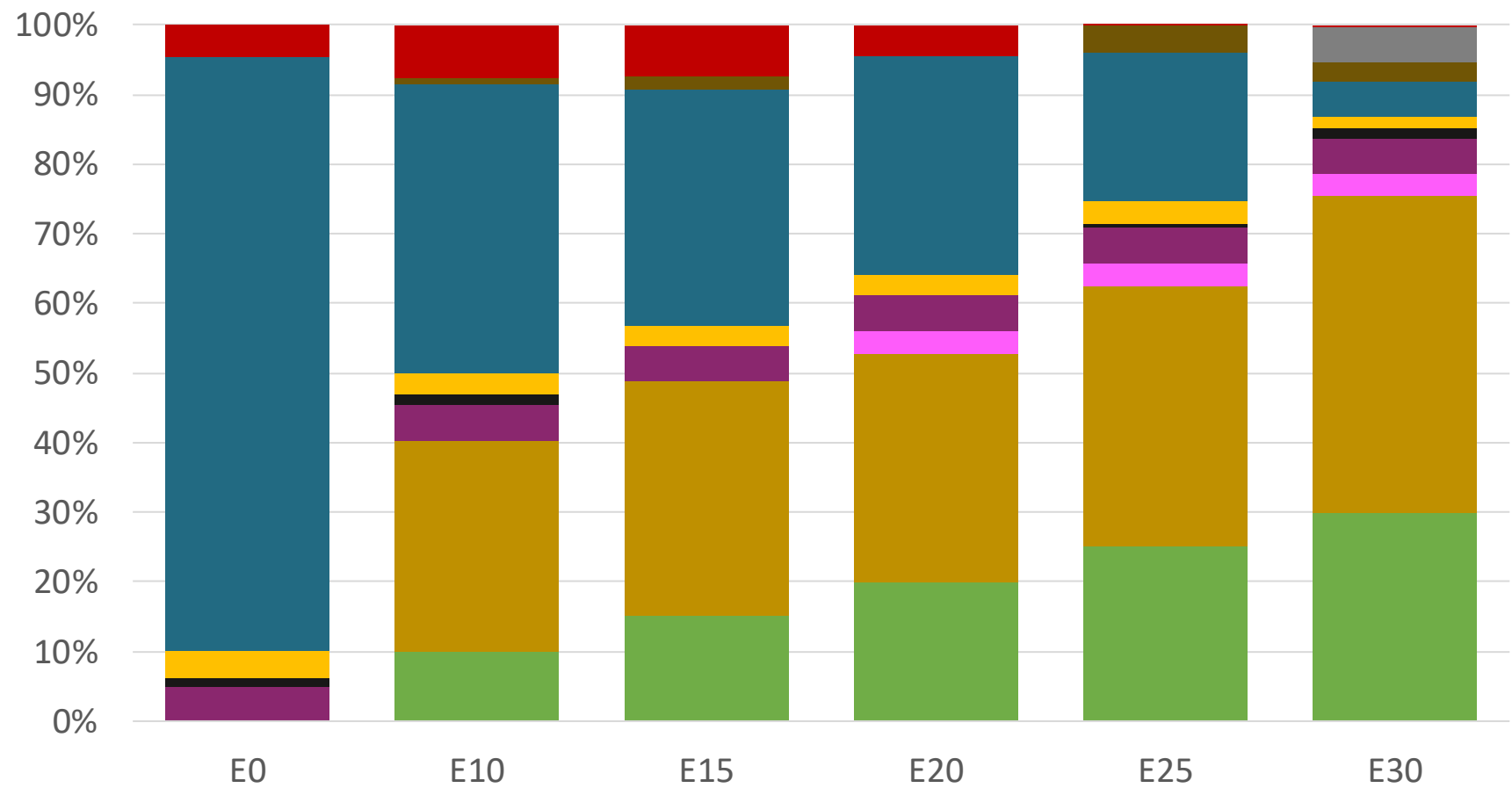
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 1.946	\$ 1.674	\$ 1.523	\$ 1.381	\$ 1.237	\$ 1.266

Nicaragua – Premium – Constant Octane



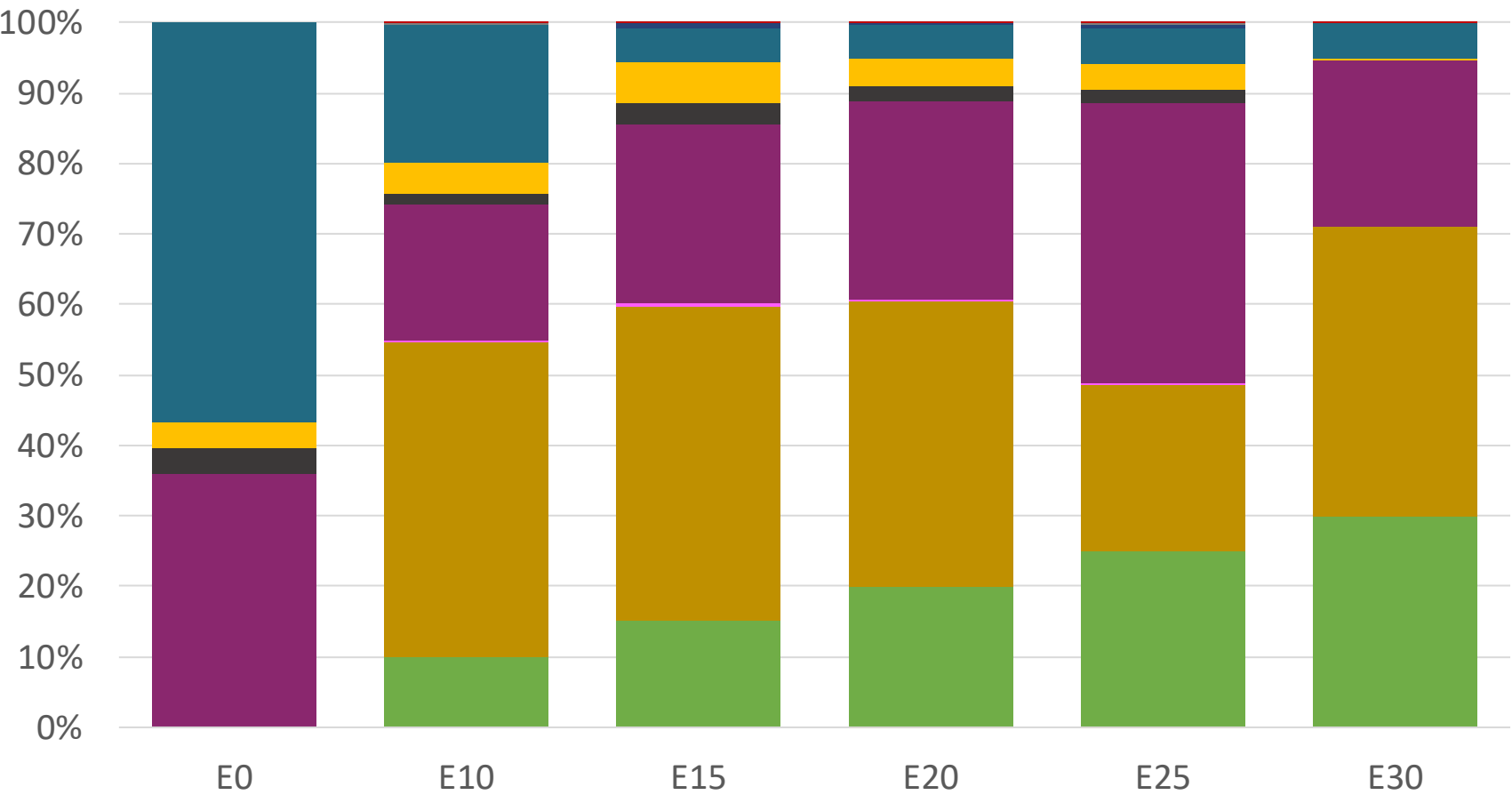
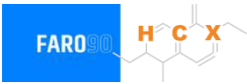
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.326	\$ 2.023	\$ 1.888	\$ 1.746	\$ 1.597	\$ 1.448

Nicaragua – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946	\$ 1.946

Nicaragua – Premium – Increased Octane



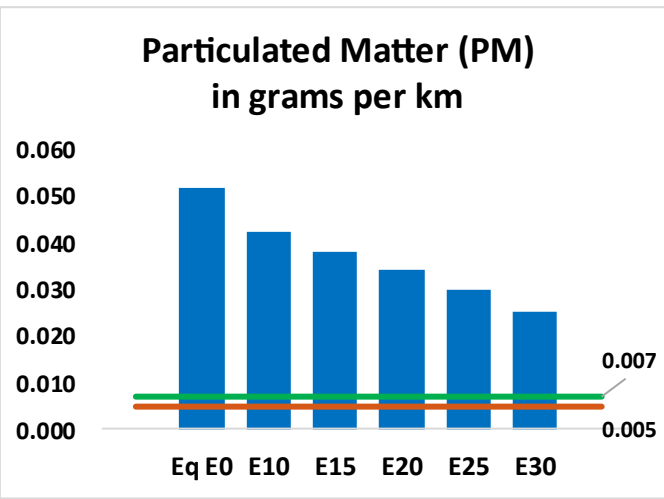
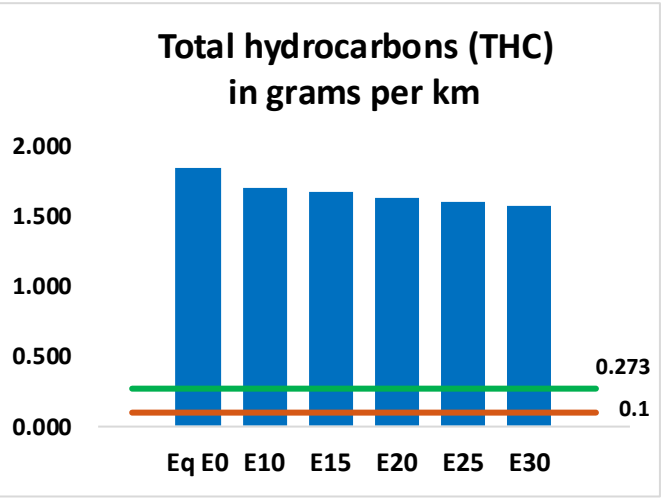
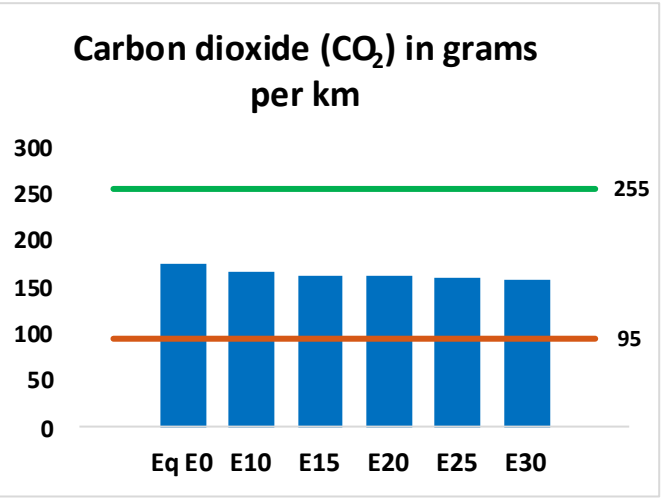
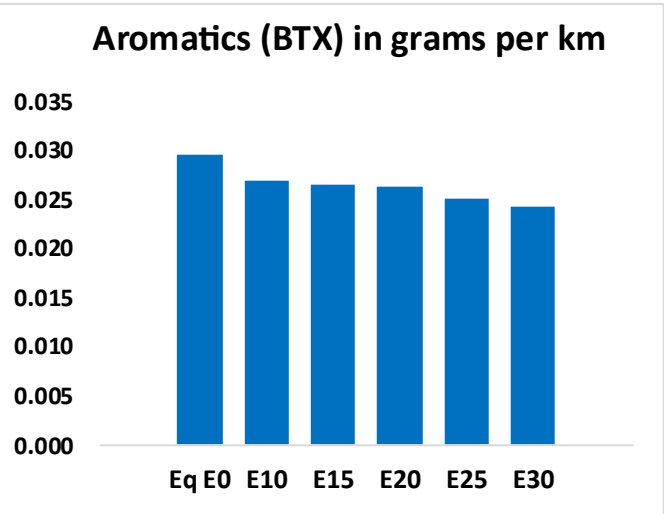
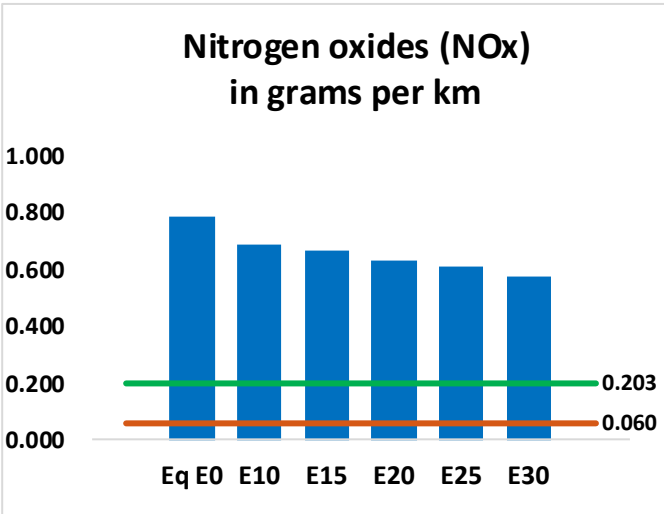
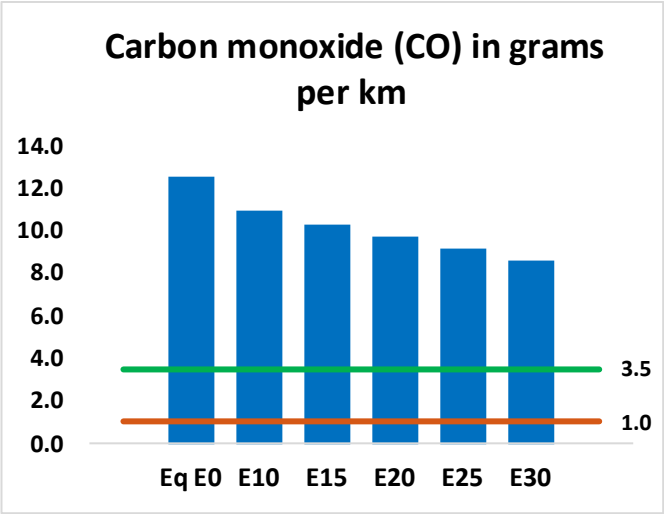
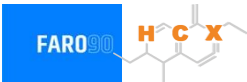
Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326

Vehicular Fleet a gasolina en Nicaragua

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles					
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	3,110					
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	10,906					
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	76,779					
		Euro III		> 8 years, > 160 tkm	50,761					
		Euro IV		> 8 years, > 160 tkm	33,980					
		Euro V / Euro 6		> 8 years, > 160 tkm	20,722					
				4-8 years, 80-160 tkm	11,153					
		< 4 years, < 80 tkm		5,762						
	Electric			Any	111					
		Carburator	3-Ways	> 8 years, > 160 tkm	6,337					
Medium automobiles	Single FI-Pt	3-Ways	PCV Tank	> 8 years, > 160 tkm	55,625					
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	122,643					
		Euro III		> 8 years, > 160 tkm	52,413					
		Euro IV		> 8 years, > 160 tkm	30,359					
		Euro V / Euro 6		> 8 years, > 160 tkm	22,640					
				4-8 years, 80-160 tkm	21,967					
		< 4 years, < 80 tkm		31,268						
	Electric			Any	179					
	Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	10,569				
			Euro II		> 8 years, > 160 tkm	12,856				
Euro III			> 8 years, > 160 tkm		138,325					
Euro IV			> 8 years, > 160 tkm		30,294					
Euro V			4-8 years, 80-160 tkm		103,676					
			4-8 years, 80-160 tkm		42,994					
		< 4 years, < 80 tkm	309,233							
Electric		Any	556							
		Carburator	3-Ways	> 8 years, > 160 tkm	3,917					
Heavy		FI	3-Ways	PCV	> 8 years, > 160 tkm	55,910				
	Euro III		> 8 years, > 160 tkm		17,999					
	Euro IV		> 8 years, > 160 tkm		8,673					
	Euro V / Euro 6		> 8 years, > 160 tkm		5,015					
			4-8 years, 80-160 tkm		19,823					
			< 4 years, < 80 tkm		12,569					
	Electric		Any		219					

Nicaragua - Vehicular Emissions

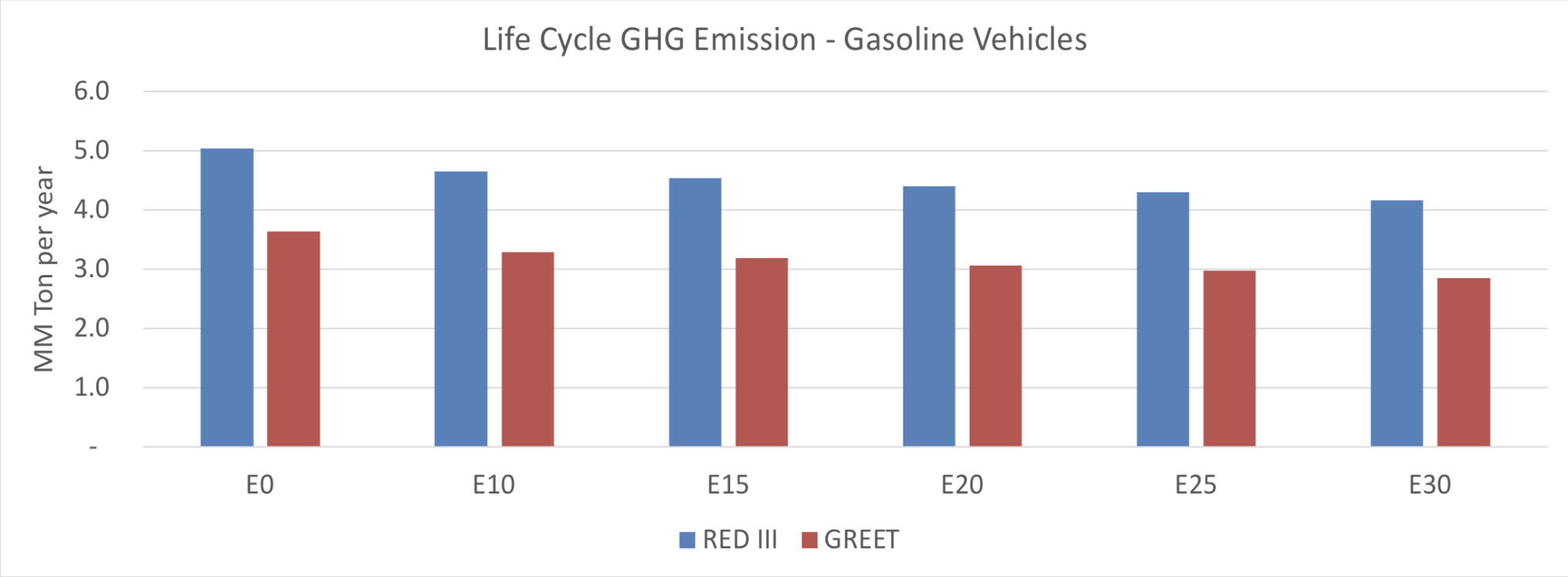
United States
Euro 6



Nicaragua - Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	270,015	252,688	235,362	221,833	208,527	198,388	185,823	-13%	-23%
CO2	3,775,039	3,677,967	3,586,287	3,514,184	3,478,505	3,469,136	3,405,190	-5%	-8%
VOC	39,766	38,306	36,846	35,962	35,215	34,684	33,823	-7%	-11%
NOx	16,992	15,690	14,818	14,385	13,676	13,125	12,474	-13%	-20%
SOx	44	41	37	35	32	29	26	-15%	-28%
NH3	1,513	1,498	1,483	1,490	1,507	1,518	1,528	-2%	0%
N2O	68	67	67	68	69	70	70	-1%	2%
CH4	8,884	8,884	8,884	9,017	9,213	9,382	9,541	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	157	147	137	130	122	116	108	-13%	-22%
Acetaldehyde	395	453	664	1,012	1,379	1,575	1,861	68%	249%
Formaldehyde	1,488	1,446	1,686	1,980	2,074	2,295	2,498	13%	39%
Benzene	639	614	582	573	568	543	526	-9%	-11%
PM2.5	225	204	184	166	148	129	110	-18%	-34%
PM10	893	812	730	659	589	514	436	-18%	-34%

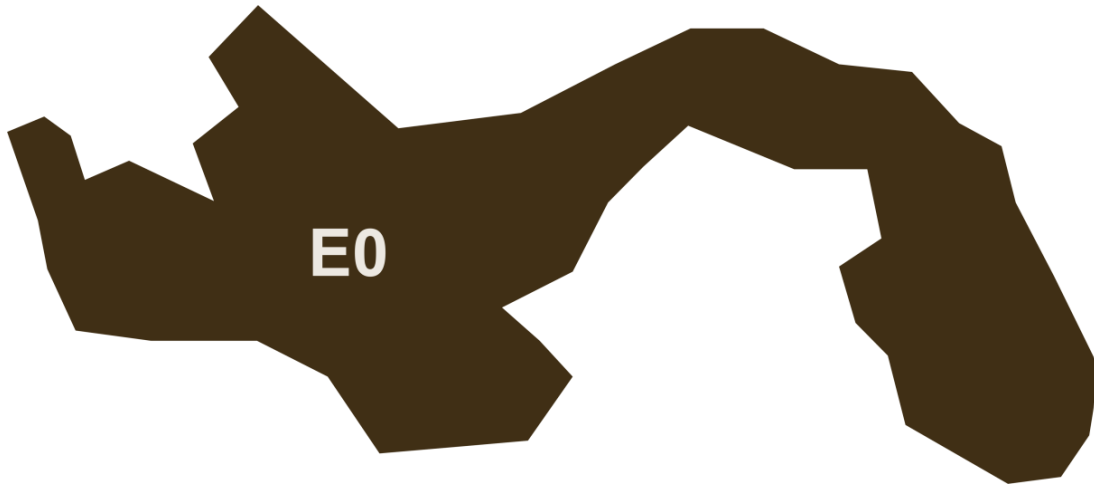
Nicaragua – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	41.1
2035 Target (MMT/year)	28.8
Delta	12.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.9%	12.6%	15.9%	18.4%	21.9%
RED II/III	0.0%	7.7%	10.0%	12.5%	14.6%	17.3%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	2.9%	3.7%	4.7%	5.4%	6.5%
RED II/III	0.0%	3.1%	4.1%	5.1%	6.0%	7.0%

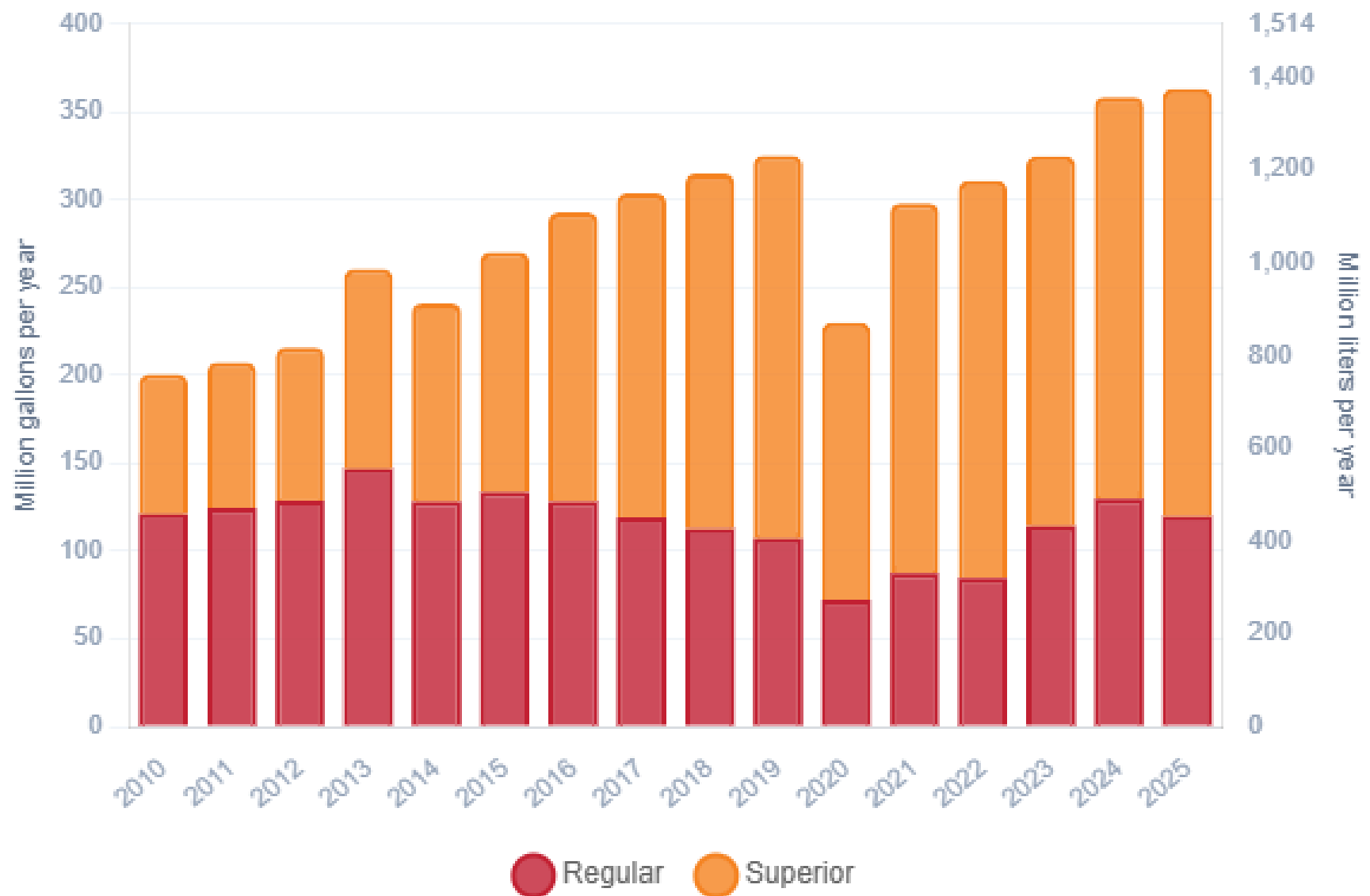
Ethanol Blending in Gasoline - Panama



En 2025 demand was 363 million gallons (1374 million liters). Regular gasoline (RON 91) represented 33% of total demand and Superior gasoline (RON 95) 67%. Gasoline is supplied only with imports, mainly from United States, the Netherlands and Germany.

In 2011, 2% v/v ethanol blends with gasoline were authorized, with a planned gradual increase until reaching 10% v/v by 2016. However, in 2013 it was limited to 5% v/v and by 2014 blending stopped because the only national producer announced its production cease due to political dispute. A new mandate implementation has been approved for E10 and is expected to be implemented between 2026 and 2027.

Gasoline Demand in Panama



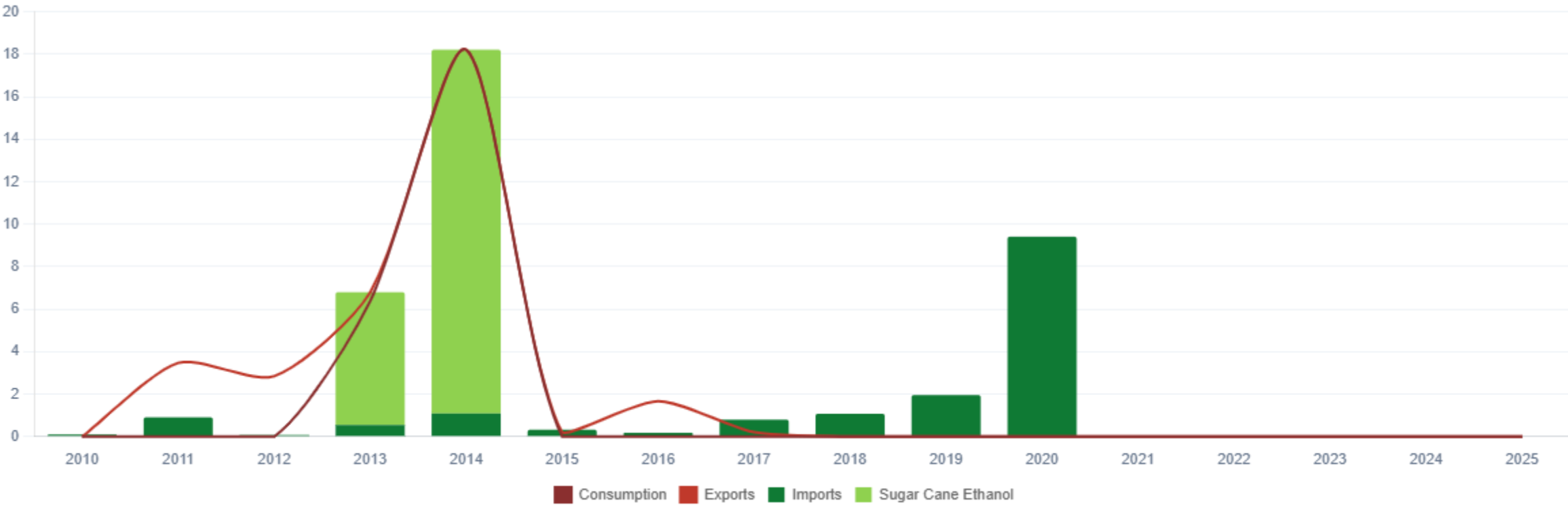
Source: Secretaría de Energía - Panama

Gasoline Imports to Supply Demand in Panama



Source: Secretaría de Energía - Panama

Ethanol Balance in Panama



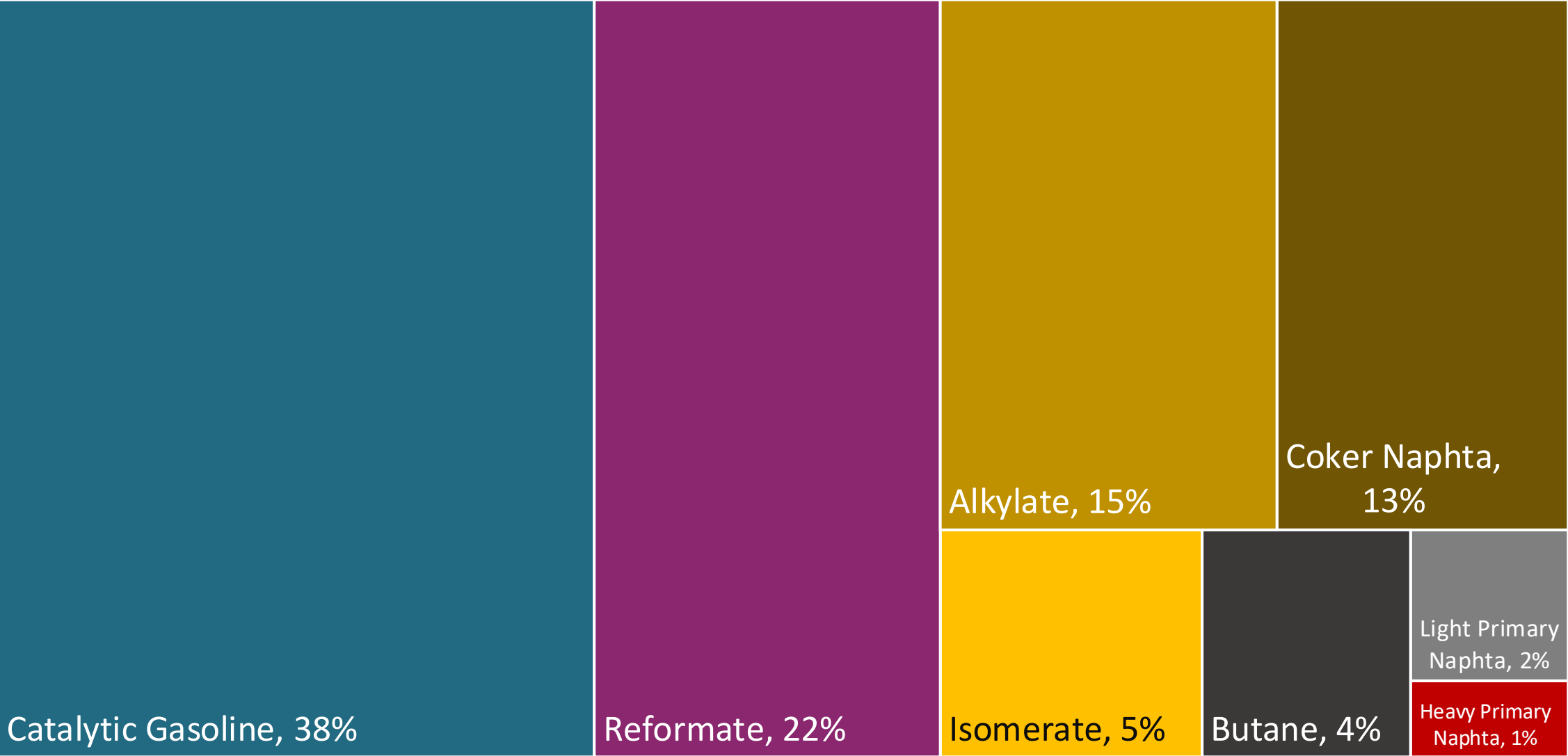
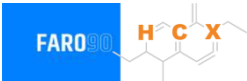
Source: Secretaría de Energía - Panama

Gasoline Quality in Panama

Name	DGNTI-COPANIT 83-2013/ Resolution 425/2020		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2014/2021		2017			
Applicability	Whole country	Whole country	All countries			
Selected Grade	RON 91	RON 95	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 5% v/v / <1,5% v/v	< 5% v/v / <1,5% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 50% v/v	< 50% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 30% v/v	< 30% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	-	-	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 91	> 95	> 95	> 95	> 98	> 98
MON	-	-	> 85	> 88	> 85	> 88
AKI						
Sulfur Content	< 500 mg/kg / < 150 mg/kg	< 500 mg/kg / < 150 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	-/ < 0,7% v/v	-/ < 0,7% v/v	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	10% v/v	10% v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 69 kPa (< 76 kPa if ethanol is added)	< 69 kPa (< 76 kPa if ethanol is added)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)						
RVP 37.8°C (Transition)						
MTBE	0% v/v if ethanol is added	0% v/v if ethanol is added	-	-	-	-
Ethers 5 or more C Atoms	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

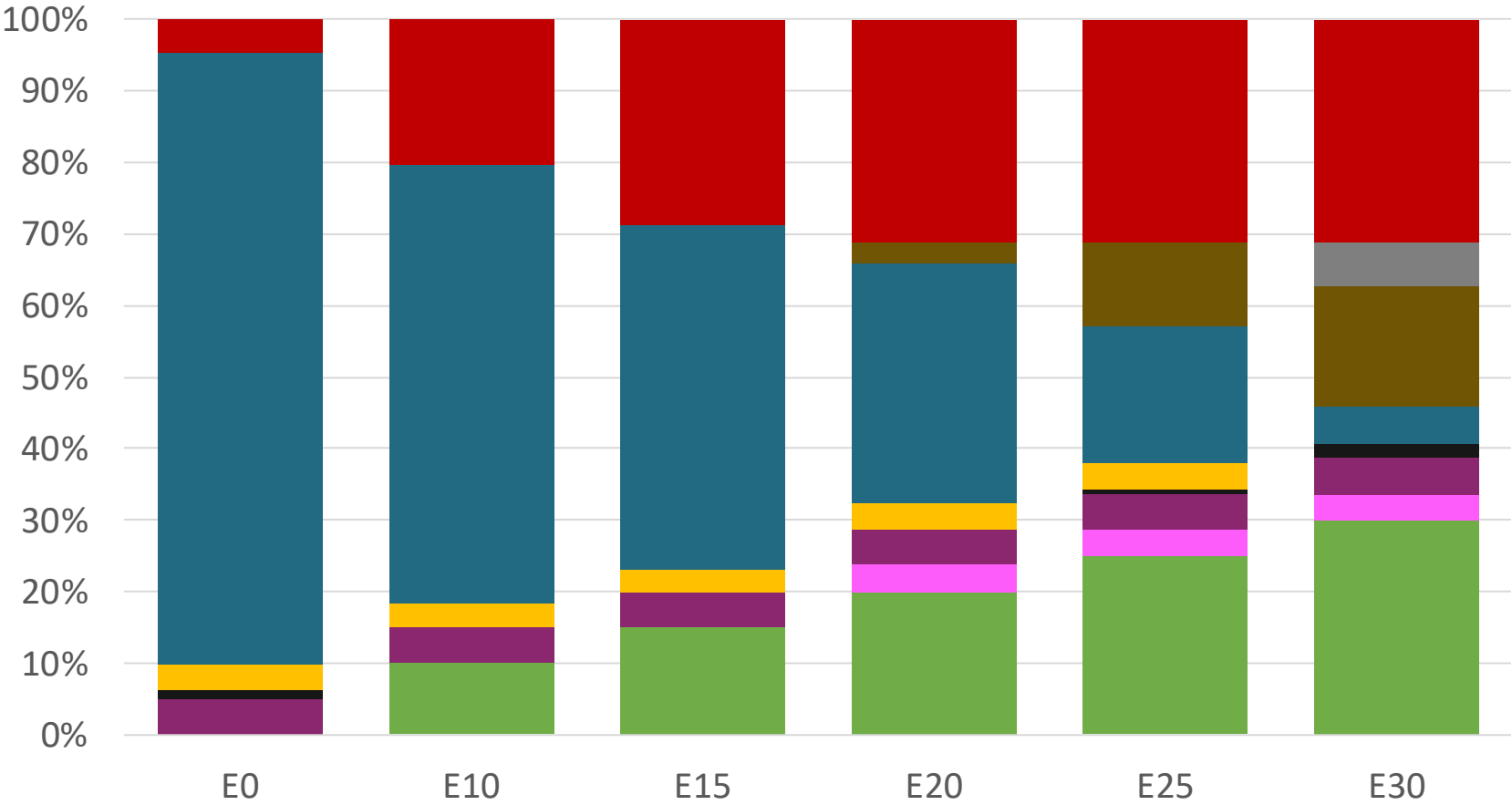
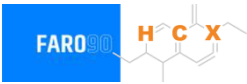
Source: RTCA

Blending Components - Panama



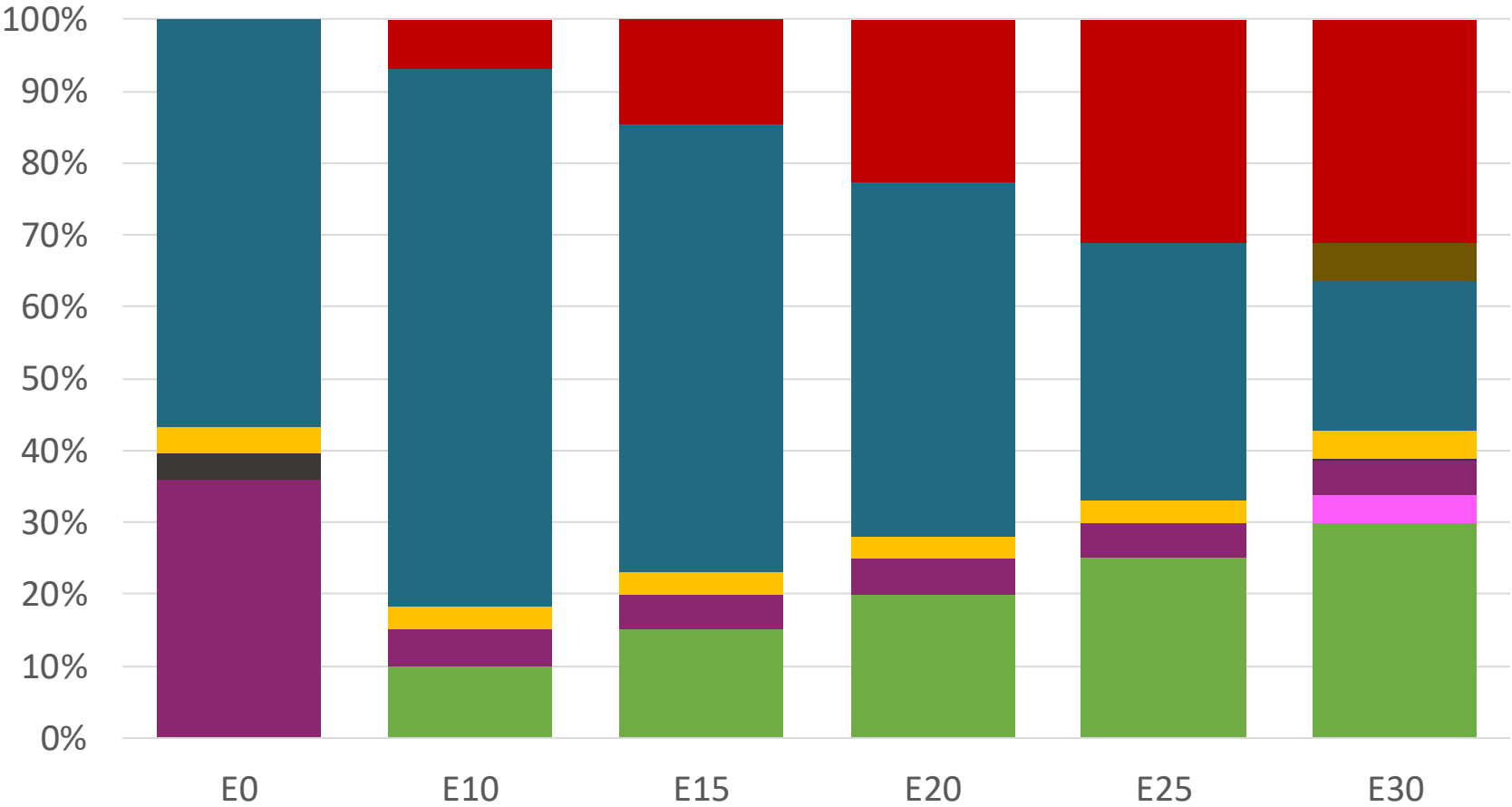
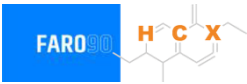
Source: Faro90

Panama – Regular – Constant Octane



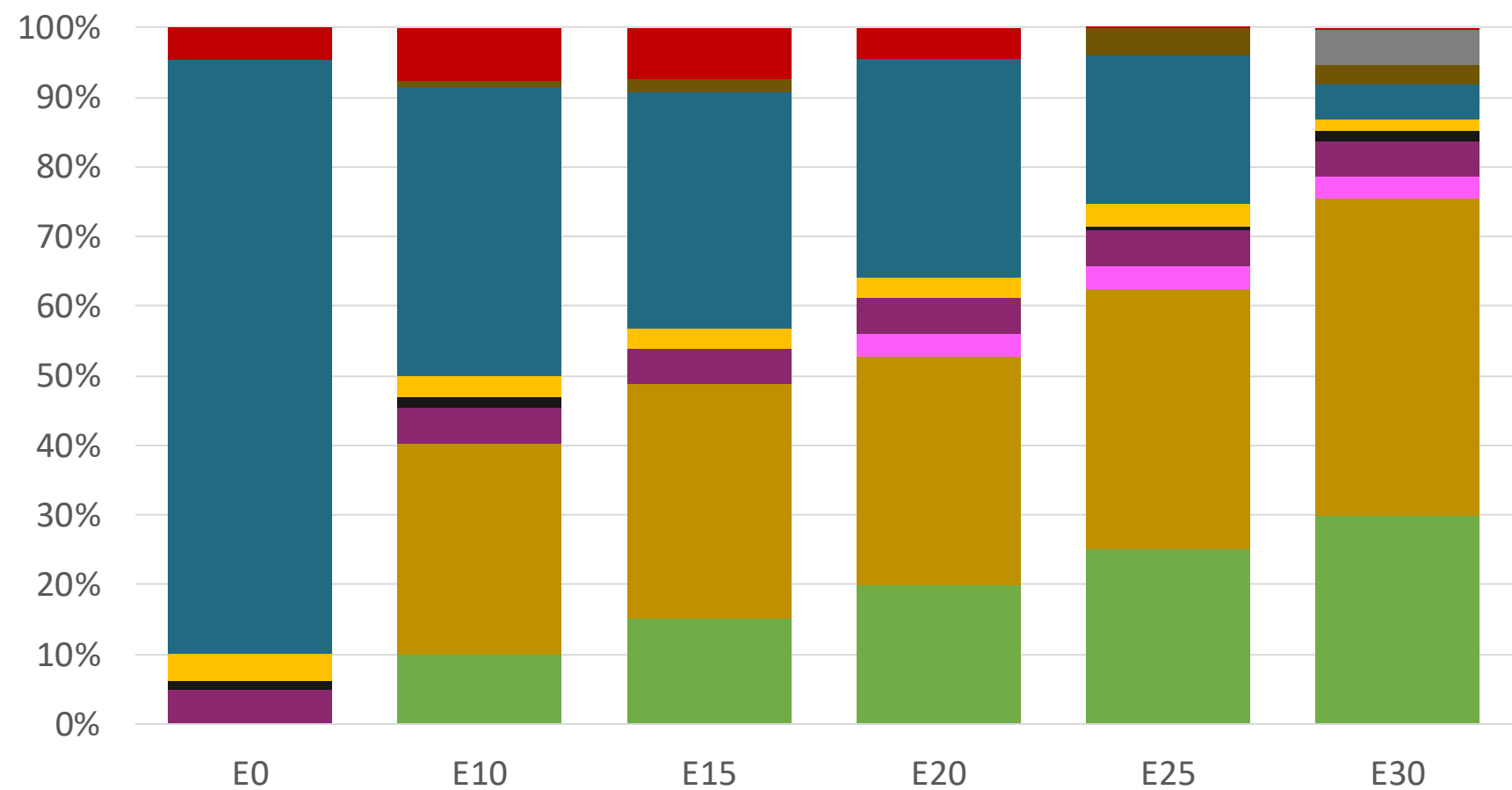
Octane (RON)	91.0	91.0	91.0	91.0	91.0	91.0
Price (USD/gal)	\$ 2.076	\$ 1.817	\$ 1.673	\$ 1.526	\$ 1.375	\$ 1.257

Panama – Premium – Constant Octane



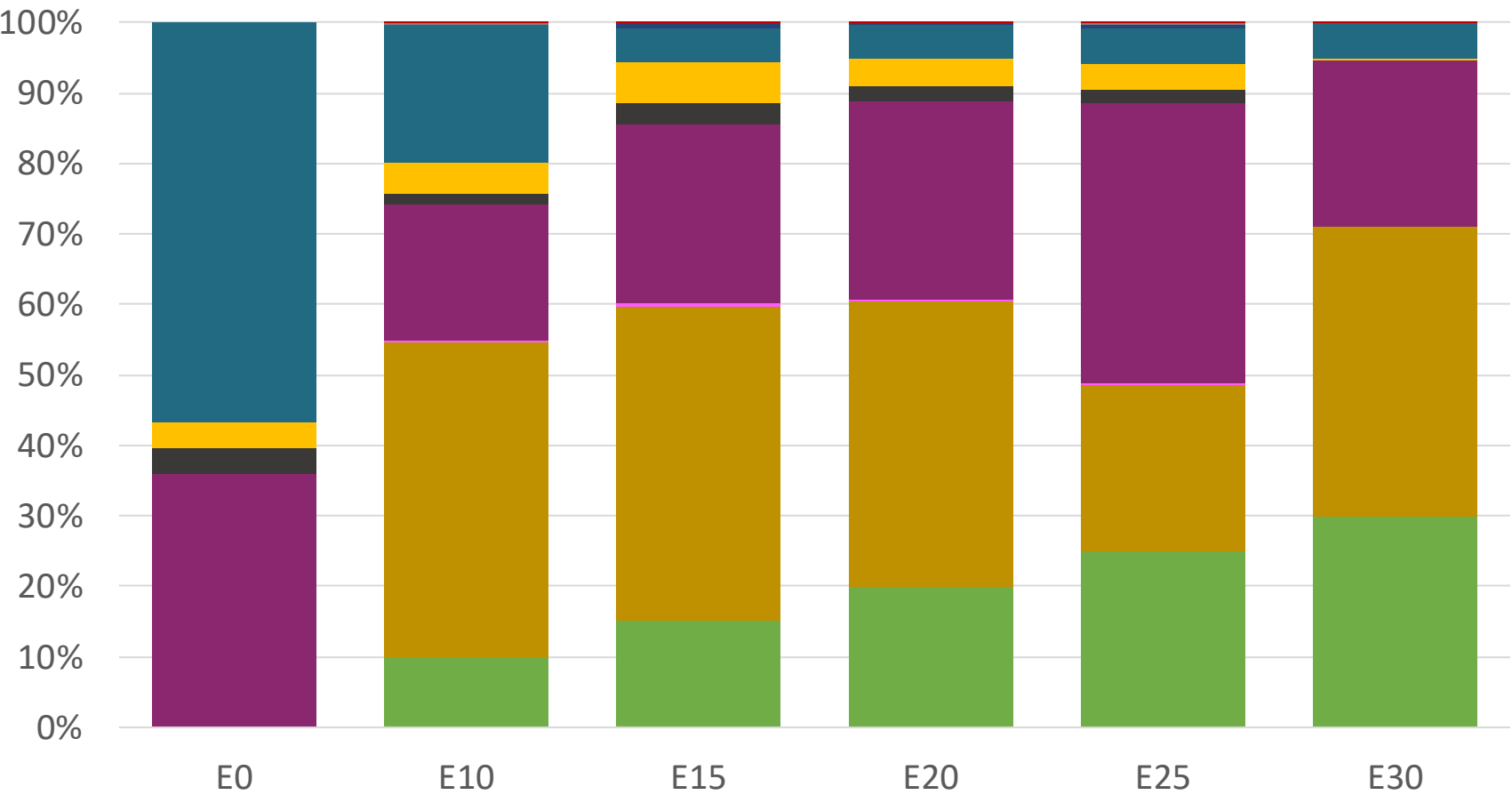
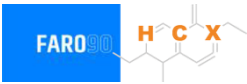
Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (USD/gal)	\$ 2.326	\$ 2.023	\$ 1.888	\$ 1.746	\$ 1.597	\$ 1.448

Panama – Regular – Increased Octane



Octane (RON)	91.0	94.6	96.6	99.1	101.2	102.4
Price (USD/gal)	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076	\$ 2.076

Panama – Premium – Increased Octane



Octane (RON)	95.0	98.0	100.3	102.6	104.9	105.8
Price (USD/gal)	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326	\$ 2.326

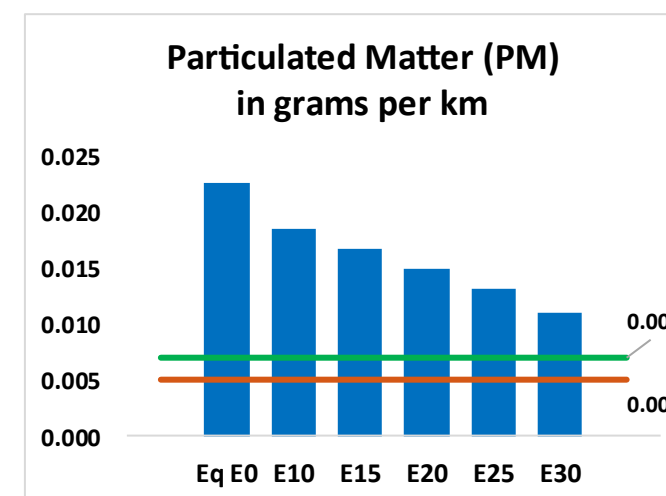
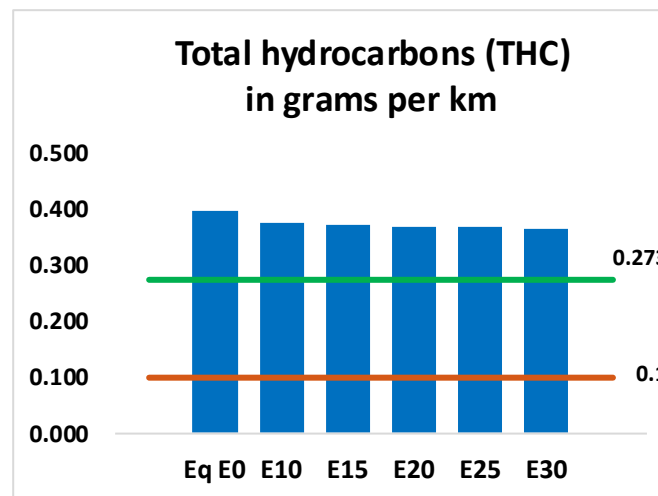
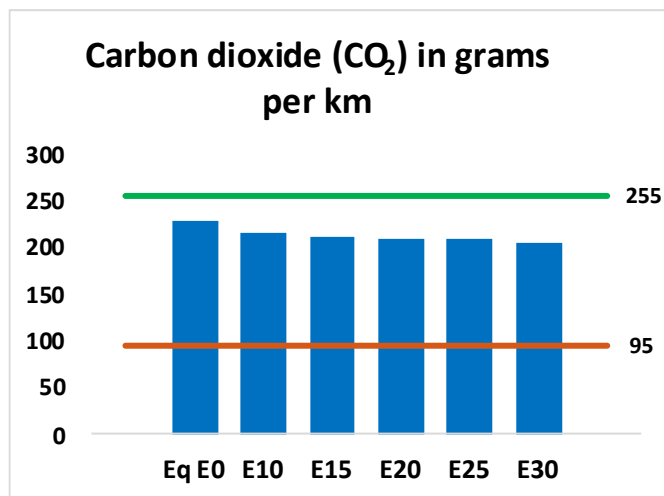
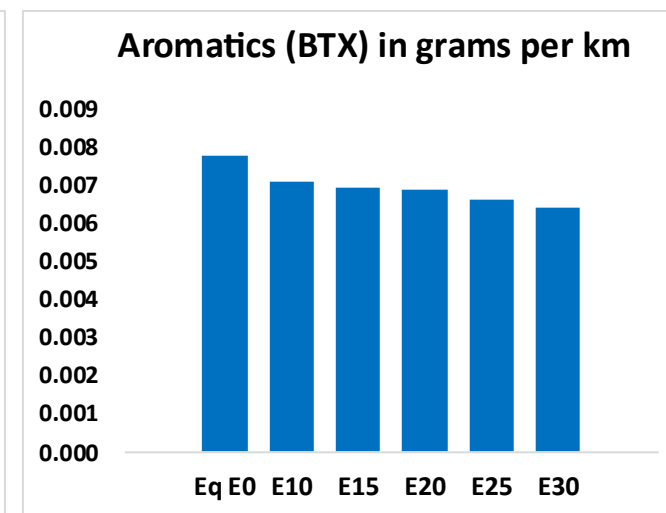
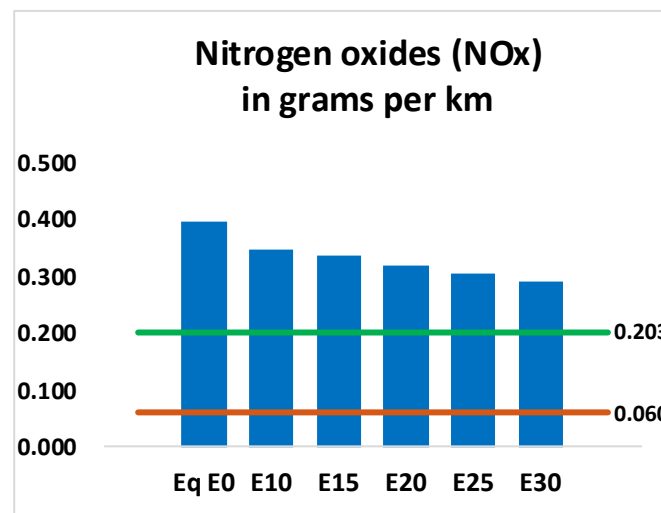
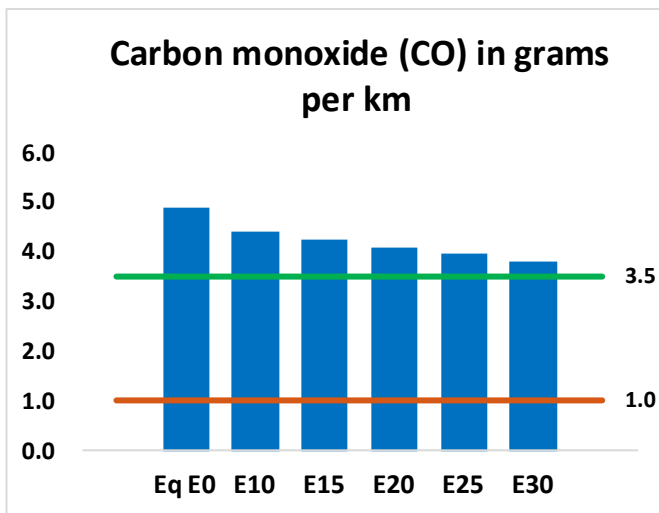
Gasoline Vehicular Fleet in Panama

Type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Small automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	764				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	5,313				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm		69,297			
		Euro III		> 8 years, > 160 tkm		36,491			
		Euro IV		> 8 years, > 160 tkm		53,388			
		Euro V / Euro 6		> 8 years, > 160 tkm		88,980			
				4-8 years, 80-160 tkm		60,428			
		< 4 years, < 80 tkm			49,672				
	Hybrid			Any	2,427				
Electric			Any	148					
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	4,777				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm		17,469			
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm		82,743			
		Euro III		> 8 years, > 160 tkm		32,962			
		Euro IV		> 8 years, > 160 tkm		53,136			
		Euro V / Euro 6		> 8 years, > 160 tkm		88,560			
				4-8 years, 80-160 tkm		97,068			
		< 4 years, < 80 tkm			123,287				
	Hybrid			Any	3,330				
Electric			Any	202					
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	8,716				
		Euro II		> 8 years, > 160 tkm	3,140				
		Euro III		> 8 years, > 160 tkm		19,498			
		Euro IV		> 8 years, > 160 tkm	3,828				
		Euro V		4-8 years, 80-160 tkm	6,380				
				4-8 years, 80-160 tkm	3,562				
				< 4 years, < 80 tkm	7,123				
	Electric			Any					
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	169				
	FI	3-Ways		> 8 years, > 160 tkm		23,019			
		Euro III		> 8 years, > 160 tkm		11,110			
		Euro IV		> 8 years, > 160 tkm		17,322			
		Euro V / Euro 6		> 8 years, > 160 tkm		28,870			
				4-8 years, 80-160 tkm		16,958			
		< 4 years, < 80 tkm			10,820				
	Hybrid			Any					
Electric			Any						

Vehicle Fleet: **916,582**
Average Age: **11.5 years**
Motorcycles: **5%**
Source: INEC, analysis Faro 90

Vehicle Emissions in Panama

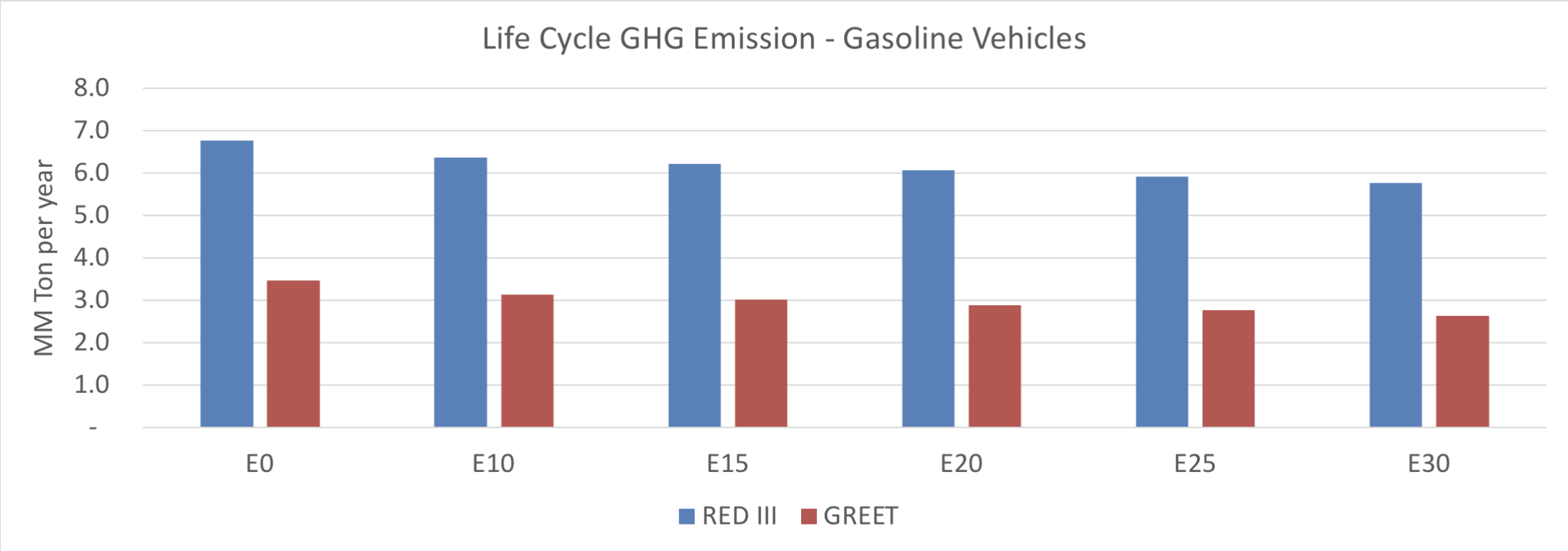
United States
Euro 6 Standard



Panama – Vehicle Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	101,418	96,500	91,582	88,113	84,881	82,474	79,107	-10%	-16%
CO2	4,716,772	4,595,483	4,480,933	4,390,843	4,346,262	4,324,891	4,245,171	-5%	-8%
VOC	8,233	8,023	7,814	7,725	7,673	7,645	7,567	-5%	-7%
NOx	8,260	7,627	7,195	6,976	6,622	6,345	6,019	-13%	-20%
SOx	56	52	48	44	40	37	33	-15%	-28%
NH3	1,545	1,530	1,514	1,521	1,539	1,551	1,561	-2%	0%
N2O	99	98	98	98	100	101	103	-1%	2%
CH4	1,793	1,793	1,793	1,820	1,859	1,893	1,926	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	99	97	96	95	95	95	95	-4%	-4%
Acetaldehyde	179	205	301	459	625	714	844	68%	249%
Formaldehyde	533	518	604	710	743	822	895	13%	39%
Benzene	161	155	146	144	143	137	132	-9%	-11%
PM2.5	262	238	214	193	173	151	128	-18%	-34%
PM10	209	190	171	154	138	120	102	-18%	-34%

Panama – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	23.8
2035 Target (MMT/year)	16.7
Delta	7.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.6%	13.3%	16.9%	20.1%	24.2%
RED II/III	0.0%	5.9%	8.2%	10.5%	12.5%	14.9%
% 2035 Target	E0	E10	E15	E20	E25	E30
2025	0.0%	4.7%	6.4%	8.2%	9.8%	11.8%
RED II/III	0.0%	5.6%	7.8%	9.9%	11.8%	14.1%

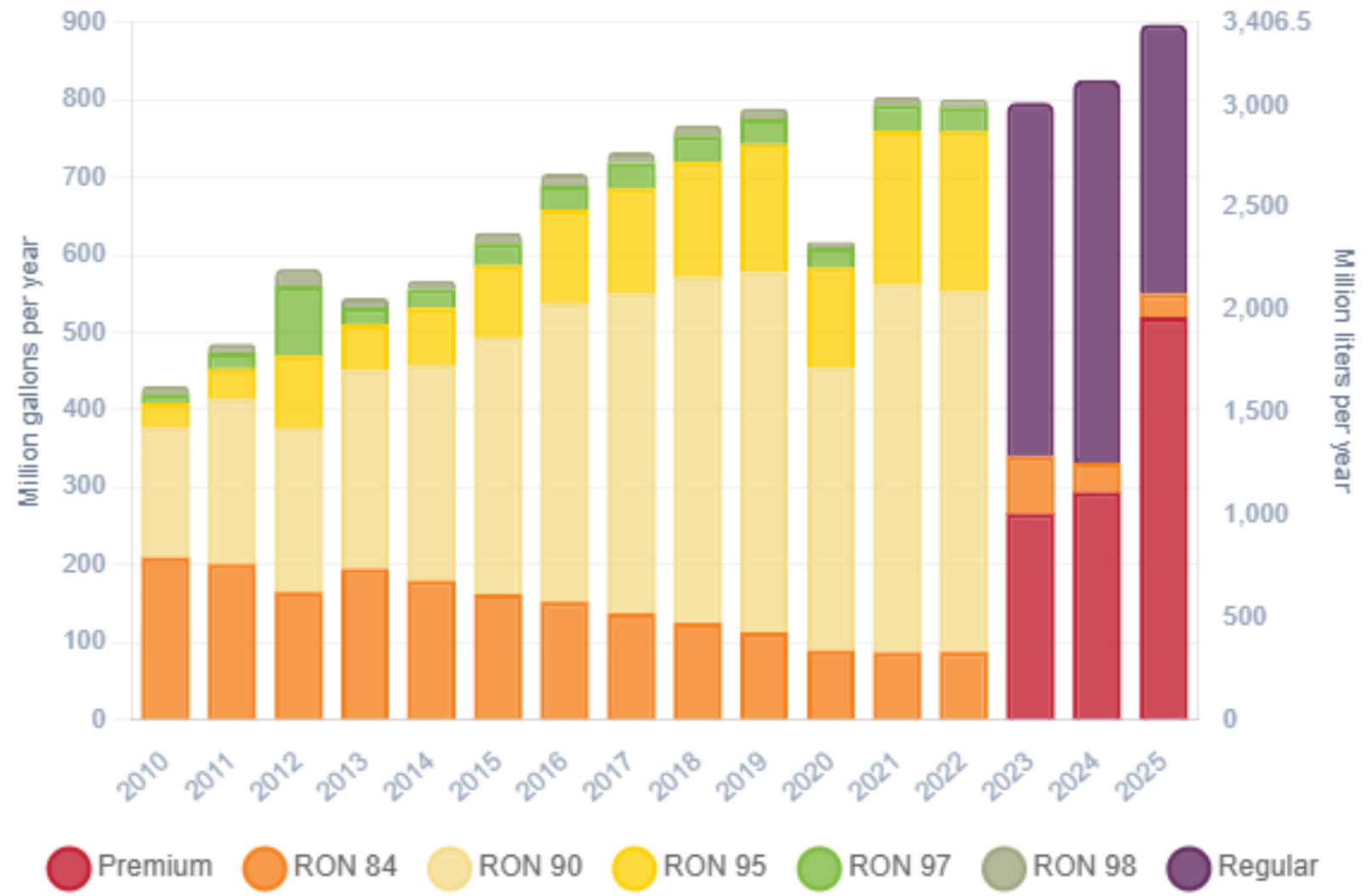
Ethanol Blending in Gasoline - Peru



In 2025, Peru gasoline consumption is 897 million gallons (3,397 millions liters). Starting in 2023 a program to increase octane in gasoline was implemented; RON 84 and RON 90 gasoline are commercialized as Regular gasoline (RON 91) with a 42% market participation. RON 95, RON 97 and RON 98 gasolines are now marketed as Premium gasoline (RON 96) with a 58% market share. Gasoline with ethanol (gasohol) is required to meet one additional octane number (RON 91 and RON 96 respectively). Local gasoline supply is done with production from Petroperú, Repsol and Primax. Gasoline imports account for 10 to 15% of demand, and are supplied by local markets, United states and Europe.

Blends with 7.8% v/v are allowed (Decreto Supremo N.° 021-2007-EM: MINEM). Peru produces ethanol from sugarcane; however, its destination is mainly the EU market. Ethanol imports from the United States and to a lesser degree from Brazil are used for the national blend.

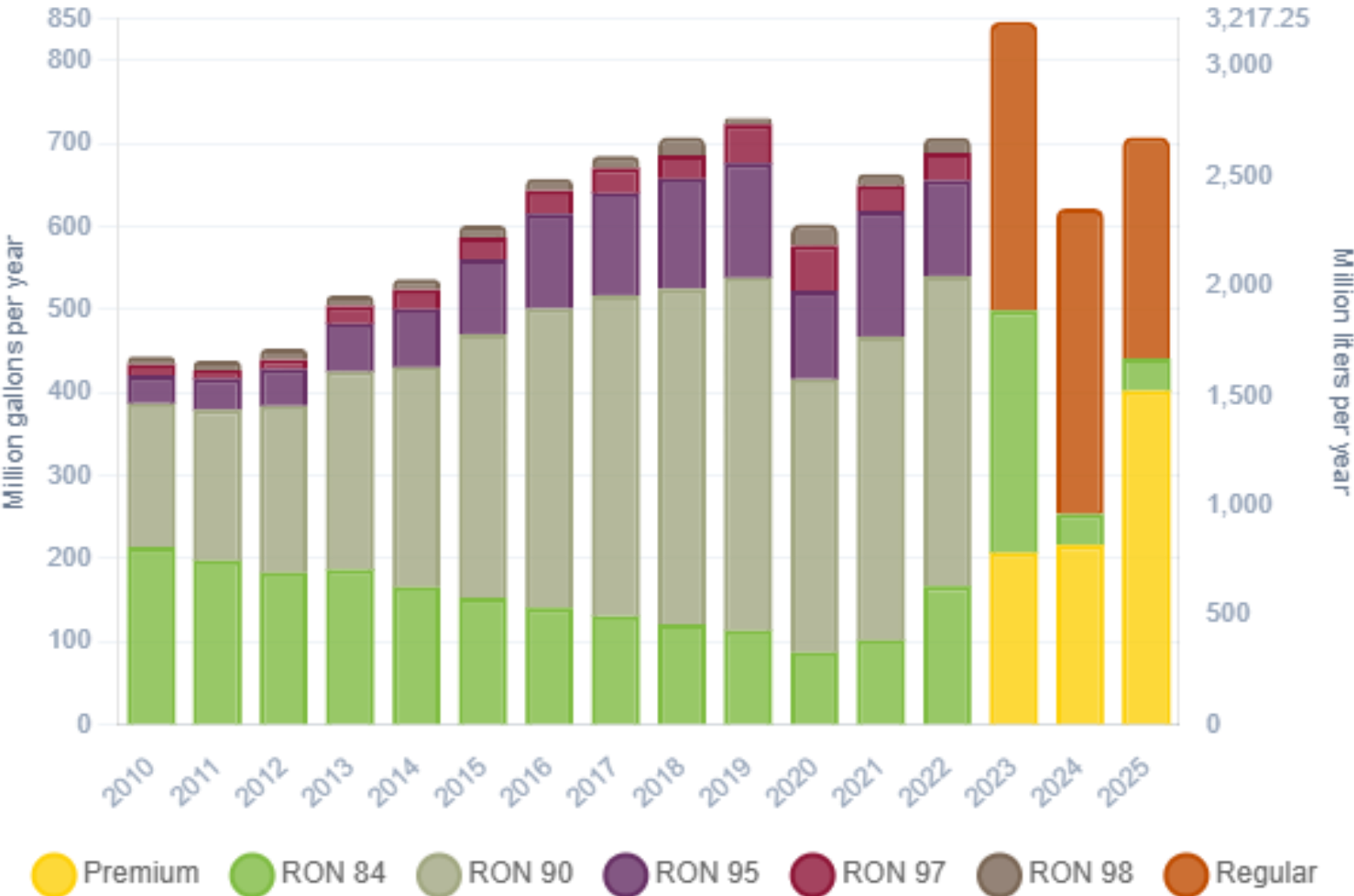
Gasoline Demand in Peru



Since 2023, RON 90 is commercialized as Regular Gasoline (RON 90-91) and RON 95, RON 97 and RON 98 are now marketed as Premium Gasoline (RON 95-96).

Source: MINEM, 2024

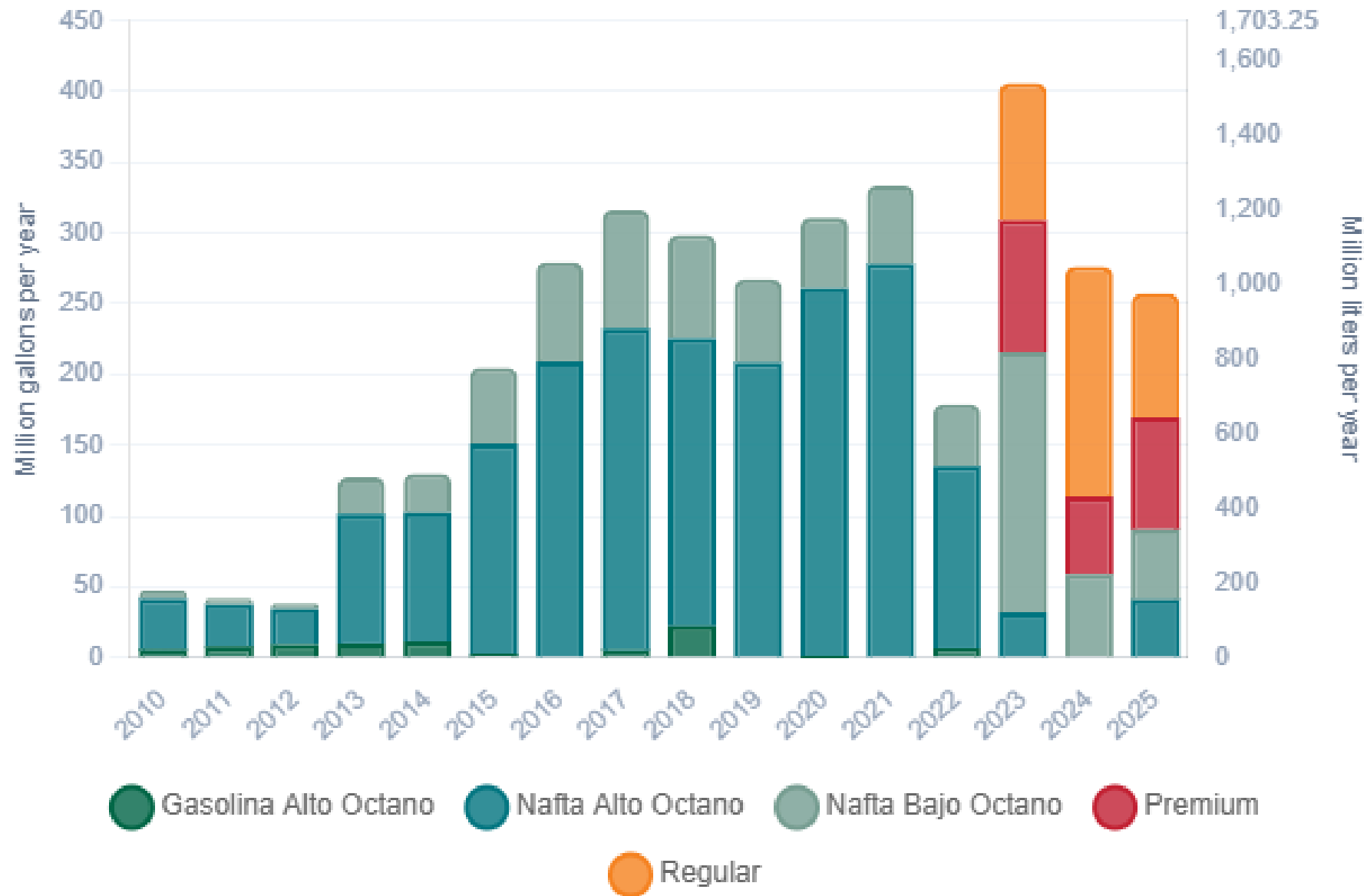
Gasoline Production in Peru



Since 2023, RON 90 is commercialized as Regular Gasoline (RON 90-91) and RON 95, RON 97 and RON 98 are now marketed as Premium Gasoline (RON 95-96).

Source: MINEM, 2024

Gasoline Imports in Peru



Source: MINEM, 2024

Ethanol Balance in Peru



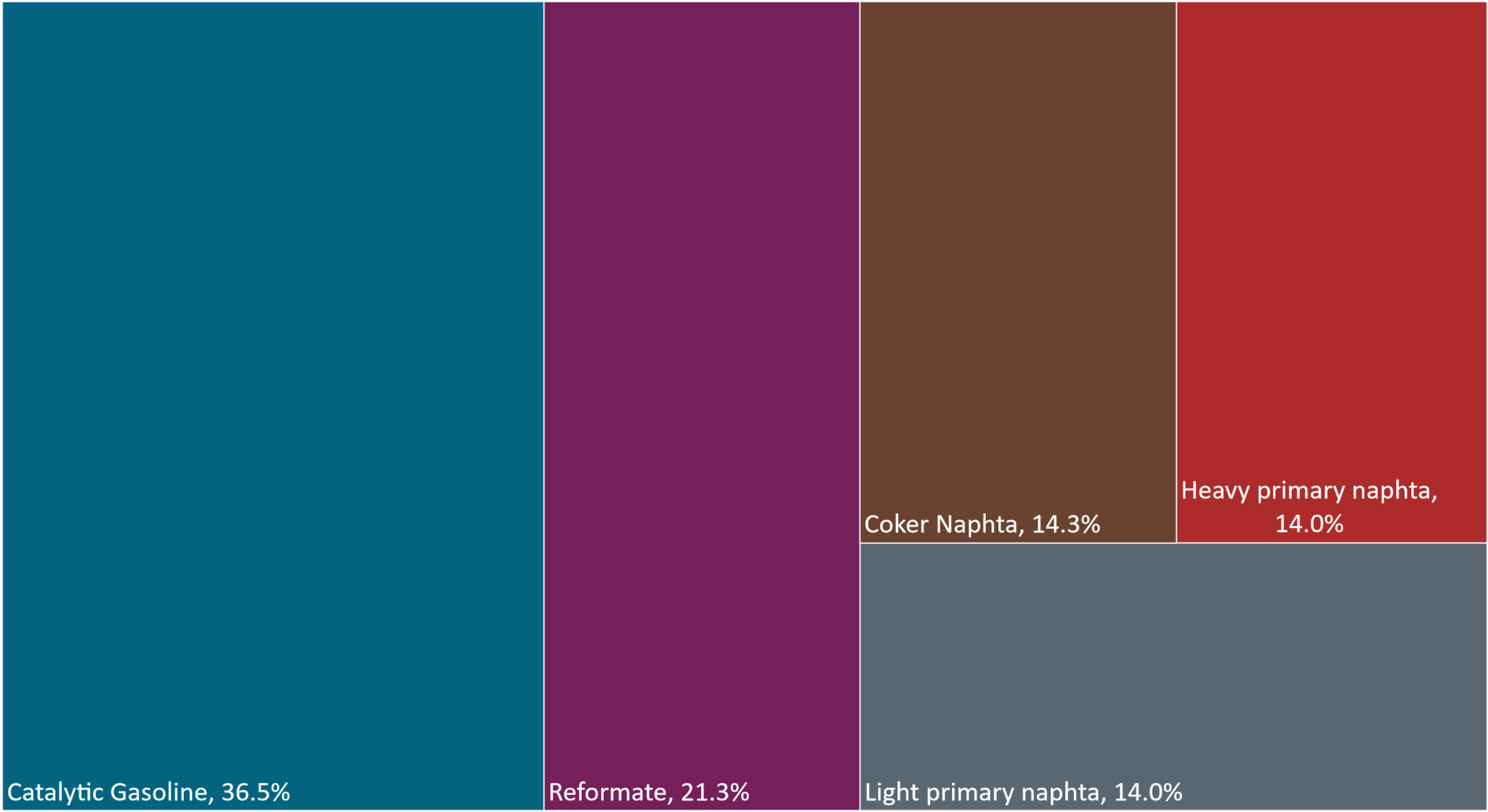
Source: MINEM, 2024

Gasoline Quality in Peru

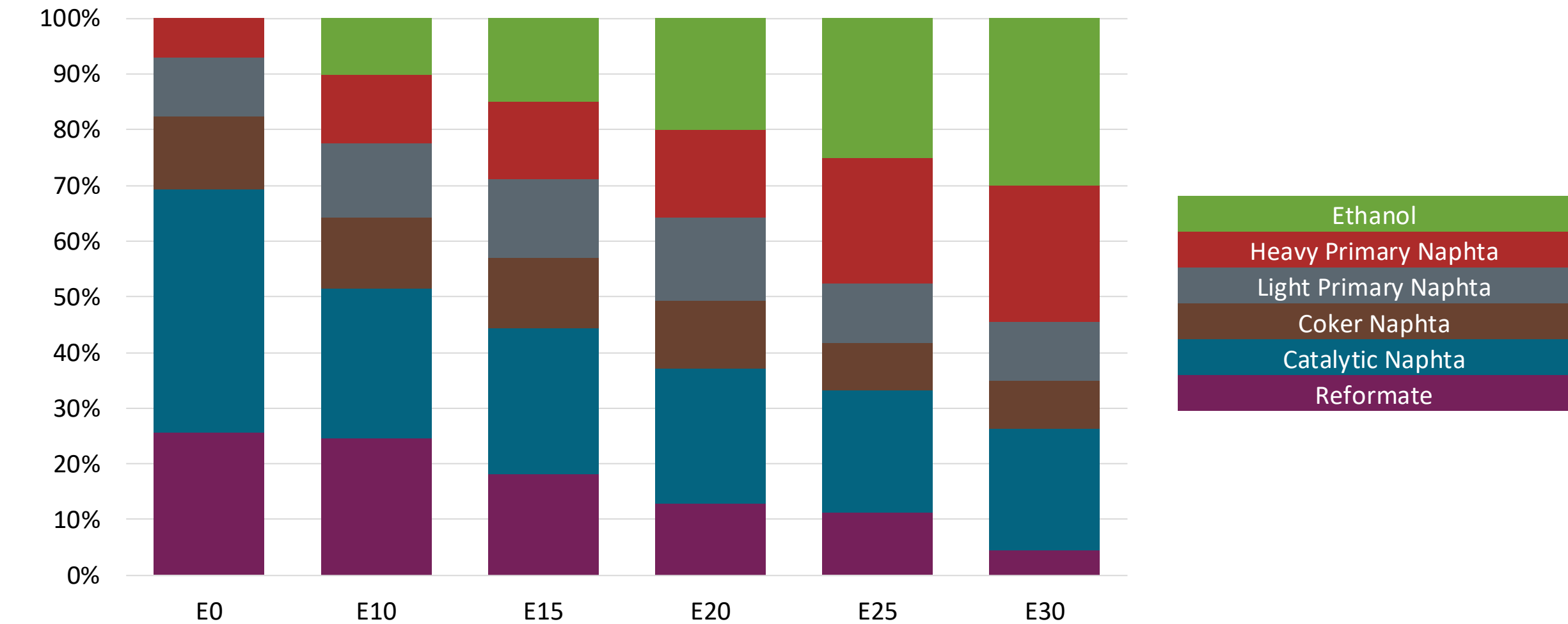
Name	Ministerial Resolution N° 469-2021-MINEM/DM				EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2021				2017			
Applicability	Whole contry	Whole contry	Whole contry	Whole contry	All countries			
Selected Grade	Gasolina Regular	Gasolina Premium	Gasohol Regular	Gasohol Premium	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 2% v/v	< 2% v/v	< 2% v/v	< 2% v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 45% v/v	< 45% v/v	< 45% v/v	< 45% v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 25% v/v	< 25% v/v	< 25% v/v	< 25% v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 0,25 mg/l	< 0,25 mg/l	< 0,25 mg/l	< 0,25 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 90	> 95	> 91	> 96	> 95	> 95	> 98	> 98
MON	-	-	-	-	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 50 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	Reportar	Reportar	< 3,45 %m/m	< 3,45 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)			< 7,8 %v/v	< 7,8 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	< 76 kPa	< 76 kPa	< 76 kPa	< 76 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)								
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: MINEM, 2023

Available Blending Components



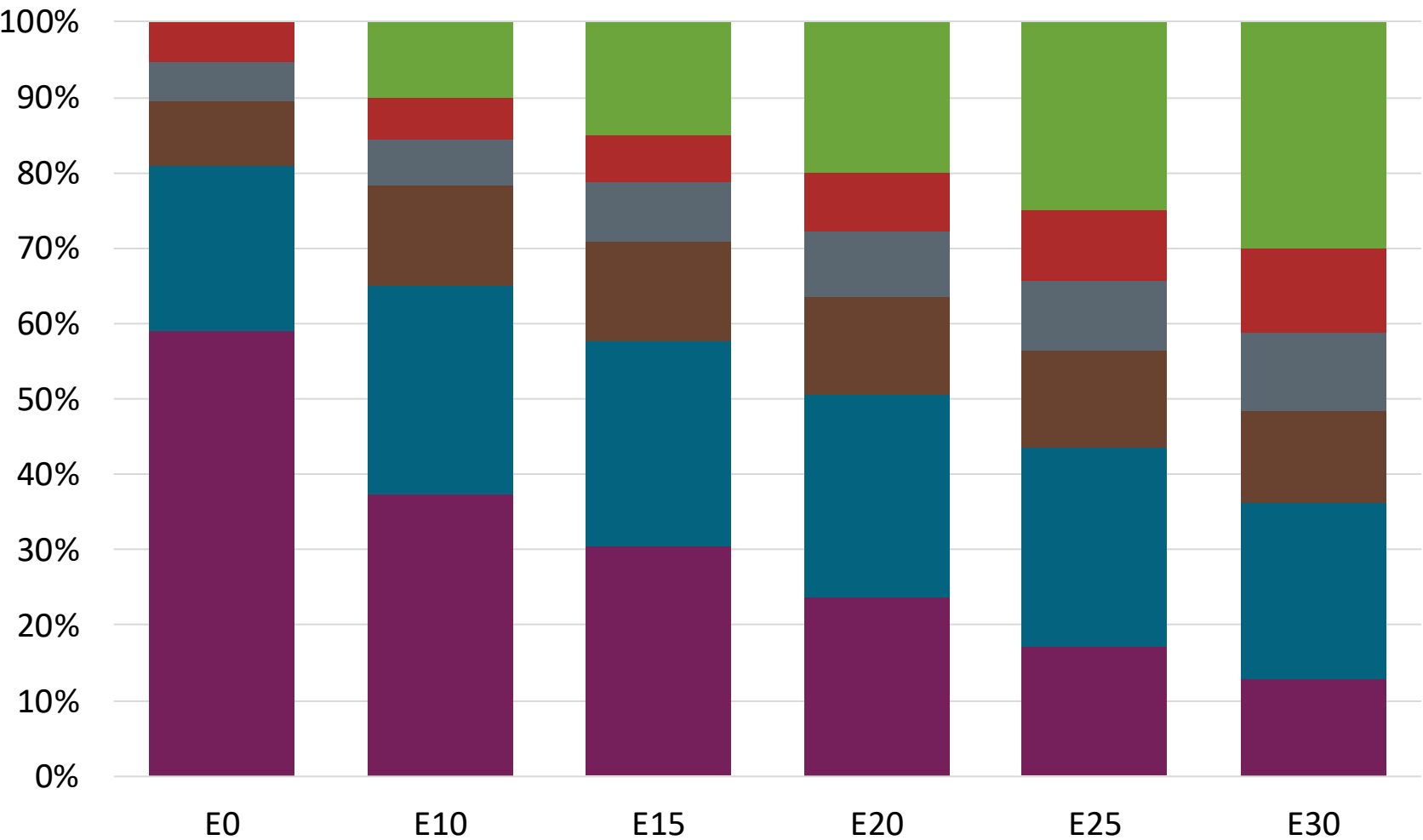
Peru – Regular – Constant Octane



Octane (RON)	90.0		90.0		90.0	
Price (US\$/gal)	\$ 2.201	\$ 2.031	\$ 1.945	\$ 1.860	\$ 1.784	\$ 1.697

Source: Faro90

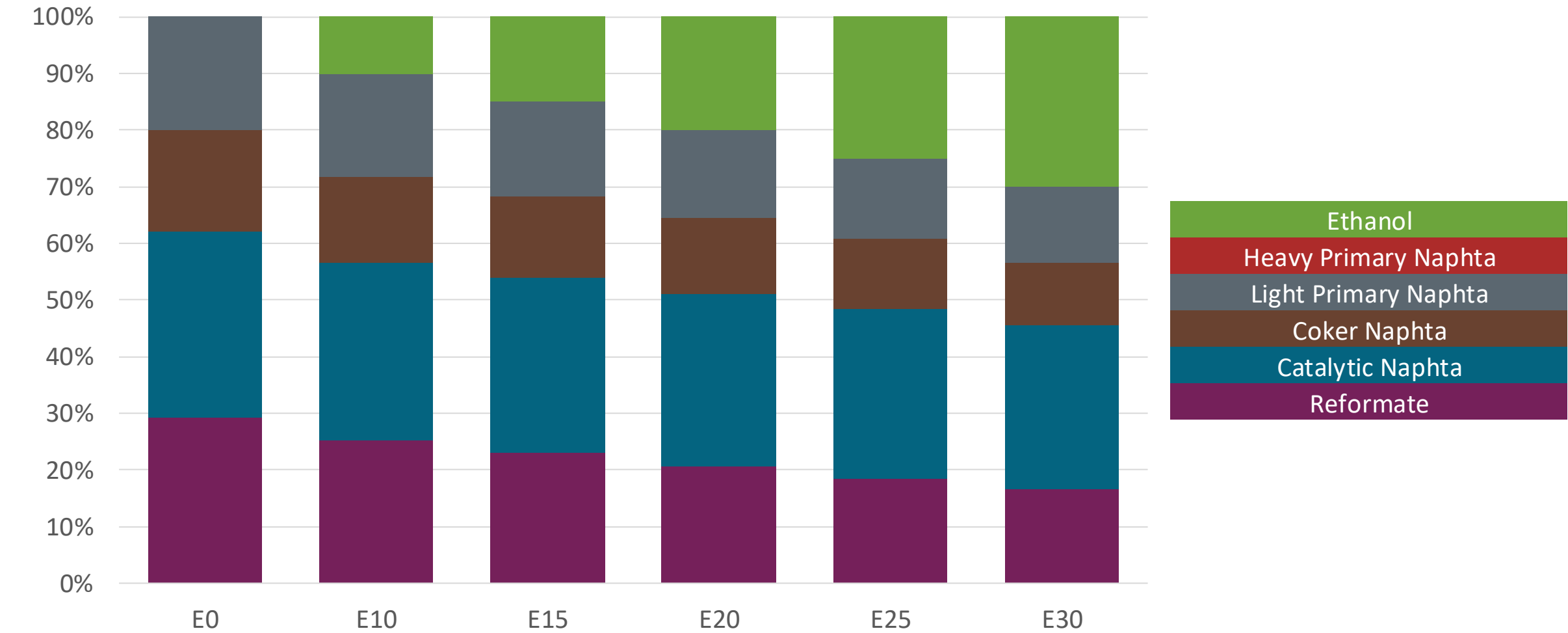
Peru – Premium – Constant Octane



Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.0
Price (US\$/gal)	\$ 2.461	\$ 2.272	\$ 2.185	\$ 2.098	\$ 2.012	\$ 1.927

Source: Faro90

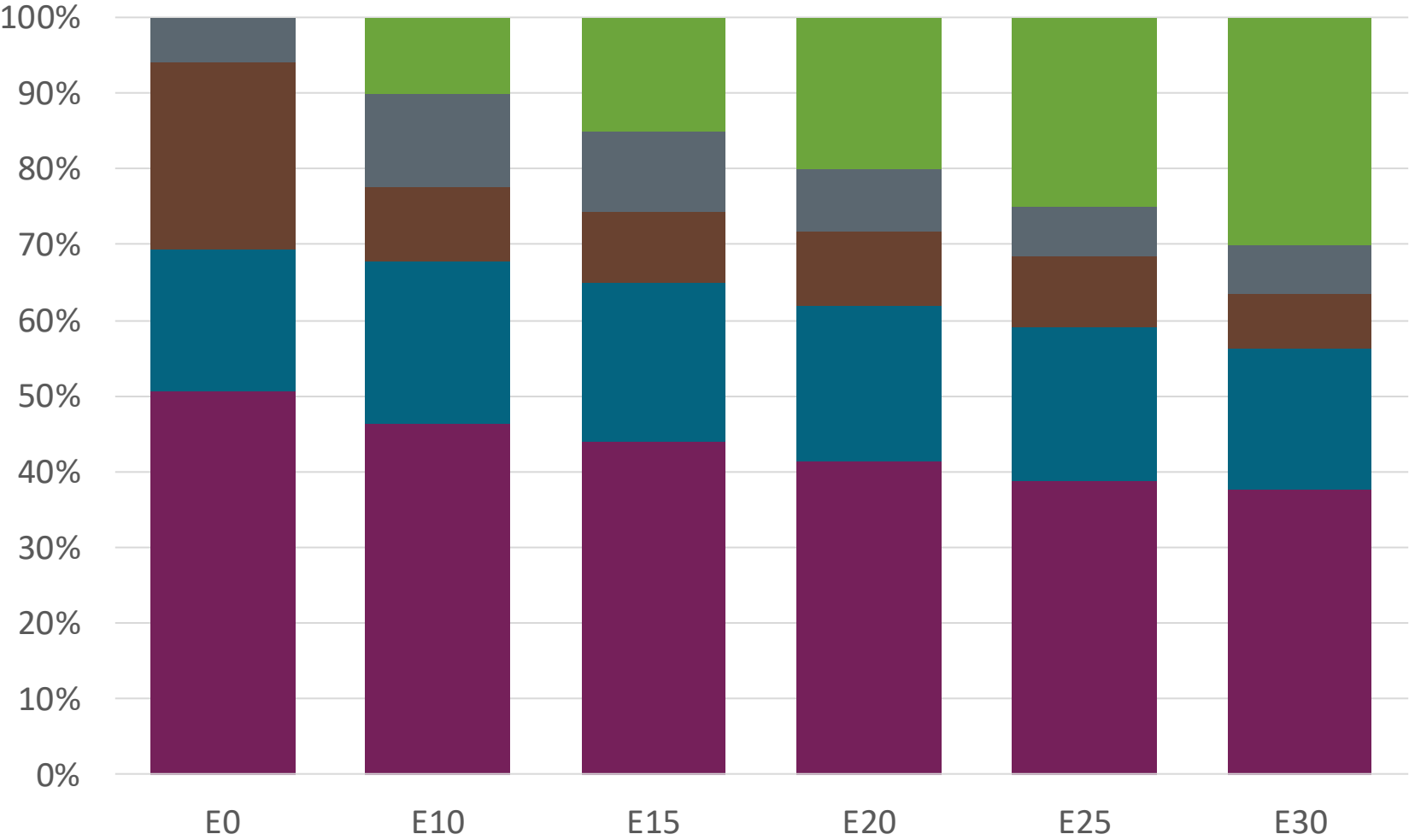
Peru – Regular – Increased Octane



Octane (RON)	E0	E10	E15	E20	E25	E30
	90.0	93.5	95.2	96.9	98.4	105.4
Price (US\$/gal)	\$ 2.199	\$ 2.199	\$ 2.199	\$ 2.199	\$ 2.199	\$ 2.199

Source: Faro90

Peru – Premium – Increased Octane



Octane (RON)	95.0	96.4	97.8	99.4	100.8	102.1
Price (US\$/gal)	\$ 2.412	\$ 2.412	\$ 2.412	\$ 2.412	\$ 2.412	\$ 2.412

Source: Faro90

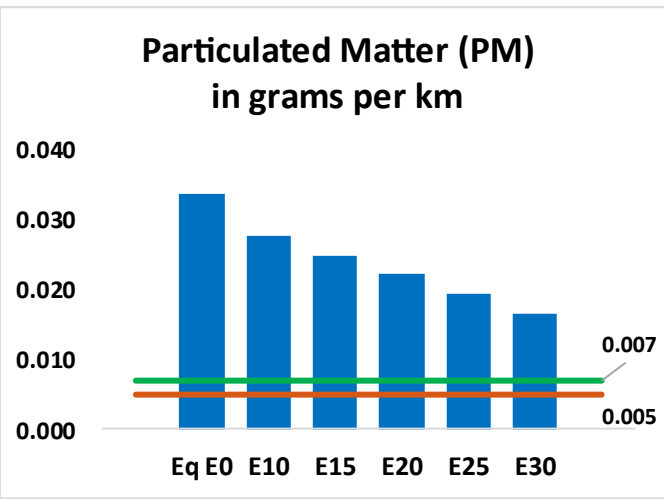
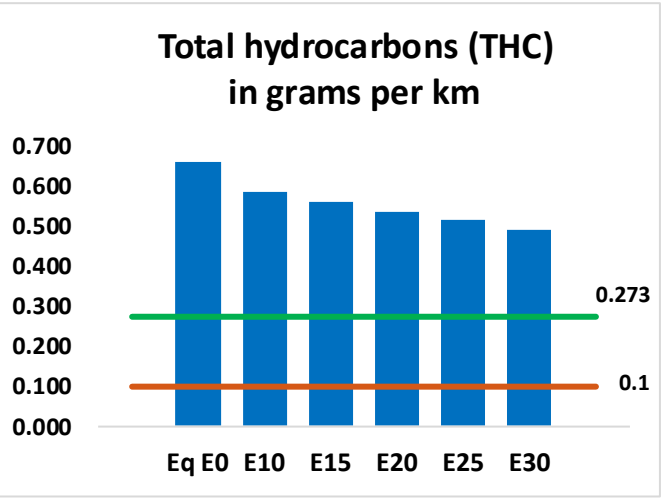
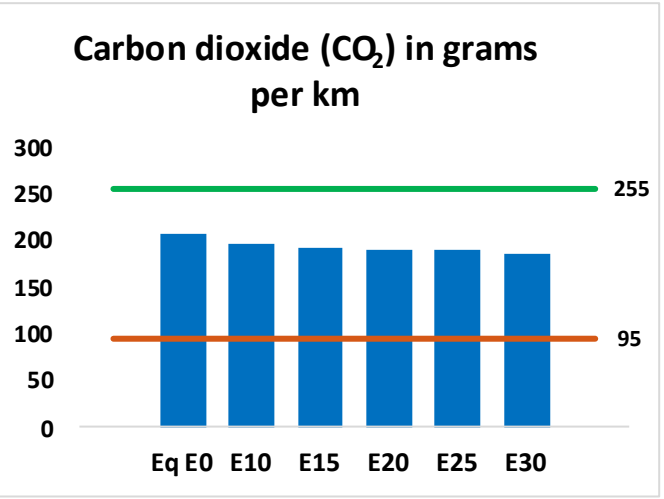
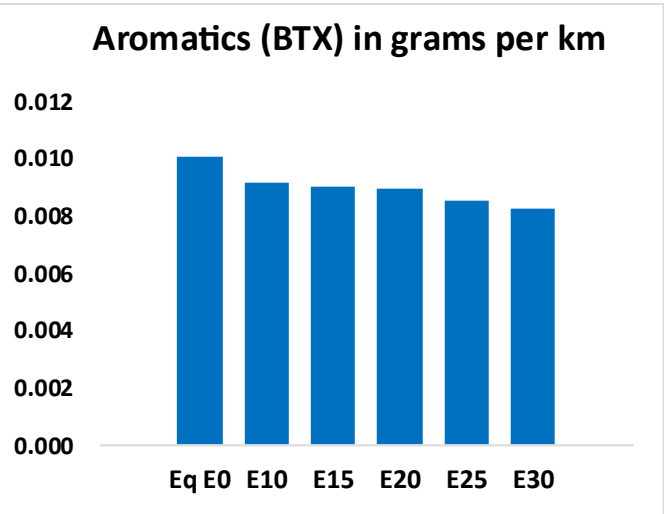
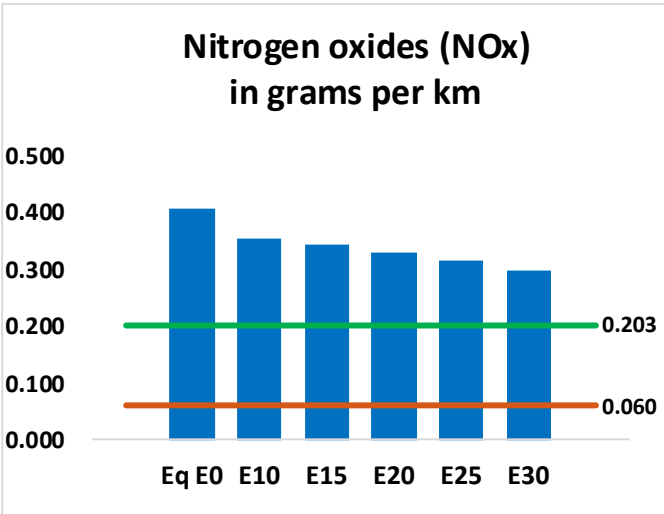
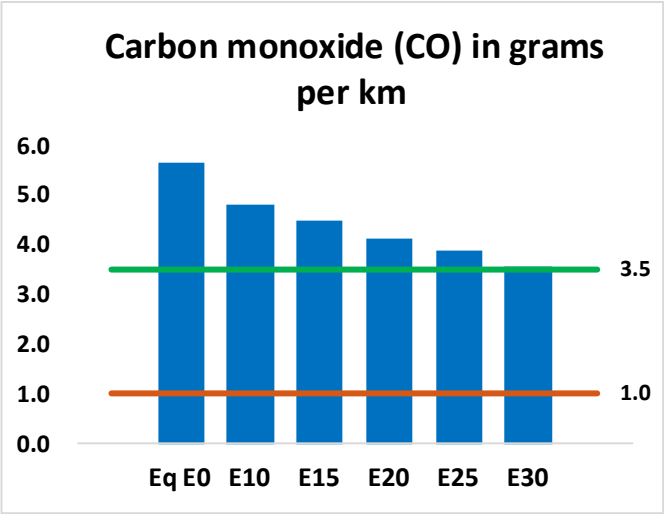
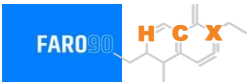
Peru – Gasoline Vehicle Fleet

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	20,555				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	117,359				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	358,008				
		Euro III		> 8 years, > 160 tkm	226,059				
		Euro IV		> 8 years, > 160 tkm	164,186				
		Euro V / Euro 6		> 8 years, > 160 tkm	215,648				
				4-8 years, 80-160 tkm	312,744				
	Hybrid			< 4 years, < 80 tkm	500,733				
	Electric			Any	208				
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	13,073				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	74,642				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	227,700				
		Euro III		> 8 years, > 160 tkm	143,778				
		Euro IV		> 8 years, > 160 tkm	104,425				
		Euro V / Euro 6		> 8 years, > 160 tkm	137,156				
				4-8 years, 80-160 tkm	198,911				
	Hybrid			< 4 years, < 80 tkm	318,475				
	Electric			Any	133				
Pequeños motores	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	89,447				
		Euro II		> 8 years, > 160 tkm	23,575				
		Euro III		> 8 years, > 160 tkm	93,546				
		Euro IV		> 8 years, > 160 tkm	25,582				
				4-8 years, 80-160 tkm	29,796				
		Euro V		4-8 years, 80-160 tkm	36,099				
	Electric			< 4 years, < 80 tkm	105,503				
	Camiones y pesados	Carburator		3-Ways	PCV	> 8 years, > 160 tkm	5,495		
FI		3-Ways	> 8 years, > 160 tkm	127,082					
		Euro III	> 8 years, > 160 tkm	4,331					
		Euro IV	> 8 years, > 160 tkm	24,727					
		Euro V / Euro 6	> 8 years, > 160 tkm	75,432					
			4-8 years, 80-160 tkm	47,630					
Hybrid			< 4 years, < 80 tkm	34,594					
Electric			Any	10					
			Any	4,061					

Gasoline vehicles: **3,537,001**
Average Age: **13.2 years**
Motorcycles: 21%

Peru – Emissions

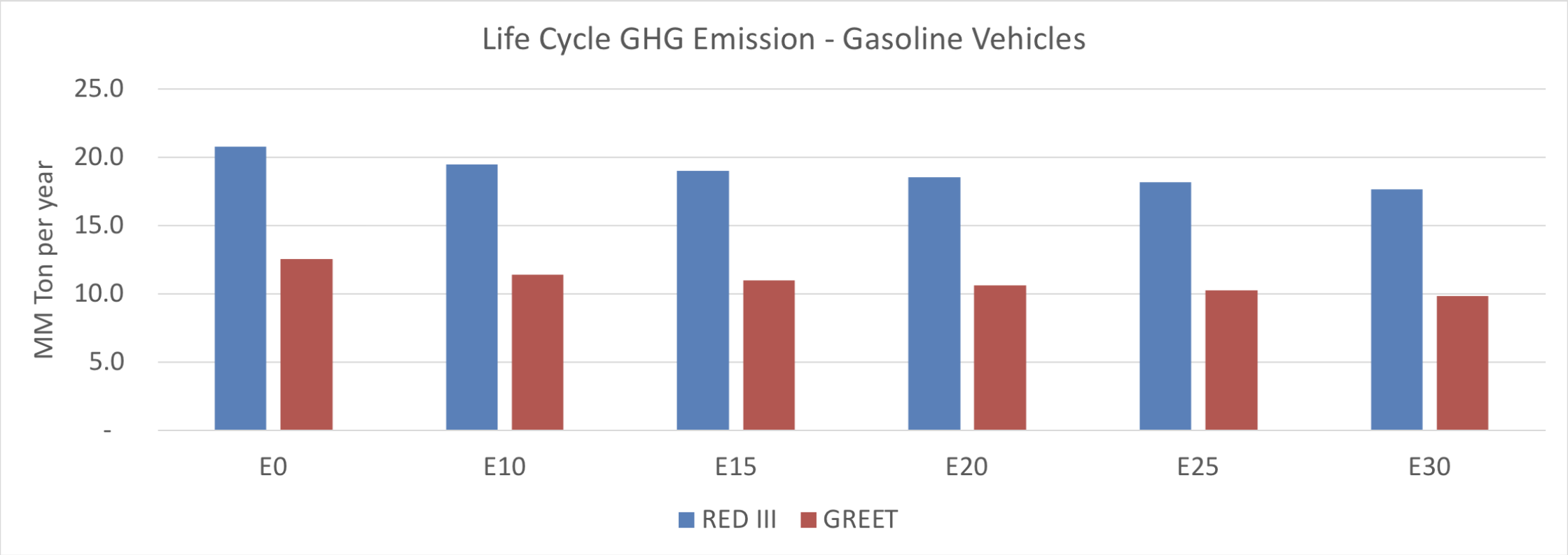
United States
Euro 6



Peru – Gasoline Vehicle Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	490,598	454,656	418,714	389,607	360,476	338,123	310,616	-15%	-27%
CO2	18,012,219	17,549,048	17,111,608	16,767,575	16,597,334	16,507,451	16,203,171	-5%	-8%
VOC	57,394	54,284	51,174	48,880	46,688	45,038	42,711	-11%	-19%
NOx	35,553	32,827	31,019	30,131	28,663	27,529	26,184	-13%	-19%
SOx	206	191	175	162	149	136	121	-15%	-28%
NH3	6,338	6,275	6,212	6,242	6,312	6,362	6,403	-2%	0%
N2O	315	314	312	314	320	324	327	-1%	2%
CH4	12,438	12,438	12,438	12,625	12,898	13,135	13,358	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	482	462	442	428	416	407	394	-8%	-14%
Acetaldehyde	1,133	1,300	1,908	2,905	3,958	4,523	5,344	68%	249%
Formaldehyde	3,944	3,834	4,469	5,248	5,498	6,082	6,621	13%	39%
Benzene	877	843	798	786	779	745	721	-9%	-11%
PM2.5	1,044	949	854	771	689	601	510	-18%	-34%
PM10	1,894	1,722	1,550	1,398	1,250	1,091	925	-18%	-34%

Peru – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2025 (MMT/year)	111.2
2035 Target (MMT/year)	96.1
Delta	15.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	9.1%	12.2%	15.4%	18.1%	21.6%
RED II/III	0.0%	6.3%	8.5%	10.7%	12.6%	15.0%
% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2025	0.0%	7.5%	10.1%	12.7%	14.9%	17.9%
RED II/III	0.0%	8.6%	11.7%	14.7%	17.3%	20.6%

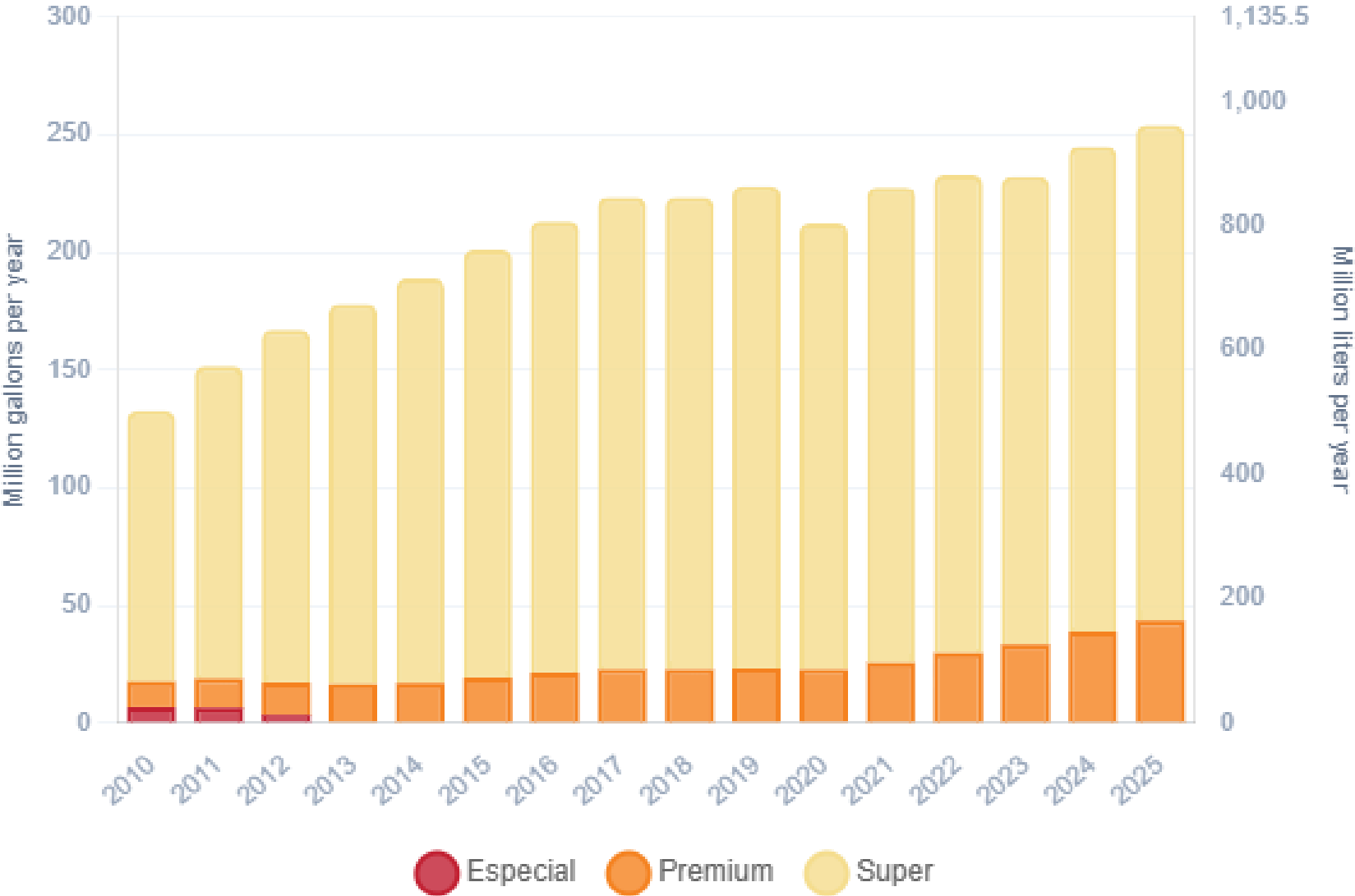
Ethanol Blending in Gasoline - Uruguay



In 2025, gasoline demand was 254 million gallons (960 million liters). Super Gasoline with RON 95 represented 83% of total demand and Premium Gasoline with RON 97 the rest. Gasoline production was 210 millions gallons (almost 800 millions liters) and covers 83% of local demand.

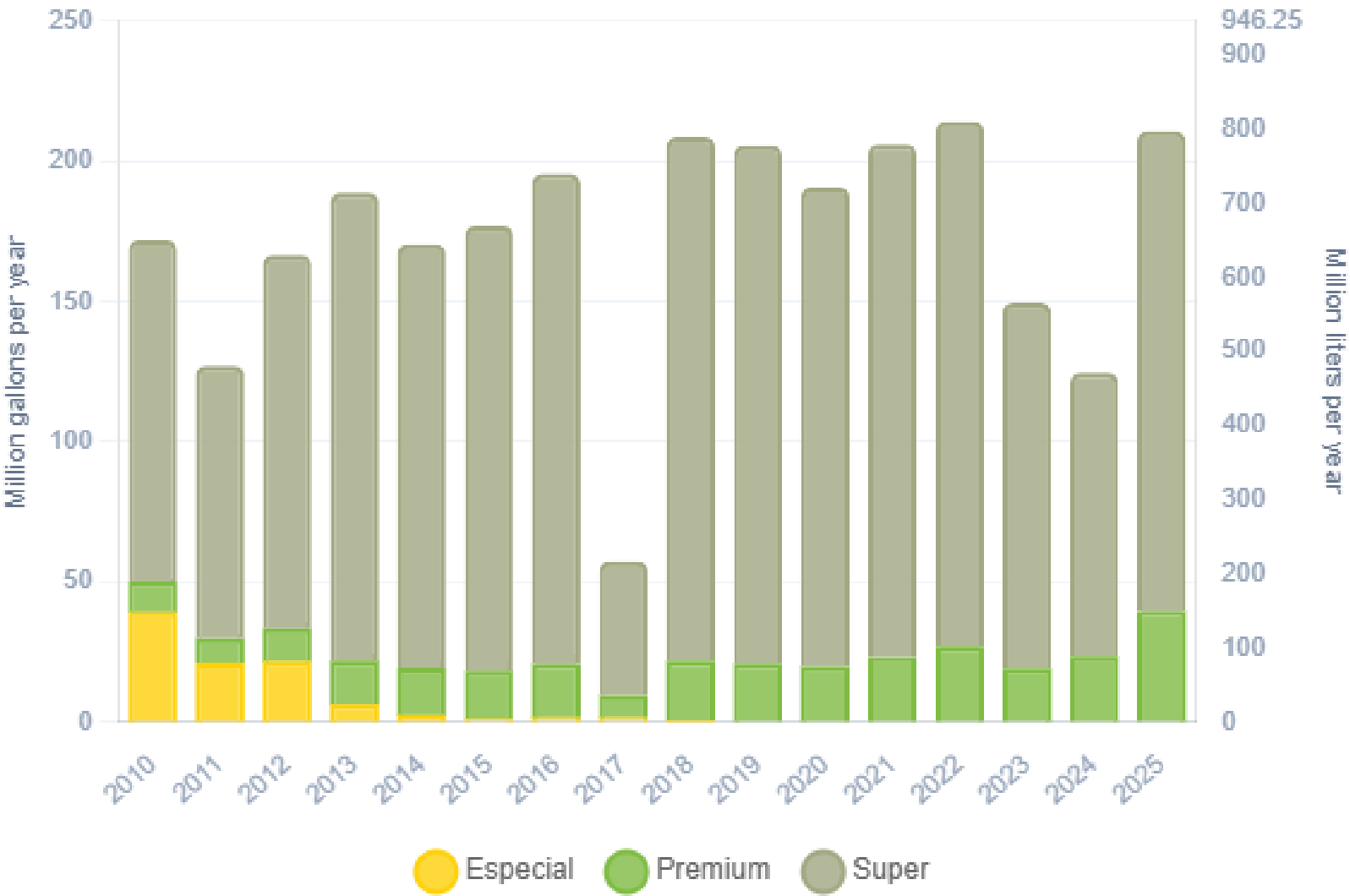
Gasoline Blends with up to 10% v/v are allowed. In Uruguay there are two ethanol plants with an overall capacity of 100 million liters per year, raw material commonly used are sugar cane, sorghum, and corn. In 2025, ethanol consumption was 25 million gallons (94 million liters).

Gasoline Demand in Uruguay



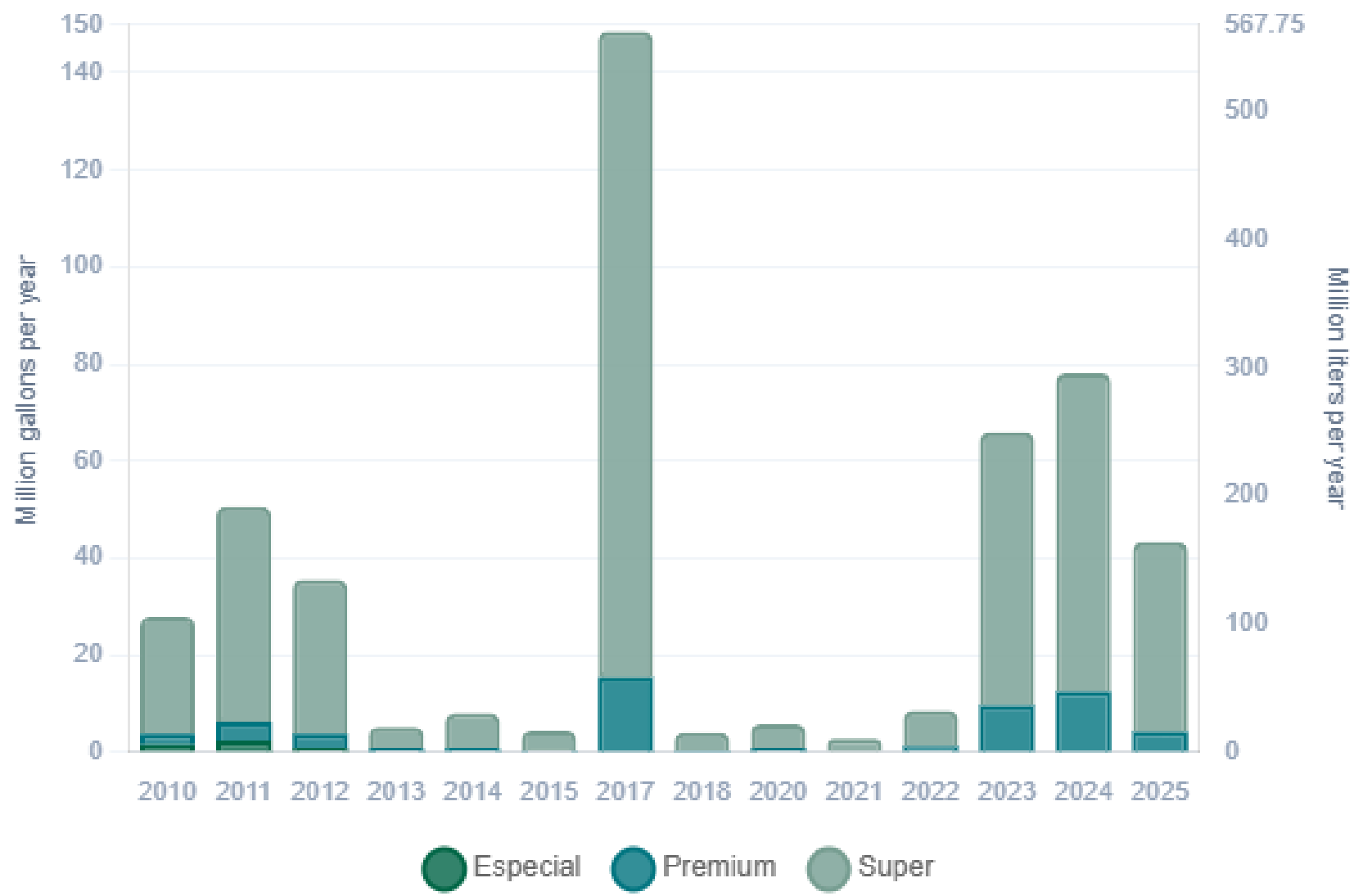
Source: ANCAP

Gasoline Production in Uruguay



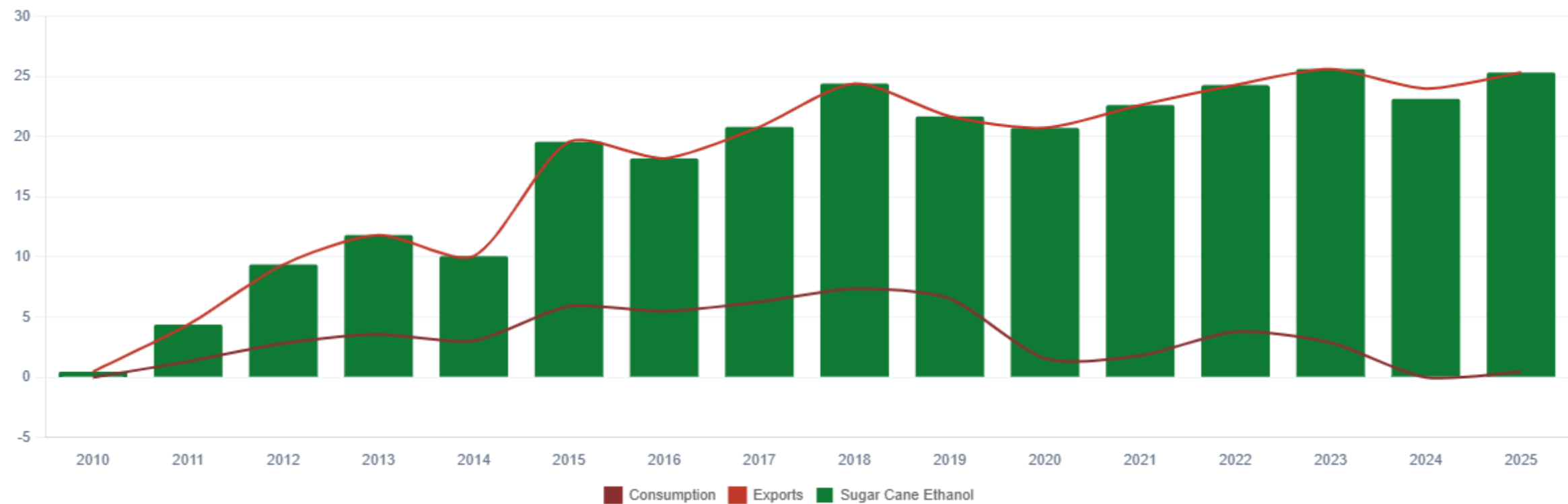
Source: ANCAP

Gasoline Imports in Uruguay



Source: ANCAP

Ethanol Balance in Uruguay



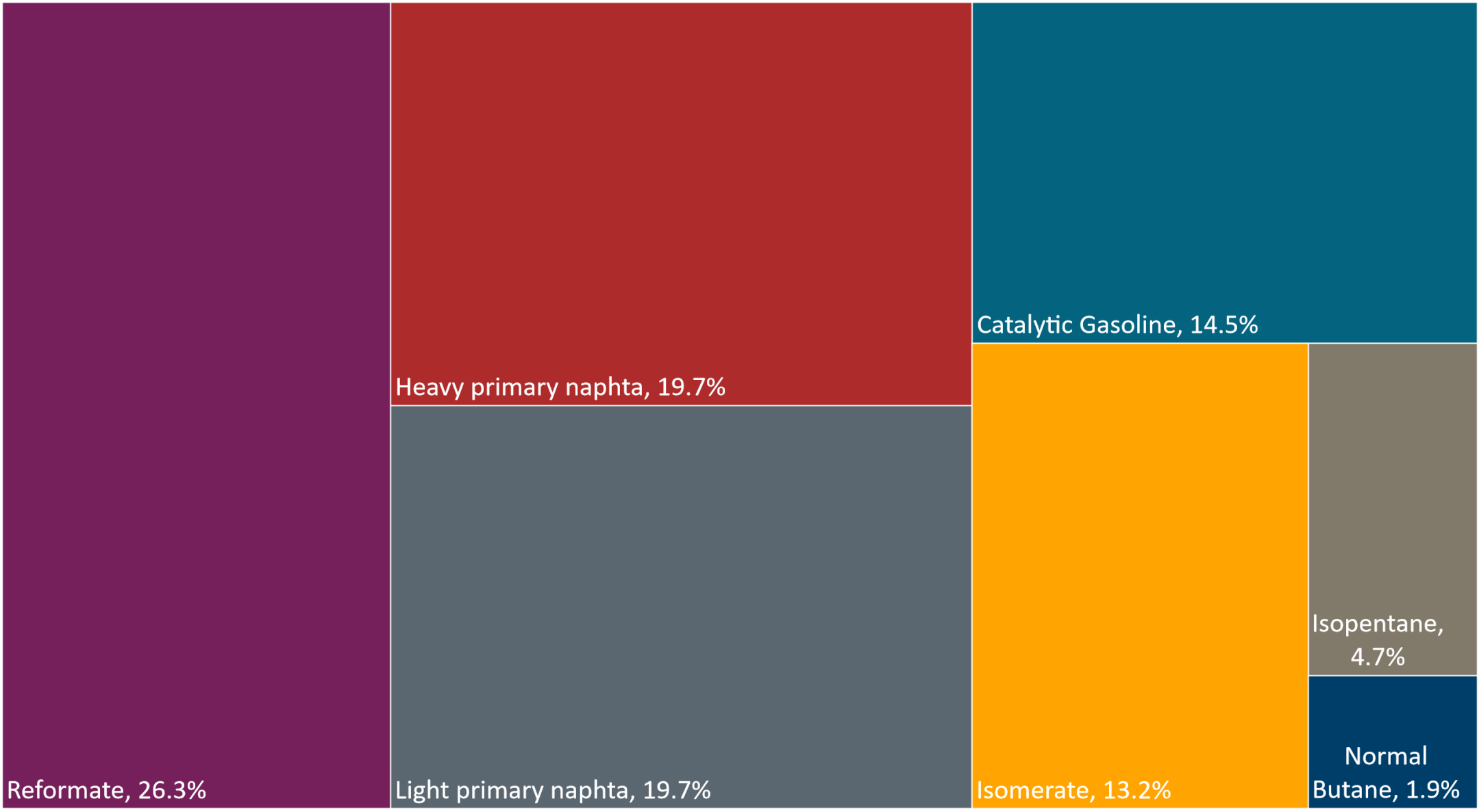
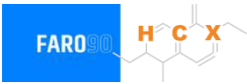
Source: ANCAP

Gasoline Quality in Uruguay

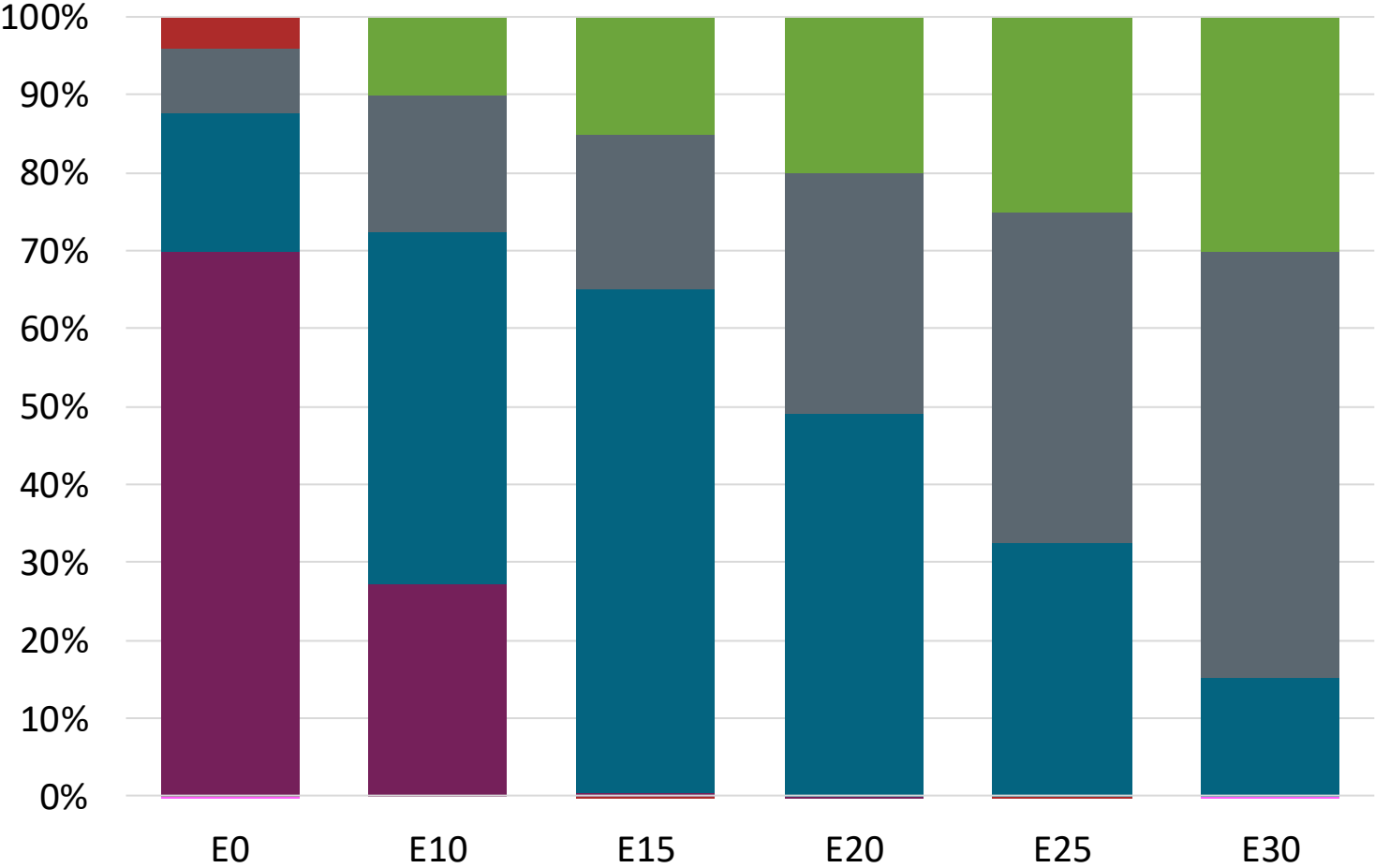
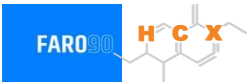
Name	Resolution 110/2014		ANCAP		EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2019		2020		2017			
Applicability	Whole country	Whole country	Whole country	Whole country	All countries			
Selected Grade	Super 95 30-S	Premium 97 30-S	Super 95 30-S	Premium 97 30-S	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 40 %v/v	< 40 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	< 20 %v/v	< 20 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,005 g/l	< 0,005 g/l	< 0,005 g/l	< 0,005 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	< 2,5 mg/l	< 2,5 mg/l	< 2,5 mg/l	< 2,5 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	> 95	> 97	> 95	> 97	> 95	> 95	> 98	> 98
MON	> 82	> 84	> 85	> 85	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 30 mg/kg	< 30 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	< 2,7 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	< > 10 %v/v	< > 10 %v/v			<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 72 kPa	<> 72 kPa	<> 50-80 kPa	<> 50-80 kPa	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 83 kPa	<> 83 kPa	<> 45-67 kPa	<> 45-67 kPa				
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: ANCAP

Available Blending Components

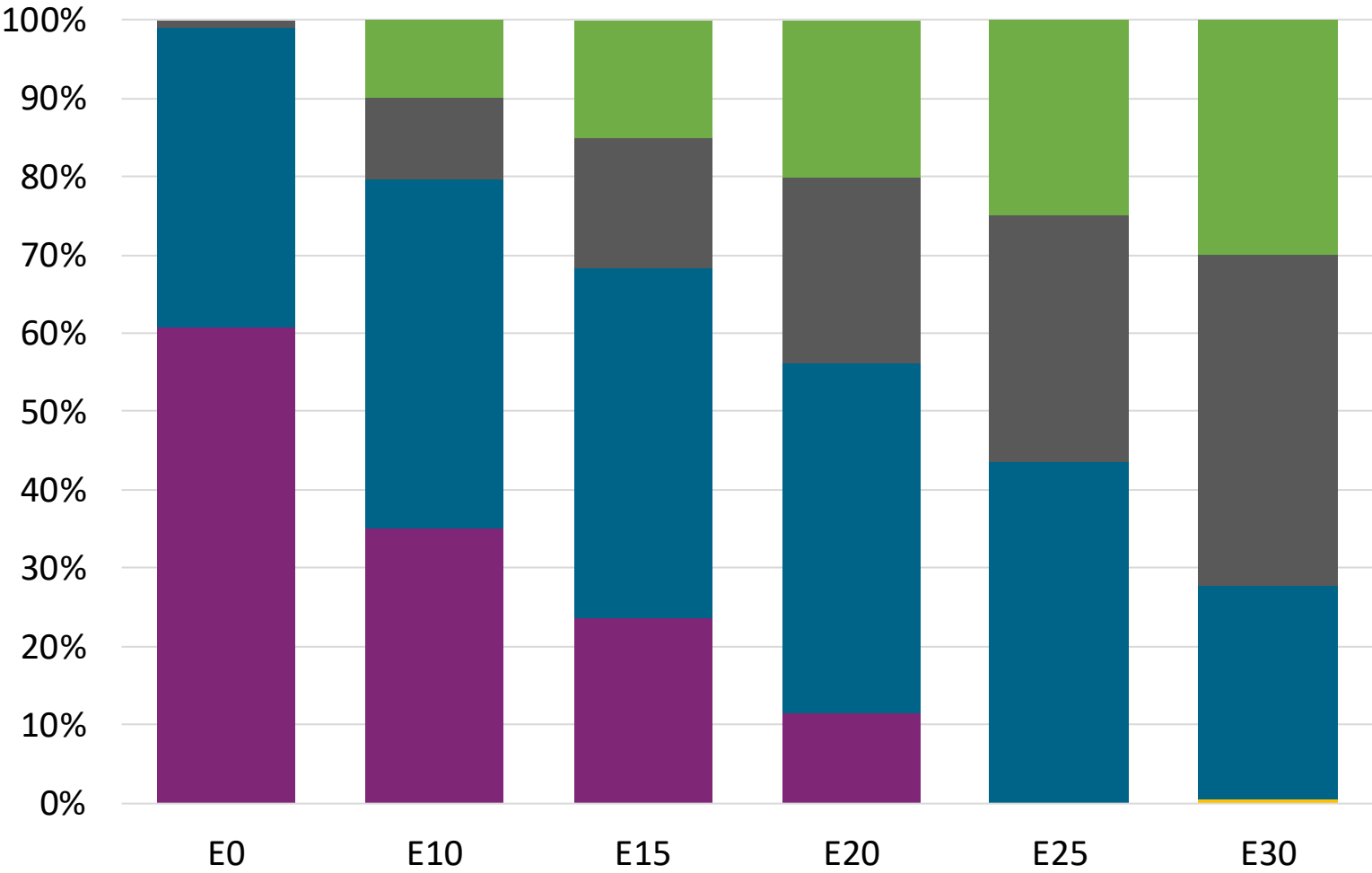
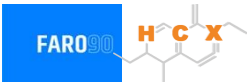


Super Gasoline – Constant Octane



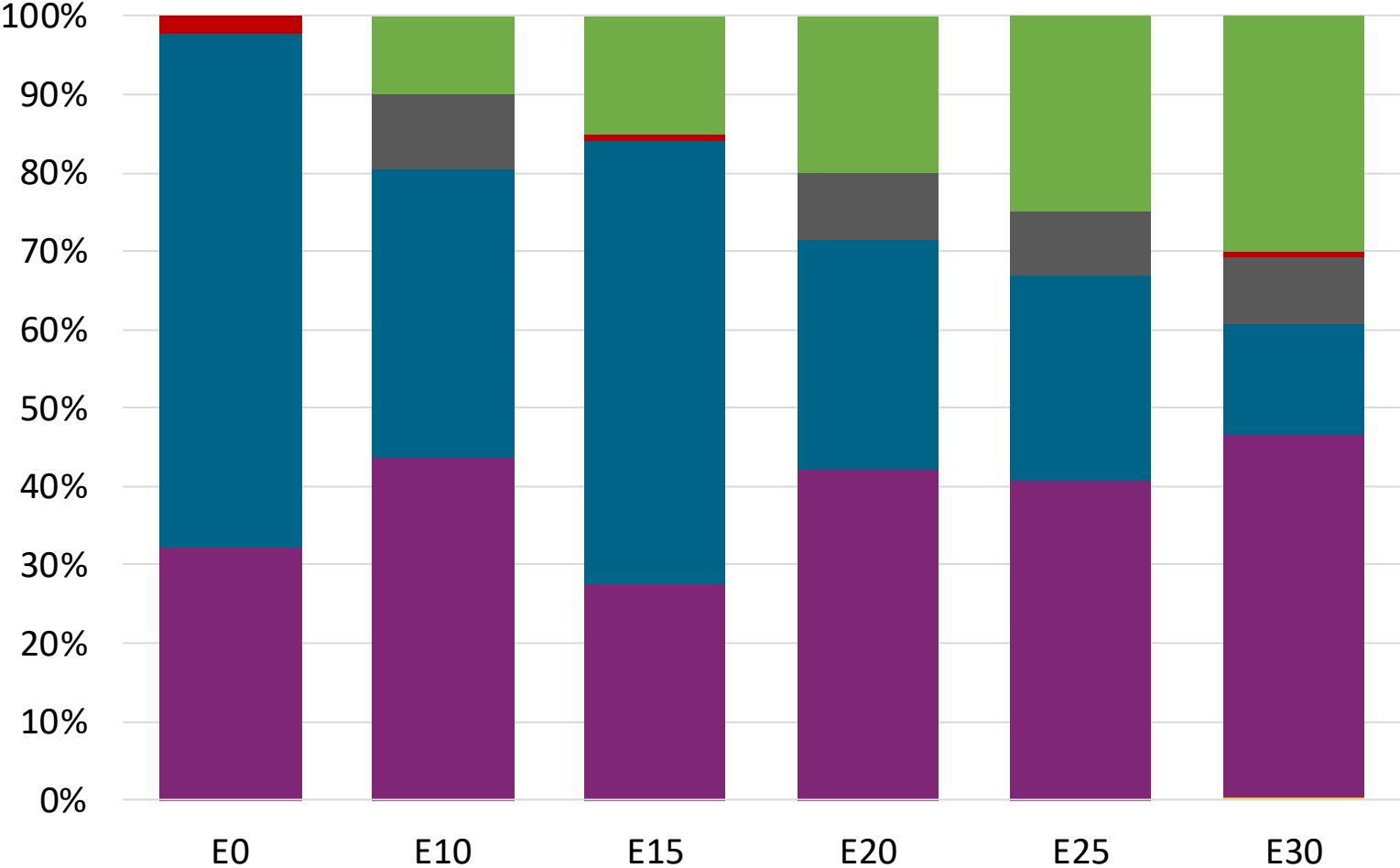
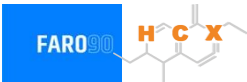
Octane (RON)	94.7	95.5	95.5	95.5	95.5	95.5
Price (US\$/gal)	\$ 2.222	\$ 2.028	\$ 1.898	\$ 1.802	\$ 1.702	\$ 1.596

Premium Gasoline – Constant Octane



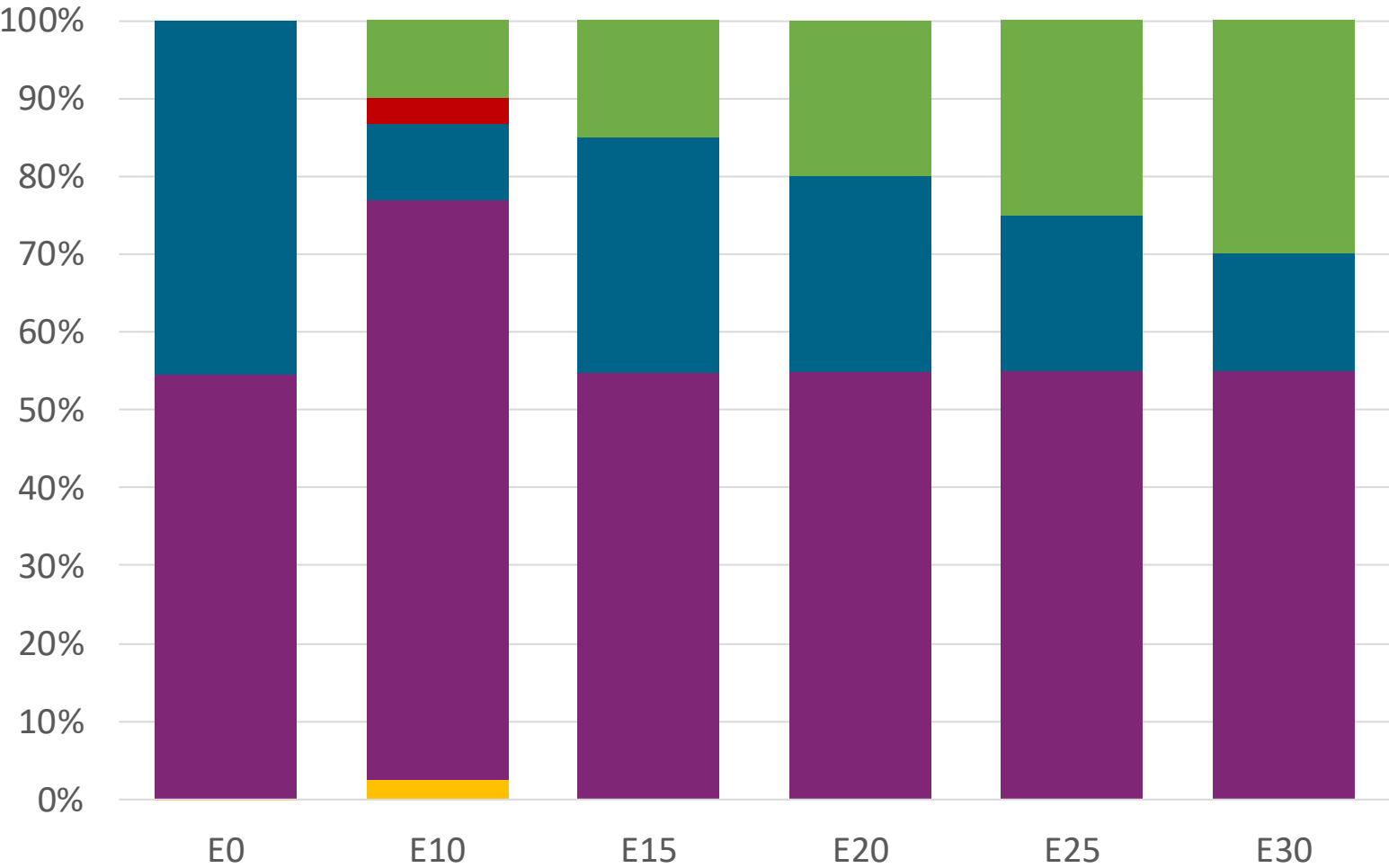
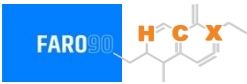
Octane (RON)	97.0	97.5	97.5	97.5	97.5	97.5
Price (US\$/gal)	\$ 2.308	\$ 2.122	\$ 2.018	\$ 1.909	\$ 1.797	\$ 1.696

Super Gasoline – Increased Octane



Octane (RON)	94.5	98.2	100.8	102.1	103.9	105.5
Price (US\$/gal)	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164

Premium Gasoline – Increased Octane



Octane (RON)	96.8	100.5	102.7	104.5	106.2	107.7
Price (US\$/gal)	\$ 2.290	\$ 2.290	\$ 2.290	\$ 2.290	\$ 2.290	\$ 2.290

Gasoline Vehicular Fleet in Uruguay

Vehicle type	Engine	Exhaust	Recovery	Age	Number of vehicles				
Light automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	75				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	728				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	52,279				
		Euro III		> 8 years, > 160 tkm	65,358				
		Euro IV		> 8 years, > 160 tkm	129,214				
		Euro V / Euro 6		> 8 years, > 160 tkm	134,024				
				> 8 years, > 160 tkm	111,792				
	Hybrid			< 4 years, < 80 tkm	146,785				
	Electric			Any	2,691				
				Any	9,241				
Medium automobiles	Carburator	3-Ways	PCV Tank	> 8 years, > 160 tkm	26				
	Single FI-Pt	3-Ways		> 8 years, > 160 tkm	328				
	Multi FI-Pt	3-Ways		> 8 years, > 160 tkm	28,414				
		Euro III		> 8 years, > 160 tkm	40,145				
		Euro IV		> 8 years, > 160 tkm	84,729				
		Euro V / Euro 6		> 8 years, > 160 tkm	93,083				
				> 8 years, > 160 tkm	84,516				
	Hybrid			< 4 years, < 80 tkm	132,139				
	Electric			Any	9,683				
				Any	10,724				
Small Engines	FI 4-cycles	None	PCV	> 8 years, > 160 tkm	2,337				
		Euro II		> 8 years, > 160 tkm	2,384				
		Euro III		> 8 years, > 160 tkm	126,919				
		Euro IV		> 8 years, > 160 tkm	47,086				
		Euro V		4-8 years, 80-160 tkm	41,414				
				4-8 years, 80-160 tkm	43,102				
	Electric			< 4 years, < 80 tkm	132,139				
			Any	5,528					
Heavy	Carburator	3-Ways	PCV	> 8 years, > 160 tkm	144				
	FI	3-Ways		> 8 years, > 160 tkm	53,007				
		Euro III		> 8 years, > 160 tkm	65,358				
		Euro IV		> 8 years, > 160 tkm	129,214				
		Euro V / Euro 6		> 8 years, > 160 tkm	134,024				
				> 8 years, > 160 tkm	111,792				
	Hybrid			< 4 years, < 80 tkm	146,785				
	Electric			Any	179				
				Any	553				

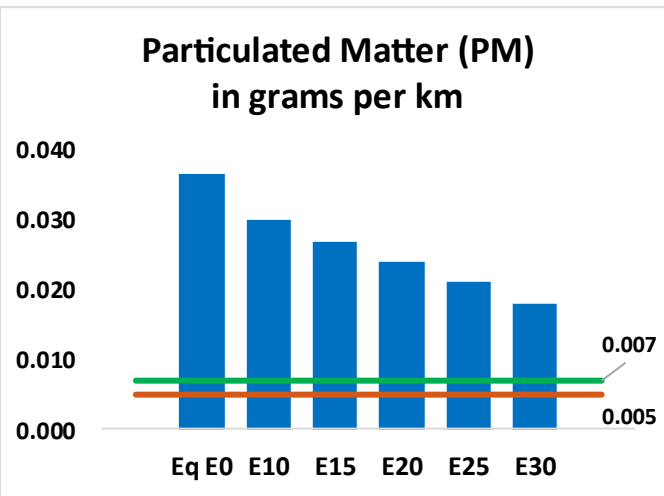
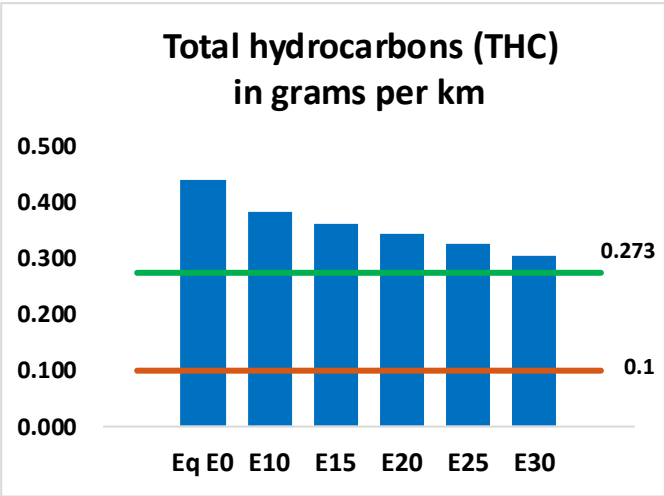
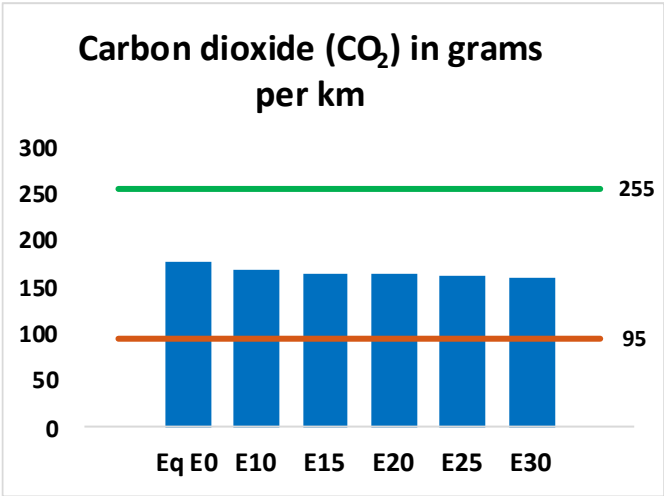
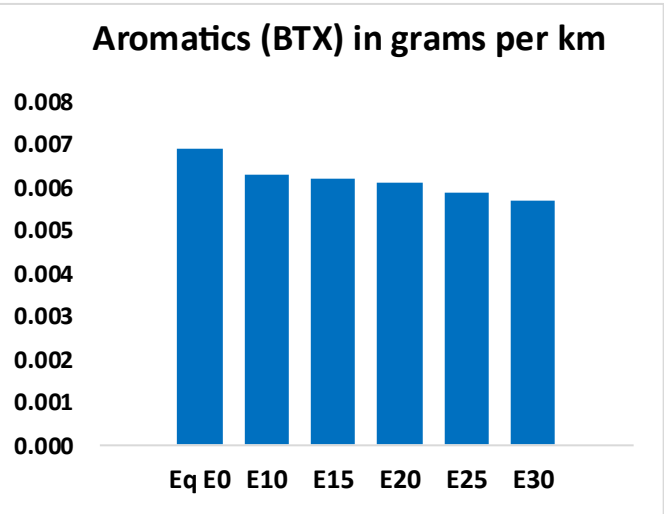
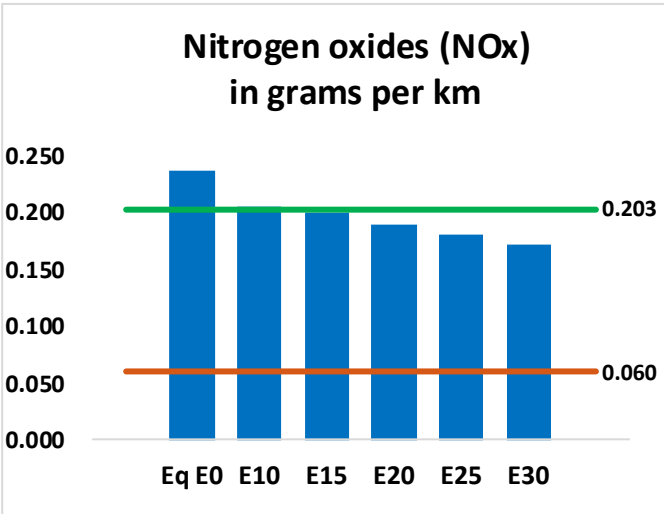
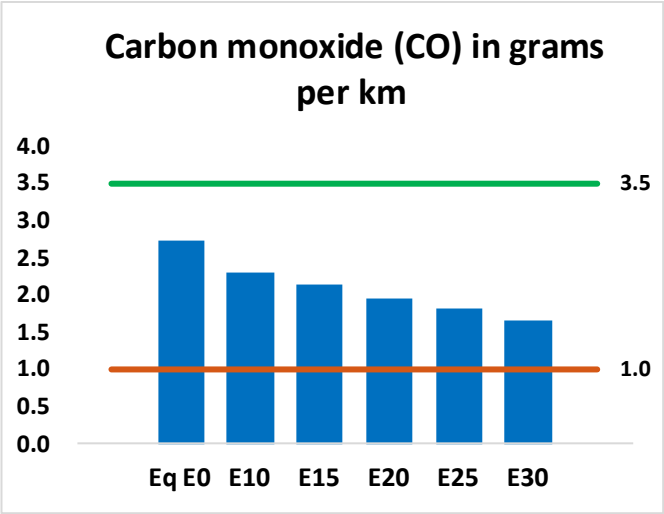
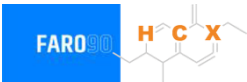
Gasoline Vehicular Fleet: 1,499,0171,499,01

Average Age: 8 years

Source: MIEM, 2025.

Vehicular Emissions in Uruguay

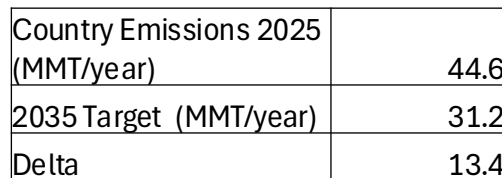
United States
Euro 6



Vehicular Emissions in Uruguay

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	102,382	94,381	86,380	79,798	73,162	68,056	62,169	-16%	-29%
CO2	6,641,625	6,470,840	6,309,543	6,182,688	6,119,915	6,086,937	5,974,737	-5%	-8%
VOC	16,424	15,400	14,375	13,582	12,806	12,215	11,404	-12%	-22%
NOx	8,889	8,207	7,733	7,489	7,099	6,791	6,430	-13%	-20%
SOx	76	70	65	60	55	50	45	-15%	-28%
NH3	2,426	2,402	2,378	2,389	2,416	2,435	2,450	-2%	0%
N2O	93	92	92	93	94	96	97	-1%	2%
CH4	4,356	4,356	4,356	4,421	4,517	4,600	4,678	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	143	135	126	120	114	109	103	-12%	-20%
Acetaldehyde	371	426	625	952	1,297	1,482	1,751	68%	249%
Formaldehyde	1,436	1,396	1,628	1,911	2,002	2,215	2,412	13%	39%
Benzene	258	248	235	232	230	220	213	-9%	-11%
PM2.5	407	370	333	301	269	235	199	-18%	-34%
PM10	958	871	784	707	632	552	468	-18%	-34%

The FARO90 logo is shown in a blue box, with the text "FARO90" in white. To the right of the logo is a chemical structure of a substituted cyclohexadiene, with the letters "H", "C", and "X" in orange highlighting specific atoms in the ring.



Source: Greet, RED II/III, climateactiontracker.org, F90